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GOVERNMENT OF TAMIL NADU

HIGHER SECONDARY FIRST YEAR

PHYSICS

VOLUME - 2

A publication under Free Textbook Programme of Government of Tamil Nadu

Department of School Education

Untouchability is Inhuman and a Crime

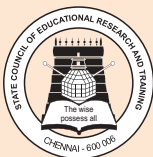


Government of Tamil Nadu

First Edition - 2018
Revised Edition - 2019, 2020,
2022, 2023
Reprint - 2021, 2024
(Published Under New Syllabus)

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Content Creation



State Council of Educational
Research and Training

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Printing & Publishing



Tamil NaduTextbook and Educational
Services Corporation

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E-book



Assessment

HOW TO USE THE BOOK?

Scope of Physics

- Awareness on higher learning - courses, institutions and required competitive exams
- Financial assistance possible to help students to climb academic ladder



Learning Objectives:

- Overview of the unit
- Gives clarity on the goals and intention of the topics



- Additional facts related to the topics covered to facilitate curiosity driven learning

Example problems

- To ensure understanding, problems/illustrations are given at every stage before advancing to next level



- Visual representation of concepts with illustrations
- Videos, animations, and tutorials



ICT

- To harness the digital skills to class room learning and experimenting

Summary

- Recap of salient points of the lesson

Concept Map

- Schematic outline of salient learning of the unit

Evaluation

- Evaluate students' understanding and get them acquainted with the application of physical concepts to numerical and conceptual questions

Books for Reference

- List of relevant books for further reading

Solved examples

- Solutions to exercise problems are accessible here. In addition, a few solved examples are given to facilitate students to apply the concepts learnt.

Competitive Exam corner

- Model Questions - To motivate students aspiring to take up competitive examinations such as NEET, JEE, Physics Olympiad, JIPMER etc

Practical

- List of practical and the description of each is appended for easy access.

Glossary

- Scientific terms frequently used with their Tamil equivalents

Back wrapper: Solvay Conference 1927, Belgium

Photograph of the attendees of the most famous fifth solvay international conference held in october 1927. 17 of the 29 scientists found in this photograph are Nobel laureates (shown in bold).

Front row (L to R) : I. Langmuir, M. Planck, Marie Curie, H.A. Lorentz, A. Einstein, P. Langevin, Ch.E. Guye, C.T.R. Wilson, O.W. Richardson

Middle row (L to R): P. Debye, M. Knudsen, W.L. Bragg, H.A. Kramers, P.A.M. Dirac, A.H. Compton, L.de Broglie, M. Born, N. Bohr

Back row (L to R) : A. Piccard, E. Henriot, P. Ehrenfest, Ed. Herzen, Th.De Donder, E. Schrödinger, E.Verschaffelt, W. Pauli, W. Heisenberg, R.H. Fowler, L. Brillouin

—Photographie by Benjamin Couprie

Scope of Physics - Higher Education



Exams

- JEE-Joint Entrance Examination
- Physics Olympiad Exam
- NEET- National Eligibility and Entrance Test
- NEST- National Entrance Screening Test
- AIEEE- All India Engineering Entrance Exam
- AIIMS- All India Institute of Medical Science (Entrance Examination)
- JIPMER- Jawaharlal Institute of Postgraduate Medical Education and Research (Entrance Examination)
- KVPY- Kishore Vaigyanik Protsahan Yojana
- JAM- Joint Admission Test
- TIFR GS- Tata Institute of Fundamental Research Graduate School Admissions Examination
- JEST- Joint Entrance Screening Test
- NET- National Eligibility Test (CSIR and UGC)
- GATE-Graduate Aptitude Test in Engineering
- ICAR -AIEEA-Indian Council of Agricultural Research All India Entrance Examination



After completing +2

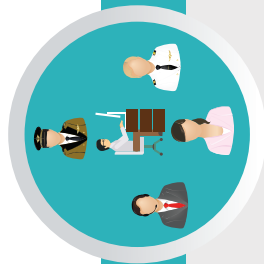
- B.Sc (Physics)
- Integrated M.Sc (Physics) (Central Universities)
- Integrated M.Sc (in Central Research Institutes through NEST and KVPY with stipend)
- B.Sc./B.S./B.Stat./B.Math./M.S. in Mathematics, Chemistry and Biology. (KVPY)
- B.E/B.Tech/ B.Arch (JEE, AIEEE in IITs and NITs)
- MBBS/ B.D.S/B.Pharm (NEET, JIPMER, AIIMS)
- B.Sc. (Agriculture) (ICAR -AIEEA)
- Dual Degree Program BS & MS (JEE, JEST in IITs and IISERs)
- B.Sc (Hospitality administration)
- B.Sc (Optoelectronics)
- B.Sc (Optometry)
- B. Tech (Optics and Optoelectronics)



After completing undergraduate course in Physics (B.Sc Physics)

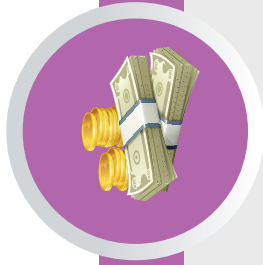
- M.Sc. (Physics) (In Central and State Universities and Colleges)
- M.Sc. Physics (JAM in IISc ,IITs and NITs)
- M.Sc. (In State and Central Universities)
- Medical Physics
- Materials Science
- Energy
- Earth Sciences
- Space science
- Oceanography
- Remote sensing
- Electronics
- Photonics
- Optoelectronics
- Acoustics
- Applied electronics
- Astronomy and Astrophysics
- Nanoscience and Nanotechnology
- Biostatistics
- Bio informatics
- Vacuum sciences

Opportunities after B.Sc. Physics



Jobs in Government Sector

- Indian Forest Services
- Scientist Job in ISRO, DRDO, CSIR labs
- Union Public Service Commission
- Staff Selection Commission
- Indian Defence Services etc.
- Public Sector Bank
- State PCS
- Grade III & Compiler Post
- Tax Assistant
- Statistical Investigator
- Combined Higher Secondary
- Combined Graduate Level Exam etc.

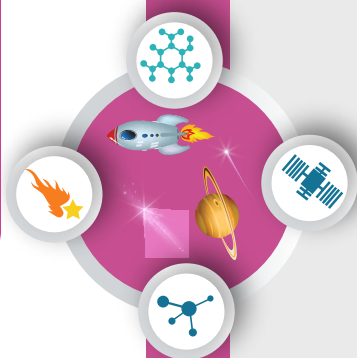


Financial assistance to pursue higher education

Scholarships for graduate and post graduate courses

- International Olympiad: for getting stipend for Higher Education in Science and Mathematics
- DST – INSPIRE Scholarships (for UG and PG)
- DST – INSPIRE Fellowships (for Ph.D)
- UGC National Fellowship (for Ph.D)
- Indira Gandhi Fellowship for Single Girl Child (for UG and PG)
- Maulana Azad Fellowship for Minorities (for Ph.D)
- In addition various fellowships for SC/ST/PWD, OBC etc. are available.
- Visit website of University Grants Commission (UGC) and Department of Science and Technology (DST)

Institutes in india to pursue research in physics



After completing Post Graduate course in Physics (M.Sc Physics)

Topics of Research

- Quantum Physics and Quantum Optics
- Astrophysics, Astronomy
- String theory, Quantum Gravity
- Mathematical Physics, Statistical Mechanics
- Quantum Field Theory
- Particle Physics and Quantum Thermodynamics
- Quantum Information Theory
- Condensed Matter Physics, Materials Science
- Electro Magnetic Theory
- Black Holes, Cosmology
- Crystal Growth, Crystallography
- Spectroscopy, Atomic, Molecular and Optical Physics
- Nano Science and Nanotechnology
- Energy and Environment Studies
- Biophysics, Medical Physics
- Cryptography, Spintronics
- Optics and Photonics
- Meteorology and Atmospheric Science



Research Institutions in various areas of science		
Name of the Institution		Website
Indian Institute of Science (IISc) Bangalore		www.iisc.ac.in
Raman Research Institute (RRI) Bangalore		www.rri.res.in
Institute of Mathematical Sciences (IMSc) Chennai		www.imsc.res.in
Indian Association for Cultivation of Science (IACS) Calcutta		www.iacs.res.in
Chennai Mathematical Institute (CMI) Chennai		www.cmi.ac.in
Tata Institute of Fundamental Research (TIFR) Mumbai		www.tifr.res.in
Bhaba Atomic Research Centre (BARC) Mumbai		www.barc.gov.in
SN Bose centre Basic Natural science Calcutta		www.bose.res.in
Indian Institute of Space Science and Technology (IIST) Trivandrum		www.iist.ac.in
Physics Research Laboratory (PRL) Ahmedabad		www.prl.res.in
Indian Institute of Astrophysics (IIA) Bangalore		www.iia.res.in
Institute of Physics (IOP) Bhubaneswar		www.iopb.res.in
Institute for Plasma Research (IPR) Gujarat		www.ipr.res.in
Inter University Centre for Astronomy and Astrophysics (IUCAA) Pune		www.iucaa.in
Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam		www.igcar.gov.in
Hyderabad Central University, Hyderabad		www.uohyd.ac.in
Delhi University, Delhi		www.du.ac.in
Mumbai University, Mumbai		www.mu.ac.in
SavitribaiPhule Pune university, Pune		www.unipune.ac.in
National Institute of Science Education and Research (NISER), Bhubaneswar		www.niser.ac.in
IISER Educational Institutions		www.iiseradmission.in
Indian Institute of Technology in various places (IIT's)		www.iitm.ac.in
National Institute of Technology (NITs)		www.nit.edu
Jawaharlal Nehru University (JNU)		www.jnu.ac.in
Central Universities		www.ugc.ac.in
State Universities		https://www.ugc.ac.in
CSIR – Academy (National laboratories, Delhi, Hyderabad, Trivandrum, Chennai, Calcutta etc)		



UNIT 6

GRAVITATION

“The most remarkable discovery in all of astronomy is that the stars are made up of atoms of same kind as those in the Earth” – Richard Feynman



LEARNING OBJECTIVES

In this unit, the student is exposed to

- Kepler's laws for planetary motion
- Newton's law of gravitation
- connection between Kepler's laws and law of gravitation
- calculation of gravitational field and potential
- calculation of variation of acceleration due to gravity
- calculation of escape speed and energy of satellites
- concept of weightlessness
- advantage of heliocentric system over geocentric system
- measurement of the radius of Earth using Eratosthenes method
- recent developments in gravitation and astrophysics



6.1

INTRODUCTION

We are amazed looking at the glittering sky; we wonder how the Sun rises in the East and sets in the West, why there are comets or why stars twinkle. The sky has been an object of curiosity for human beings from time immemorial. We have always wondered about the motion of stars, the Moon, and the planets. From Aristotle to Stephen Hawking, great minds have tried to understand the movement of celestial objects in space and what causes their motion.

The 'Theory of Gravitation' was developed by Newton in the late 17th century to explain the motion of celestial

objects and terrestrial objects and answer most of the queries raised. In spite of the study of gravitation and its effect on celestial objects, spanning last three centuries, "gravitation" is still one of the active areas of research in physics today. In 2017, the Nobel Prize in Physics was given for the detection of 'Gravitational waves' which was theoretically predicted by Albert Einstein in the year 1915. Understanding planetary motion, the formation of stars and galaxies, and recently massive objects like black holes and their life cycle have remained the focus of study for the past few centuries in physics.

Geocentric Model of Solar System

In the second century, Claudius Ptolemy, a famous Greco-Roman astronomer,

developed a theory to explain the motion of celestial objects like the Sun, the Moon, Mars, Jupiter etc. This theory was called the geocentric model. According to the geocentric model, the Earth is at the center of the universe and all celestial objects including the Sun, the Moon, and other planets orbit the Earth. Ptolemy's model closely matched with the observations of the sky with our naked eye. But later, astronomers found that even though Ptolemy's model successfully explained the motion of the Sun and the Moon up to a certain level, the motion of Mars and Jupiter could not be explained effectively.

Heliocentric Model of Nicholas Copernicus

In the 15th century, a Polish astronomer, Nicholas Copernicus (1473-1543) proposed a new model called the 'Heliocentric model' in which the Sun was considered to be at the center of the solar system and all planets including the Earth orbited the Sun in circular orbits. This model successfully explained the motion of all celestial objects.

Around the same time, Galileo, a famous Italian physicist discovered that all objects close to Earth were accelerated towards the Earth at the same rate. Meanwhile, a noble man called Tycho Brahe (1546-1601) spent his entire lifetime in recording the observations of the stellar and planetary positions with his naked eye. The data that he compiled were analyzed later by his assistant Johannes Kepler (1571-1630) and eventually the analysis led to the deduction of the laws of the planetary motion. These laws are termed as 'Kepler's laws of planetary motion'.

6.1.1 Kepler's Laws of Planetary Motion

Kepler's laws are stated as follows:

1. Law of orbits:

Each planet moves around the Sun in an elliptical orbit with the Sun at one of the foci.

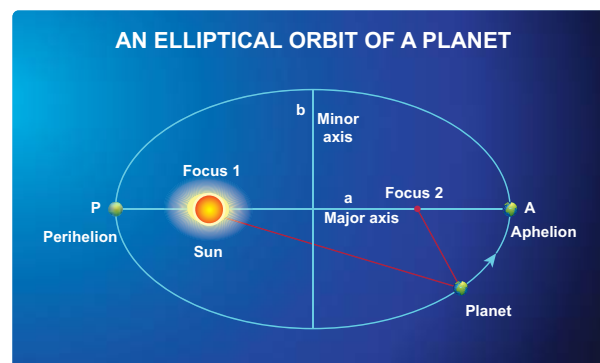


Figure 6.1 An ellipse traced out by a planet around the Sun.

The closest point of approach of the planet to the Sun 'P' is called perihelion and the farthest point 'A' is called aphelion (Figure 6.1). The semi-major axis is 'a' and semi-minor axis is 'b'. In fact, both Copernicus and Ptolemy considered planetary orbits to be circular, but Kepler discovered that the actual orbits of the planets are elliptical.



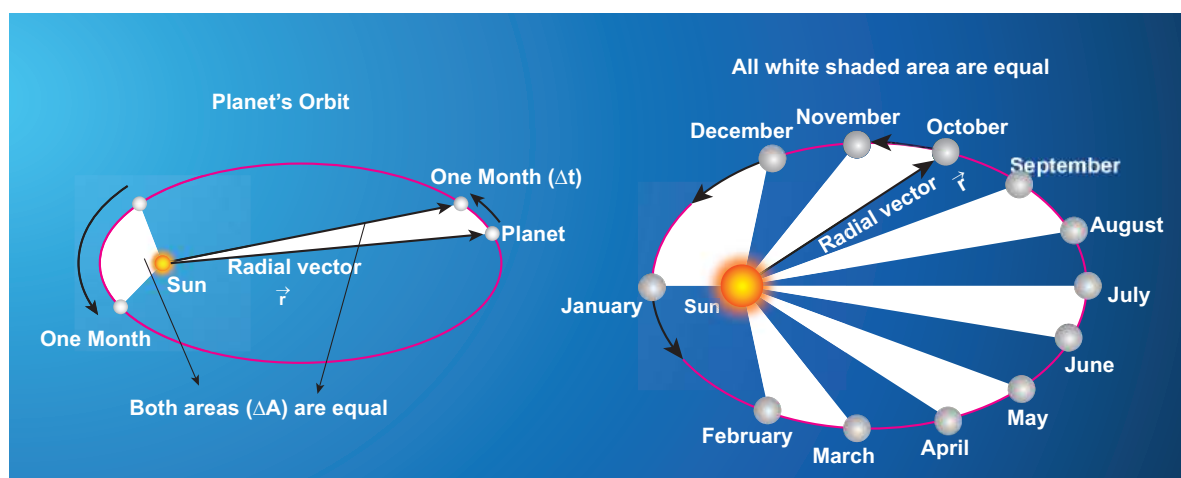


Figure 6.2 Motion of a planet around the Sun depicting 'law of area'.

2. Law of area:

The radial vector (line joining the Sun to a planet) sweeps equal areas in equal intervals of time.

In Figure 6.2, the white shaded portion is the area ΔA swept in a small interval of time Δt , by a planet around the Sun. Since the Sun is not at the center of the ellipse, *the planets travel faster when they are nearer to the Sun and slower when they are farther from it, to cover equal area in equal intervals of time.* Kepler discovered the law of area by carefully noting the variation in the speed of planets.

3. Law of period:

The square of the time period of revolution of a planet around the Sun in its elliptical orbit is directly proportional to the cube of the semi-major axis of the ellipse. It can be written as:

$$T^2 \propto a^3 \quad (6.1)$$

$$\frac{T^2}{a^3} = \text{constant} \quad (6.2)$$

where, T is the time period of revolution for a planet and a is the semi-major axis. Physically this law implies that as the distance of the planet from the Sun increases, the time period also increases but not at the same rate.

In Table 6.1, the time period of revolution of planets around the Sun along with their semi-major axes are given. From column four, we can realize that $\frac{T^2}{a^3}$ is nearly a constant endorsing Kepler's third law.

Table 6.1 The time period of revolution of the planets revolving around the Sun and their semi-major axes.

Planet	a ($10^{10} m$)	T (years)	$\frac{T^2}{a^3}$
Mercury	5.79	0.24	2.95
Venus	10.8	0.615	3.00
Earth	15.0	1	2.96
Mars	22.8	1.88	2.98
Jupiter	77.8	11.9	3.01
Saturn	143	29.5	2.98
Uranus	287	84	2.98
Neptune	450	165	2.99



Points to Contemplate

DATA			PROBLEM
Planet	a	T	
A	1	3	What is
B	2	6	the law
C	4	18	connecting
			a and T?

Comment on the relation between a and T for these imaginary planets

6.1.2 Universal Law of Gravitation

Even though Kepler's laws were able to explain the planetary motion, they failed to explain the forces responsible for it. It was Isaac Newton who analyzed Kepler's laws, Galileo's observations and deduced the law of gravitation.

Newton's law of gravitation states that a particle of mass M_1 attracts any other particle of mass M_2 in the universe with an attractive force. The strength of this force of attraction was found to be directly proportional to the product of their masses and is inversely proportional to the square of the distance between them. In mathematical form, it can be written as:

$$\vec{F} = -\frac{GM_1M_2}{r^2}\hat{r} \quad (6.3)$$

where $\hat{r}$ is the unit vector from M_1 towards M_2 as shown in Figure 6.3, and G is the Gravitational constant that has the value of $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$, and r is the distance between the two masses M_1 and M_2 . In Figure 6.3, the vector $\vec{F}$ denotes the gravitational force experienced by M_2 due to M_1 . Here the negative sign indicates that the gravitational force is always attractive in nature and the direction of the force is along the line joining the two masses.

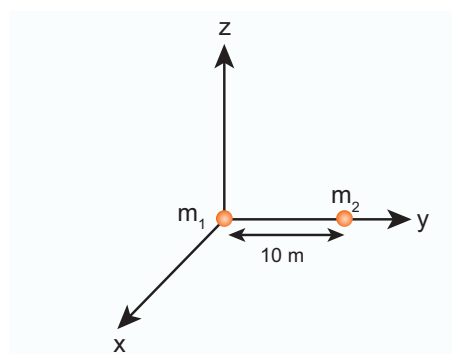


Figure 6.3 Attraction of two masses towards each other.

In cartesian coordinates, the square of the distance is expressed as $r^2 = (x^2 + y^2 + z^2)$. This is dealt in unit 2.

EXAMPLE 6.1

Consider two point masses m_1 and m_2 which are separated by a distance of 10 meter as shown in the following figure. Calculate the force of attraction between them and draw the directions of forces on each of them. Take $m_1 = 1 \text{ kg}$ and $m_2 = 2 \text{ kg}$



Solution

The force of attraction is given by

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r}$$

From the figure, $r = 10$ m.

First, we can calculate the magnitude of the force

$$\begin{aligned} F &= \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 1 \times 2}{100} \\ &= 13.34 \times 10^{-13} \text{ N} \end{aligned}$$

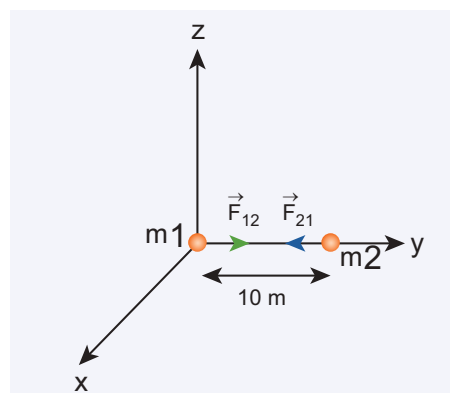
It is to be noted that this force is very small. This is the reason we do not feel the gravitational force of attraction between each other. The small value of G plays a very crucial role in deciding the strength of the force.

The force of attraction ($\vec{F}_{21}$) experienced by the mass m_2 due to m_1 is in the negative 'y' direction i.e., $\hat{r} = -\hat{j}$. According to Newton's third law, the mass m_2 also exerts equal and opposite force on m_1 . So the force of attraction ($\vec{F}_{12}$) experienced by m_1 due to m_2 is in the direction of positive 'y' axis i.e., $\hat{r} = \hat{j}$.

$$\vec{F}_{21} = -13.34 \times 10^{-13} \hat{j}$$

$$\vec{F}_{12} = 13.34 \times 10^{-13} \hat{j}$$

The direction of the force is shown in the figure,



Gravitational force of attraction between m_1 and m_2

$\vec{F}_{12} = -\vec{F}_{21}$ which confirms Newton's third law.

Important features of gravitational force:

- As the distance between two masses increases, the strength of the force tends to decrease because of inverse dependence on r^2 . Physically it implies that the planet Uranus experiences less gravitational force from the Sun than the Earth since Uranus is at larger distance from the Sun compared to the Earth.

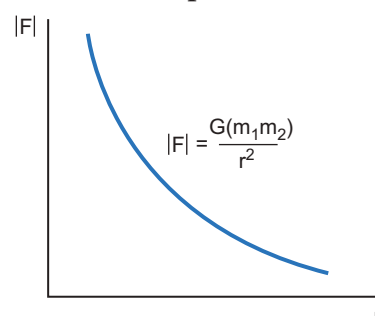


Figure 6.4 Variation of gravitational force with distance

- The gravitational forces between two particles always constitute an action-reaction pair. It implies that the gravitational force exerted by the Sun on the Earth is always towards the Sun. The reaction-force is exerted by the Earth on the Sun. The direction of this reaction force is towards Earth.



- The torque experienced by the Earth due to the gravitational force of the Sun is given by

$$\vec{\tau} = \vec{r} \times \vec{F} = \vec{r} \times \left(-\frac{GM_s M_E}{r^2} \hat{r} \right) = 0$$

$$\text{Since } \vec{r} = r \hat{r}, (\hat{r} \times \hat{r}) = 0$$

So $\vec{\tau} = \frac{d\vec{L}}{dt} = 0$. It implies that angular momentum $\vec{L}$ is a constant vector. The angular momentum of the Earth about the Sun is constant throughout the motion. It is true for all the planets. In fact, this constancy of angular momentum leads to the Kepler's second law.

- The expression $\vec{F} = -\frac{GM_1 M_2}{r^2} \hat{r}$ has one inherent assumption that both M_1 and M_2 are treated as point masses. When it is said that Earth orbits around the Sun due to Sun's gravitational force, we assumed Earth and Sun to be point masses. This assumption is a good approximation because the distance between the two bodies is very much larger than their diameters. For some irregular and extended objects separated by a small distance, we cannot directly use the equation (6.3). Instead, we have to invoke separate mathematical treatment which will be brought forth in higher classes.
- However, this assumption about point masses holds even for small distance for one special case. To calculate force of attraction between a hollow sphere of mass M with uniform density and point mass m kept outside the hollow sphere, we can replace the hollow sphere of mass M as equivalent to a point mass M located at the center of the hollow sphere. The force of attraction between the hollow sphere of mass M and point mass m can be calculated by treating the

hollow sphere also as another point mass. Essentially the entire mass of the hollow sphere appears to be concentrated at the center of the hollow sphere. It is shown in the Figure 6.5(a).

- There is also another interesting result. Consider a hollow sphere of mass M . If we place another object of mass ' m ' inside this hollow sphere as in Figure 6.5(b), the force experienced by this mass ' m ' will be zero. This calculation will be dealt with in higher classes.

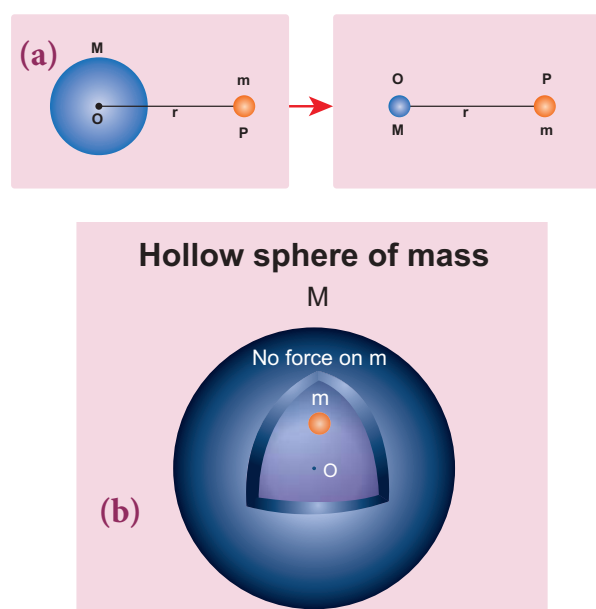


Figure 6.5 A mass placed in a hollow sphere.

- The triumph of the law of gravitation is that it concludes that the mango that is falling down and the Moon orbiting the Earth are due to the same gravitational force.

Newton's inverse square Law:

Newton considered the orbits of the planets as circular. For circular orbit of radius r , the centripetal acceleration towards the center is

$$a = -\frac{v^2}{r} \quad (6.4)$$

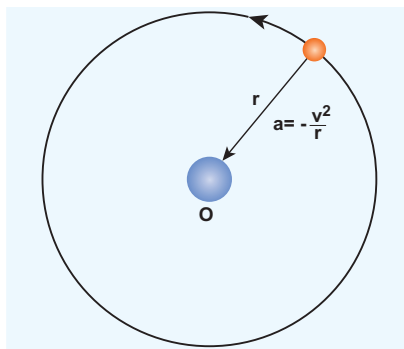


Figure 6.6 Point mass orbiting in a circular orbit.

Here v is the velocity and r , the distance of the planet from the center of the orbit (Figure 6.6).

The velocity in terms of known quantities r and T , is

$$v = \frac{2\pi r}{T} \quad (6.5)$$

Here T is the time period of revolution of the planet. Substituting this value of v in equation (6.4) we get,

$$a = -\frac{\left(\frac{2\pi r}{T}\right)^2}{r} = -\frac{4\pi^2 r}{T^2} \quad (6.6)$$

Substituting the value of ' a ' from (6.6) in Newton's second law, $F = ma$, where ' m ' is the mass of the planet.

$$F = -\frac{4\pi^2 mr}{T^2} \quad (6.7)$$

From Kepler's third law,

$$\frac{r^3}{T^2} = k(\text{constant}) \quad (6.8)$$

$$\frac{r}{T^2} = \frac{k}{r^2} \quad (6.9)$$

By substituting equation 6.9 in the force expression, we can arrive at the law of gravitation.

$$F = -\frac{4\pi^2 mk}{r^2} \quad (6.10)$$

Here negative sign implies that the force is attractive and it acts towards the center. In equation (6.10), mass of the planet ' m ' comes explicitly. But Newton strongly felt that according to his third law, if Earth is attracted by the Sun, then the Sun must also be attracted by the Earth with the same magnitude of force. So he felt that the Sun's mass (M) should also occur explicitly in the expression for force (6.10). From this insight, he equated the constant $4\pi^2 k$ to GM which turned out to be the law of gravitation.

$$F = -\frac{GMm}{r^2}$$

Again the negative sign in the above equation implies that the gravitational force is attractive.

In the above discussion we assumed that the orbit of the planet to be circular which is not true as the orbit of the planet around the Sun is elliptical. But this circular orbit assumption is justifiable because planet's orbit is very close to being circular and there is only a very small deviation from the circular shape.



Points to Contemplate

If Kepler's third law was " $r^3 T^2 = \text{constant}$ " instead of " $\frac{r^3}{T^2} = \text{constant}$ " what would be the new law of gravitation? Would it still be an inverse square law? How would the gravitational force change with distance? In this new law of gravitation, will Neptune experience greater gravitational force or lesser gravitational force when compared to the Earth?

EXAMPLE 6.2

Moon and an apple are accelerated by the same gravitational force due to Earth. Compare the acceleration of the two.

The gravitational force experienced by the apple due to Earth

$$F = -\frac{GM_E M_A}{R^2}$$

Here M_A – Mass of the apple, M_E – Mass of the Earth and R – Radius of the Earth.

Equating the above equation with Newton's second law,

$$M_A a_A = -\frac{GM_E M_A}{R^2}.$$

Simplifying the above equation we get,

$$a_A = -\frac{GM_E}{R^2}$$

Here a_A is the acceleration of apple that is equal to 'g'.

Similarly the force experienced by Moon due to Earth is given by

$$F = -\frac{GM_E M_m}{R_m^2}.$$

Here R_m – distance of the Moon from the Earth, M_m – Mass of the Moon

The acceleration experienced by the Moon is given by

$$a_m = -\frac{GM_E}{R_m^2}.$$

The ratio between the apple's acceleration to Moon's acceleration is given by

$$\frac{a_A}{a_m} = \frac{R_m^2}{R^2}.$$

From the Hipparchus measurement, the distance to the Moon is 60 times that of Earth radius. $R_m = 60R$.

$$a_A / a_m = \frac{(60R)^2}{R^2} = 3600.$$

The apple's acceleration is 3600 times the acceleration of the Moon.

The same result was obtained by Newton using his gravitational formula. The apple's acceleration is measured easily and it is 9.8 m s^{-2} . Moon orbits the Earth once in 27.3 days and by using the centripetal acceleration formula, (Refer unit 3).

$$\frac{a_A}{a_m} = \frac{9.8}{0.00272} = 3600$$

which is exactly what he got through his law of gravitation.

Note

The above calculation depends on knowing the distance between the Earth and the Moon and the radius of the Earth. The radius of the Earth was measured by Greek librarian Eratosthenes and distance between the Earth and the Moon was measured by Greek astronomer Hipparchus 2400 years ago. It is very interesting to note that in order to measure these distances he used only high school geometry and trigonometry. These details are discussed in the astronomy section (6.5).

6.1.3 Gravitational Constant

In the law of gravitation, the value of gravitational constant G plays a very important role. The value of G explains why the gravitational force between the Earth and the Sun is so great while the same force between two small objects (for example between two human beings) is negligible.

The force experienced by a mass 'm' which is on the surface of the Earth (Figure 6.7) is given by

$$F = -\frac{GM_E m}{R_E^2} \quad (6.11)$$

M_E -mass of the Earth, m - mass of the object, R_E - radius of the Earth.

Equating Newton's second law, $F = -mg$, to equation (6.11) we get,

$$\begin{aligned} -mg &= -\frac{GM_E m}{R_E^2} \\ g &= \frac{GM_E}{R_E^2} \end{aligned} \quad (6.12)$$

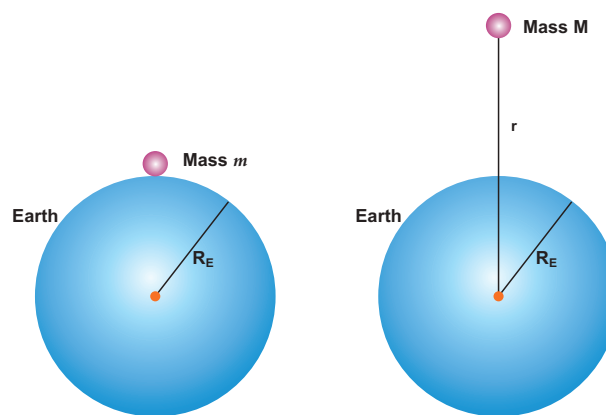


Figure 6.7 Force experienced by a mass on the (i) surface of the Earth (ii) at a distance from the centre of the Earth

Now the force experienced by some other object of mass M at a distance r from the center of the Earth is given by,

$$F = -\frac{GM_E M}{r^2}$$

Using the value of g in equation (6.12), the force F will be,

$$F = -gM \frac{R_E^2}{r^2} \quad (6.13)$$

From this it is clear that the force can be calculated simply by knowing the value of g . It is to be noted that in the above calculation G is not required.



In the year 1798, Henry Cavendish experimentally determined the value of gravitational constant 'G' by using a torsion balance. He calculated the value of 'G' to be equal to $6.75 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$. Using modern techniques a more accurate value of G could be measured. The currently accepted value of G is $6.67259 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.

6.2

GRAVITATIONAL FIELD AND GRAVITATIONAL POTENTIAL

6.2.1 Gravitational field

Force is basically due to the interaction between two particles. Depending upon the type of interaction we can have two kinds of forces: Contact forces and Non-contact forces (Figure 6.8).

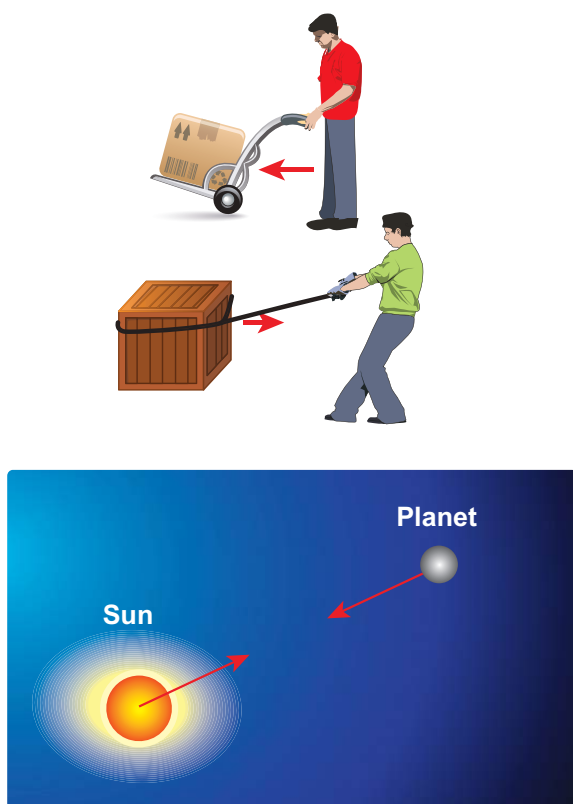


Figure 6.8 Depiction of contact and non-contact forces

Contact forces are the forces applied where one object is in physical contact with the other. The movement of the object is caused by the physical force exerted through the contact between the object and the agent which exerts force.

Consider the case of Earth orbiting around the Sun. Though the Sun and the

Earth are not physically in contact with each other, there exists an interaction between them. This is because of the fact that the Earth experiences the gravitational force of the Sun. This gravitational force is a non-contact force.

It sounds mysterious that the Sun attracts the Earth despite being very far from it and without touching it. For contact forces like push or pull, we can calculate the strength of the force since we can feel or see. But how do we calculate the strength of non-contact force at different distances? To understand and calculate the strength of non-contact forces, the concept of 'field' is introduced.

The gravitational force on a particle of mass ' m_2 ' due to a particle of mass ' m_1 ' is

$$\vec{F}_{21} = -\frac{Gm_1m_2}{r^2}\hat{r} \quad (6.14)$$

where $\hat{r}$ is a unit vector that points from m_1 to m_2 along the line joining the masses m_1 and m_2 .

The gravitational field intensity $\vec{E}_1$ (here after called as gravitational field) at a point which is at a distance r from m_1 is defined as the gravitational force experienced by unit mass placed at that point. It is given by the ratio $\frac{\vec{F}_{21}}{m_2}$ (where m_2 is the mass of the object on which $\vec{F}_{21}$ acts)

Using $\vec{E}_1 = \frac{\vec{F}_{21}}{m_2}$ in equation (6.14) we get,

$$\vec{E}_1 = -\frac{Gm_1}{r^2}\hat{r} \quad (6.15)$$

$\vec{E}_1$ is a vector quantity that points towards the mass m_1 and is independent of mass m_2 . Here m_2 is taken to be of unit magnitude. The unit $\hat{r}$ is along the line between m_1 and the point in question. The field $\vec{E}_1$ is due to the mass m_1 .

In general, the gravitational field intensity due to a mass M at a distance r is given by

$$\vec{E} = -\frac{GM}{r^2} \hat{r} \quad (6.16)$$

Now in the region of this gravitational field, a mass 'm' is placed at a point P (Figure 6.9). Mass 'm' interacts with the field $\vec{E}$ and experiences an attractive force due to M as shown in Figure 6.9. The gravitational force experienced by 'm' due to 'M' is given by

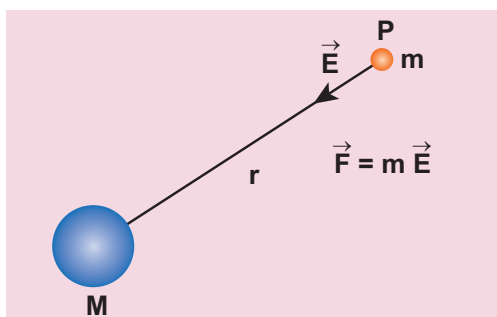


Figure 6.9 Gravitational Field intensity measured with an object of unit mass

$$\vec{F}_m = m\vec{E} \quad (6.17)$$

Now we can equate this with Newton's second law $\vec{F} = m\vec{a}$

$$m\vec{a} = m\vec{E} \quad (6.18)$$

$$\vec{a} = \vec{E} \quad (6.19)$$

In other words, equation (6.18) implies that the gravitational field at a point is equivalent to the acceleration experienced

by a particle at that point. However, it is to be noted that $\vec{a}$ and $\vec{E}$ are separate physical quantities that have the same magnitude and direction. The gravitational field $\vec{E}$ is the property of the source and acceleration $\vec{a}$ is the effect experienced by the test mass (unit mass) which is placed in the gravitational field $\vec{E}$. The non-contact interaction between two masses can now be explained using the concept of "Gravitational field".

Points to be noted:

- i) The strength of the gravitational field decreases as we move away from the mass M as depicted in the Figure 6.10. The magnitude of $\vec{E}$ decreases as the distance r increases.

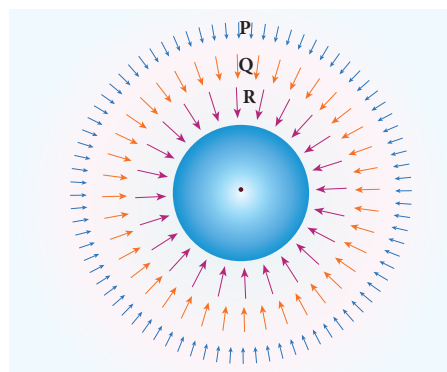


Figure 6.10 Strength of the Gravitational field lines decreases with distance

Figure 6.10 shows that the strength of the gravitational field at points P, Q, and R is given by $|\vec{E}_P| < |\vec{E}_Q| < |\vec{E}_R|$. It can be understood by comparing the length of the vectors at points P, Q, and R.

- ii) The "field" concept was introduced as a mathematical tool to calculate gravitational interaction. Later it was found that field is a real physical quantity and it carries energy and momentum in

space. The concept of field is inevitable in understanding the behaviour of charges.

- iii) The unit of gravitational field is newton per kilogram (N/kg) or m s^{-2} .

6.2.2 Superposition principle for Gravitational field

Consider 'n' particles of masses $m_1, m_2, \dots, m_n$, distributed in space at positions $\vec{r}_1, \vec{r}_2, \vec{r}_3 \dots$ etc, with respect to point P. The total gravitational field at a point P due to all the masses is given by the vector sum of the gravitational field due to the individual masses (Figure 6.11). This principle is known as superposition of gravitational fields.

$$\begin{aligned}\vec{E}_{\text{total}} &= \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n \\ &= -\frac{Gm_1}{r_1^2} \hat{r}_1 - \frac{Gm_2}{r_2^2} \hat{r}_2 - \dots - \frac{Gm_n}{r_n^2} \hat{r}_n \\ &= -\sum_{i=1}^n \frac{Gm_i}{r_i^2} \hat{r}_i.\end{aligned}\quad (6.20)$$

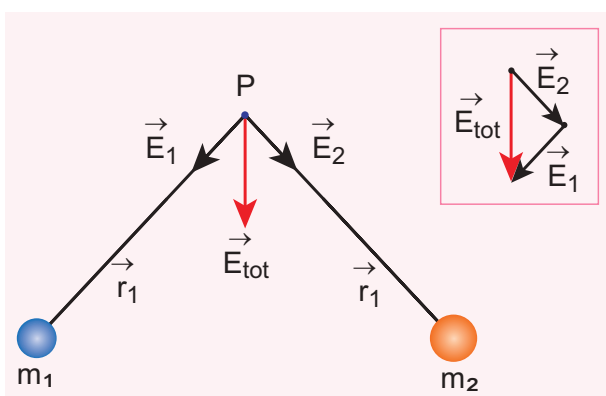
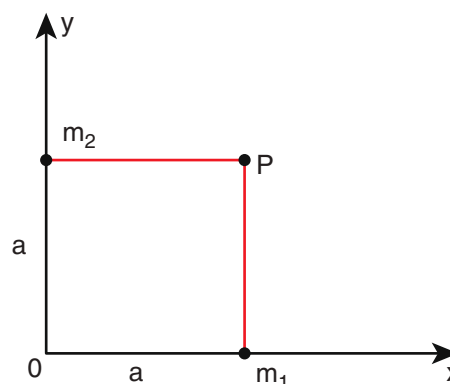


Figure 6.11 Superposition of two gravitational field intensities giving resultant field.

Instead of discrete masses, if we have continuous distribution of a total mass M , then the gravitational field at a point P is calculated using the method of integration.

EXAMPLE 6.3

- (a) Two particles of masses m_1 and m_2 are placed along the x and y axes respectively at a distance 'a' from the origin. Calculate the gravitational field at a point P shown in figure below.



Solution

Gravitational field due to m_1 at a point P is given by,

$$\vec{E}_1 = -\frac{Gm_1}{a^2} \hat{j}$$

Gravitational field due to m_2 at the point p is given by,

$$\vec{E}_2 = -\frac{Gm_2}{a^2} \hat{i}$$

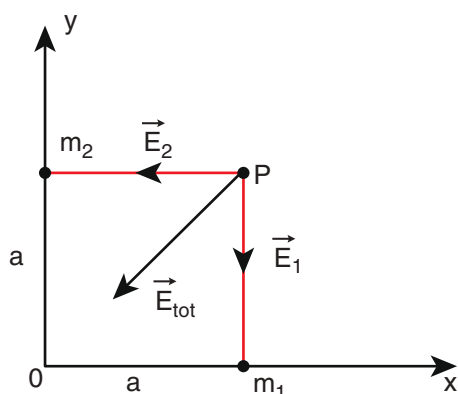
$$\begin{aligned}\vec{E}_{\text{total}} &= -\frac{Gm_1}{a^2} \hat{j} - \frac{Gm_2}{a^2} \hat{i} \\ &= -\frac{G}{a^2} (m_1 \hat{j} + m_2 \hat{i})\end{aligned}$$

The direction of the total gravitational field is determined by the relative value of m_1 and m_2 .

When $m_1 = m_2 = m$

$$\vec{E}_{total} = -\frac{Gm}{a^2}(\hat{i} + \hat{j})$$

($\hat{i} + \hat{j} = \hat{j} + \hat{i}$ as vectors obeys commutation law).

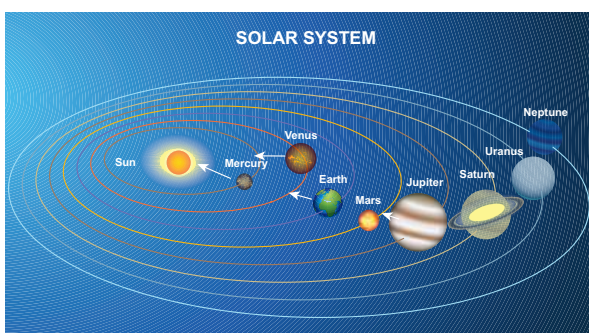


$\vec{E}_{total}$ points towards the origin of the co-ordinate system and the magnitude of $\vec{E}_{total}$ is $\sqrt{2} \frac{Gm}{a^2}$.

EXAMPLE 6.4

Qualitatively indicate the gravitational field of Sun on Mercury, Earth, and Jupiter shown in figure.

Since the gravitational field decreases as distance increases, Jupiter experiences a weak gravitational field due to the Sun. Since Mercury is the nearest to the Sun, it experiences the strongest gravitational field.



Solar System

6.2.3 Gravitational Potential Energy

The concept of potential energy and its physical meaning were dealt in unit 4. The gravitational force is a conservative force and hence we can define a gravitational potential energy associated with this conservative force field.

Two masses m_1 and m_2 are initially separated by a distance r' . Assuming m_1 to be fixed in its position, work must be done on m_2 to move the distance from r' to r as shown in Figure 6.12(a)

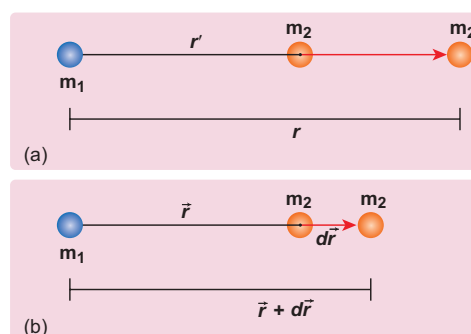


Figure 6.12 Two distant masses changing the linear distance

To move the mass m_2 through an infinitesimal displacement $d\vec{r}$ from $\vec{r}$ to $\vec{r} + d\vec{r}$ (shown in the Figure 6.12(b)), work has to be done externally. This infinitesimal work is given by

$$dW = \vec{F}_{ext} \cdot d\vec{r} \quad (6.21)$$

The work is done against the gravitational force, therefore,

$$|\vec{F}_{ext}| = |\vec{F}_G| = \frac{Gm_1m_2}{r^2} \quad (6.22)$$

Substituting Equation (6.22) in 6.21, we get

$$dW = \frac{Gm_1m_2}{r^2} \hat{r} \cdot d\vec{r} \quad (6.23)$$

Also we know,

$$d\vec{r} = dr \hat{r} \quad (6.24)$$

$$\Rightarrow dW = \frac{Gm_1m_2}{r^2} \hat{r} \cdot (dr \hat{r}) \quad (6.25)$$

$$\hat{r} \cdot \hat{r} = 1 \text{ (since both are unit vectors)}$$

$$\therefore dW = \frac{Gm_1m_2}{r^2} dr \quad (6.26)$$

Thus the total work done for displacing the particle from r' to r is

$$W = \int_{r'}^r dW = \int_{r'}^r \frac{Gm_1m_2}{r^2} dr \quad (6.27)$$

$$W = - \left(\frac{Gm_1m_2}{r} \right)_{r'}$$

$$W = - \frac{Gm_1m_2}{r} + \frac{Gm_1m_2}{r'} \quad (6.28)$$

$$W = U(r) - U(r')$$

$$\text{where } U(r) = \frac{-Gm_1m_2}{r}$$

This work done W gives the gravitational potential energy difference of the system of masses m_1 and m_2 when the separation between them are r and r' respectively.

Case 1: If $r < r'$

Since gravitational force is attractive, m_2 is attracted by m_1 . Then m_2 can move from r' to r without any external work (Figure 6.13(a)). Here work is done by the system spending its internal energy and hence the work done is said to be negative.

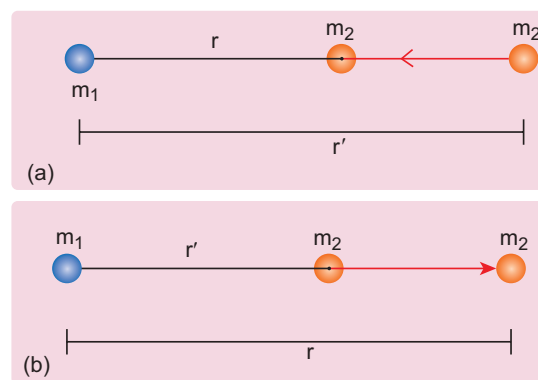


Figure 6.13 Cases for calculation of work done by gravity

Case 2: If $r > r'$

Work has to be done against gravity to move the object from r' to r (Figure 6.13(b)). Therefore work is done on the body by external force and hence work done is positive.

It is to be noted that only potential energy difference has physical significance. Now gravitational potential energy can be discussed by choosing one point as the reference point.

Let us choose $r' = \infty$. Then the second term in the equation (6.28) becomes zero.

$$W = - \frac{Gm_1m_2}{r} + 0 \quad (6.29)$$

Now we can define gravitational potential energy of a system of two masses m_1 and m_2 separated by a distance r as the amount of work done to take the mass m_2 from a distance r to infinity assuming m_1 to be fixed in its position and is written as $U(r) = - \frac{Gm_1m_2}{r}$. It is to be noted that the gravitational potential energy of the system consisting of two masses m_1 and m_2 separated by a distance r , is the gravitational potential energy difference of the system when the masses are

separated by an infinite distance and by distance r . $U(r) = U(r) - U(\infty)$. Here we choose $U(\infty) = 0$ as the reference point. The gravitational potential energy $U(r)$ is always negative because when two masses come together slowly from infinity, work is done by the system.

The unit of gravitational potential energy $U(r)$ is Joule and it is a scalar quantity. The gravitational potential energy depends upon the two masses and the distance between them.

6.2.4 Gravitational potential energy near the surface of the Earth

It is already discussed in chapter 4 that when an object of mass m is raised to a height h , the potential energy stored in the object is mgh (Figure 6.14). This can be derived using the general expression for gravitational potential energy.

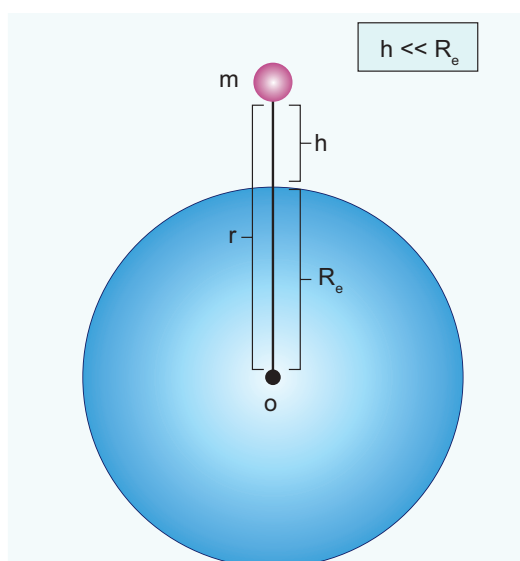


Figure 6.14 Mass placed at a distance r from the center of the Earth

Consider the Earth and mass system, with r , the distance between the mass m and the Earth's centre. Then the gravitational potential energy,

$$U = -\frac{GM_e m}{r} \quad (6.30)$$

Here $r = R_e + h$, where R_e is the radius of the Earth. h is the height above the Earth's surface

$$U = -G \frac{M_e m}{(R_e + h)} \quad (6.31)$$

If $h \ll R_e$, equation (6.31) can be modified as

$$U = -G \frac{M_e m}{R_e (1 + h/R_e)}$$

$$U = -G \frac{M_e m}{R_e} (1 + h/R_e)^{-1} \quad (6.32)$$

By using Binomial expansion and neglecting the higher order terms, we get

$$U = -G \frac{M_e m}{R_e} \left(1 - \frac{h}{R_e} \right). \quad (6.33)$$

We know that, for a mass m on the Earth's surface,

$$G \frac{M_e m}{R_e} = mgR_e \quad (6.34)$$

Substituting equation (6.34) in (6.33) we get,

$$U = -mgR_e + mgh \quad (6.35)$$

It is clear that the first term in the above expression is independent of the height h . For example, if the object is taken from

height h_1 to h_2 then the potential energy at h_1 is

$$U(h_1) = -mgR_e + mgh_1 \quad (6.36)$$

and the potential energy at h_2 is

$$U(h_2) = -mgR_e + mgh_2 \quad (6.37)$$

The potential energy difference between h_1 and h_2 is

$$U(h_2) - U(h_1) = mg(h_2 - h_1). \quad (6.38)$$

The term mgR_e in equations (6.36) and (6.37) plays no role in the result. Hence in the equation (6.35) the first term can be omitted or taken to zero. Thus it can be stated that The gravitational potential energy stored in the particle of mass m at a height h from the surface of the Earth is $U = mgh$. On the surface of the Earth, $U = 0$, since h is zero.

It is to be noted that mgh is the work done on the particle when we take the mass m from the surface of the Earth to a height h . This work done is stored as a gravitational potential energy in the mass m . Even though mgh is gravitational potential energy of the system (Earth and mass m), we can take mgh as the gravitational potential energy of the mass m since Earth is stationary when the mass moves to height h .

6.2.5 Gravitational potential $V(r)$

It is explained in the previous sections that the gravitational field $\vec{E}$ depends only on the source mass which creates the field. It is a vector quantity. We can also define a scalar quantity called “gravitational potential” which depends only on the source mass.

The gravitational potential at a distance r due to a mass is defined as the amount of work required to take unit mass from the distance r to infinity and it is denoted as $V(r)$. In other words, the gravitational potential at distance r is equivalent to gravitational potential energy per unit mass at the same distance r . It is a scalar quantity and its unit is J kg^{-1} .

We can determine gravitational potential from gravitational potential energy. Consider two masses m_1 and m_2 separated by a distance r which has gravitational potential energy $U(r)$ (Figure 6.15). The gravitational potential due to mass m_1 at a point P which is at a distance r from m_1 is obtained by making m_2 equal to unity ($m_2 = 1\text{kg}$). Thus the gravitational potential $V(r)$ due to mass m_1 at a distance r is

$$V(r) = -\frac{Gm_1}{r} \quad (6.39)$$

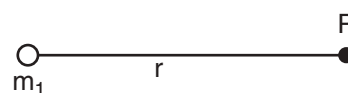


Figure 6.15 Point mass placed at a distance

Gravitational field and gravitational force are vector quantities whereas the gravitational potential and gravitational potential energy are scalar quantities. The motion of particles can be easily analyzed using scalar quantities than vector quantities. Consider the example of a falling apple:

Figure 6.16 shows an apple which falls on Earth due to Earth's gravitational force. This can be explained using the concept of gravitational potential $V(r)$ as follows.

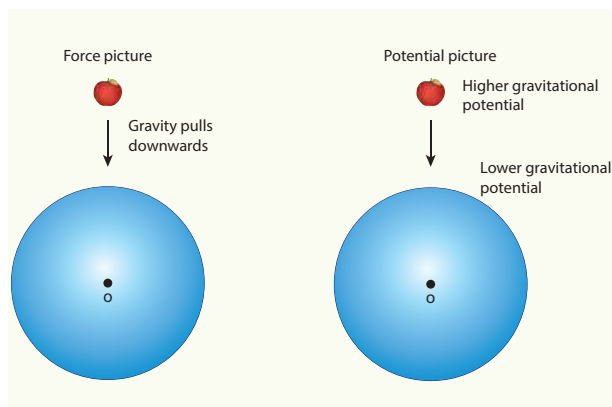


Figure 6.16 Apple falling freely under gravity

The gravitational potential $V(r)$ at a point of height h from the surface of the Earth is given by,

$$V(r = R + h) = -\frac{GM_e}{(R + h)} \quad (6.40)$$

The gravitational potential $V(r)$ on the surface of Earth is given by,

$$V(r = R) = -\frac{GM_e}{R} \quad (6.41)$$

Thus we see that

$$V(r = R) < V(r = R + h). \quad (6.42)$$

It is already discussed in the previous section that the gravitational potential energy near the surface of the Earth at height h is mgh . The gravitational potential at this point is simply $V(h) = U(h)/m = gh$. In fact, the gravitational potential on the surface of the Earth is zero since h is zero. So the apple falls from a region of a higher gravitational potential to a region of lower gravitational potential. In general, the mass will move from a region of higher gravitational potential to a region of lower gravitational potential.

EXAMPLE 6.5

Water falls from the top of a hill to the ground. Why?

This is because the top of the hill is a point of higher gravitational potential than the surface of the Earth i.e. $V_{\text{hill}} > V_{\text{ground}}$

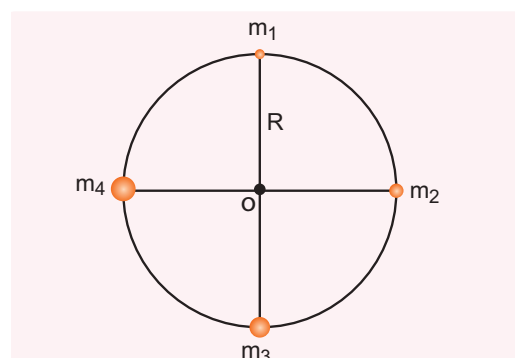


Water falling from hill top

The motion of particles can be analyzed more easily using scalars like $U(r)$ or $V(r)$ than vector quantities like $\vec{F}$ or $\vec{E}$. In modern theories of physics, the concept of potential plays a vital role.

EXAMPLE 6.6

Consider four masses m_1 , m_2 , m_3 , and m_4 arranged on the circumference of a circle as shown in figure below





Calculate

- (a) The gravitational potential energy of the system of 4 masses shown in figure.
- (b) The gravitational potential at the point O due to all the 4 masses.

Solution

The gravitational potential energy $U(r)$ can be calculated by finding the sum of gravitational potential energy of each pair of particles.

$$U = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \frac{Gm_1m_4}{r_{14}} - \frac{Gm_2m_3}{r_{23}} - \frac{Gm_2m_4}{r_{24}} - \frac{Gm_3m_4}{r_{34}}$$

Here $r_{12}, r_{13} \dots$ are distance between pair of particles

$$r_{14}^2 = R^2 + R^2 = 2R^2$$

$$r_{14} = \sqrt{2}R = r_{12} = r_{23} = r_{34}$$

$$r_{13} = r_{24} = 2R$$

$$U = -\frac{Gm_1m_2}{\sqrt{2}R} - \frac{Gm_1m_3}{2R} - \frac{Gm_1m_4}{\sqrt{2}R} - \frac{Gm_2m_3}{\sqrt{2}R} - \frac{Gm_2m_4}{2R} - \frac{Gm_3m_4}{\sqrt{2}R}$$

$$U = -\frac{G}{R} \left[\frac{m_1m_2}{\sqrt{2}} + \frac{m_1m_3}{2} + \frac{m_1m_4}{\sqrt{2}} + \frac{m_2m_3}{\sqrt{2}} + \frac{m_2m_4}{2} + \frac{m_3m_4}{\sqrt{2}} \right]$$

If all the masses are equal, then $m_1 = m_2 = m_3 = m_4 = M$

$$U = -\frac{GM^2}{R} \left[\frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{\sqrt{2}} \right]$$

$$U = -\frac{GM^2}{R} \left[1 + \frac{4}{\sqrt{2}} \right]$$

$$U = -\frac{GM^2}{R} [1 + 2\sqrt{2}]$$

The gravitational potential $V(r)$ at a point O is equal to the sum of the gravitational potentials due to individual mass. Since potential is a scalar, the net potential at point O is the algebraic sum of potentials due to each mass.

$$V_o(r) = -\frac{Gm_1}{R} - \frac{Gm_2}{R} - \frac{Gm_3}{R} - \frac{Gm_4}{R}$$

$$\text{If } m_1 = m_2 = m_3 = m_4 = M$$

$$V_o(r) = -\frac{4GM}{R}$$

6.3

ACCELERATION DUE TO GRAVITY OF THE EARTH

When objects fall on the Earth, the acceleration of the object is towards the Earth. From Newton's second law, an object is accelerated only under the action of a force. In the case of Earth, this force is the gravitational pull of Earth. This force produces a constant acceleration near the Earth's surface in all bodies, irrespective of their masses. The gravitational force exerted by Earth on the mass m near the surface of the Earth is given by

$$\vec{F} = -\frac{GmM_e}{R_e^2} \hat{r}$$

Now equating Gravitational force to Newton's second law,



$$m\vec{a} = -\frac{GmM_e}{R_e^2}\hat{r}$$

hence, acceleration is,

$$\vec{a} = -\frac{GM_e}{R_e^2}\hat{r} \quad (6.43)$$

The acceleration experienced by the object near the surface of the Earth due to its gravity is called acceleration due to gravity. It is denoted by the symbol g . The magnitude of acceleration due to gravity is

$$|g| = g = \frac{GM_e}{R_e^2} \quad (6.44)$$

It is to be noted that the acceleration experienced by any object is independent of its mass. The value of g depends only on the mass and radius of the Earth. Infact, Galileo arrived at the same conclusion 400 years ago that *all objects fall towards the Earth with the same acceleration* through various quantitative experiments. The acceleration due to gravity g is found to be 9.8 m s^{-2} on the surface of the Earth near the equator.

6.3.1 Variation of g with altitude, depth and latitude

Consider an object of mass m at a height h from the surface of the Earth. Acceleration experienced by the object due to Earth is

$$g' = \frac{GM}{(R_e + h)^2} \quad (6.45).$$

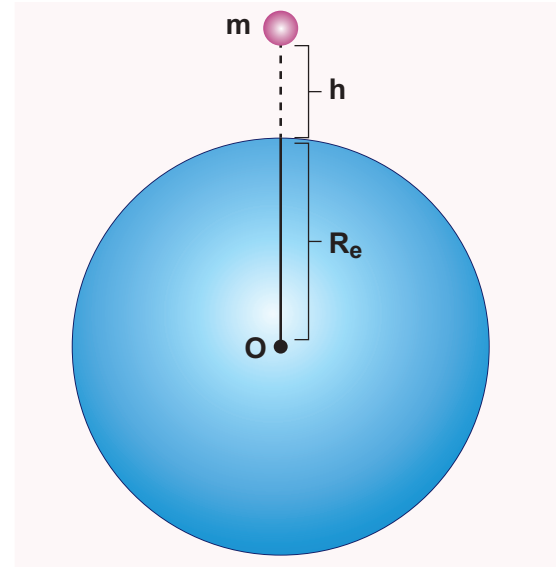


Figure 6.17(a) Mass at a height h from the center of the Earth

$$g' = \frac{GM}{R_e^2 \left(1 + \frac{h}{R_e}\right)^2}$$

$$g' = \frac{GM}{R_e^2} \left(1 + \frac{h}{R_e}\right)^{-2}$$

If $h \ll R_e$

We can use Binomial expansion. Taking the terms upto first order

$$g' = \frac{GM}{R_e^2} \left(1 - 2\frac{h}{R_e}\right)$$

$$g' = g \left(1 - 2\frac{h}{R_e}\right) \quad (6.46)$$

We find that $g' < g$. This means that as altitude h increases the acceleration due to gravity g decreases.

EXAMPLE 6.7

1. Calculate the value of g in the following two cases:

- (a) If a mango of mass $\frac{1}{2}$ kg falls from a tree from a height of 15 meters, what is the acceleration due to gravity when it begins to fall?

Solution

$$g' = g \left(1 - 2 \frac{h}{R_e} \right)$$

$$g' = 9.8 \left(1 - \frac{2 \times 15}{6400 \times 10^3} \right)$$

$$g' = 9.8 (1 - 0.469 \times 10^{-5})$$

$$\text{But } 1 - 0.00000469 \cong 1$$

$$\text{Therefore } g' = g$$

- (b) Consider a satellite orbiting the Earth in a circular orbit of radius 1600 km above the surface of the Earth. What is the acceleration experienced by the satellite due to Earth's gravitational force?

Solution

$$g' = g \left(1 - 2 \frac{h}{R_e} \right)$$

$$g' = g \left(1 - \frac{2 \times 1600 \times 10^3}{6400 \times 10^3} \right)$$

$$g' = g \left(1 - \frac{2}{4} \right)$$

$$g' = g \left(1 - \frac{1}{2} \right) = g / 2$$

The above two examples show that the acceleration due to gravity is a constant near the surface of the Earth.



Note Can we substitute $h = R_e$ in the equation 6.46? No. To get equation 6.46 we assumed that $h \ll R_e$. However $h = R_e$ can be substituted in equation 6.45.

Variation of g with depth:

Consider a particle of mass m which is in a deep mine on the Earth. (Example: coal mines in Neyveli). Assume the depth of the mine as d . To calculate g' at a depth d , consider the following points.

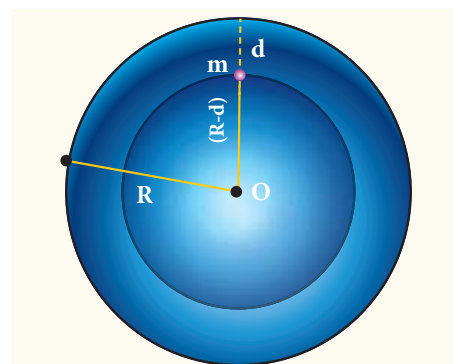


Figure 6.17(b) Particle in a mine

The part of the Earth which is above the radius $(R_e - d)$ do not contribute to the acceleration. The result is proved earlier and is given as

$$g' = \frac{GM'}{(R_e - d)^2} \quad (6.47)$$

Here M' is the mass of the Earth of radius $(R_e - d)$

Assuming the density of Earth ρ to be constant,

$$\rho = \frac{M}{V} \quad (6.48)$$

where M is the mass of the Earth and V its volume, Thus,

$$\rho = \frac{M'}{V'}$$

$$\frac{M'}{V'} = \frac{M}{V} \text{ and } M' = \frac{M}{V} V'$$

$$M' = \left(\frac{M}{\frac{4}{3}\pi R_e^3} \right) \left(\frac{4}{3}\pi (R_e - d)^3 \right)$$

$$M' = \frac{M}{R_e^3} (R_e - d)^3 \quad (6.49)$$

Substituting equation (6.49) in equation (6.47)

$$g' = G \frac{M}{R_e^3} (R_e - d)^3 \cdot \frac{1}{(R_e - d)^2}$$

$$g' = GM \frac{R_e \left(1 - \frac{d}{R_e} \right)}{R_e^3}$$

$$g' = GM \frac{\left(1 - \frac{d}{R_e} \right)}{R_e^2}$$

Thus

$$g' = g \left(1 - \frac{d}{R_e} \right) \quad (6.50)$$

Here also $g' < g$. As depth increases, g' decreases. It is very interesting to know that acceleration due to gravity is maximum on the surface of the Earth but decreases when we go either upward or downward.

Variation of g with latitude:

Whenever we analyze the motion of objects in rotating frames [explained in chapter 3] we must take into account the centrifugal force. Even though we treat the Earth as an inertial frame, it is not exactly correct because the Earth spins about its own axis. So when an object is on the surface of the Earth, it experiences a centrifugal force that depends on the latitude of the object on Earth. If the Earth were not spinning, the force on the object would have been mg . However, the object experiences an additional centrifugal force due to spinning of the Earth.

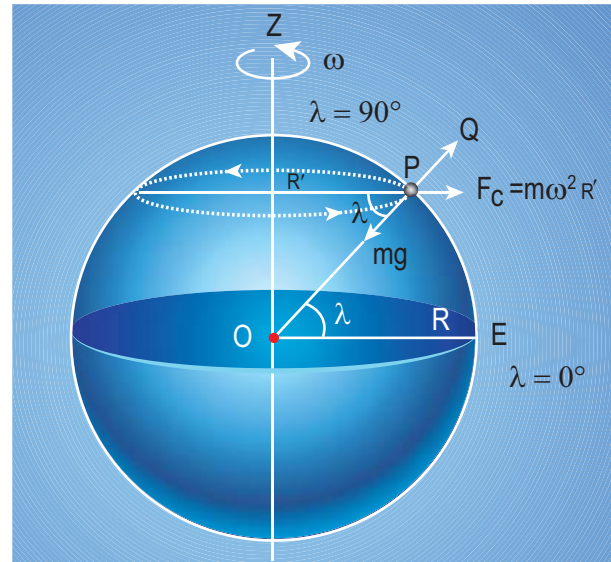


Figure 6.18 Variation of g with latitude

This centrifugal force is given by $m\omega^2 R'$.

$$R' = R \cos \lambda \quad (6.51)$$

where λ is the latitude. The component of centrifugal acceleration experienced by the object in the direction opposite to g is

$$a_{PQ} = \omega^2 R' \cos \lambda = \omega^2 R \cos^2 \lambda$$

since $R' = R \cos \lambda$

Therefore,

$$g' = g - \omega^2 R \cos^2 \lambda \quad (6.52)$$

From the expression (6.52), we can infer that at equator, $\lambda = 0$; $g' = g - \omega^2 R$. The acceleration due to gravity is minimum. At poles $\lambda = 90^\circ$; $g' = g$, it is maximum. At the equator, g' is minimum.

EXAMPLE 6.8

Find out the value of g' in your school laboratory?

Solution

Calculate the latitude of the city or village where the school is located. The information is available in Google search. For example, the latitude of Chennai is approximately 13 degree.

$$g' = g - \omega^2 R \cos^2 \lambda$$

Here $\omega^2 R = (2\pi \times 3.14 / 86400)^2 \times (6400 \times 10^3) = 3.4 \times 10^{-2} \text{ m s}^{-2}$.

It is to be noted that the value of λ should be in radian and not in degree. 13 degree is equivalent to 0.2268 rad.

$$g' = 9.8 - (3.4 \times 10^{-2}) \times (\cos 0.2268)^2$$

$$g' = 9.7677 \text{ m s}^{-2}$$

Points to Contemplate

Suppose you move towards east-west along the same latitude. Will the value of g' change?

6.4

ESCAPE SPEED AND ORBITAL SPEED

Hydrogen and helium are the most abundant elements in the universe but Earth's atmosphere consists mainly of nitrogen and oxygen. The following discussion brings forth the reason why hydrogen and helium are not found in abundance on the Earth's atmosphere. When an object is thrown up with some initial speed it will reach a certain height after which it will fall back to Earth. If the same object is thrown again with a higher speed, it reaches a greater height than the previous one and falls back to Earth. This leads to the question of what should be the speed of an object thrown vertically up such that it escapes the Earth's gravity and would never come back. This speed is called "Escape speed" and it is defined as "the minimum speed required for a body to escape from the earth's gravitational pull".

Consider an object of mass M on the surface of the Earth. When it is thrown up with an initial speed v_i , the initial total energy of the object is

$$E_i = \frac{1}{2} M v_i^2 - \frac{G M M_E}{R_E} \quad (6.53)$$

where, M_E is the mass of the Earth and R_E the radius of the Earth. The term $-\frac{G M M_E}{R_E}$ is the potential energy of the mass M .

When the object reaches a height far away from Earth and hence treated as approaching infinity, the gravitational potential energy becomes zero [$U(\infty) = 0$] and the kinetic energy becomes zero as well. Therefore the final total energy of the

object becomes zero. This is for minimum energy and for minimum speed to escape. Otherwise Kinetic energy can be nonzero.

$$E_f = 0$$

According to the law of energy conservation,

$$E_i = E_f \quad (6.54)$$

Substituting (6.53) in (6.54) we get,

$$\begin{aligned} \frac{1}{2} M v_i^2 - \frac{G M M_E}{R_E} &= 0 \\ \frac{1}{2} M v_i^2 &= \frac{G M M_E}{R_E} \end{aligned} \quad (6.55)$$

Consider the escape speed, the minimum speed required by an object to escape Earth's gravitational field, hence replace v_i with v_e . i.e,

$$\begin{aligned} \frac{1}{2} M v_e^2 &= \frac{G M M_E}{R_E} \\ v_e^2 &= \frac{G M M_E}{R_E} \cdot \frac{2}{M} \\ v_e^2 &= \frac{2 G M_E}{R_E} \end{aligned}$$

Using $g = \frac{G M_E}{R_E^2}$,

$$\begin{aligned} v_e^2 &= 2 g R_E \\ v_e &= \sqrt{2 g R_E} \end{aligned} \quad (6.56)$$

From equation (6.56) the escape speed depends on two factors: acceleration due to gravity and radius of the Earth. It is completely independent of the mass of the object. By substituting the values of g (9.8 m s^{-2}) and $R_e = 6400 \text{ km}$, the escape speed of the Earth is $v_e = 11.2 \text{ km s}^{-1}$. The escape speed is independent of the direction

in which the object is thrown. Irrespective of whether the object is thrown vertically up, radially outwards or tangentially it requires the same initial speed to escape Earth's gravity. It is shown in Figure 6.19

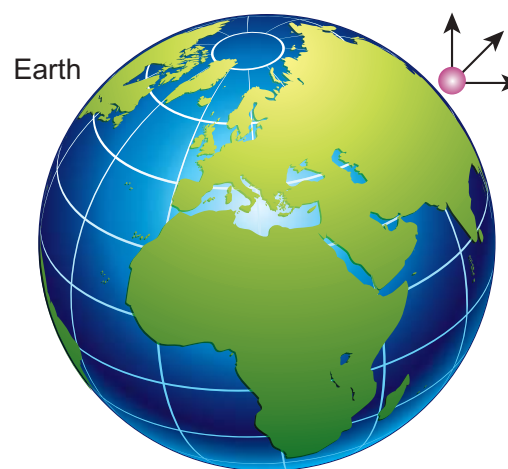


Figure 6.19 Escape speed independent of angle

Lighter molecules such as hydrogen and helium have enough speed to escape from the Earth, unlike the heavier ones such as nitrogen and oxygen. (The average speed of hydrogen and helium atoms compared with the escape speed of the Earth, is presented in the kinetic theory of gases, unit 9).

6.4.1 Satellites, orbital speed and time period

We are living in a modern world with sophisticated technological gadgets and are able to communicate to any place on Earth. This advancement was made possible because of our understanding of solar system. Communication mainly depends on the satellites that orbit the Earth (Figure 6.20). Satellites revolve around the Earth just like the planets revolve around the Sun. Kepler's laws are applicable to man-made satellites also.

For a satellite of mass M to move in a circular orbit, centripetal force must be acting on the satellite. This centripetal force is provided by the Earth's gravitational force.

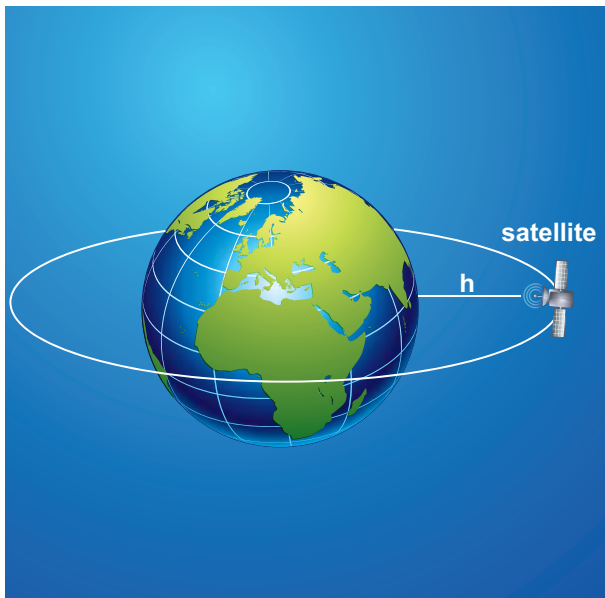


Figure 6.20 Satellite revolving around the Earth.

$$\frac{Mv^2}{(R_E + h)} = \frac{GMM_E}{(R_E + h)^2} \quad (6.57)$$

$$v^2 = \frac{GM_E}{(R_E + h)}$$

$$v = \sqrt{\frac{GM_E}{(R_E + h)}} \quad (6.58)$$

As h increases, the speed of the satellite decreases.

Time period of the satellite:

The distance covered by the satellite during one rotation in its orbit is equal to $2\pi(R_E + h)$ and time taken for it is the time period, T . Then

$$\text{Speed } v = \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{2\pi(R_E + h)}{T}$$

From equation (6.58)

$$\sqrt{\frac{GM_E}{(R_E + h)}} = \frac{2\pi(R_E + h)}{T} \quad (6.59)$$

$$T = \frac{2\pi}{\sqrt{GM_E}}(R_E + h)^{3/2} \quad (6.60)$$

Squaring both sides of the equation (6.60), we get

$$T^2 = \frac{4\pi^2}{GM_E}(R_E + h)^3$$

$$\frac{4\pi^2}{GM_E} = \text{constant say } c$$

$$T^2 = c(R_E + h)^3 \quad (6.61)$$

Equation (6.61) implies that a satellite orbiting the Earth has the same relation between time and distance as that of Kepler's law of planetary motion. For a satellite orbiting near the surface of the Earth, h is negligible compared to the radius of the Earth R_E . Then,

$$T^2 = \frac{4\pi^2}{GM_E} R_E^3$$

$$T^2 = \frac{4\pi^2}{GM_E / R_E^2} R_E$$

$$T^2 = \frac{4\pi^2}{g} R_E$$

$$\text{since } GM_E / R_E^2 = g$$

$$T = 2\pi \sqrt{\frac{R_E}{g}} \quad (6.62)$$

By substituting the values of $R_E = 6.4 \times 10^6 \text{ m}$ and $g = 9.8 \text{ m s}^{-2}$, the orbital time period is obtained as $T \cong 85 \text{ minutes}$.

EXAMPLE 6.9

Moon is the natural satellite of Earth and it takes 27 days to go once around its orbit. Calculate the distance of the Moon from the surface of the Earth assuming the orbit of the Moon as circular.

Solution

We can use Kepler's third law,

$$\begin{aligned} T^2 &= c(R_E + h)^3 \\ T^{2/3} &= c^{1/3}(R_E + h) \\ \left(\frac{T^2}{c}\right)^{1/3} &= (R_E + h) \\ \left(\frac{T^2 GM_E}{4\pi^2}\right)^{1/3} &= (R_E + h); \\ c &= \frac{4\pi^2}{GM_E} \\ h &= \left(\frac{T^2 GM_E}{4\pi^2}\right)^{1/3} - R_E \end{aligned}$$

Here h is the distance of the Moon from the surface of the Earth. Here,

$$\begin{aligned} R_E &- \text{radius of the Earth} = 6.4 \times 10^6 \text{ m} \\ M_E &- \text{mass of the Earth} = 6.02 \times 10^{24} \text{ kg} \\ G &- \text{Universal gravitational} \\ &\text{constant} = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \end{aligned}$$

By substituting these values, the distance to the Moon from the surface of the Earth is calculated to be $3.77 \times 10^5 \text{ km}$.

6.4.2 Energy of an Orbiting Satellite

The total energy of a satellite orbiting the Earth at a distance h from the surface of Earth is calculated as follows; The total energy of the satellite is the sum of its kinetic energy and the gravitational potential energy. The potential energy of the satellite is,

$$U = -\frac{GM_s M_E}{(R_E + h)} \quad (6.63)$$

Here M_s - mass of the satellite, M_E - mass of the Earth, R_E - radius of the Earth.

The Kinetic energy of the satellite is

$$K.E = \frac{1}{2} M_s v^2 \quad (6.64)$$

Here v is the orbital speed of the satellite and is equal to

$$v = \sqrt{\frac{GM_E}{(R_E + h)}} \quad (6.65)$$

Substituting the value of v in (6.64), the kinetic energy of the satellite becomes,

$$K.E = \frac{1}{2} \frac{GM_E M_s}{(R_E + h)}$$

Therefore the total energy of the satellite is

$$E = \frac{1}{2} \frac{GM_E M_s}{(R_E + h)} - \frac{GM_s M_E}{(R_E + h)}$$

$$E = -\frac{GM_s M_E}{2(R_E + h)} \quad (6.66)$$

The negative sign in the total energy implies that the satellite is bound to the Earth and it cannot escape from the Earth.

As h approaches ∞ , the total energy tends to zero. Its physical meaning is that the satellite is completely free from the influence of Earth's gravity and is not bound to Earth at large distances.

EXAMPLE 6.10

Calculate the energy of the (i) Moon orbiting the Earth and (ii) Earth orbiting the Sun.

Solution

Assuming the orbit of the Moon to be circular, the energy of Moon is given by,

$$E_m = -\frac{GM_E M_m}{2R_m}$$

where M_E is the mass of Earth 6.02×10^{24} kg; M_m is the mass of Moon 7.35×10^{22} kg; and R_m is the distance between the Moon and the center of the Earth 3.84×10^5 km

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}.$$

$$E_m = -\frac{6.67 \times 10^{-11} \times 6.02 \times 10^{24} \times 7.35 \times 10^{22}}{2 \times 3.84 \times 10^5 \times 10^3}$$

$$E_m = -38.42 \times 10^{-19} \times 10^{46}$$

$$E_m = -38.42 \times 10^{27} \text{ joule}$$

The negative energy implies that the Moon is bound to the Earth.

Same method can be used to prove that the energy of the Earth is also negative.

6.4.3 Geo-stationary and polar satellite

The satellites orbiting the Earth have different time periods corresponding to different orbital radii. Can we calculate the orbital radius of a satellite if its time period is 24 hours?

Kepler's third law is used to find the radius of the orbit.

$$T^2 = \frac{4\pi^2}{GM_E} (R_E + h)^3$$

$$(R_E + h)^3 = \frac{GM_E T^2}{4\pi^2}$$

$$R_E + h = \left(\frac{GM_E T^2}{4\pi^2} \right)^{1/3}$$

Substituting for the time period (24 hrs = 86400 seconds), mass, and radius of the Earth, h turns out to be 36,000 km. Such satellites are called "geo-stationary satellites", since they appear to be stationary when seen from Earth.

India uses the INSAT group of satellites that are basically geo-stationary satellites for the purpose of telecommunication. Another type of satellite which is placed at a distance

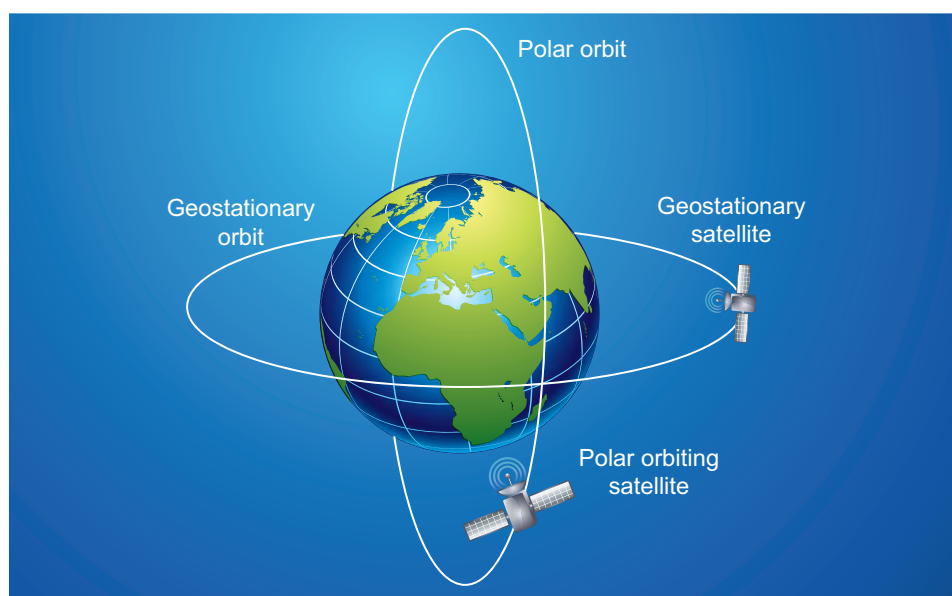


Figure 6.21 Polar orbit and geostationary satellite

of 500 to 800 km from the surface of the Earth orbits the Earth from north to south direction. This type of satellite that orbits Earth from North Pole to South Pole is called a polar satellite. The time period of a polar satellite is nearly 100 minutes and the satellite completes many revolutions in a day. A Polar satellite covers a small strip of area from pole to pole during one revolution. In the next revolution it covers a different strip of area since the Earth would have moved by a small angle. In this way polar satellites cover the entire surface area of the Earth.

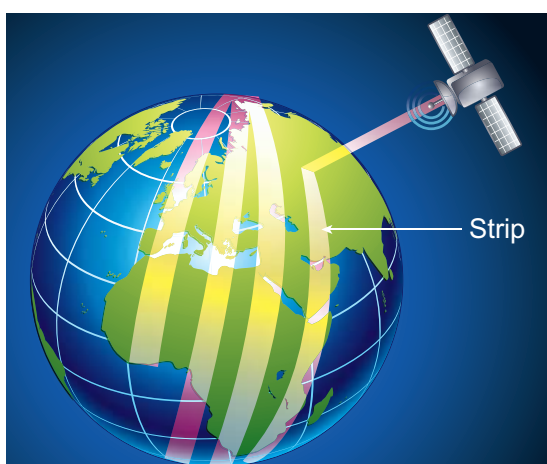


Figure 6.22 Strip of communication region, covered by a polar satellite.

6.4.4 Weightlessness

Weight of an object

Objects on Earth experience the gravitational force of Earth. The gravitational force acting on an object of mass m is mg . This force always acts downwards towards the center of the Earth. When we stand on the floor, there are two forces acting on us. One is the gravitational force, acting downwards and the other is the normal force exerted by the floor upwards on us to keep us at rest. The weight of an object $\vec{W}$ is defined as the downward force whose magnitude W is equal to that of upward force that must be applied to the object to hold it at rest or at constant velocity relative to the earth. The direction of weight is in the direction of gravitational force. So the magnitude of

weight of an object is denoted as, $W=N=mg$. Note that even though magnitude of weight is equal to mg , it is not same as gravitational force acting on the object.

Apparent weight in elevators

Everyone who used an elevator would have felt a jerk when the elevator takes off or stops. Why does it happen? Understanding the concept of weight is crucial for explaining this effect. Let us consider a man inside an elevator in the following scenarios.

When a man is standing in the elevator, there are two forces acting on him.

1. Gravitational force which acts downward. If we take the vertical direction as positive y direction, the gravitational force acting on the man is $\vec{F}_G = -mg\hat{j}$
2. The normal force exerted by floor on the man which acts vertically upward, $\vec{N} = N\hat{j}$

Case (i) When the elevator is at rest

The acceleration of the man is zero. Therefore the net force acting on the man is zero. With respect to inertial frame (ground), applying Newton's second law on the man,

$$\begin{aligned}\vec{F}_G + \vec{N} &= 0 \\ -mg\hat{j} + N\hat{j} &= 0\end{aligned}$$

By comparing the components, we can write

$$N - mg = 0 \text{ (or) } N = mg \quad (6.67)$$

Since weight, $W = N$, the apparent weight of the man is equal to his actual weight.

Case (ii) When the elevator is moving uniformly in the upward or downward direction

In uniform motion (constant velocity), the net force acting on the man is still zero.

Hence, in this case also the apparent weight of the man is equal to his actual weight. It is shown in Figure 6.23(a)

Case (iii) When the elevator is accelerating upwards

If an elevator is moving with upward acceleration ($\vec{a} = a\hat{j}$) with respect to inertial frame (ground), applying Newton's second law on the man,

$$\vec{F}_G + \vec{N} = m\vec{a}$$

Writing the above equation in terms of unit vector in the vertical direction,

$$-mg\hat{j} + N\hat{j} = ma\hat{j}$$

By comparing the components,

$$N = m(g + a) \quad (6.68)$$

Therefore, apparent weight of the man is greater than his actual weight. It is shown in Figure 6.23(b)

Case (iv) When the elevator is accelerating downwards

If the elevator is moving with downward acceleration ($\vec{a} = -a\hat{j}$), by applying Newton's second law on the man, we can write

$$\vec{F}_G + \vec{N} = m\vec{a}$$

Writing the above equation in terms of unit vector in the vertical direction,

$$-mg\hat{j} + N\hat{j} = -ma\hat{j}$$

By comparing the components,

$$N = m(g - a) \quad (6.69)$$

Therefore, apparent weight $W = N = m(g-a)$ of the man is lesser than his actual weight. It is shown in Figure 6.23(c)

Weightlessness of freely falling bodies

Freely falling objects experience only gravitational force. As they fall freely, they are not in contact with any surface (by neglecting air friction). The normal force acting on the object is zero. The downward acceleration is equal to the acceleration due to the gravity of the Earth. i.e ($a = g$). From equation (6.69) we get.

$$a = g \quad \therefore N = m(g - g) = 0.$$

This is called the state of weightlessness. When the lift falls (when the lift wire cuts) with downward acceleration $a = g$, the person inside the elevator is in the state of weightlessness or free fall. It is shown in Figure 6.23(d)

When the apple was falling from the tree it was weightless. As soon as it hit Newton's head, it gained weight! and Newton gained physics!

Weightlessness in satellites:

There is a wrong notion that the astronauts in satellites experience no gravitational force because they are far away from the Earth. Actually the Earth satellites that orbit very close to Earth experience only gravitational force. The astronauts inside the satellite also experience the same gravitational force. Because of this, they cannot exert any force on the floor of the satellite. Thus, the floor of the satellite also cannot exert any normal force on the astronaut. Therefore, the astronauts inside a satellite are in the state of weightlessness. Not only the astronauts, but all the objects in the satellite will be in the state of weightlessness which is similar to that of a free fall. It is shown in the Figure 6.24.

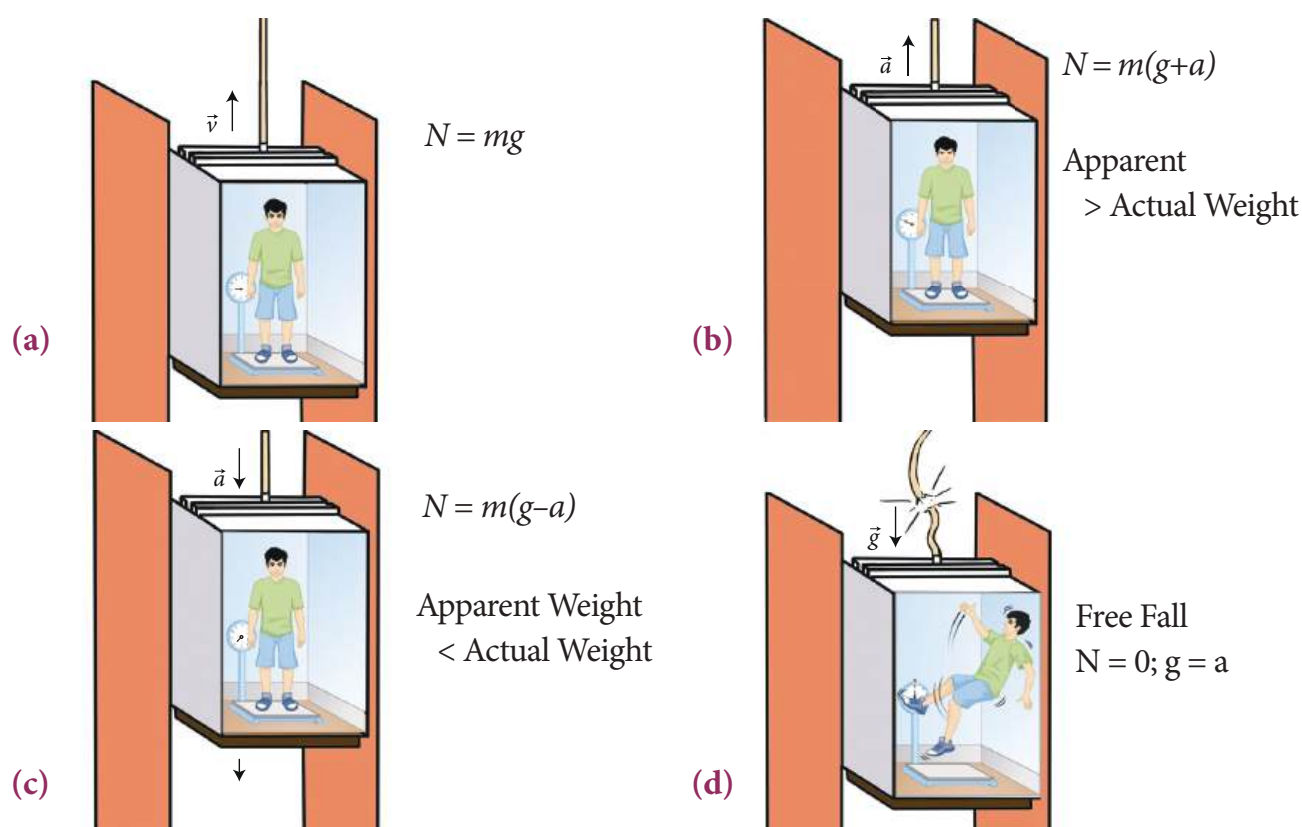


Figure 6.23 Apparent weight in the lift



Figure 6.24 The well known scientist Stephen Hawking in the state of weightlessness.
https://www.youtube.com/watch?v=OCsuHvv_D0s

6.5

ELEMENTARY IDEAS OF ASTRONOMY

Astronomy is one of the oldest sciences in the history of mankind. In the olden days, astronomy was an inseparable part of physical science. It contributed a lot to the development of physics in the 16th century. In fact Kepler's laws and Newton's theory of gravitation were formulated and verified using astronomical observations and data accumulated over the centuries by famous astronomers like Hipparchus, Aristarchus, Ptolemy, Copernicus and Tycho Brahe. Without Tycho Brahe's astronomical observations, Kepler's laws would not have emerged. Without Kepler's laws, Newton's theory of gravitation would not have been formulated.

It was mentioned in the beginning of this chapter that Ptolemy's geocentric model was replaced by Copernicus' heliocentric model. It is important to analyze and explain the

shortcoming of the geocentric model over heliocentric model.

6.5.1 Heliocentric system over geocentric system

When the motion of the planets are observed in the night sky by naked eyes over a period of a few months, it can be seen that the planets move eastwards and reverse their motion for a while and return to eastward motion again. This is called "retrograde motion" of planets.

Figure 6.25 shows the retrograde motion of the planet Mars. Careful observation for a period of a year clearly shows that Mars initially moves eastwards (February to June), then reverses its path and moves backwards (July, August, September). It changes its direction of motion once again and continues its forward motion (October onwards). In olden days, astronomers recorded the retrograde motion of all

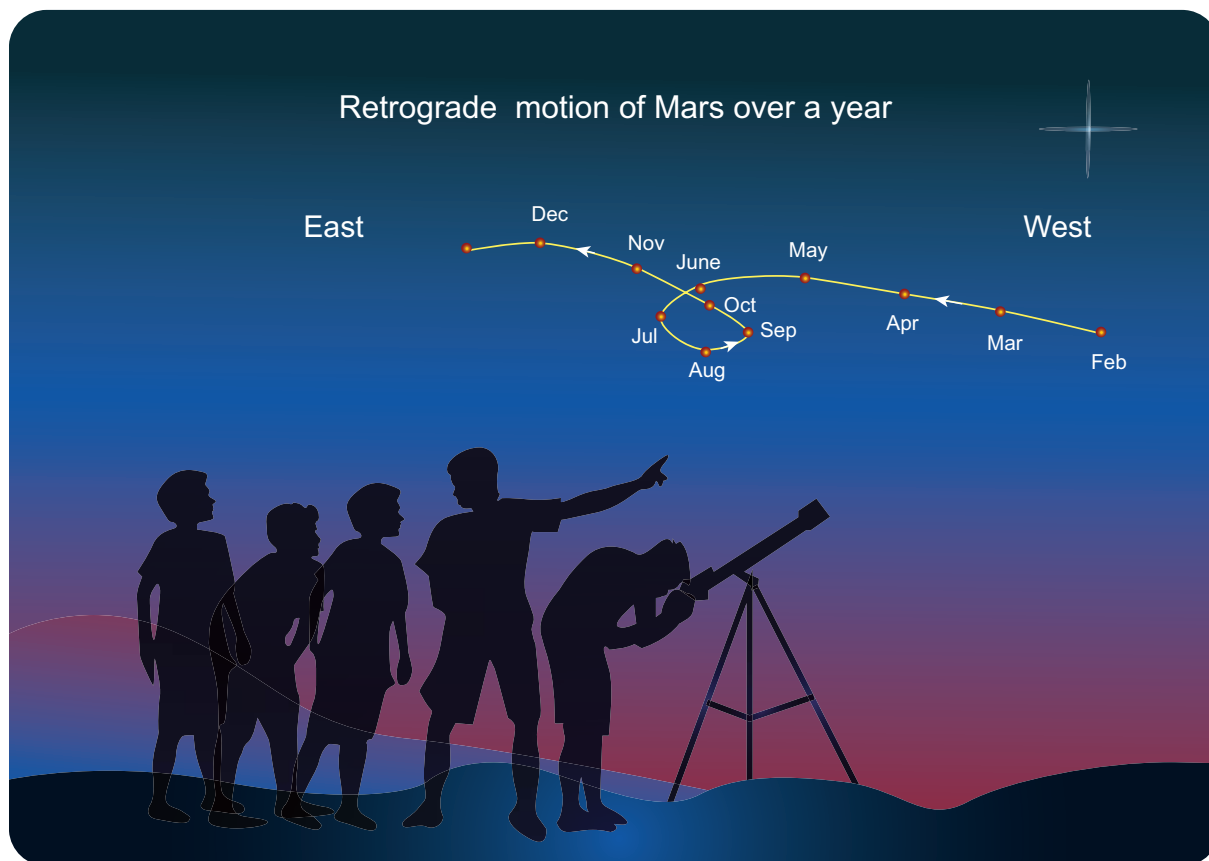


Figure 6.25 Retrograde motion of planets

visible planets and tried to explain the motion. According to Aristotle, the other planets and the Sun move around the Earth in the circular orbits. If it was really a circular orbit it was not known how the planet could reverse its motion for a brief interval. To explain this retrograde motion, Ptolemy introduced the concept of “epicycle” in his geocentric model. According to this theory, while the planet orbited the Earth, it also underwent another circular motion termed as “epicycle”. A combination of epicycle and circular motion around the Earth gave rise to retrograde motion of the planets with respect to Earth (Figure 6.26). Essentially Ptolemy retained the Earth centric idea of Aristotle and added the epicycle motion to it.

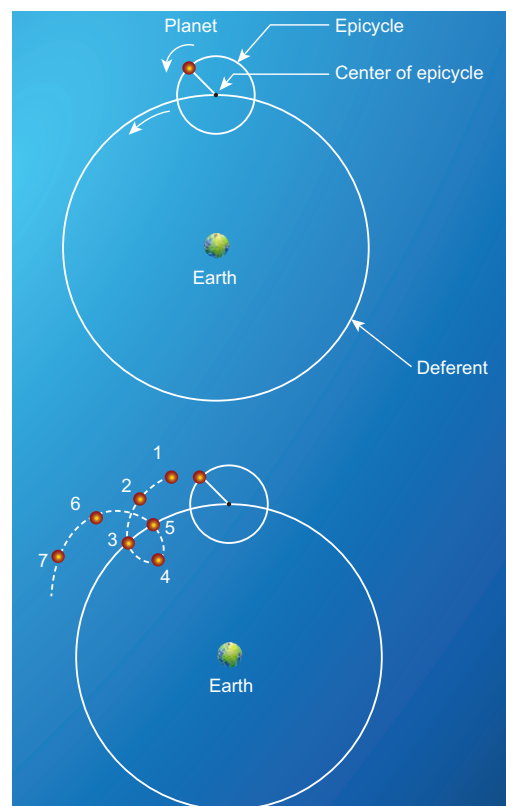


Figure 6.26 “Epicycle” motion of planetary objects around Earth, depicted with respect to months of observation.



But Ptolemy's model became more and more complex as every planet was found to undergo retrograde motion. In the 15th century, the Polish astronomer Copernicus proposed the heliocentric model to explain this problem in a simpler manner. According to this model, the Sun is at the center of the solar system and all planets orbited the Sun. The retrograde motion of planets with respect to Earth is because of the relative motion of the planet with respect to Earth. The retrograde motion from the heliocentric point of view is shown in Figure 6.27.

Figure 6.27 shows that the Earth orbits around the Sun faster than Mars. Because of the relative motion between Mars and Earth, Mars appears to move backwards from July to October. In the same way the retrograde motion of all other planets was explained successfully by the Copernicus model. It was because of its simplicity, the heliocentric model slowly replaced the geocentric model. Historically, if any natural phenomenon has one or more explanations, the simplest one is usually accepted. Though this was not the only reason to disqualify the geocentric model, a detailed discussion

on correctness of the Copernicus model over to Ptolemy's model can be found in astronomy books.



ACTIVITY

Students are encouraged to observe the motion of the planet Mars by naked eye and identify its retrograde motion. As mentioned above, to observe the retrograde motion six to seven months are required. So students may start their observation of Mars from the month of June and continue till April next year. Mars is the little bright planet with reddish color. The position of the planet Mars in the sky can be easily taken from 'Google'.

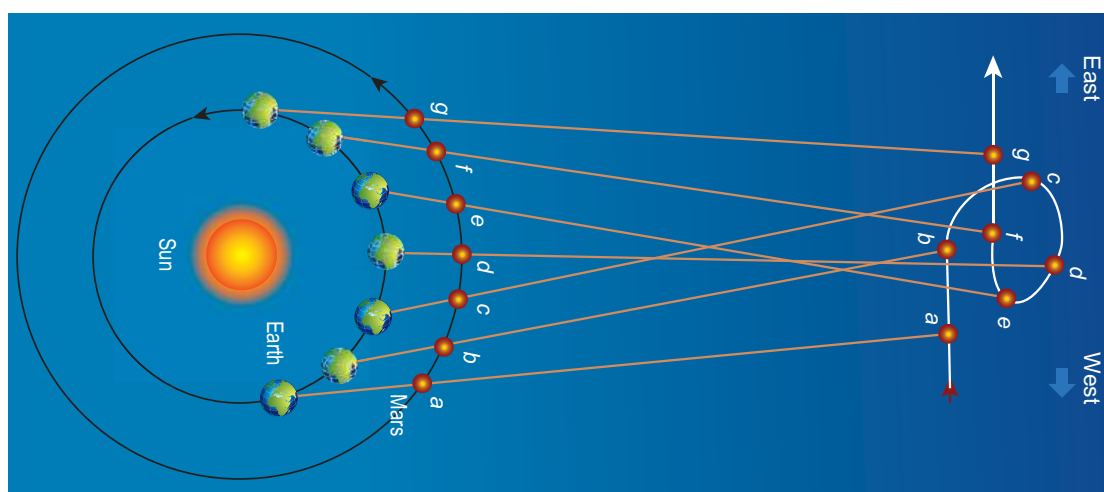


Figure 6.27 'Retrograde motion' in heliocentric model



6.5.2 Kepler's Third Law and The Astronomical Distance

When Kepler derived his three laws, he strongly relied on Tycho Brahe's astronomical observation. In his third law, he formulated the relation between the distance of a planet from the Sun to the time period of revolution of the planet. Astronomers cleverly used geometry and trigonometry to calculate the distance of a planet from the Sun in terms of the distance between Earth and Sun. Here we can see how the distance of Mercury and Venus from the Sun were measured. The Venus and Mercury, being inner planets with respect to Earth, the maximum angular distance they can subtend at a point on Earth with respect to the Sun is 46 degree for Venus and 22.5 degree for Mercury. It is shown in the Figure 6.28

Figure 6.29 shows that when Venus is at maximum elongation (i.e., 46 degree) with respect to Earth, Venus makes 90 degree to Sun. This allows us to find the distance between Venus and Sun. The distance between Earth and Sun is taken as one Astronomical unit (1 AU).

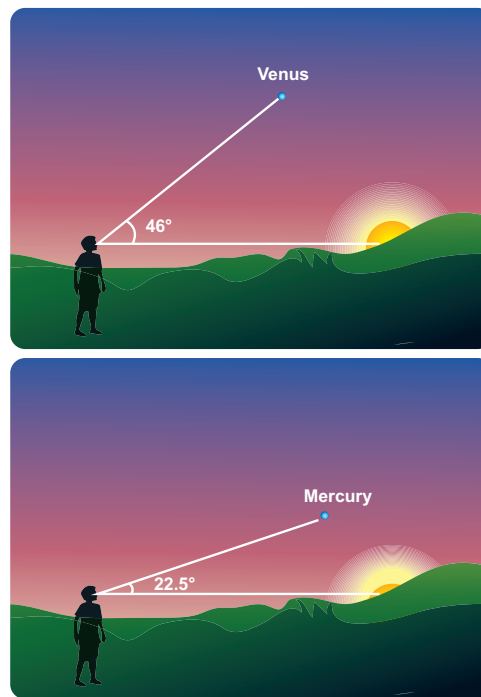


Figure 6.28 Angle of elevation for Venus and Mercury from horizon

The trigonometric relation satisfied by this right angled triangle is shown in Figure 6.29.

$$\sin \theta = \frac{r}{R}$$

where $R = 1 \text{ AU}$.

$$r = R \sin \theta = (1 \text{ AU})(\sin 46^\circ)$$

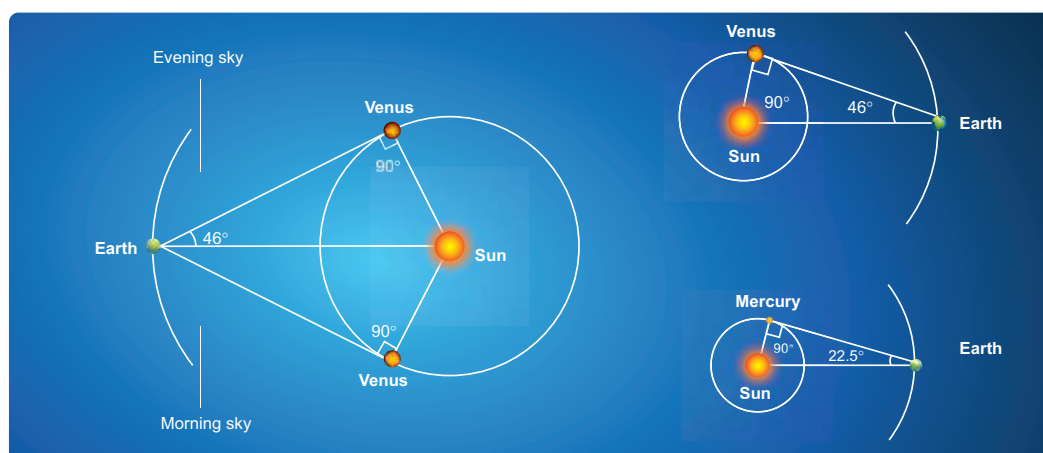


Figure 6.29 Angle of elevation for Mercury from horizon

Here $\sin 46^\circ = 0.72$. Hence Venus is at a distance of 0.72 AU from Sun. Similarly, the distance between Mercury (θ is 22.5 degree) and Sun is calculated as 0.38 AU. To find the distance of exterior planets like Mars and Jupiter, a slightly different method is used. The distances of planets from the Sun is given in the table below.

Table 6.2 a^3/T^2 for different planets

Planet	semi major axis of the orbit(a)	Period T (days)	a^3/T^2
Mercury	0.389 AU	87.77	7.64
Venus	0.724 AU	224.70	7.52
Earth	1.000 AU	365.25	7.50
Mars	1.524 AU	686.98	7.50
Jupiter	5.200 AU	4332.62	7.49
Saturn	9.510 AU	10,759.20	7.40

It is to be noted that to verify the Kepler's law we need only high school level geometry and trigonometry.



ACTIVITY

Venus can be observed with the naked eye. We can see Venus during sunrise or sunset. Students are encouraged to observe the motion of Venus and verify that the maximum elevation is at 46 degree and calculate the distance of Venus from the Sun. As pointed out already Google or Stellarium will be helpful in locating the position of Venus in the sky.

6.5.3 Measurement of radius of the Earth

Around 225 B.C a Greek librarian "Eratosthenes" who lived at Alexandria measured the radius of the Earth with a small error when compared with results using modern measurements. The technique he used involves lower school geometry and

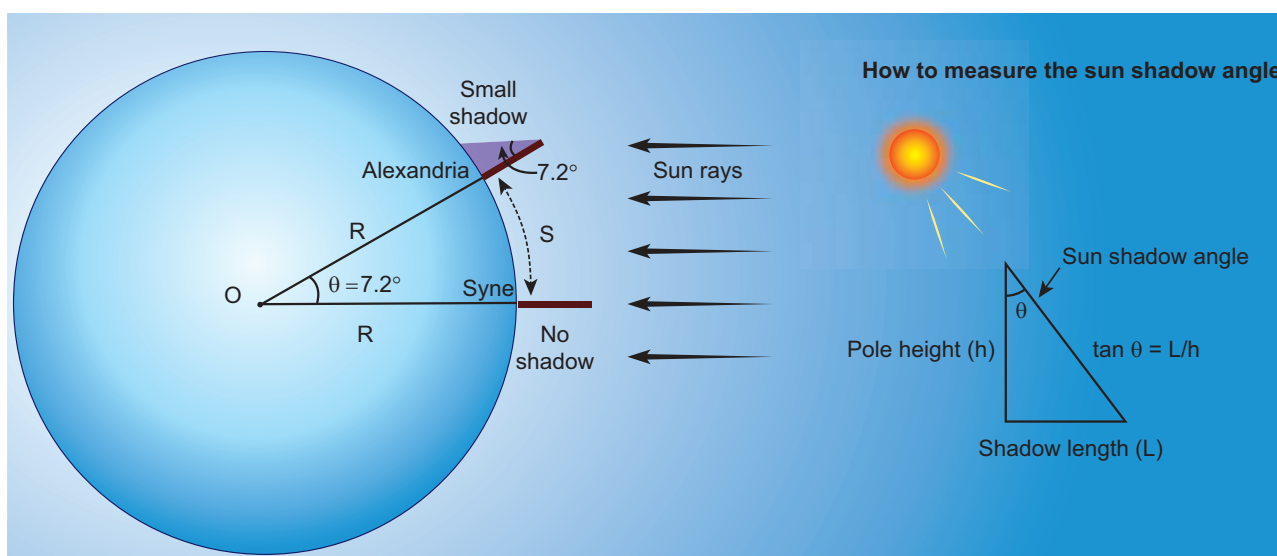


Figure 6.30 Measuring radius of The Earth



brilliant insight. He observed that during noon time of summer solstice the Sun's rays cast no shadow in the city Syene which was located 500 miles away from Alexandria. At the same day and same time he found that in Alexandria the Sun's rays made 7.2 degree with local vertical as shown in the Figure 6.30. He realized that this difference of 7.2 degree was due to the curvature of the Earth.

The angle 7.2 degree is equivalent to $\frac{1}{8}$ radian. So $\theta = \frac{1}{8}$ rad;

If S is the length of the arc between the cities of Syene and Alexandria, and if R is radius of Earth, then

$$S = R\theta = 500 \text{ miles,}$$

so radius of the Earth

$$R = \frac{500}{\theta} \text{ miles}$$

$$R = \frac{500}{\left(\frac{1}{8}\right)} \text{ miles}$$

$$R = 4000 \text{ miles}$$

1 mile is equal to 1.609 km. So, he measured the radius of the Earth to be equal to $R = 6436$ km, which is amazingly close to the correct value of 6378 km.

The distance of the Moon from Earth was measured by a famous Greek astronomer Hipparchus in the 3rd century BC.



ACTIVITY

To measure the radius of the Earth, choose two different places (schools) that are separated by at least 500 km. It is important to note that these two places have to be along the same longitude of the Earth (For example Hosur and Kanyakumari lie along the same longitude of 77.82° E). Take poles of known length (h) and fix them vertically in the ground (it may be in the school playgrounds) at both the places. At exactly noon in both the places the length of the shadow (L) cast by each pole has to be noted down. Draw the picture like in Figure 6.30. By using the equation $\tan \theta = \frac{L}{h}$, the angle in radian can be found at each place. The difference in angle (θ') is due to the curvature of the Earth. Now the distance between the two schools can be obtained from 'Google maps'. Divide the distance with the angle (θ' in radians) which will give the radius of the Earth.

6.5.4 Interesting Astronomical Facts

1. Lunar eclipse and measurement of shadow of Earth

On January 31, 2018 there was a total lunar eclipse which was observed from various places including Tamil Nadu. It is possible to measure the radius of shadow of the Earth at the point where the Moon crosses. Figure 6.31 illustrates this.

When the Moon is inside the umbra shadow, it appears red in color. As soon as the Moon exits from the umbra

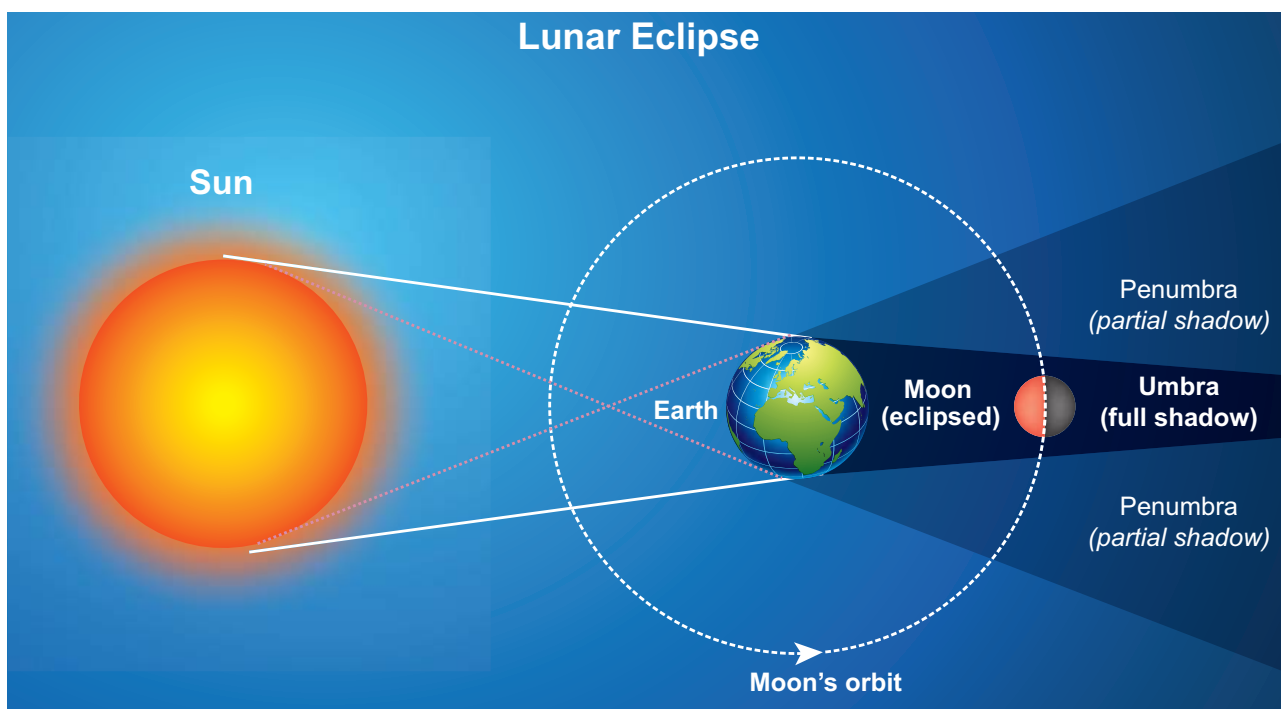


Figure 6.31 Total lunar eclipse

shadow, it appears in crescent shape. Figure 6.32 is the photograph taken by digital camera during Moon's exit from the umbra shadow.



Figure 6.32 Image of the Moon when it exits from umbra shadow

By finding the apparent radii of the Earth's umbra shadow and the Moon, the ratio of these radii can be calculated. This is shown in Figures 6.33 and 6.34.

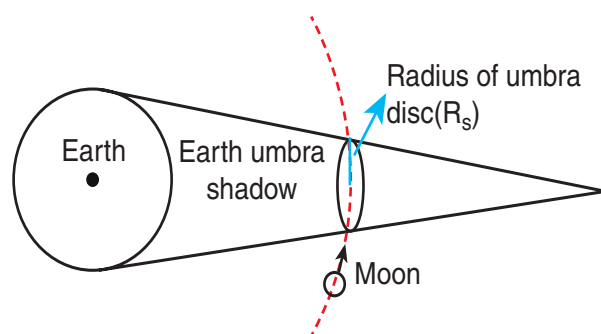


Figure 6.33 Schematic diagram of umbra disk radius

The apparent radius of Earth's umbra shadow = $R_s = 13.2$ cm

The apparent radius of the Moon = $R_m = 5.15$ cm

The ratio $\frac{R_s}{R_m} \approx 2.56$

The radius of the Earth's umbra shadow is $R_s = 2.56 \times R_m$

The radius of Moon $R_m = 1737$ km

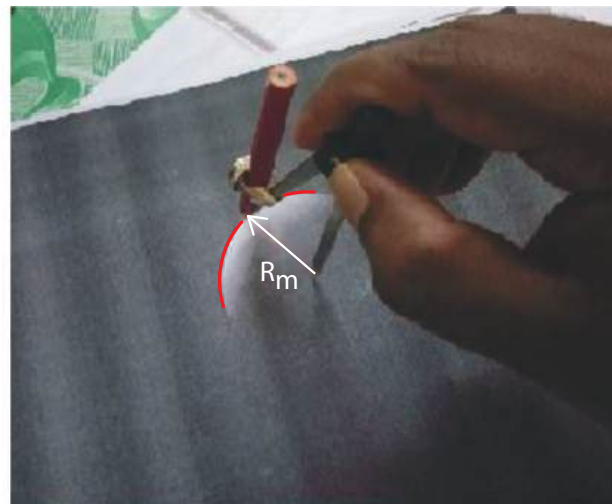
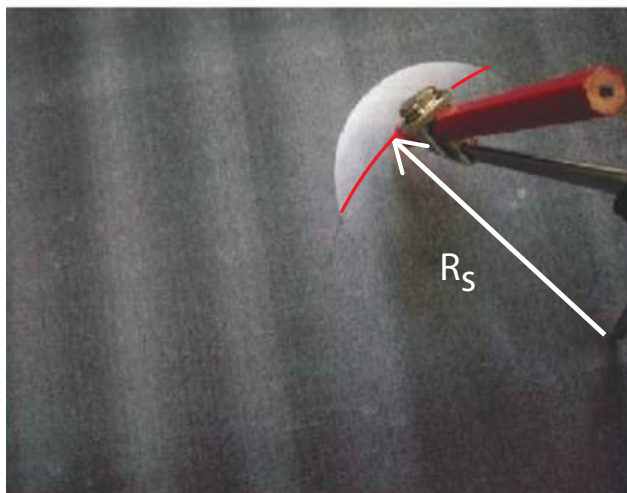


Figure 6.34 Calculation of umbra radius

The radius of the Earth's umbra shadow is $R_s = 2.56 \times 1737 \text{ km} \cong 4446 \text{ km}$.

The correct radius is 4610 km.

The percentage of error in the calculation

$$= \frac{4610 - 4446}{4610} \times 100 = 3.5\%.$$

The error will reduce if the pictures taken using a high quality telescope are used. It is to be noted that this calculation is done using very simple mathematics.

Early astronomers proved that Earth is spherical in shape by looking at the shape of the shadow cast by Earth on the Moon during lunar eclipse.

2. Why there are no lunar eclipse and solar eclipse every month?

If the orbits of the Moon and Earth lie on the same plane, during full Moon of every month, we can observe lunar eclipse. If this is so during new Moon we can

observe solar eclipse. But Moon's orbit is tilted 5° with respect to Earth's orbit. Due to this 5° tilt, only during certain periods of the year, the Sun, Earth and Moon align in straight line leading to either lunar eclipse or solar eclipse depending on the alignment. This is shown in Figure 6.35

3. Why do we have seasons on Earth?

The common misconception is that 'Earth revolves around the Sun, so when the Earth is very far away, it is winter and when the Earth is nearer, it is summer'. Actually, the seasons in the Earth arise due to the rotation of Earth around the Sun with 23.5° tilt. This is shown in Figure 6.36

Due to this 23.5° tilt, when the northern part of Earth is farther to the Sun, the southern part is nearer to the Sun. So when it is summer in the southern hemisphere, the northern hemisphere experience winter.

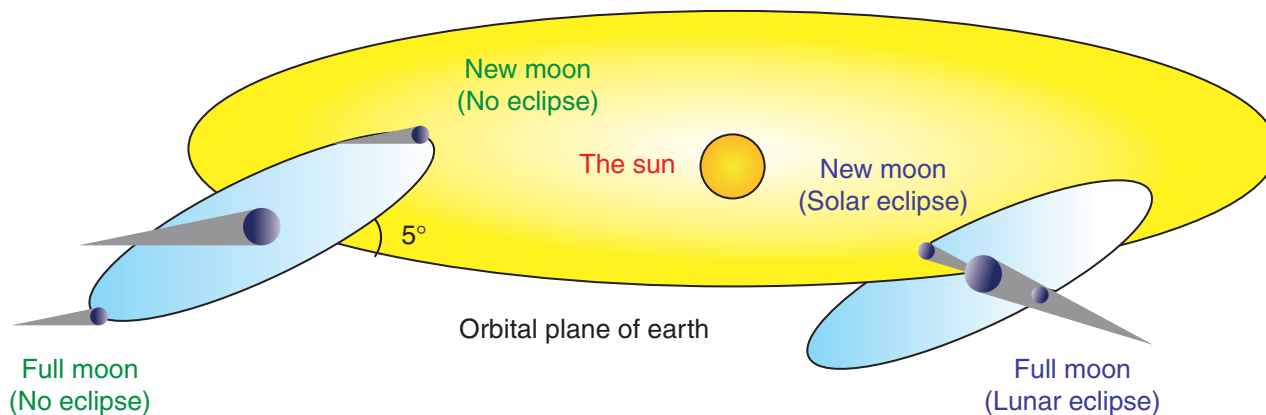


Figure 6.35 Orbital tilt of the Moon

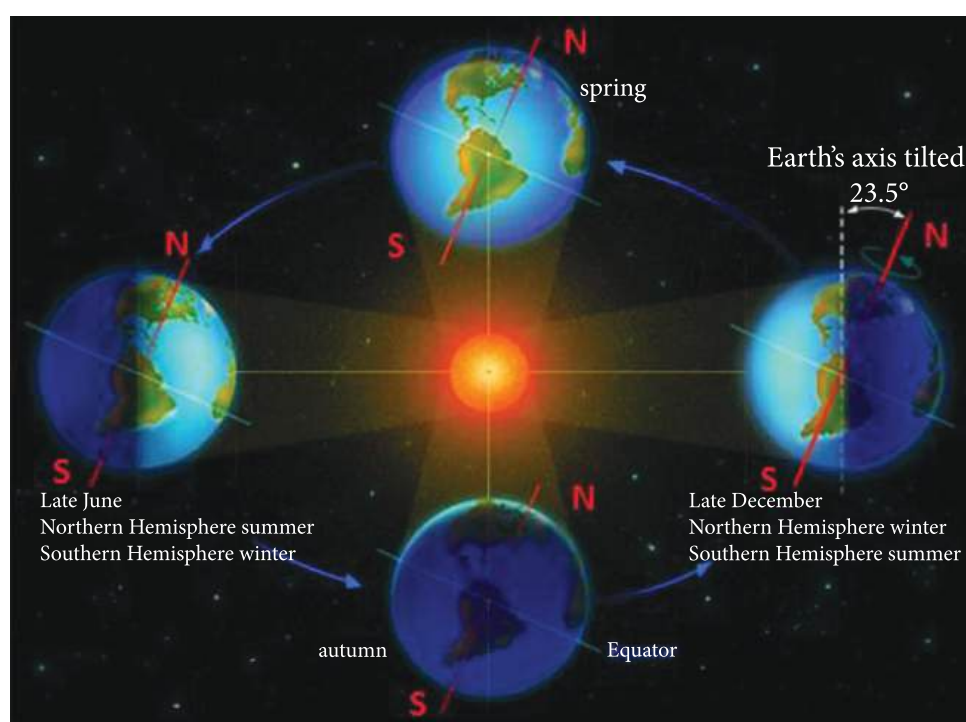


Figure 6.36 Seasons on Earth

4. Star's apparent motion and spinning of the Earth

The Earth's spinning motion can be proved by observing star's position over a night. Due to Earth's spinning motion, the stars in sky appear to move in circular motion about the pole star as shown in Figure 6.37



Pole star is a star located exactly above the Earth's axis of rotation, hence it appears to be stationary. The Star Polaris is our pole star.

Point to ponder

Using Sun rays and shadows, How will you prove that the Earth's tilt is 23.5° ?

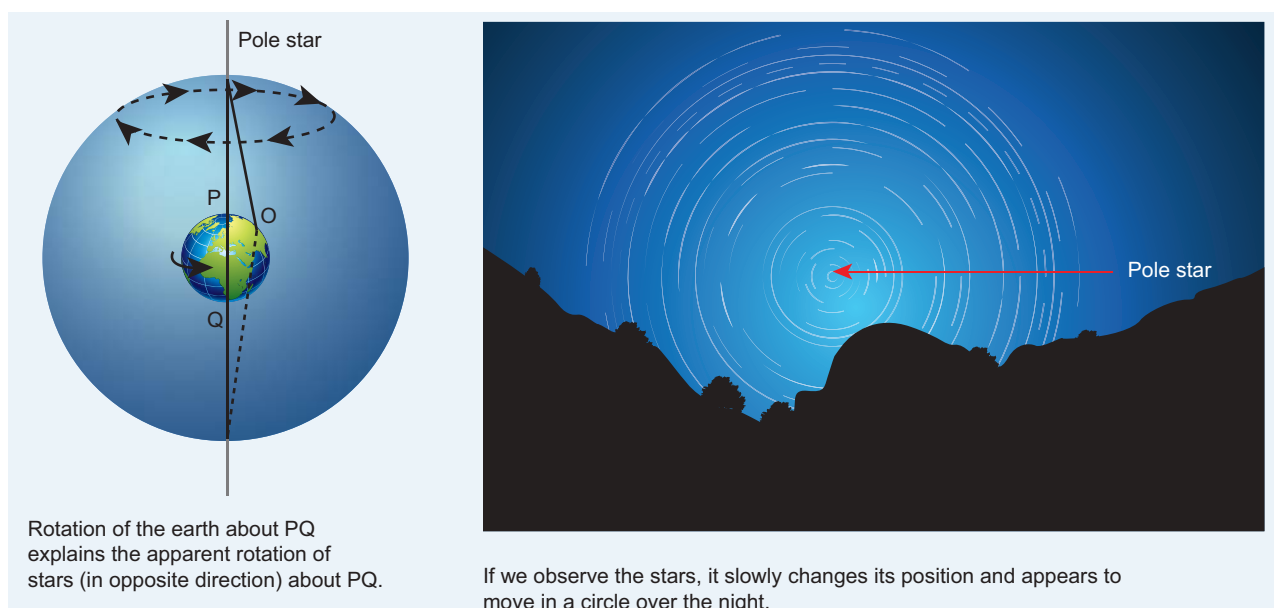


Figure 6.37 Star's apparent circular motion due to Earth's rotation.

6.5.5 Recent developments of astronomy and gravitation

Till the 19th century astronomy mainly depended upon either observation with the naked eye or telescopic observation. After the discovery of the electromagnetic spectrum at the end of the 19th century, our understanding of the universe increased enormously. Because of this development in the late 19th century it was found that Newton's law of gravitation could not explain certain phenomena and showed some discrepancies. Albert Einstein formulated his 'General theory of relativity' which was one of the most successful theories of 20th century in the field of gravitation.

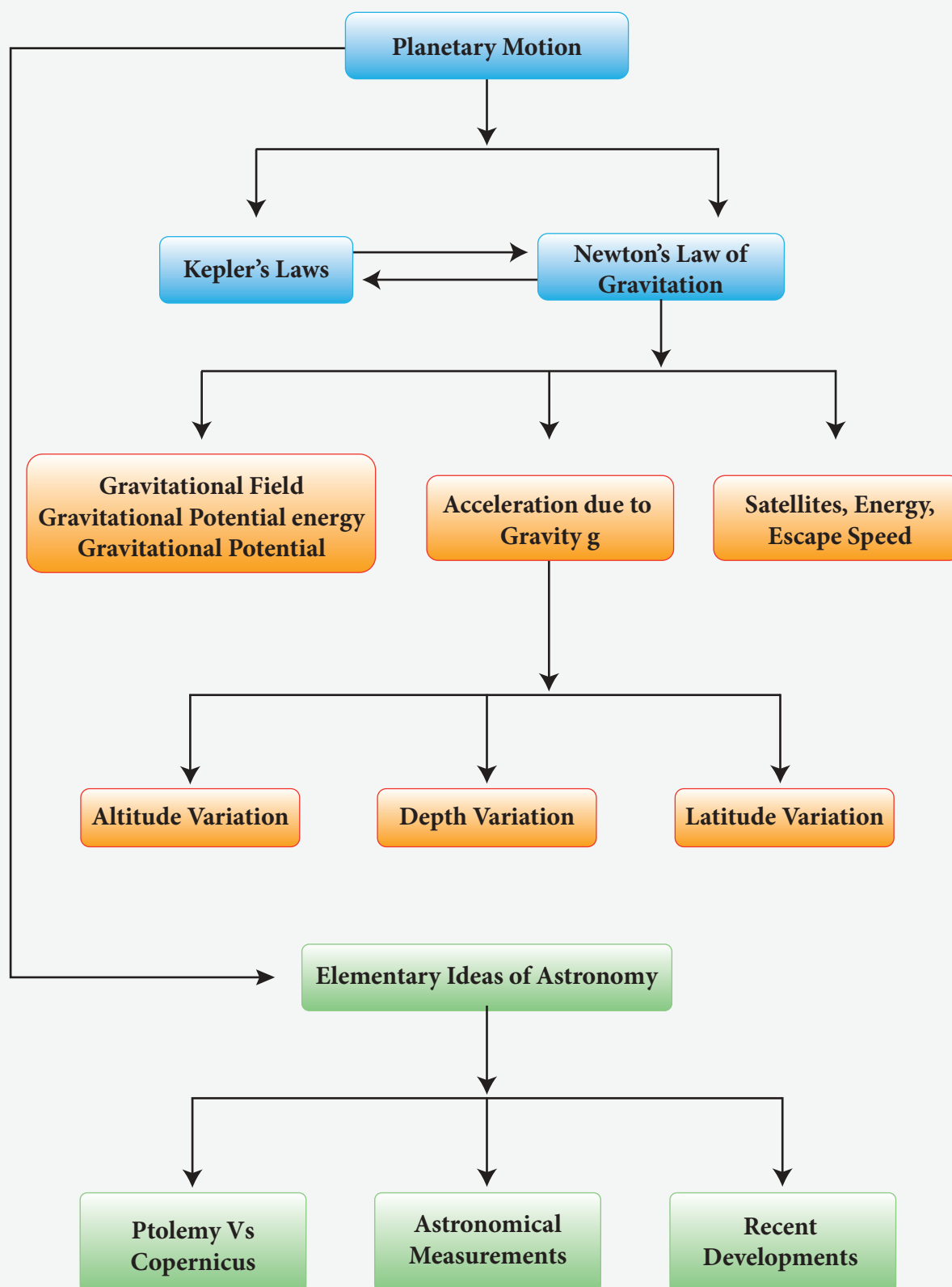
In the twentieth century both astronomy and gravitation merged together and have grown in manifold. The birth and death of stars were more clearly understood. Many Indian physicists made very important contributions to the field of astrophysics and gravitation.

Subramanian Chandrasekar formulated the theory of black holes and explained the life of stars. These studies brought him the Nobel prize in the year 1983. Another very notable Indian astrophysicist Meghnad Saha discovered the ionization formula which was useful in classifying stars. This formula is now known as "Saha ionization formula". In the field of gravitation Amal Kumar Raychaudhuri solved an equation now known as "Raychaudhuri equation" which was a very important contribution. Another notable Indian Astrophysicist Jayant V Narlikar made pioneering contribution in the field of astrophysics and has written interesting books on astronomy and astrophysics. IUCAA (Inter University Center for Astronomy and Astrophysics) is one of the important Indian research institutes where active research in astrophysics and gravitation are conducted. The institute was founded by Prof. J.V. Narlikar. Students are encouraged to read more about the recent developments in these fields.

SUMMARY

- The motion of planets can be explained using Kepler's laws.
- **Kepler's first law:** All the planets in the solar system orbit the Sun in elliptical orbits with the Sun at one of the foci.
- **Kepler's second law:** The radial vector line joining the Sun to a planet sweeps equal areas in equal intervals of time.
- **Kepler's third law:** The ratio of the square of the time period of planet to the cubic power of semi major axis is constant for all the planets in the solar system.
- **Newton's law of gravitation** states that the gravitational force between two masses is directly proportional to product of masses and inversely proportional to square of the distance between the masses. In vector form it is given by $\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r}$
- Gravitational force is a central force.
- Kepler's laws can be derived from Newton's law of gravitation.
- The gravitational field due to a mass m at a point which is at a distance r from mass m is given by $\vec{E} = -\frac{Gm}{r^2}\hat{r}$. It is a vector quantity.
- The gravitational potential energy of two masses is given by $U = -\frac{Gm_1m_2}{r}$. It is a scalar quantity.
- The gravitational potential at a point which is at a distance r from mass m is given by $V = -\frac{Gm}{r}$. It is a scalar quantity.
- The acceleration due to Earth's gravity decreases as altitude increases and as depth increases.
- Due to rotation of the Earth, the acceleration due to gravity is maximum at poles and minimum at Earth's equator.
- The (escape) speed of any object required to escape from the Earth's gravitational field is $v_e = \sqrt{2gR_e}$. It is independent of mass of the object.
- The energy of the satellite is negative. It implies that the satellite is bound to Earth's gravitational force.
- Copernicus model explained that retrograde motion is due to relative motion between planets. This explanation is simpler than Ptolemy's epicycle explanation which is complicated
- Copernicus and Kepler measured the distance between a planet and the Sun using simple geometry and trigonometry.
- 2400 years ago, Eratosthenes measured the radius of the Earth using simple geometry and trigonometry.

CONCEPT MAP



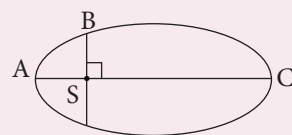


EVALUATION



I. Multiple Choice Questions

- The linear momentum and position vector of the planet is perpendicular to each other at
 - perihelion and aphelion
 - at all points
 - only at perihelion
 - no point
- If the masses of the Earth and Sun suddenly double, the gravitational force between them will
 - remain the same
 - increase 2 times
 - increase 4 times
 - decrease 2 times
- A planet moving along an elliptical orbit is closest to the Sun at distance r_1 and farthest away at a distance of r_2 . If v_1 and v_2 are linear speeds at these points respectively. Then the ratio $\frac{v_1}{v_2}$ is
(NEET 2016)
 - $\frac{r_2}{r_1}$
 - $\left(\frac{r_2}{r_1}\right)^2$
 - $\frac{r_1}{r_2}$
 - $\left(\frac{r_1}{r_2}\right)^2$
- The time period of a satellite orbiting Earth in a circular orbit is independent of.
 - Radius of the orbit
 - The mass of the satellite
 - Both the mass of the satellite and radius of the orbit
 - Neither the mass of the satellite nor the radius of its orbit
- If the distance between the Earth and Sun were to be doubled from its present value, the number of days in a year would be
 - 64.5
 - 1032
 - 182.5
 - 730
- According to Kepler's second law, the radial vector to a planet from the Sun sweeps out equal areas in equal intervals of time. This law is a consequence of
 - conservation of linear momentum
 - conservation of angular momentum
 - conservation of energy
 - conservation of kinetic energy
- The gravitational potential energy of the Moon with respect to Earth is
 - always positive
 - always negative
 - can be positive or negative
 - always zero
- The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A , K_B and K_C respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then
(NEET 2018)
 - $K_A > K_B > K_C$
 - $K_B < K_A < K_C$
 - $K_A < K_B < K_C$
 - $K_B > K_A > K_C$





9. The work done by the Sun's gravitational force on the Earth is
(a) always zero
(b) always positive
(c) can be positive or negative
(d) always negative
10. If the mass and radius of the Earth are both doubled, then the acceleration due to gravity g'
(a) remains same (b) $\frac{g}{2}$
(c) $2g$ (d) $4g$
11. The magnitude of the Sun's gravitational field as experienced by Earth is
(a) same over the year
(b) decreases in the month of January and increases in the month of July
(c) decreases in the month of July and increases in the month of January
(d) increases during day time and decreases during night time.
12. If a person moves from Chennai to Trichy, his weight
(a) increases
(b) decreases
(c) remains same
(d) increases and then decreases
13. An object of mass 10 kg is hanging on a spring scale which is attached to the roof of a lift. If the lift is in free fall, the reading in the spring scale is
(a) 98 N (b) zero
(c) 49 N (d) 9.8 N
14. If the acceleration due to gravity becomes 4 times its original value, then escape speed
(a) remains same
(b) 2 times of original value

- (c) becomes halved
(d) 4 times of original value

15. The kinetic energy of the satellite orbiting around the Earth is
(a) equal to potential energy
(b) less than potential energy
(c) greater than kinetic energy
(d) zero

Answers

- 1) a 2) c 3) a 4) b 5) b
6) b 7) b 8) a 9) c 10) b
11) c 12) b 13) b 14) b 15) b

II. Short Answer Questions

1. State Kepler's three laws.
2. State Newton's Universal law of gravitation.
3. Will the angular momentum of a planet be conserved? Justify your answer.
4. Define the gravitational field. Give its unit.
5. What is meant by superposition of gravitational field?
6. Define gravitational potential energy.
7. Is potential energy the property of a single object? Justify.
8. Define gravitational potential.
9. What is the difference between gravitational potential and gravitational potential energy?
10. What is meant by escape speed in the case of the Earth?
11. Why is the energy of a satellite (or any other planet) negative?
12. What are geostationary and polar satellites?
13. Define weight



14. Why is there no lunar eclipse and solar eclipse every month?
15. How will you prove that Earth itself is spinning?

III. Long Answer Questions

1. Discuss the important features of the law of gravitation.
2. Explain how Newton arrived at his law of gravitation from Kepler's third law.
3. Explain how Newton verified his law of gravitation.
4. Derive the expression for gravitational potential energy.
5. Prove that at points near the surface of the Earth, the gravitational potential energy of the object is $U = mgh$
6. Explain in detail the idea of weightlessness using lift as an example.
7. Derive an expression for escape speed.
8. Explain the variation of g with latitude.
9. Explain the variation of g with altitude.
10. Explain the variation of g with depth from the Earth's surface.
11. Derive the time period of satellite orbiting the Earth.
12. Derive an expression for energy of satellite.
13. Explain in detail the geostationary and polar satellites.
14. Explain how geocentric theory is replaced by heliocentric theory using the idea of retrograde motion of planets.
15. Explain in detail the Eratosthenes method of finding the radius of Earth.

16. Describe the measurement of Earth's shadow (umbra) radius during total lunar eclipse

IV. Exercises

1. An unknown planet orbits the Sun with distance twice the semi major axis distance of the Earth's orbit. If the Earth's time period is T_1 , what is the time period of this unknown planet?

$$\text{Ans: } T_2 = 2\sqrt{2}T_1$$

2. Assume that you are in another solar system and provided with the set of data given below consisting of the planets' semi major axes and time periods. Can you infer the relation connecting semi major axis and time period?

Planet (imaginary)	Time period (T) (in year)	Semi major axis (a) (in AU)
Kurinji	2	8
Mullai	3	18
Marutham	4	32
Neithal	5	50
Paalai	6	72

$$\text{Ans: } a \propto 2T^2$$

3. If the masses and mutual distance between the two objects are doubled, what is the change in the gravitational force between them?

$$\text{Ans: No change}$$

4. Two bodies of masses m and $4m$ are placed at a distance r . Calculate the gravitational potential at a point on the line joining them where the gravitational field is zero.

$$\text{Ans: } V = -\frac{9Gm}{r}$$



5. If the ratio of the orbital distance of two planets $\frac{d_1}{d_2} = 2$, what is the ratio of gravitational field experienced by these two planets?

Ans: $E_2 = 4 E_1$

6. The Moon Io orbits Jupiter once in 1.769 days. The orbital radius of the Moon Io is 421700 km. Calculate the mass of Jupiter?

Ans: $1.898 \times 10^{27} \text{ kg}$

7. If the angular momentum of a planet is given by $\vec{L} = 5t^2\hat{i} - 6t\hat{j} + 3\hat{k}$. What is the torque experienced by the planet? Will the torque be in the same direction as that of the angular momentum?

Ans: $\vec{\tau} = 10t\hat{i} - 6\hat{j}$

8. Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. Calculate the speed of each particle

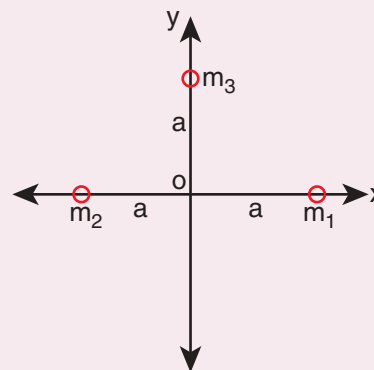
Ans: $\frac{1}{2} \sqrt{\frac{GM}{R}(1+2\sqrt{2})}$

9. Suppose unknowingly you wrote the universal gravitational constant value as $G = 6.67 \times 10^{11}$ instead of the correct value $G = 6.67 \times 10^{-11}$, what is the acceleration due to gravity g' for this incorrect G ? According to this new acceleration due to gravity, what will be your weight W' ?

Ans: $g' = 10^{22} g, W' = 10^{22} W$

10. Calculate the gravitational field at point O due to three masses m_1, m_2 and m_3 whose positions are given by the following figure. If the masses m_1

and m_2 are equal what is the change in gravitational field at the point O ?



Ans: $\vec{E} = \frac{G}{a^2} [(m_1 - m_2)\hat{i} + m_3\hat{j}]$

if $m_1 = m_2, \vec{E} = \frac{G}{a^2} [m_3\hat{j}]$

11. What is the gravitational potential energy of the Earth and Sun? The Earth to Sun distance is around 150 million km. The mass of the Earth is $5.9 \times 10^{24} \text{ kg}$ and mass of the Sun is $1.9 \times 10^{30} \text{ kg}$.

Ans: $U = -49.84 \times 10^{32} \text{ Joule}$

12. Earth revolves around the Sun at 30 km s^{-1} . Calculate the kinetic energy of the Earth. In the previous example you calculated the potential energy of the Earth. What is the total energy of the Earth in that case? Is the total energy positive? Give reasons.

Ans: $K.E = 26.5 \times 10^{32} \text{ J}$

$E = -23.29 \times 10^{32} \text{ J}$

(-) ve implies that Earth is bounded with Sun

13. An object is thrown from Earth in such a way that it reaches a point at infinity with non-zero kinetic energy

$\left[\text{K.E}(r = \infty) = \frac{1}{2} M v_{\infty}^2 \right]$, with what velocity should the object be thrown from Earth?

$$\text{Ans: } v_e = \sqrt{v_{\infty}^2 + 2gR_E}$$

14. Suppose we go 200 km above and below the surface of the Earth, what are the g values at these two points? In which case, is the value of g small?

$$\begin{aligned} \text{Ans: } g_{\text{down}} &= 0.96 \text{ g} \\ g_{\text{up}} &= 0.94 \text{ g} \end{aligned}$$

15. Calculate the change in g value in your district of Tamil nadu. (Hint: Get the latitude of your district of Tamil nadu from the Google). What is the difference in g values at Chennai and Kanyakumari?

$$\begin{aligned} \text{Ans: } g_{\text{chennai}} &= 9.7677 \text{ m s}^{-2} \\ g_{\text{Kanyakumari}} &= 9.7667 \text{ m s}^{-2} \\ \Delta g &= 0.001 \text{ m s}^{-2} \end{aligned}$$

V. Conceptual Questions

- In the following, what are the quantities which that are conserved?
 - Linear momentum of planet
 - Angular momentum of planet
 - Total energy of planet
 - Potential energy of a planet
- The work done by Sun on Earth in one year will be
 - Zero
 - Non zero
 - positive
 - negative
- The work done by Sun on Earth at any finite interval of time is

- positive, negative or zero
- Strictly positive
- Strictly negative
- It is always zero

- If a comet suddenly hits the Moon and imparts energy which is more than the total energy of the Moon, what will happen?
- If the Earth's pull on the Moon suddenly disappears, what will happen to the Moon?
- If the Earth has no tilt, what happens to the seasons of the Earth?
- A student was asked a question 'why are there summer and winter for us? He replied as 'since Earth is orbiting in an elliptical orbit, when the Earth is very far away from the Sun(aphelion) there will be winter, when the Earth is nearer to the Sun(perihelion) there will be winter'. Is this answer correct? If not, what is the correct explanation for the occurrence of summer and winter?
- The following photographs are taken from the recent lunar eclipse which occurred on January 31, 2018. Is it possible to prove that Earth is a sphere from these photographs?





BOOKS FOR REFERENCE

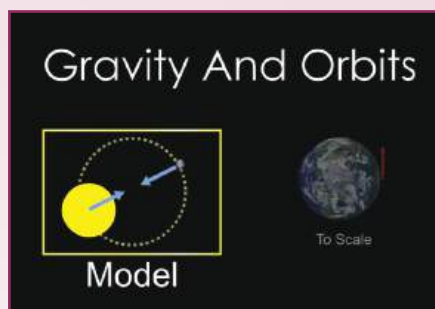
1. Mechanics by Charles Kittel, Walter Knight, Malvin Ruderman, Carl Helmholtz and Moyer
2. Newtonian Mechanics by A.P. French
3. Introduction to Mechanics by Daniel Kepler and Robert Kolenkow
4. Mechanics by Somnath Datta
5. Concepts of Physics volume 1 and Volume 2 by H.C. Verma
6. Physics for Scientist and Engineers with Modern physics by Serway and Jewett
7. Physics for Scientist and Engineers by Paul Tipler and Gene Mosca
8. Physics for the Inquiring Mind by Eric Rogers
9. Fundamental laws of Mechanics by Irodov.
10. Question and Problems in School Physics by Tarasov and Tarasova



ICT CORNER

Gravitation

Through this activity you will be able to learn about the gravitational force and orbital paths.



STEPS:

- Click the URL or scan the QR code to launch the activity page. Click on the box labelled “Model” to start the activity.
- In the activity window, a diagram of sun and earth is given. Click the play icon to see the motion of earth.
- We can change the objects by selecting objects from the table given in the right side window.
- The path of the gravity, velocity and the object in motion can be viewed. Check on the relevant boxes given in the table.

Step1



Step2



Step3



Step4



URL:

https://phet.colorado.edu/sims/html/gravity-and-orbits/latest/gravity-and-orbits_en.html

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.



UNIT 7

PROPERTIES OF MATTER

Many of the greatest advances that have been made from the beginning of the world to the present time have been made in the earnest desire to turn the knowledge of the properties of matter to some purpose useful to mankind— Lord Kelvin

LEARNING OBJECTIVES

In this unit, the student is exposed to

- inter atomic or intermolecular forces in matter
- stress, strain and elastic modulus
- surface tension
- viscosity
- properties of fluids and their applications



7.1

INTRODUCTION

One of the oldest dams in the world is Kallanai (கல்லணை) located at Trichy. Kallanai was built across river Kaveri for the purpose of irrigation. During heavy floods, the velocity of water is generally very high in river Kaveri. The stability and utility of

Kallanai dam reveal the intuitive scientific understanding of Tamils who designed this dam as early as 2nd century AD. The other example known for insightful constructions of the past is the pyramids in Egypt. The flyovers and overbridges are common worldwide today. Heavy vehicles ply over and hence, the bridges are always under stress. Without effective design using suitable materials, the bridges and flyovers will not



Kallanai (கல்லணை)



be stable. The growth of human civilization is due to the understanding of various forms of matter (solid, liquid, and gas).

The study of properties of matter is very essential in selecting a material for a specific application. For example, in technology, the materials used for space applications should be of lightweight but should be strong. Materials used for artificial human organ replacements should be biocompatible. Artificial body fluids are used as tissue substitute for radiotherapy analysis in medicine. Fluids used as lubricants or fuel should possess certain properties. Such salient macroscopic properties are decided by the microscopic phenomena within matter. This unit deals with the properties of solids and fluids and the laws governing the behaviour of matter.

7.2

MICROSCOPIC UNDERSTANDING OF VARIOUS STATES OF MATTER

Even though various forms of matter such as solid food, liquids like water, and the air that we breathe are familiar in the day – to – day lifestyle for the past several thousand years, the microscopic understanding of solids, liquids, and gases was established only in 20th century. In the universe, everything is made up of atoms. If so, why the same materials exist in three states? For example, water exists in three forms as solid ice, liquid water, and gaseous steam. Interestingly ice, water, and steam are made up of same water molecules; two hydrogen atoms and one oxygen atom form a water molecule. Physics helps us to explore this beauty of nature at the microscopic level. The distance between

the atoms or molecules determines whether it exists in solid, liquid or gaseous state.

Solids

In solids, atoms or molecules are tightly fixed. In the solid formation, atoms get bound together through various types of bonding. Due to the interaction between the atoms, they position themselves at a particular interatomic distance. This position of atoms in this bound condition is called their mean positions.

Liquids

When the solid is not given any external energy such as heat, it will remain as a solid due to the bonding between atoms. When heated, atoms of the solid receive thermal energy and vibrate about their mean positions. When the solid is heated above its melting point, the heat energy will break the bonding between atoms and eventually the atoms receive enough energy and wander around. Here also the intermolecular (or interatomic) forces are important, but the molecules will have enough energy to move around, which makes the structure mobile.

Gases

When a liquid is heated at constant pressure to its boiling point or when the pressure is reduced at a constant temperature it will convert to a gas. This process of a liquid changing to a gas is called evaporation. The gas molecules have either very weak bonds or no bonds at all. This enables them to move freely and quickly. Hence, the gas will conform to the shape of its container and also will expand to fill the container. The transition from solid to liquid to gaseous states with the variation in external energy is schematically shown in Figure 7.1.

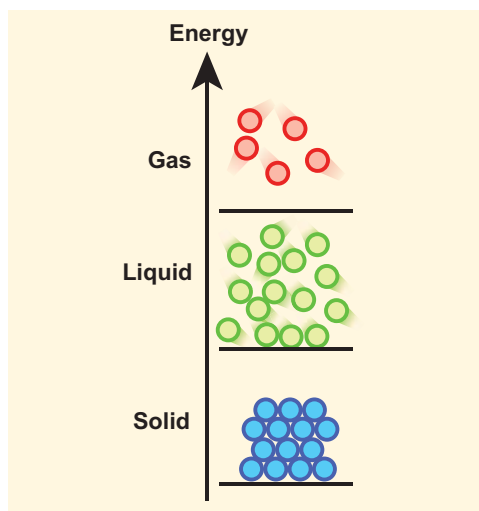


Figure 7.1 Schematic representations of the transition from solid to liquid to gaseous states with a change in external energy



In addition to the three physical states of matter (solid, liquid, and gas), in extreme environments, matter can exist in other states such as plasma, Bose-Einstein condensates. Additional states, such as quark-gluon plasmas are also believed to be possible. A major part of the atomic matter of the universe is hot plasma in the form of rarefied interstellar medium and dense stars.

In the study of Newtonian mechanics (Volume 1), we assumed the objects to be either as point masses or perfect rigid bodies (collection of point masses). Both these are idealized models. In rigid bodies, changes in the shape of the bodies are so small that they are neglected. In real materials, when a force is applied on the objects, there could be some deformations due to the applied force. It is very important to know how materials behave when a deforming force is applied.

7.2.1 Elastic behaviour of materials

In a solid, interatomic forces bind two or more atoms together and the atoms occupy the positions of stable equilibrium. When a deforming force is applied on a body, its atoms are pulled apart or pushed closer. When the deforming force is removed, interatomic forces of attraction or repulsion restore the atoms to their equilibrium positions. **If a body regains its original shape and size after the removal of deforming force, it is said to be elastic and the property is called elasticity.** The force which changes the size or shape of a body is called a deforming force.

Examples: Rubber, metals, steel ropes.

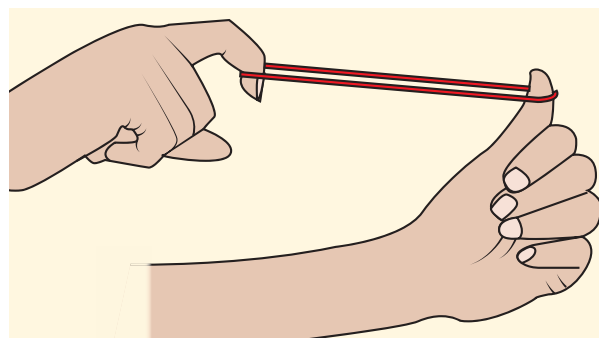


Figure 7.2 Elasticity

Plasticity:

If a body does not regain its original shape and size after removal of the deforming force, it is said to be a plastic body and the property is called plasticity.

Example: Glass

7.2.2 Stress and strain

(a) Stress:

When a force is applied, the size or shape or both may change due to the change in relative positions of atoms or molecules. This deformation may not be noticeable to

our naked eyes but it exists in the material itself. When a body is subjected to such a deforming force, internal force is developed in it, called as restoring force. The force per unit area is called as stress.

$$\text{Stress, } \sigma = \frac{\text{Force}}{\text{Area}} = \frac{F}{A} \quad (7.1)$$

The SI unit of stress is N m^{-2} or pascal (Pa) and its dimension is $[\text{ML}^{-1}\text{T}^{-2}]$. Stress is a tensor.

(i) Longitudinal stress and shearing stress:

Let us consider a body as shown in Figure 7.3. When many forces act on the system (body), the center of mass (defined in unit 5) remains at rest. However, the body gets deformed due to these forces and so the internal forces appear. Let ΔA be the cross sectional area of the body. The parts of the body on two sides of ΔA exert internal forces $\vec{F}$ and $-\vec{F}$ on each other which is due to deformation. The force can be resolved in two components; F_n normal to the surface ΔA (perpendicular to the surface) and F_t tangential to the surface ΔA (tangent to the surface). The normal stress or longitudinal stress (σ_n) over the area is defined as

$$\sigma_n = \frac{F_n}{\Delta A}$$

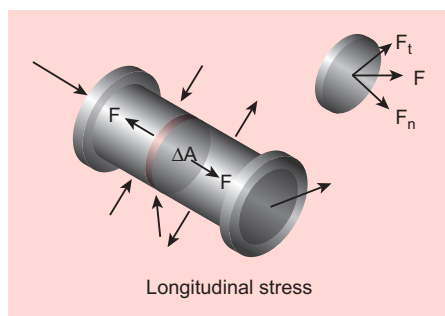


Figure 7.3 Longitudinal stress

Similarly, the tangential stress or shearing stress σ_t over the area is defined as

$$\sigma_t = \frac{F_t}{\Delta A}$$

Longitudinal stress can be classified into two types, tensile stress and compressive stress.

Tensile stress

Internal forces on the two sides of ΔA may pull each other, i.e., it is stretched by equal and opposite forces. Then, the longitudinal stress is called tensile stress.

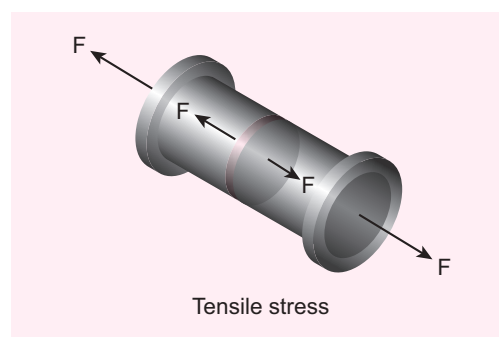


Figure 7.4 Tensile stress

Compressive stress

When forces acting on the two sides of ΔA push each other, ΔA is pushed by equal and opposite forces at the two ends. In this case, ΔA is said to be under compression. Then, the longitudinal stress is called compressive stress.

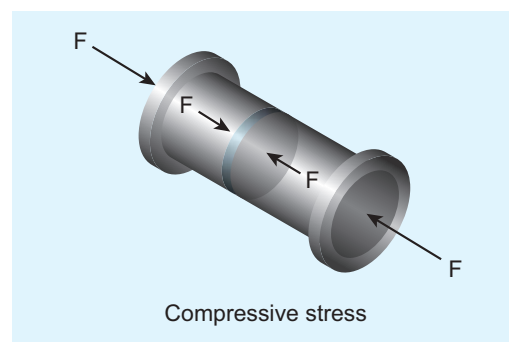


Figure 7.5 Compressive stress

(ii) Volume stress

This happens when a body is acted by forces everywhere on the surface such that the force at any point is normal to the surface and the magnitude of the force on a small surface area is proportional to the area. For instance, when a solid is immersed in a fluid, the pressure at the location of the solid is P , the force on any area ΔA is

$$F = P \Delta A$$

Where, F is perpendicular to the area. Thus, force per unit area is called volume stress.

$$\sigma_v = \frac{F}{A}$$

which is the same as the pressure.

(b) Strain:

Strain measures how much an object is stretched or deformed when a force is applied. Strain deals with the fractional change in the size of the object, in other words, strain measures the degree of deformation. As an example, in one dimension, consider a rod of length L when it stretches to a new length ΔL then

$$\text{Strain, } \epsilon = \frac{\text{Change in size}}{\text{Original size}} = \frac{\Delta L}{L} \quad (7.2)$$

ϵ is a dimensionless quantity and has no unit. Strain is classified into three types.

(1) Longitudinal strain

When a rod of length L is pulled by equal and opposite forces, the longitudinal strain is defined as

$$\epsilon_l = \frac{\text{Increase in length of the rod}}{\text{Original or natural length of the rod}} = \frac{\Delta L}{L} \quad (7.3)$$

Longitudinal strain can be classified into two types

(i) **Tensile strain:** If the length is increased from its natural length then it is known as tensile strain.

(ii) **Compressive strain:** If the length is decreased from its natural length then it is known as compressive strain.

(2) Shearing strain

Consider a cuboid as shown in Figure 7.6. Let us assume that the body remains in translational and rotational equilibrium. Let us apply the tangential force F along AD such that the cuboid deforms as shown in Figure 7.6. Hence, shearing strain or shear is (ϵ_s)

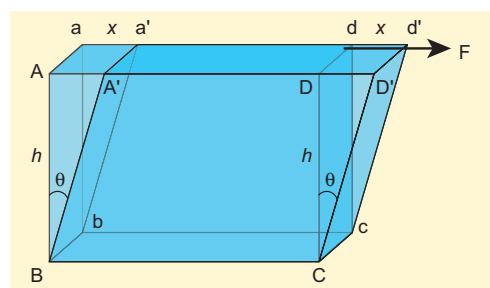


Figure 7.6 Shearing strain

$$\epsilon_s = \frac{AA'}{BA} = \frac{x}{h} = \tan \theta \quad (7.4)$$

For small angle, $\tan \theta \approx \theta$

Therefore, shearing strain or shear,

$$\epsilon_s = \frac{x}{h} = \theta = \text{Angle of shear}$$

(3) Volume strain

If the body is subjected to a volume stress, the volume will change. Let V be the original volume of the body before stress and $V + \Delta V$ be the change in volume due to stress. The volume strain which measures the fractional change in volume is

$$\text{Volume strain, } \epsilon_v = \frac{\Delta V}{V} \quad (7.5)$$

Elastic Limit

The maximum stress within which the body regains its original size and shape after the

removal of deforming force is called the elastic limit.

If the deforming force exceeds the elastic limit, the body acquires a permanent deformation. For example, rubber band loses its elasticity if pulled apart too much. It changes its size and becomes misfit to be used again.

7.2.3 Hooke's law and its experimental verification

Hooke's law states that for a small deformation within the elastic limit, the strain produced in a body is directly proportional to the stress that produces it.

It can be verified in a simple way by stretching a thin straight wire (stretches like spring) of length L and uniform cross-sectional area A suspended from a fixed point O . A pan and a pointer are attached at the free end of the wire as shown in Figure 7.7 (a).

The extension produced on the wire is measured using a vernier scale arrangement. The experiment shows that for a given load, the corresponding stretching force is F and the elongation produced on the wire is ΔL . It is directly proportional to the original length L and inversely proportional to the area of cross section A . A graph is plotted using F on the X- axis and ΔL on the Y- axis. This graph is a straight line passing through the origin as shown in Figure 7.7 (b).

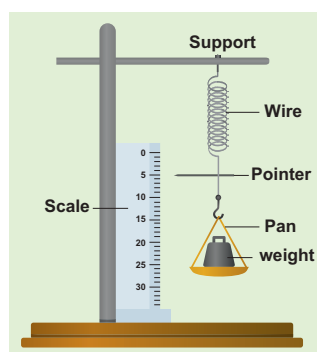


Figure 7.7 (a) Experimental verification of Hooke's law

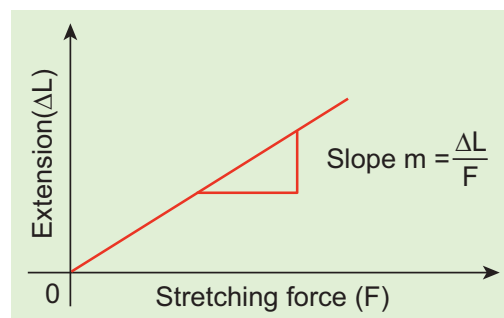


Figure 7.7 (b) Variation of ΔL with F

Therefore,

$$\Delta L = (\text{slope})F$$

Multiplying and dividing by volume,

$$V = A L,$$

$$F (\text{slope}) = \frac{AL}{AL} \Delta L$$

Rearranging, we get

$$\frac{F}{A} = \left(\frac{L}{A(\text{slope})} \right) \frac{\Delta L}{L}$$

$$\text{Therefore, } \frac{F}{A} \propto \left(\frac{\Delta L}{L} \right)$$

Comparing with equation (7.1) and equation (7.2), we get equation (7.5) as

$$\sigma \propto \varepsilon$$

i.e., the stress is proportional to the strain in the elastic limit.

Stress – Strain profile curve:

The stress versus strain profile is a plot in which stress and strain are noted for each load and a graph is drawn taking strain along the X-axis and stress along the Y-axis. The elastic characteristics of the materials can be analyzed from the stress-strain profile.

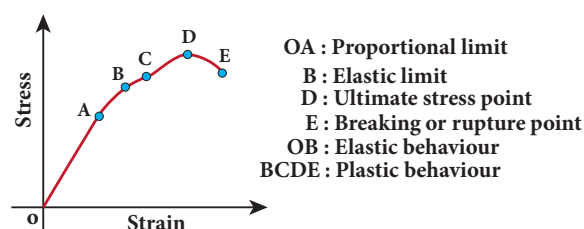


Figure 7.8 Stress-Strain profile



(a) Portion OA:

In this region, stress is very small such that stress is proportional to strain, which means Hooke's law is valid. The point A is called *limit of proportionality* because above this point Hooke's law is not valid. The slope of the line OA gives the Young's modulus of the wire.

(b) Portion AB:

This region is reached if the stress is increased by a very small amount. In this region, stress is not proportional to the strain. But once the stretching force is removed, the wire will regain its original length. This behaviour ends at point B and hence, the point B is known as *yield point* (elastic limit). The elastic behaviour of the material (here wire) in stress-strain curve is OAB.

(c) Portion BC:

If the wire is stretched beyond the point B (elastic limit), stress increases and the wire will not regain its original length after the removal of stretching force.

(d) Portion CD:

With further increase in stress (beyond the point C), the strain increases rapidly and reaches the point D. Beyond D, the strain increases even when the load is removed and breaks (ruptures) at the point E. Therefore, the maximum stress (here D) beyond which the wire breaks is called *breaking stress* or *tensile strength*. The corresponding point D is known as *fracture point*. The region BCDE represents the plastic behaviour of the material of the wire.

7.2.4 Moduli of elasticity

From Hooke's law,

$$\text{stress} \propto \text{strain}$$

$\frac{\text{stress}}{\text{strain}}$ = a constant, known as modulus of elasticity. Its SI unit is Nm^{-2} or pascal and its dimensional formula is $\text{ML}^{-1}\text{T}^{-2}$.

In this section, we shall define the elastic modulus of a given material. There are three types of elastic modulus.

- (a) Young's modulus
- (b) Rigidity modulus (or Shear modulus)
- (c) Bulk modulus

Young's modulus:

When a wire is stretched or compressed, then the ratio between tensile stress (or compressive stress) and tensile strain (or compressive strain) is defined as Young's modulus.

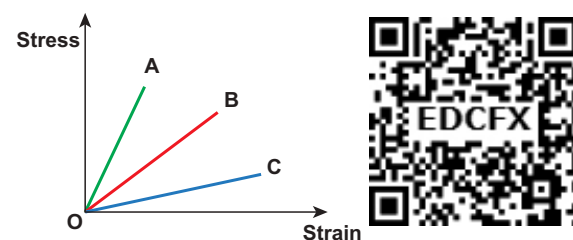
Young modulus of a material

$$= \frac{\text{Tensile stress or compressive stress}}{\text{Tensile strain or compressive strain}}$$

$$Y = \frac{\sigma_t}{\epsilon_t} \quad \text{or} \quad Y = \frac{\sigma_c}{\epsilon_c} \quad (7.6)$$

EXAMPLE 7.1

Within the elastic limit, the stretching strain produced in wires A, B, and C due to stress is shown in the figure. Assume the load applied are the same and discuss the elastic property of the material.



Write down the elastic modulus in ascending order.

Solution

Here, the elastic modulus is Young modulus and due to stretching, stress is tensile stress and strain is tensile strain.



Within the elastic limit, stress is proportional to strain (obey Hooke's law). Therefore, it shows a straight line behaviour. So, Young modulus can be computed by taking slope of these straight lines. Hence, calculating the slope for the straight line, we get

Slope of A > Slope of B > Slope of C

Which implies,

Young modulus of C < Young modulus of B < Young modulus of A

Notice that larger the slope, lesser the strain (fractional change in length). So, the material is much stiffer. Hence, the elasticity of wire A is greater than wire B which is greater than C. From this example, we have understood that Young's modulus measures the resistance of solid to a change in its length.

EXAMPLE 7.2

A wire 10 m long has a cross-sectional area $1.25 \times 10^{-4} \text{ m}^2$. It is subjected to a load of 5 kg. If Young's modulus of the material is $4 \times 10^{10} \text{ N m}^{-2}$, calculate the elongation produced in the wire. Take $g = 10 \text{ ms}^{-2}$.

Solution

We know that $\frac{F}{A} = Y \times \frac{\Delta L}{L}$

$$\Delta L = \left(\frac{F}{A} \right) \left(\frac{L}{Y} \right)$$
$$= \left(\frac{50}{1.25 \times 10^{-4}} \right) \left(\frac{10}{4 \times 10^{10}} \right) = 10^{-4} \text{ m}$$

Bulk modulus:

The bulk modulus is defined as the ratio of the volume stress to the volume strain.

Bulk modulus, K =

$$\frac{\text{Normal (Perpendicular) stress or Pressure}}{\text{Volume strain}}$$

The normal stress or pressure is

$$\sigma_n = \frac{F_n}{\Delta A} = \Delta p$$

The volume strain is

$$\epsilon_v = \frac{\Delta V}{V}$$

Therefore, Bulk modulus is

$$K = - \frac{\sigma_n}{\epsilon_v} = - \frac{\Delta P}{\frac{\Delta V}{V}} \quad (7.7)$$

The negative sign in the equation (7.7) means that when pressure is applied on the body, its volume decreases. Further, the equation (7.7) implies that a material can be easily compressed if it has a small value of bulk modulus. In other words, bulk modulus measures the resistance of solids to change in their volume.

For an example, we know that gases can be easily compressed than solids. This means that gases have small value of bulk modulus compared to solids.

Compressibility

The reciprocal of the bulk modulus is called compressibility. It is defined as the fractional change in volume per unit increase in pressure.

From equation (7.7), we can say that the compressibility

$$C = \frac{1}{K} = - \frac{\epsilon_v}{\sigma_n} = - \frac{\frac{\Delta V}{V}}{\Delta P} \quad (7.8)$$

Since gases have small value of bulk modulus than solids, their values of compressibility is very high.



After pumping the air in the cycle tyre, usually we press the cycle tyre to check whether it has

enough air. What is checked here is essentially the compressibility of air. The tyre should be less compressible for its easy rolling



In fact the rear tyre is less compressible than front tyre for a smooth ride.

EXAMPLE 7.3

A metallic cube of side 100 cm is subjected to a uniform force acting normal to the whole surface of the cube. The pressure is 10^6 pascal. If the volume changes by $1.5 \times 10^{-5} \text{ m}^3$, calculate the bulk modulus of the material.

Solution

$$\text{By definition, } K = \frac{\frac{F}{A}}{\frac{\Delta V}{V}} = P \frac{V}{\Delta V}$$

$$K = \frac{10^6 \times 1}{1.5 \times 10^{-5}} = 6.67 \times 10^{10} \text{ N m}^{-2}$$

Rigidity modulus or shear modulus:

The rigidity modulus is defined as the ratio of the shearing stress to the shearing strain.

$$\eta_R = \frac{\text{Shearing stress}}{\text{Angle of shear or shearing strain}}$$

The shearing stress is

$$\sigma_s = \frac{\text{Tangential force}}{\text{Area over which it is applied}} = \frac{F_t}{\Delta A}$$

The angle of shear or shearing strain

$$\epsilon_s = \frac{x}{h} = \theta$$

Therefore, Rigidity modulus is

$$\eta_R = \frac{\sigma_s}{\epsilon_s} = \frac{\frac{F_t}{\Delta A}}{\frac{x}{h}} = \frac{F_t}{\Delta A \theta} \quad (7.9)$$

Further, the equation (7.9) implies that a material can be easily twisted if it has small value of rigidity modulus. For example, consider a wire twisted through an angle θ , a restoring torque (τ) developed is

$$\tau \propto \theta$$

This means that for a larger torque, wire will twist by a larger amount (angle of shear θ is large). Since the rigidity modulus is inversely proportional to angle of shear, the modulus of rigidity is small.

For the best understanding, the elastic coefficients of some of the important materials are listed in Table 7.1.

Table 7.1 Elastic coefficient of some materials

Material	Young's modulus (Y) (10^{10} N m^{-2})	Bulk modulus (K) (10^{10} N m^{-2})	Shear modulus (η_R) (10^{10} N m^{-2})
Steel	20.0	15.8	8.0
Aluminium	7.0	7.0	2.5
Copper	12.0	12.0	4.0
Iron	19.0	10	6.4
Glass	7.0	3.6	2.6

EXAMPLE 7.4

A metal cube of side 0.20 m is subjected to a shearing force of 4000 N. The top surface is displaced through 0.50 cm with respect to the bottom. Calculate the shear modulus of elasticity of the metal.

Solution

Here, $L = 0.20$ m, $F = 4000$ N, $x = 0.50$ cm = 0.005 m and Area $A = L^2 = 0.04$ m²

Therefore,

$$\eta_R = \frac{F}{A} \times \frac{L}{x} = \frac{4000}{0.04} \times \frac{0.20}{0.005} = 4 \times 10^6 \text{ N m}^{-2}$$

7.2.5 Poisson's ratio

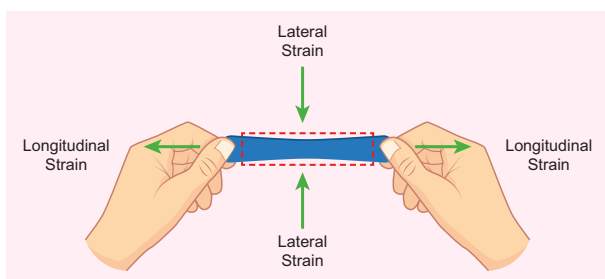


Figure 7.9 Lateral strain versus longitudinal strain

Suppose we stretch a wire, its length increases (elongation) but its diameter decreases (contraction). Similarly, when we stretch a rubber band (elongation), it becomes noticeably thinner (contraction). That is, deformation of the material in one direction produces deformation in another direction. To quantify this, French Physicist S.D. Poisson proposed a ratio, known as Poisson's ratio.

It is defined as the ratio of relative contraction (lateral strain) to relative expansion (longitudinal strain). It is denoted by the symbol μ .

$$\text{Poisson's ratio, } \mu = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} \quad (7.10)$$

Consider a wire of length L with diameter D . Due to applied force, wire stretches and let the increase in length be l and decrease in diameter be d . Then

$$\mu = -\frac{\frac{d}{D}}{\frac{l}{L}} = -\frac{L}{l} \times \frac{d}{D}$$

Negative sign indicates the elongation is along longitudinal and the contraction along lateral dimension. Further, notice that it is the ratio between quantities of the same dimension. So, Poisson's ratio has no unit and no dimension (dimensionless number). The Poisson's ratio values of some of the materials are listed in Table 7.2.

Table 7.2 Poisson's ratio of some of the materials

Material	Poisson's ratio
Rubber	0.4999
Gold	0.42 -0.44
Copper	0.33
Stainless steel	0.30-0.31
Steel	0.27-0.30
Cast iron	0.21-0.26
Concrete	0.1-0.2
Glass	0.18-0.3
Foam	0.10-0.50
Cork	0.0

7.2.6 Elastic energy

When a body is stretched, work is done against the restoring force (internal force). This work done is stored in the body in the form of elastic energy.

Consider a wire whose un-stretch length is L and area of cross section is A . Let a force produce an extension l and further assume that the elastic limit of the wire has not been exceeded and there is no loss in energy. Then, the work done by the force F is equal to the energy gained by the wire.

The work done in stretching the wire by dl , $dW = F dl$

The total work done in stretching the wire from 0 to l is

$$W = \int_0^l F dl \quad (7.11)$$

From Young's modulus of elasticity,

$$Y = \frac{F}{A} \times \frac{L}{l} \Rightarrow F = \frac{YAl}{L} \quad (7.12)$$

Substituting equation (7.12) in equation (7.11), we get

$$W = \int_0^l \frac{YAl}{L} dl$$

Since l is the dummy variable in the integration, we can change l to l' (not in limits). Therefore

$$W = \int_0^l \frac{YAl'}{L} dl' = \frac{YAl}{L} \left(\frac{l'^2}{2} \right) \Big|_0^l = \frac{YAl}{L} \frac{l^2}{2} = \frac{1}{2} \left(\frac{YAl}{L} \right) l = \frac{1}{2} Fl$$

$$W = \frac{1}{2} Fl = \text{Elastic potential energy}$$

The energy per unit volume is called energy density which is given by,

$$u = \frac{\text{Elastic potential energy}}{\text{Volume}} = \frac{\frac{1}{2} Fl}{AL}$$

$$= \frac{1}{2} \frac{F}{A} \frac{l}{L} = \frac{1}{2} \times \text{Stress} \times \text{Strain} \quad (7.13)$$

EXAMPLE 7.5

A wire of length 2 m with the area of cross-section 10^{-6} m^2 is used to suspend a load of 980 N. Calculate i) the stress developed in the wire ii) the strain and iii) the energy stored.

Given: $Y = 12 \times 10^{10} \text{ N m}^{-2}$.

Solution

$$(i) \text{ stress} = \frac{F}{A} = \frac{980}{10^{-6}} = 98 \times 10^7 \text{ N m}^{-2}$$

$$(ii) \text{ strain} = \frac{\text{stress}}{Y} = \frac{98 \times 10^7}{12 \times 10^{10}} = 8.17 \times 10^{-3} \quad (\text{no unit})$$

$$(iii) \text{ Since volume} = 2 \times 10^{-6} \text{ m}^3,$$

$$\begin{aligned} \text{Energy} &= \frac{1}{2} (\text{stress} \times \text{strain}) \times \text{volume} \\ &= \frac{1}{2} (98 \times 10^7) \times (8.17 \times 10^{-3}) \times 2 \times 10^{-6} = 8 \text{ J} \end{aligned}$$

7.2.7 Applications of elasticity

The mechanical properties of materials play a very vital role in everyday life. The elastic behaviour is one such property which especially decides the structural design of the columns and beams of a building.

As far as the structural engineering is concerned, the amount of stress that the design could withstand is a primary safety factor. A bridge has to be designed in such a way that it should have the capacity to withstand the load of the flowing traffic, the force of winds, and even its own weight. The elastic behaviour or in other words, the bending of beams is a major concern over the stability of the buildings or bridges.

For an example, to reduce the bending of a beam for a given load, one should use the material with a higher value of Young's modulus of elasticity. It is obvious from Table 7.1 that the Young's modulus of steel is greater than aluminium or copper. Iron comes next to steel. This is the reason why steel is mostly preferred in the design of heavy duty machines and iron rods in the construction of buildings.



Which one is more elastic? Rubber or steel? Steel is more elastic than rubber. If an equal stress is applied to both steel and rubber, the steel produces less strain. So the Young's modulus is higher for steel than rubber. The object which has higher Young's modulus is more elastic.

7.3 FLUIDS

7.3.1 Introduction

Fluids are found everywhere in the world. Earth's surface has about two-thirds of water and one-third of land. Fluids are different from solids. Unlike solid, fluid has no defined shape of its own. As far as fluid is concerned, liquid has fixed volume whereas gas fills the entire volume of the container.

Pressure of a fluid:

Fluid is a substance which begins to flow when an external force is applied on it. It offers a very small resistance to the applied force. If the force acts on a smaller area, then the impact will be more and vice versa. This particular idea decides yet another quantity called '*pressure*'.

Assume that an object is submerged in a fluid (say water) at rest. In this case, the fluid will exert a force on the surface of the object. This force is always normal to the surface of the object. If F is the magnitude of the normal force acting on the surface area A , then the pressure is defined as the '*force acting per unit area*'.

$$P = \frac{F}{A} \quad (7.14)$$

Pressure is a scalar quantity. Its S.I. unit and dimensions are N m^{-2} or pascal (Pa) and $[\text{ML}^{-1}\text{T}^{-2}]$ respectively.

Another common unit of pressure is atmosphere which is abbreviated as 'atm'. It is defined as the pressure exerted by the atmosphere at sea level.

i.e., $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$ or N m^{-2} .

Apart from pressure, there are two more parameters such as density and relative density (or specific gravity) which are used to describe the nature of fluids.

Density of a fluid:

The density of a fluid is defined as its mass per unit volume. For a fluid of mass m occupying volume V , the density $\rho = \frac{m}{V}$. The dimensions and S.I unit of ρ are $[\text{ML}^{-3}]$ and kg m^{-3} , respectively. It is a positive scalar quantity.

Mostly, a liquid is largely incompressible and hence its density is nearly constant at ambient pressure (i.e. at 1 atm. pressure). In the case of gases, there are variations in densities with reference to pressure.

Relative density or specific gravity:

The relative density of a substance is defined as the ratio of the density of a substance to the density of water at 4°C . It is a dimensionless positive scalar quantity.

For example, the density of mercury is $13.6 \times 10^3 \text{ kg m}^{-3}$. Its relative density is equal to $\frac{13.6 \times 10^3 \text{ kg m}^{-3}}{1.0 \times 10^3 \text{ kg m}^{-3}} = 13.6$.

[The density of water at 4°C is 1000 kg m^{-3}]

EXAMPLE 7.6

A solid sphere has a radius of 1.5 cm and a mass of 0.038 kg. Calculate the specific gravity or relative density of the sphere.



Solution

Radius of the sphere $R = 1.5 \text{ cm}$

mass $m = 0.038 \text{ kg}$

Volume of the sphere $V = \frac{4}{3} \pi R^3$
 $= \frac{4}{3} \times (3.14) \times (1.5 \times 10^{-2})^3 = 1.413 \times 10^{-5} \text{ m}^3$

Therefore, density

$$\rho = \frac{m}{V} = \frac{0.038 \text{ kg}}{1.413 \times 10^{-5} \text{ m}^3} = 2690 \text{ kg m}^{-3}$$

Hence, the specific gravity of the sphere

$$= \frac{2690}{1000} = 2.69$$

7.3.2 Pressure due to fluid column at rest

A mountaineer climbing the mountain is able to experience a decrease in pressure with altitude. A person jumping into the swimming pool always realizes an increase in pressure with depth below the water surface. In both the cases, the pressure encountered by the mountaineer and diver is usually due to the hydrostatic pressure, because they are due to fluids that are static.

In order to understand the increase in pressure with depth below the water surface, consider a water sample of cross sectional area A in the form of a cylinder. Let h_1 and h_2 be the depths from the air-water interface to level 1 and level 2 of the cylinder respectively as shown in Figure 7.10(a). Let F_1 be the force acting downwards on level 1 and F_2 be the force acting upwards on level 2 such that $F_1 = P_1 A$ and $F_2 = P_2 A$. Let us assume the mass of the sample to be m and under equilibrium condition, the total upward force (F_2) is balanced by the total downward force ($F_1 + mg$). In other words, the gravitational force will act downward which is being exactly balanced by the difference between the force $F_2 - F_1$.

$$F_2 - F_1 = mg = F_G \quad (7.15)$$

Let ρ be the density of the water, then the mass of water available in the sample element is

$$m = \rho V = \rho A (h_2 - h_1)$$
$$V = A (h_2 - h_1)$$

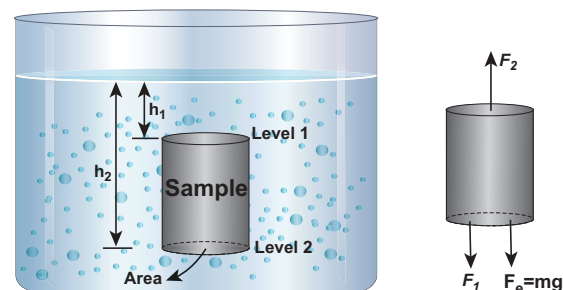


Figure 7.10 (a) A sample of water with base area A in a static fluid with its forces in equilibrium

Hence gravitational force,

$$F_G = \rho A (h_2 - h_1) g$$

On substituting the value of F_G in equation (7.15)

$$F_2 = F_1 + mg$$
$$P_2 A = P_1 A + \rho A (h_2 - h_1) g$$

Cancelling out A on both sides,

$$P_2 = P_1 + \rho (h_2 - h_1) g \quad (7.16)$$

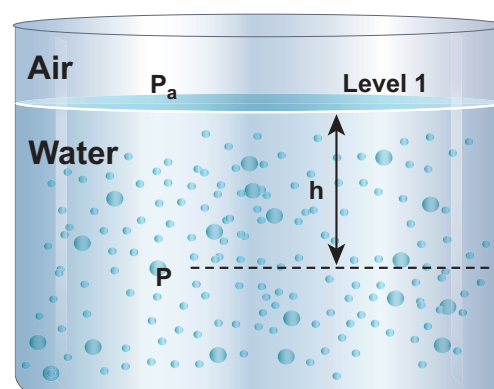


Figure 7.10 (b) The pressure (P) at a depth (h) below the water surface



If we choose the level 1 at the surface of the liquid (i.e., air-water interface) and the level 2 at a depth 'h' below the surface (as shown in Figure 7.10(b)), then the value of h_1 becomes zero ($h_1 = 0$) and in turn P_1 assumes the value of atmospheric pressure (say P_a). In addition, the pressure (P_2) at a depth becomes P . Substituting these values in equation (7.16), we get

$$P = P_a + \rho gh \quad (7.17)$$

Therefore, the pressure at a depth h is greater than the pressure on the surface of the liquid.

If the atmospheric pressure is neglected or ignored, then the pressure at a depth h is

$$P = \rho gh \quad (7.18)$$

For a given liquid, ρ is fixed and g is also constant, then the pressure due to the fluid column is directly proportional to vertical distance or height of the fluid column. This implies that the height of the fluid column is more important to decide the pressure and not the cross sectional or base area or even the shape of the container.

Hydrostatic Paradox

When we talk about liquid at rest, the liquid pressure is the same at all points at the same horizontal level (or same depth). This statement can be demonstrated by an experiment called 'hydrostatic paradox'. Let us consider three vessels of different shapes A, B, and C as shown in Figure 7.11. These vessels are connected at the bottom by a horizontal pipe. When they are filled with a liquid (say water), it occupies the same level even though the vessels hold different amounts of water. It is true because the liquid at the bottom of each section of the vessel experiences the same pressure.

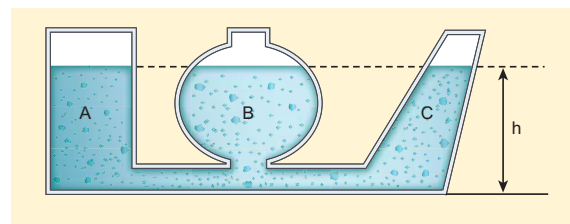


Figure 7.11 Illustration of hydrostatic paradox



The atmospheric pressure at a place is the gravitational force exerted by air above that place per unit surface area. It changes with height and weather conditions (i.e. density of air). In fact, the atmospheric pressure decreases with increasing elevation.

The decrease of atmospheric pressure with altitude has an unwelcome consequence in daily life. For example, it takes longer time to cook at higher altitudes. Nose bleeding is another common experience at higher altitude because of larger difference in atmospheric pressure and blood pressure. Its value on the surface of the Earth at sea level is 1atm.

$$1 \text{ atm pressure} = 1.013 \times 10^5 \text{ pa}$$

ACTIVITY

Take a metal container with an opening. Connect a vacuum pump to the opening. Evacuate the air from inside the container. Why the shape of the metal container gets crumbled?

Inference:

Due to the force of atmospheric pressure acting on its outer surface, the shape of the container crumbles.



ACTIVITY

Take a glass tumbler. Fill it with water to the brim. Slide a cardboard on its rim so that no air remains in between the cardboard and the tumbler. Invert the tumbler gently. The water does not fall down.

Inference:

This is due to the fact that the weight of water in the tumbler is balanced by the upward thrust caused due to the atmospheric pressure acting on the lower surface of the cardboard that is exposed to air.

7.3.3 Pascal's law and its applications

The French scientist Blaise Pascal observed that the pressure in a fluid at rest is the same at all points if they are at the same height. Pascal's law states that *if the pressure in a liquid is changed at a particular point, the change is transmitted to the entire liquid without being diminished in magnitude.*

Application of Pascal's law Hydraulic lift

A practical application of Pascal's law is the hydraulic lift which is used to lift a heavy load with a small force. It is a force multiplier. It consists of two cylinders A and B connected to each other by a horizontal pipe, filled with a liquid (Figure 7.12). They are fitted with

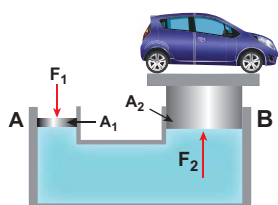


Figure 7.12 Hydraulic lift

frictionless pistons of cross sectional areas A_1 and A_2 ($A_2 > A_1$).

Suppose a downward force F is applied on the smaller piston, the pressure of the liquid under this piston increases to P (where, $P = \frac{F_1}{A_1}$). According to Pascal's law,

this increased pressure P is transmitted undiminished in all directions. Therefore, a pressure is exerted on piston B. Upward force on piston B is

$$F_2 = P \times A_2 = \frac{F_1}{A_1} \times A_2$$
$$F_2 = \frac{A_2}{A_1} \times F_1 \quad (7.19)$$

Therefore by changing the force on the smaller piston A, the force on the piston B can be increased by the factor $\frac{A_2}{A_1}$ and this factor is called the mechanical advantage of the lift.

EXAMPLE 7.7

Two pistons of a hydraulic lift have diameters of 60 cm and 5 cm. What is the force exerted by the larger piston when 50 N is placed on the smaller piston?

Solution

Since, the diameter of the pistons are given, we can calculate the radius of the piston

$$r = \frac{D}{2}$$

$$\text{Area of smaller piston, } A_1 = \pi \left(\frac{5}{2} \right)^2 = \pi (2.5)^2$$

$$\text{Area of larger piston, } A_2 = \pi \left(\frac{60}{2} \right)^2 = \pi (30)^2$$

$$F_2 = \frac{A_2}{A_1} \times F_1 = (50 \text{ N}) \times \left(\frac{30}{2.5} \right)^2 = 7200 \text{ N}$$

This means that with the force of 50 N, the force of 7200 N can be lifted.

7.3.4 Buoyancy

When a body is partially or fully immersed in a fluid, it displaces a certain amount of fluid. The displaced fluid exerts an upward force on the body. The upward force exerted by a fluid that opposes the weight of an immersed object in a fluid is called *upthrust* or *buoyant force* and the phenomenon is called *buoyancy*.

Archimedes principle:

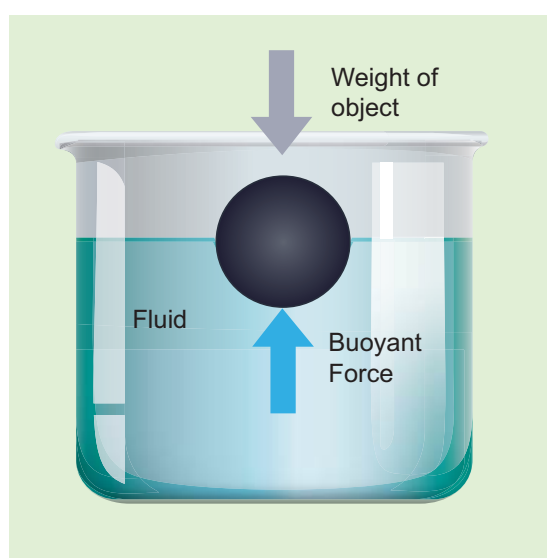


Figure 7.13 Archimedes principle

It states that when a body is partially or wholly immersed in a fluid, it experiences an upward thrust equal to the weight of the fluid displaced by it and its upthrust acts through the centre of gravity of the liquid displaced.

upthrust or buoyant force = weight of liquid displaced.

Law of floatation

It is well-known that boats, ships, and some wooden objects move on the upper part of the water. We say that they float. Floatation can be defined as the tendency of an object to rise up to the upper levels

of the fluid or to stay on the surface of the fluid.

The law of floatation states that a body will float in a liquid if the weight of the liquid displaced by the immersed part of the body equals the weight of the body. For example, a wooden object weighs 300 kg (about 3000 N) floats in water displaces 300 kg (about 3000 N) of water.

Note

If an object floats, the volume of fluid displaced is equal to the volume of the object submerged and the percentage of the volume of the object submerged is equal to the relative density of an object with respect to the density of the fluid in which it floats.

For example, if an ice cube of density 0.9 g cm^{-3} floats in the fresh water of density 1.0 g cm^{-3} then the percentage volume of an object submerged in fresh water is, $\frac{0.9 \text{ g cm}^{-3}}{1.0 \text{ g cm}^{-3}} \times 100\% = 90\%$.

Conversely, if the same ice cube floats in sea water of density 1.3 g cm^{-3} , then the percentage volume of the object submerged in seawater would be $\frac{0.9 \text{ g cm}^{-3}}{1.3 \text{ g cm}^{-3}} \times 100\% = 69.23\%$ only.

EXAMPLE 7.8

A cube of wood floating in water supports a 300 g mass at the centre of its top face. When the mass is removed, the cube rises by 3 cm. Determine the volume of the cube.

Solution

Let each side of the cube be l . The volume occupied by 3 cm depth of cube,

$$V = (3\text{cm}) \times l^2 = 3l^2\text{cm}$$

According to the principle of floatation, we have

$$V\rho g = mg \Rightarrow V\rho = m$$

$$\rho \text{ is density of water} = 1000 \text{ kg m}^{-3}$$

$$\Rightarrow (3l^2 \times 10^{-2}\text{m}) \times (1000 \text{ kg m}^{-3}) = 300 \times 10^{-3}\text{kg}$$

$$l^2 = \frac{300 \times 10^{-3}}{3 \times 10^{-2} \times 1000} \text{m}^2 \Rightarrow l^2 = 100 \times 10^{-4} \text{m}^2$$

$$l = 10 \times 10^{-2} \text{m} = 10 \text{ cm}$$

Therefore, volume of cube $V = l^3 = 1000 \text{ cm}^3$



Submarines can sink or rise in the depth of water by controlling its buoyancy. To achieve this, the submarines have ballast tanks that can be filled with water or air alternatively. When the ballast tanks are filled with air, the overall density of the submarine becomes lesser than the surrounding water and it surfaces (positive buoyancy). If the tanks are flooded with water replacing air, the overall density becomes greater than the surrounding water and submarine begins to sink (negative buoyancy). To keep the submarine at any depth, tanks are filled with air and water (neutral buoyancy).

Examples of floating bodies:

- A person can swim in sea water more easily than in river water.
- Ice floats on water.
- The ship is made of steel but its interior is made hollow by giving it a concave shape.

7.4

VISCOSITY

7.4.1 Introduction

In section 7.3, the behaviour of fluids at rest is discussed. Successive discussions will bring out the influence of fluid motion on different properties. A fluid in motion is a complex phenomenon as it possesses potential, kinetic, and gravitational energy besides causing friction viscous forces to come into play. Therefore, it is necessary to consider the case of an ideal liquid to simplify the task. An ideal liquid is incompressible (i.e., bulk modulus is infinity) and in which no shearing forces can be maintained (i.e., the coefficient of viscosity is zero).

Most of the fluids offer resistance towards motion. A frictional force acts at the contact surface when a fluid moves relative to a solid or when two fluids move relative to each other. This resistance to fluid motion is similar to the friction produced when a solid moves on a surface. The internal friction existing between the layers of a moving fluid is viscosity. So, viscosity is defined as the property of a fluid to oppose the relative motion between its layers.

ACTIVITY

Consider three steel balls of the same size, dropped simultaneously in three tall jars each filled with air, water, and oil respectively. It moves easily in air, but not so easily in water. Moving in oil would be even more difficult. There is a relative motion produced between the different layers of the liquid by the falling ball, which causes a viscous force. This frictional force depends

on the density of the liquid. This property of a moving fluid to oppose the relative motion between its layers is called viscosity.

Cause of viscosity:

Consider a liquid flowing over a horizontal surface with two neighboring layers. The upper layer tends to accelerate the lower layer and in turn, the lower layer tends to retard the upper layer. As a result, a backward tangential force is set-up. This tends to destroy the relative motion. This accounts for the viscous behaviour of fluids.

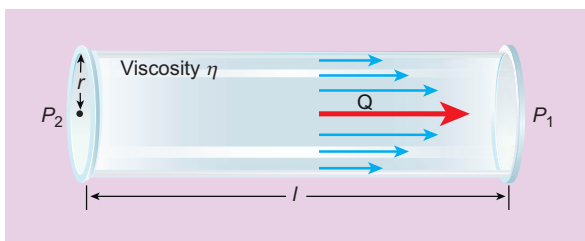


Figure 7.14 Viscosity

Coefficient of viscosity:

Consider a liquid flowing steadily over a horizontal fixed layer (Figure 7.15). The velocities of the layers increase uniformly as we move away from the fixed layer. Consider any two parallel layers A and B. Let v and $v + dv$ be the velocities of the neighboring layers at distances x and $x + dx$ respectively from the fixed layer.

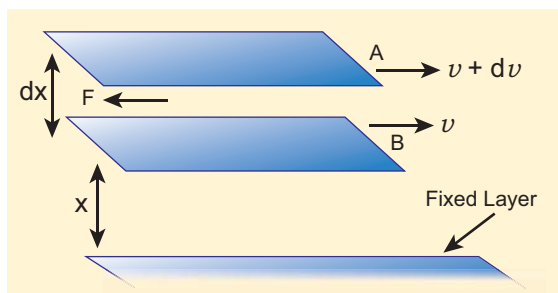


Figure 7.15 Flow of liquid over the horizontal layers

The force of viscosity F acting tangentially between two layers is given by Newton's First

law. This force is proportional to (i) area A of the liquid and (ii) the velocity gradient $\frac{dv}{dx}$

$$F \propto A \text{ and } F \propto \frac{dv}{dx}$$

$$\Rightarrow F = -\eta A \frac{dv}{dx} \quad (7.20)$$

Where the constant of proportionality η is called the coefficient of viscosity of the liquid and the negative sign implies that the force is frictional and it opposes the relative motion. The dimensional formula for coefficient of viscosity is $[ML^{-1}T^{-1}]$.



Note

Viscosity is similar to friction. The kinetic energy of the substance is dissipated as heat energy.

EXAMPLE 7.9

A metal plate of area $2.5 \times 10^{-4} \text{ m}^2$ is placed on a $0.25 \times 10^{-3} \text{ m}$ thick layer of castor oil. If a force of 2.5 N is needed to move the plate with a velocity $3 \times 10^{-2} \text{ m s}^{-1}$, calculate the coefficient of viscosity of castor oil.

Given: $A = 2.5 \times 10^{-4} \text{ m}^2$, $dx = 0.25 \times 10^{-3} \text{ m}$, $F = 2.5 \text{ N}$ and $dv = 3 \times 10^{-2} \text{ m s}^{-1}$

Solution

$$F = -\eta A \frac{dv}{dx}$$

$$\text{In magnitude, } \eta = \frac{F dx}{A dv}$$

$$= \frac{(2.5 \text{ N}) (0.25 \times 10^{-3} \text{ m})}{(2.5 \times 10^{-4} \text{ m}^2) (3 \times 10^{-2} \text{ m s}^{-1})}$$

$$= 0.083 \times 10^3 \text{ N m}^{-2} \text{ s}$$

7.4.2 Streamlined flow

The flow of fluids occurs in different ways. It can be a steady or streamlined flow, unsteady or turbulent flow, compressible or incompressible flow or even viscous or non-viscous flow. For example, consider a calm flow of water through a river. Careful observation reveals that the velocity of water at different locations of the river is quite different. It is almost faster at the center and slowest near the banks. However, the velocity of the particle at any point is constant. For better understanding, assume that the velocity of the particle is about 4 meter per second at the center of the river. Hence it will be of the same value for all other particles crossing through this point. In a similar way, if the velocity of the particle flowing near the bank of the river is 0.5 meter per second, then the succeeding particles flowing through it will have the same value.

When a liquid flows such that each particle of the liquid passing through a point moves along the same path with the same velocity as its predecessor then the flow of liquid is said to be a **streamlined flow**. It is also referred to as steady or laminar flow. The actual path taken by the particle of the moving fluid is called a streamline, which is a curve, the tangent to which at any point gives the direction of the flow of the fluid at that point as shown in Figure 7.16. It is named so because the flow looks like the flow of a stream or river under ideal conditions.

If we assume a bundle of streamlines having the same velocity over any cross section perpendicular to the direction of flow then such bundle is called a '**tube of**

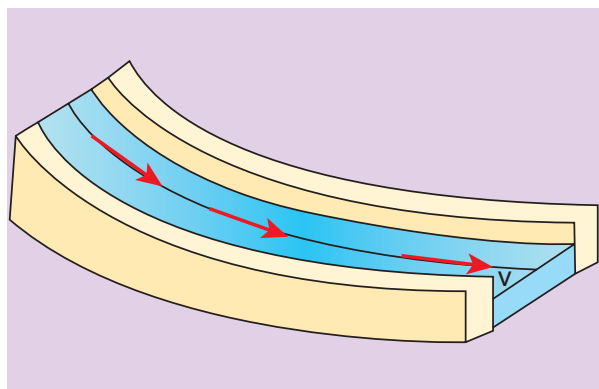


Figure 7.16 Flow is steady velocity at any point in the liquid remains constant

flow'. Thus, it is important to note that any particle in a tube of flow always remains in the tube throughout its motion and cannot mix with liquid in another tube. Always the axis of the tube of flow gives the streamline. The streamlines always represent the trajectories of the fluid particles. **The flow of fluid is streamlined up to a certain velocity called critical velocity.** This means a steady flow can be achieved at low flow speeds below the critical speed.

7.4.3 Turbulent flow

When the speed of the moving fluid exceeds the critical speed v_c , the motion becomes turbulent. In this case, the velocity changes both in magnitude and direction from particle to particle and hence the individual particles do not move in a streamlined path. Hence, the path taken by the particles in turbulent flow becomes erratic and whirlpool-like circles called eddy current or eddies (Figure 7.17 (a) and (b)). The flow of water just behind a boat or a ship and the air flow behind a moving bus are a few examples of turbulent flow.

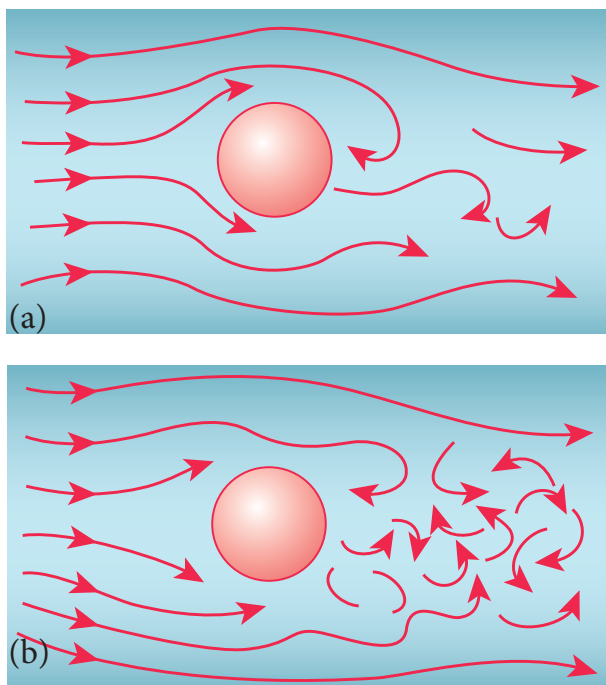


Figure 7.17 (a) turbulent flow around a sphere (when $v = v_c$) (b) turbulent flow around a sphere (when $v > v_c$)

The distinction between the two types of motion can be easily demonstrated by injecting a jet of ink axially in a wide tube through which water flows. When the velocity of the fluid is small, the ink will move in a straight line path. Conversely, when the velocity is increased beyond a certain value, the ink will spread out showing the disorderliness and hence the motion becomes turbulent. The zig-zag motion results in the formation of eddy currents and as a consequence, much energy is dissipated.

7.4.4 Reynold's number

We have learnt that the flow of a fluid becomes steady or laminar when the velocity of flow is less than the critical velocity v_c otherwise, the flow becomes turbulent. Osborne Reynolds (1842-1912) formulated an equation to find out the nature of the

flow of fluid, whether it is streamlined or turbulent.

$$R_c = \frac{\rho v D}{\eta} \quad (7.21)$$

It is a dimensionless number called '*Reynold's number*'. It is denoted by the symbol R_c or K . In the equation, ρ denotes the density of the fluid, v the velocity of the flowing fluid, D is the diameter of the pipe in which the fluid flow, and η is the coefficient of viscosity of the fluid. The value of R_c remains the same in any system of units.

Table 7.3 To understand the flow of liquid, Reynold has estimated the value of R_c as follows

S. No.	Reynold's number	Flow
1	$R_c < 1000$	Streamline
2	$1000 < R_c < 2000$	Unsteady
3	$R_c > 2000$	Turbulent

Hence, Reynold's number R_c is a critical variable which decides whether the flow of a fluid through a cylindrical pipe is streamlined or turbulent.

In fact, the critical value of R_c at which the turbulence sets is found to be the same for geometrically similar flows. For example, when two liquids (say oil and water) of different densities and viscosities flow in pipes of same shapes and sizes, the turbulence sets in at almost the same value of R_c . The above fact leads to the *Law of similarity* which states that when there are two geometrically similar flows, both are essentially equal to each other, as long as they embrace the same Reynold's number. The *Law of similarity* plays a very important role in technological applications. The

shape of ships, submarines, racing cars, and airplanes are designed in such a way that their speed can be maximized.

7.4.5 Terminal velocity

To understand terminal velocity, consider a small metallic sphere falling freely from rest through a large column of a viscous fluid.

The forces acting on the sphere are (i) gravitational force of the sphere acting vertically downwards, (ii) upthrust U due to buoyancy and (iii) viscous drag acting upwards (viscous force always acts in a direction opposite to the motion of the sphere).

Initially, the sphere is accelerated in the downward direction so that the upward force is less than the downward force. As the velocity of the sphere increases, the velocity of the viscous force also increases. A stage is reached when the net downward force balances the upward force and hence the resultant force on the sphere becomes zero. It now moves down with a constant velocity.

The maximum constant velocity acquired by a body while falling freely through a viscous medium is called the terminal velocity v_t . In the Figure 7.18, a graph is drawn with velocity along y- axis and time along x- axis. It is evident from the graph that the sphere is accelerated initially

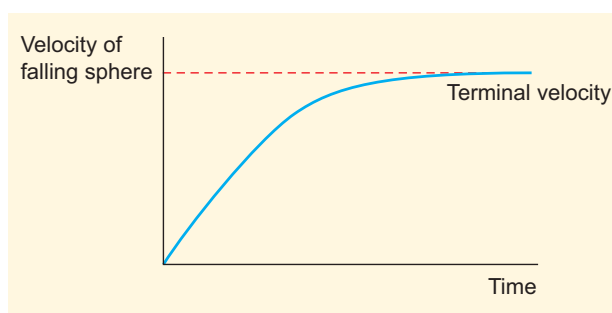


Figure 7.18 Velocity versus time graph

and in course of time it becomes constant, and attains terminal velocity (v_t).

Expression for terminal velocity:

Consider a sphere of radius r which falls freely through a highly viscous liquid of coefficient of viscosity η . Let the density of the material of the sphere be ρ and the density of the fluid be σ .

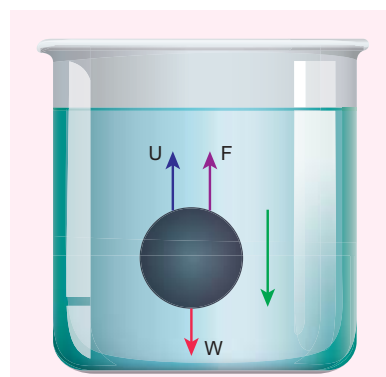


Figure 7.19 Forces acting on the sphere when it falls in a viscous liquid

Gravitational force acting on the sphere,

$$F_g = mg = \frac{4}{3}\pi r^3 \rho g \text{ (downward force)}$$

$$\text{Up thrust, } U = \frac{4}{3}\pi r^3 \sigma g \text{ (upward force)}$$

$$\text{Viscous force, } F = 6\pi\eta r v_t$$

At terminal velocity v_t ,

The net downward force = upward force

$$F_g = U + F$$

$$F_g - U = F \Rightarrow \frac{4}{3}\pi r^3 \rho g - \frac{4}{3}\pi r^3 \sigma g = 6\pi\eta r v_t$$

$$v_t = \frac{2}{9} \times \frac{r^2(\rho - \sigma)}{\eta} g \Rightarrow v_t \propto r^2 \quad (7.22)$$

Here, it should be noted that the terminal speed of the sphere is directly proportional to the square of its radius.

If σ is greater than ρ , then the term $(\rho - \sigma)$ becomes negative leading to a negative terminal velocity. That is why air bubbles

rise up through water or any fluid. This is also the reason for the clouds in the sky to move in the upward direction.

Point to ponder

- The terminal speed of a sphere is directly proportional to the square of the radius of the sphere. Hence, larger raindrops fall with greater speed as compared to the smaller raindrops.
- If the density of the material of the sphere is less than the density of the medium, then the sphere shall attain terminal velocity in the upward direction. That is why gas bubbles rise up in soda water.

7.4.6 Stoke's law and its applications

When a body falls through a viscous medium, it drags the layer of the fluid immediately in contact with it. This produces a relative motion between the different layers of the liquid. Stoke performed many experiments on the motion of small spherical bodies in different fluids and concluded that the viscous force F acting on a spherical body of radius r depends directly on

- radius (r) of the sphere
- velocity (v) of the sphere and
- coefficient of viscosity η of the liquid

Therefore $F \propto \eta^x r^y v^z \Rightarrow F = k\eta^x r^y v^z$, where k is a dimensionless constant.

Using dimensions, the above equation can be written as

$$[MLT^{-2}] = k[ML^{-1}T^{-1}]^x [L]^y [LT^{-1}]^z$$

On solving, we get $x=1$, $y=1$, and $z=1$

Therefore, $F = k\eta r v$

Experimentally, Stoke found that the value of $k = 6\pi$

$$F = 6\pi\eta r v \quad (7.23)$$

This relation is known as Stoke's law

Practical applications of Stoke's law

Since the raindrops are smaller in size and their terminal velocities are small, remain suspended in air in the form of clouds. As they grow up in size, their terminal velocities increase and they start falling in the form of rain.

This law explains the following:

- Floataction of clouds
- Larger raindrops hurt us more than the smaller ones
- A man coming down with the help of a parachute acquires constant terminal velocity.

7.4.7 Poiseuille's equation

Poiseuille analyzed the steady flow of liquid through a capillary tube. He derived an expression for the volume of the liquid flowing per second through the capillary tube.

As per the theory, the following conditions must be retained while deriving the equation.

- The flow of liquid through the tube is streamlined.
- The tube is horizontal so that gravity does not influence the flow
- The layer in contact with the wall of the tube is at rest
- The pressure is uniform over any cross section of the tube



We can derive Poiseuille's equation using dimensional analysis. Consider a liquid flowing steadily through a horizontal capillary tube. Let $v = \left(\frac{V}{t}\right)$ be the volume of the liquid flowing out per second through a capillary tube. It depends on (1) coefficient of viscosity (η) of the liquid, (2) radius of the tube (r), and (3) the pressure gradient $\left(\frac{P}{l}\right)$. Then,

$$v \propto \eta^a r^b \left(\frac{P}{l}\right)^c$$

$$v = k \eta^a r^b \left(\frac{P}{l}\right)^c \quad (7.24)$$

where, k is a dimensionless constant. Therefore,

$$[v] = \frac{\text{volume}}{\text{time}} = [L^3 T^{-1}], \left[\frac{dP}{dx}\right] = \frac{\text{Pressure}}{\text{distance}}$$

$$[ML^{-2}T^{-2}], [\eta] = [ML^{-1}T^{-1}] \text{ and } [r] = [L]$$

Substituting in equation (7.24)

$$[L^3 T^{-1}] = [ML^{-1}T^{-1}]^a [L]^b [ML^{-2}T^{-2}]^c$$

$$M^0 L^3 T^{-1} = M^{a+c} L^{-a+b-2c} T^{-a-2c}$$

So, equating the powers of M , L , and T on both sides, we get

$$a + c = 0, -a + b - 2c = 3, \text{ and } -a - 2c = -1$$

We have three unknowns a , b , and c . We have three equations, on solving, we get

$$a = -1, b = 4, \text{ and } c = 1$$

Therefore, equation (7.24) becomes,

$$v = k \eta^{-1} r^4 \left(\frac{P}{l}\right)^1$$

Experimentally, the value of k is shown to be $\frac{\pi}{8}$, we have

$$v = \frac{\pi r^4 P}{8 \eta l} \quad (7.25)$$

The above equation is known as *Poiseuille's equation* for the flow of liquid through a narrow tube or a capillary tube. This relation holds good for the fluids whose velocities are lesser than the critical velocity (v_c).

7.4.8 Applications of viscosity

The importance of viscosity can be understood from the following examples.

- (1) The oil used as a lubricant for heavy machinery parts should have a high viscous coefficient. To select a suitable lubricant, we should know its viscosity and how it varies with temperature [Note: As temperature increases, the viscosity of the liquid decreases]. Also, it helps to choose oils with low viscosity used in car engines (light machinery).
- (2) The highly viscous liquid is used to damp the motion of some instruments and is used as brake oil in hydraulic brakes.
- (3) Blood circulation through arteries and veins depends upon the viscosity of fluids.
- (4) Millikan conducted the oil drop experiment to determine the charge of an electron. He used the knowledge of viscosity to determine the charge.

7.5 SURFACE TENSION

7.5.1 Intermolecular forces

Some liquids do not mix together due to their physical properties such as density, surface tension force, etc. For example, water and kerosene do not mix together. Mercury does not wet the glass but water sticks to it. Water



risers up to the leaves through the stem. They are mostly related to the free surfaces of liquids. Liquids have no definite shape but have a definite volume. Hence they acquire a free surface when poured into a container. Therefore, the surfaces have some additional energy, called as surface energy. The phenomenon behind the above fact is called surface tension. Laplace and Gauss developed the theory of surface and motion of a liquid under various situations.

The molecules of a liquid are not rigidly fixed like in a solid. They are free to move about. The force between the like molecules which holds the liquid together is called '*cohesive force*'. When the liquid is in contact with a solid, the molecules of these solid and liquid will experience an attractive force which is called '*adhesive force*'.

These molecular forces are effective only when the distance between the molecules is very small about 10^{-9} m (i.e., $10 \text{ \AA}$). The distance through which the influence of these molecular forces can be felt in all directions constitute a range and is called *sphere of influence*. The forces outside this range are rather negligible.

Consider three different molecules A, B, and C in a given liquid as shown in Figure 7.20. Let a molecule 'A' be considered well inside the liquid within the sphere of influence. Since this molecule interacts with all other molecules in all directions, the net force experienced by A is zero. Now consider a molecule 'B' in which three-fourth lies below the liquid surface and one-fourth on the air. Since B has more molecules towards its lower side than the upper side, it experiences a net force in the downward direction. In a similar way, if another molecule 'C' is chosen on the liquid surface (i.e., upper half in air and lower half in liquid),

it experiences a maximum downward force due to the availability of more number of liquid molecules on the lower part. Hence it is obvious that all molecules of the liquid that falls within the molecular range inside the liquid interact with the molecule and hence experience a downward force.

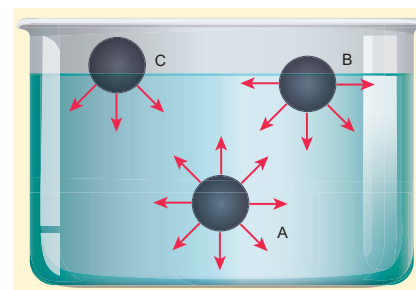


Figure 7.20 Molecules at different levels of liquid

When any molecule is brought towards the surface from the interior of the liquid, work is done against the cohesive force among the molecules of the surface. This work is stored as potential energy in molecules. So the molecules on the surface will have greater potential energy than that of molecules in the interior of the liquid. But for a system to be under stable equilibrium, its potential energy (or surface energy) must be a minimum. Therefore, in order to maintain stable equilibrium, a liquid always tends to have a minimum number of molecules. In other words, the liquid tends to occupy a minimum surface area. This behaviour of the liquid gives rise to surface tension.

Examples for surface tension.

Water bugs and water striders walk on the surface of water (Figure 7.21). The water molecules are pulled inwards and the surface of water acts like a springy or stretched membrane. This balance the weight of water bugs and enables them to walk on the

surface of water. We call this phenomenon as surface tension.

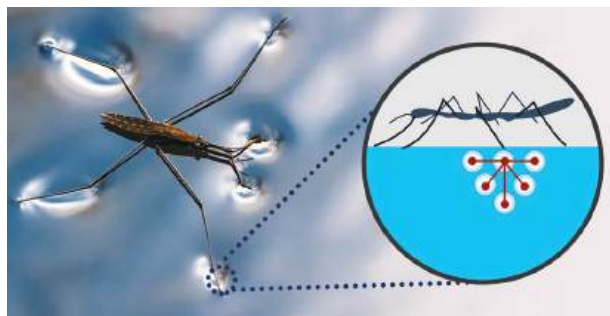


Figure 7.21 Water striders can walk on water because of the surface tension of water

The hairs of the painting brush cling together when taken out of water. This is because the water films formed on them tends to contract to a minimum area (Figure 7.22).

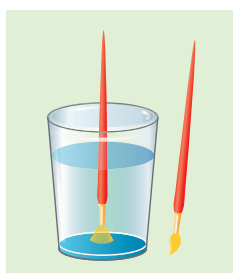


Figure 7.22 Painting brush hairs cling together due to surface tension

ACTIVITY

Needle floats on water surface

Take a greased needle of steel on a piece of blotting paper and place it gently over the water surface. Blotting paper soaks water and soon sinks down but the needle keeps floating.

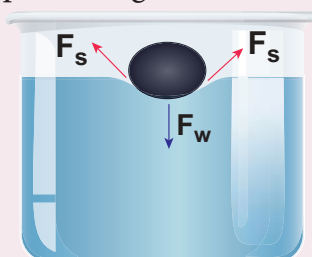


Figure 7.23 Floating needle

The floating needle causes a little depression; the forces F_s due to the surface tension of the curved surface are inclined as shown in Figure 7.23. The vertical components of these two forces support the weight of the needle. Now add liquid soap to the water and stir it. We find that the needle sinks.

ACTIVITY

Take a plastic sheet and cut out a piece in the shape of a boat (Figure 7.24). A tapering and smooth front with a notch at the back is suggested. Put a piece of camphor into the notch of the boat. Gently release the boat on the surface of the water and we find that the boat is propelled forward when the camphor dissolves. The surface tension is lowered, as the camphor dissolves and produces a difference in surface tension in the water nearby the notch. This causes the water to flow away from the back of the boat, which moves the boat forward.

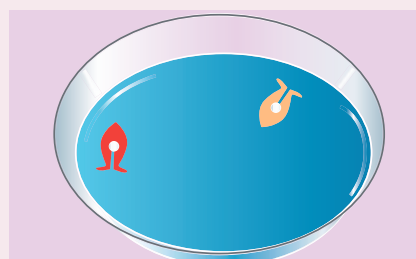


Figure 7.24 Camphor boat

7.5.2 Factors affecting the surface tension of a liquid

Surface tension for a given liquid varies in following situations

- (1) **The presence of any contamination or impurities** considerably affects the force of surface tension depending upon the degree of contamination.
- (2) **The presence of dissolved substances** can also affect the value of surface

tension. For example, a highly soluble substance like sodium chloride (NaCl) when dissolved in water (H_2O) increases the surface tension of water. But the sparingly soluble substance like phenol or soap solution when mixed in water decreases the surface tension of water.

(3) **Electrification** affects the surface tension. When a liquid is electrified, surface tension decreases. Since external force acts on the liquid surface due to electrification, area of the liquid surface increases which acts against the contraction phenomenon of the surface tension. Hence, it decreases.

(4) **Temperature** plays a very crucial role in altering the surface tension of a liquid. Obviously, the surface tension decreases linearly with the rise of temperature. For a small range of temperature, the surface tension at T_t at $t^\circ\text{C}$ is $T_t = T_0 (1 - \alpha t)$

Where, T_0 is the surface tension at temperature 0°C and α is the temperature coefficient of surface tension. It is to be noted that at the critical temperature, the surface tension is zero as the interface between liquid and vapour disappear. For example, the critical temperature of water is 374°C . Therefore, the surface tension of water is zero at that temperature. van der Waals suggested the important relation between the surface tension and the critical temperature as

$$T_t = T_0 \left(1 - \frac{t}{t_c} \right)^{\frac{3}{2}}$$

Generalizing the above relation, we get

$$T_t = T_0 \left(1 - \frac{t}{t_c} \right)^n$$

which gives more accurate value. Here, n varies for different liquids and t and t_c denote

the temperature and critical temperature in absolute scale (Kelvin scale) respectively.

7.5.3 Surface energy (S.E.) and surface tension (S.T.)

Surface Energy

Consider a sample of liquid in a container. A molecule inside the liquid is being pulled in all direction by other molecules that surround it. However, near the surface, a molecule is pulled down only by the molecules below it and there is a net downward force. As a result, the entire surface of the liquid is being pulled inward. The liquid surface thus tends to have the least surface area. To increase the surface area, some molecules are brought from the interior to the surface. For this reason, work has to be done against the forces of attraction. The amount of work done is stored as potential energy. Thus, the molecules lying on the surface possess greater potential energy than other molecules. This excess energy per unit area of the free surface of the liquid is called 'surface energy'. In other words, the work done in increasing the surface area per unit area of the liquid against the surface tension force is called the surface energy of the liquid.

$$\begin{aligned} \text{Surface energy} &= \frac{\text{work done in increasing the surface area}}{\text{increase in surface area}} \\ &= \frac{W}{\Delta A} \end{aligned} \quad (7.26)$$

It is expressed in J m^{-2} or N m^{-1} .

Surface tension

The surface tension of a liquid is defined as the force per unit length of the liquid or the energy per unit area of the surface of a liquid

$$T = \frac{F}{l} \quad (7.27)$$

The SI unit and dimensions of T are $N m^{-1}$ and $M T^{-2}$, respectively.

Relation between surface tension and surface energy:

Consider a rectangular frame of wire ABCD in a soap solution (Figure 7.25). Let AB be the movable wire. Suppose the frame is dipped in soap solution, soap film is formed which pulls the wire AB inward due to surface tension. Let F be the force due to surface tension, then

$$F = (2T)l$$

here, 2 is introduced because it has two free surfaces. Suppose AB is moved by a small distance Δx to new a position $A'B'$. Since the area increases, some work has to be done against the inward force due to surface tension.

$$\text{Work done} = \text{Force} \times \text{distance} = (2T l) (\Delta x)$$

Increase in area of the film

$$\Delta A = (2l) (\Delta x) = 2l \Delta x$$

Therefore,

$$\begin{aligned} \text{surface energy} &= \frac{\text{work done}}{\text{increase in surface area}} \\ &= \frac{2Tl \Delta x}{2l \Delta x} = T \end{aligned} \quad (7.28)$$

Hence, the surface energy per unit area of a surface is numerically equal to the surface tension.

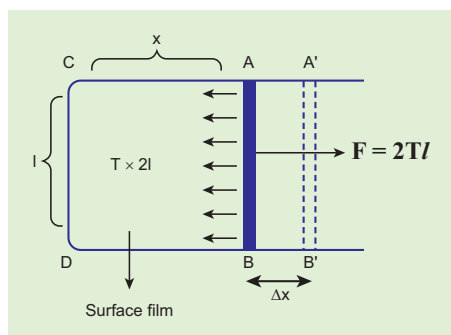


Figure 7.25 A horizontal soap film on a rectangular frame of wire ABCD



Note It should be remembered that a liquid drop has only one free surface. Therefore, the surface area of a spherical drop of radius r is equal to $4\pi r^2$, whereas, a bubble has two free surfaces and hence the surface area of a spherical bubble is equal to $2 \times 4\pi r^2$.

EXAMPLE 7.10

Let $2.4 \times 10^{-4} J$ of work is done to increase the area of a film of soap bubble from 50 cm^2 to 100 cm^2 . Calculate the value of surface tension of soap solution.

Solution:

A soap bubble has two free surfaces, therefore increase in surface area $\Delta A = A_2 - A_1 = 2(100 - 50) \times 10^{-4} \text{ m}^2 = 100 \times 10^{-4} \text{ m}^2$.

$$\begin{aligned} \text{Since, work done } W &= T \times \Delta A \Rightarrow T = \frac{W}{\Delta A} \\ &= \frac{2.4 \times 10^{-4} J}{100 \times 10^{-4} \text{ m}^2} = 2.4 \times 10^{-2} \text{ N m}^{-1} \end{aligned}$$

7.5.4 Angle of contact

When the free surface of a liquid comes in contact with a solid, then the surface of the liquid becomes curved at the point of contact. Whenever the liquid surface becomes a curve, then the angle between the two medium (solid-liquid interface) comes in the picture. For an example, when a glass plate is dipped in water with sides vertical as shown in figure, we can observe that the water is drawn up to the plate. In the same manner, instead of water the glass plate is dipped in mercury, the surface is curved but now the curve is depressed as shown in Figure 7.29



The angle between the tangent to the liquid surface at the point of contact and the solid surface inside the liquid is known as the *angle of contact between the solid and the liquid*. It is denoted by θ .

Its value is different at interfaces of different pairs of solids and liquids. In fact, it is the factor which decides whether a liquid will spread on the surface of a chosen solid or it will form droplets on it.

Let us consider three interfaces such as liquid-air, solid-air and solid-liquid with reference to the point of contact 'O' and the interfacial surface tension forces T_{sa} , T_{sl} and T_{la} on the respective interfaces as shown in Figure 7.26.

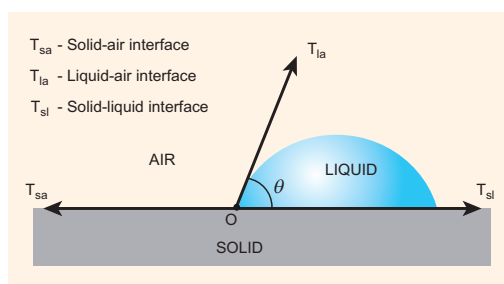


Figure 7.26 Angle of contact of a liquid

Since the liquid is stable under equilibrium, the surface tension forces between the three interfaces must also be in equilibrium. Therefore,

$$T_{sa} = T_{la} \cos \theta + T_{sl} \Rightarrow \cos \theta = \frac{T_{sa} - T_{sl}}{T_{la}} \quad (7.29)$$

From the above equation, there are three different possibilities which can be discussed as follows.

- If $T_{sa} > T_{sl}$ and $T_{sa} - T_{sl} > 0$ (water-plastic interface), then the angle of contact θ is acute angle (θ less than 90°) as $\cos \theta$ is positive.
- If $T_{sa} < T_{sl}$ and $T_{sa} - T_{sl} < 0$ (water-leaf interface), then the angle of contact is

obtuse angle (θ less than 180°) as $\cos \theta$ is negative.

- If $T_{sa} > T_{la} + T_{sl}$, then there will be no equilibrium and liquid will spread over the solid.

Therefore, the concept of *angle of contact* between the solid-liquid interface leads to some practical applications in real life. For example, soaps and detergents are wetting agents. When they are added to an aqueous solution, they will try to minimize the angle of contact and in turn penetrate well in the cloths and remove the dirt. On the other hand, water proofing paints are coated on the outer side of the building so that it will enhance the angle of contact between the water and the painted surface during the rainfall.

7.5.5 Excess of pressure inside a liquid drop, a soap bubble, and an air bubble

As it is discussed earlier, the free surface of a liquid becomes curved when it has contact with a solid. Depending upon the nature of liquid-air or liquid-gas interface, the magnitude of interfacial surface tension varies. In other words, as a consequence of surface tension, the above such interfaces have energy and for a given volume, the surface will have a minimum energy with least area. Due to this reason, the liquid drop becomes spherical (for a smaller radius).

When the free surface of the liquid is curved, there is a difference in pressure between the inner and outer the side of the surface (Figure 7.27).

- When the liquid surface is plane, the forces due to surface tension (T , T) act tangentially to the liquid surface in

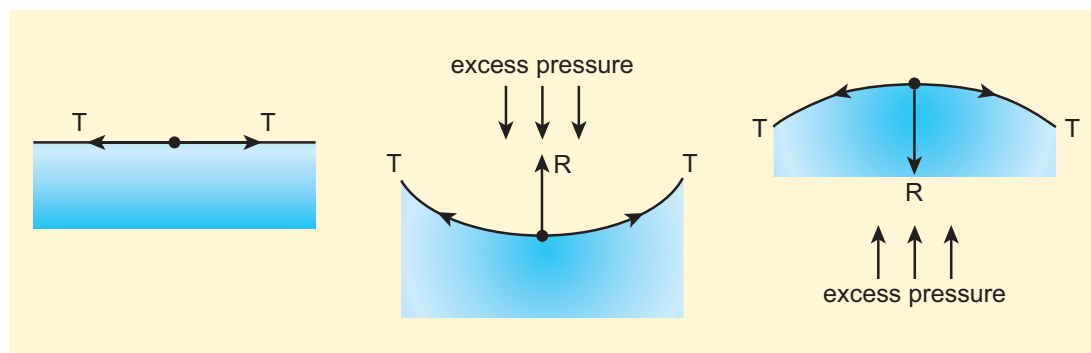


Figure 7.27 Excess of pressure across a liquid surface

opposite directions. Hence, the resultant force on the molecule is zero. Therefore, in the case of a plane liquid surface, the pressure on the liquid side is equal to the pressure on the vapour side.

- ii) When the liquid surface is curved, every molecule on the liquid surface experiences forces (F_T, F_T) due to surface tension along the tangent to the surface. Resolving these forces into rectangular components, we find that horizontal components cancel out each other while vertical components get added up. Therefore, the resultant force normal to the surface acts on the curved surface of the liquid. Similarly, for a convex surface, the resultant force is directed inwards towards the centre of curvature, whereas the resultant force is directed outwards from the centre of curvature for a concave surface. Thus, for a curved liquid surface in equilibrium, the pressure on its concave side is greater than the pressure on its convex side.

Excess of pressure inside a bubble and a liquid drop:

The small bubbles and liquid drops are spherical because of the forces of surface tension. The fact that a bubble or a liquid drop does not collapse due to the combined effect

indicates that the pressure inside a bubble or a drop is greater than that outside it.

1) Excess of pressure inside air bubble in a liquid.

Consider an air bubble of radius R inside a liquid having surface tension T as shown in Figure 7.28 (a). Let P_1 and P_2 be the pressures outside and inside the air bubble, respectively. Now, the excess pressure inside the air bubble is $\Delta P = P_2 - P_1$.

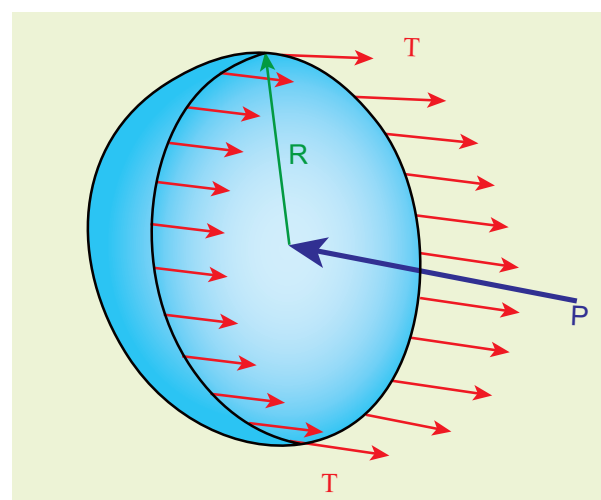


Figure 7.28. (a) Air bubble

In order to find the excess pressure inside the air bubble, let us consider the forces acting on the air bubble. For the hemispherical portion of the bubble, considering the forces acting on it, we get,



- i) The force due to surface tension acting towards right around the rim of length $2\pi R$ is $F_T = 2\pi RT$
- ii) The force due to outside pressure P_1 is to the right acting across a cross sectional area of πR^2 is $F_{P_1} = P_1 \pi R^2$
- iii) The force due to pressure P_2 inside the bubble, acting to the left is $F_{P_2} = P_2 \pi R^2$.

As the air bubble is in equilibrium under the action of these forces, $F_{P_2} = F_T + F_{P_1}$

$$P_2 \pi R^2 = 2\pi RT + P_1 \pi R^2$$

$$\Rightarrow (P_2 - P_1) \pi R^2 = 2\pi RT$$

$$\text{Excess pressure is } \Delta P = P_2 - P_1 = \frac{2T}{R} \quad (7.30)$$

2) Excess pressure inside a soap bubble

Consider a soap bubble of radius R and the surface tension of the soap bubble be T as shown in Figure 7.28 (b). A soap bubble has two liquid surfaces in contact with air, one inside the bubble and other outside the bubble. Therefore, the force on the soap bubble due to surface tension is $2 \times 2\pi RT$. The various forces acting on the soap bubble are,

- i) Force due to surface tension $F_T = 4\pi RT$ towards right
- ii) Force due to outside pressure, $F_{P_1} = P_1 \pi R^2$ towards right
- iii) Force due to inside pressure, $F_{P_2} = P_2 \pi R^2$ towards left

As the bubble is in equilibrium, $F_{P_2} = F_T + F_{P_1}$

$$P_2 \pi R^2 = 4\pi RT + P_1 \pi R^2$$

$$\Rightarrow (P_2 - P_1) \pi R^2 = 4\pi RT$$

$$\text{Excess pressure is } \Delta P = P_2 - P_1 = \frac{4T}{R} \quad (7.31)$$

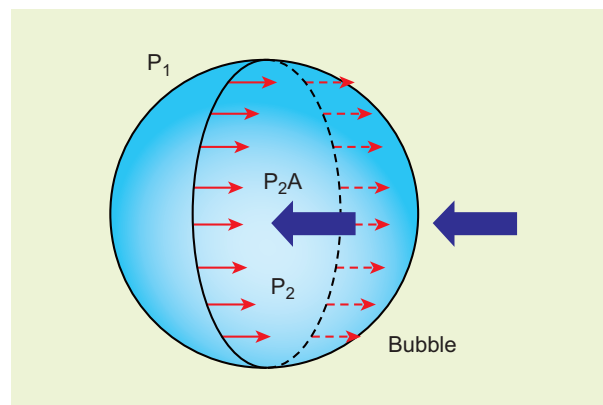


Figure 7.28 (b) Soap bubble

3) Excess pressure inside the liquid drop

Consider a liquid drop of radius R and the surface tension of the liquid is T as shown in Figure 7.28 (c).

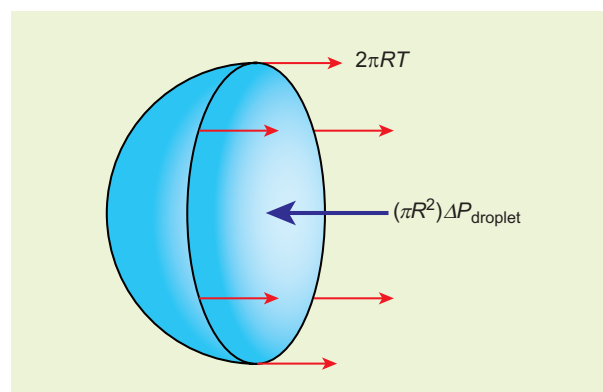


Figure 7.28 (c) Liquid drop

The various forces acting on the liquid drop are,

- i) Force due to surface tension $F_T = 2\pi RT$ towards right
- ii) Force due to outside pressure, $F_{P_1} = P_1 \pi R^2$ towards right
- iii) Force due to inside pressure, $F_{P_2} = P_2 \pi R^2$ towards left

As the drop is in equilibrium, $F_{P_2} = F_T + F_{P_1}$

$$P_2 \pi R^2 = 2\pi RT + P_1 \pi R^2$$

$$\Rightarrow (P_2 - P_1) \pi R^2 = 2\pi RT$$

$$\text{Excess pressure is } \Delta P = P_2 - P_1 = \frac{2T}{R} \quad (7.32)$$



The smaller the radius of a liquid drop, the greater is the excess of pressure inside the drop. It is due to this excess of pressure inside, the tiny fog droplets are rigid enough to behave like solids.

When an ice-skater skate over the surface of the ice, some ice melts due to the pressure exerted by the sharp metal edges of the skates, the tiny droplets of water act as rigid ball- bearings and help the skaters to run along smoothly.

EXAMPLE 7.11

If excess pressure is balanced by a column of oil (with specific gravity 0.8) 4 mm high, where $R = 2.0 \text{ cm}$, find the surface tension of the soap bubble.

Solution

The excess of pressure inside the soap bubble is $\Delta P = P_2 - P_1 = \frac{4T}{R}$

But $\Delta P = P_2 - P_1 = \rho gh \Rightarrow \rho gh = \frac{4T}{R}$
 $\Rightarrow$ Surface tension,

$$T = \frac{\rho ghR}{4} = \frac{(800)(9.8)(4 \times 10^{-3})(2 \times 10^{-2})}{4} =$$

$$T = 15.68 \times 10^{-2} \text{ N m}^{-1}$$

7.5.6 Capillarity

The word ‘capilla’ means hair in Latin. If the tubes were hair thin, then the rise would be very large. It means that the tube having a very small diameter is called a ‘capillary tube’. When a glass capillary tube open at both ends is dipped vertically in water, the water in the tube will rise above the level of water in the vessel. In case of mercury, the liquid is depressed in the tube below the level of mercury in the vessel (shown in Figure 7.29). In a liquid whose angle of contact with solid is less than 90° , suffers capillary rise. On the other hand, in a liquid whose angle of contact is greater than 90° (shown in Figure 7.30), suffers capillary fall (Table 7.4). The rise or fall of a liquid in a narrow tube is called capillarity or capillary action. Depending on the diameter of the capillary tube, liquid rises or falls to different heights.

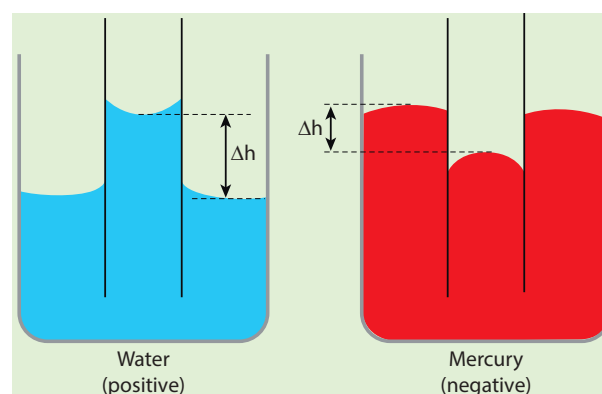


Figure 7.29 Capillary rise or fall

Table 7.4 Capillary rise and fall

Contact angle	Strength of		Degree of wetting	Meniscus	Rise or fall of liquid in the capillary tube
	Cohesive force	Adhesive force			
$\theta=0$ (A)	Weak	Strong	Perfect Wetting	Plane	Neither rises nor is depressed
$\theta<90$ (B)	Weak	Strong	High	Concave	Rise of liquid
$\theta>90$ (C)	Strong	Weak	Low	Convex	Fall of liquid

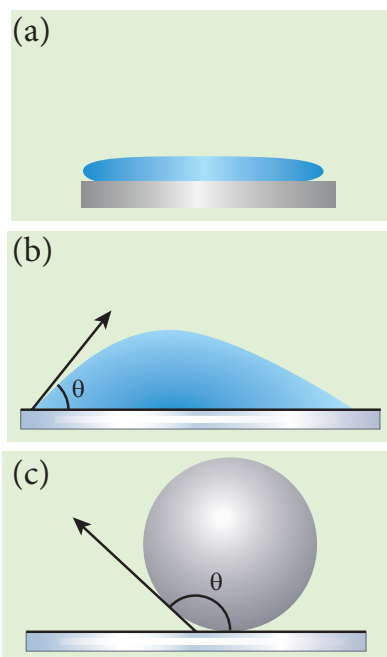


Figure 7.30 (a) water on silver surface (b) water on glass plate (c) mercury on glass plate

Practical applications of capillarity

- Due to capillary action, oil rises in the cotton within an earthen lamp. Likewise, sap rises from the roots of a plant to its leaves and branches.
- Absorption of ink by a blotting paper
- Capillary action is also essential for the tear fluid from the eye to drain constantly.
- Cotton dresses are preferred in summer because cotton dresses have fine pores which act as capillaries for sweat.

7.5.7 Surface Tension by capillary rise method

The pressure difference across a curved liquid-air interface is the basic factor behind the rising up of water in a narrow tube (influence of gravity is ignored). The capillary rise is more dominant in the case of very fine tubes. But this phenomenon is the outcome of the force of surface tension. In order to arrive a relation between the capillary rise (h) and surface tension (T),

consider a capillary tube which is held vertically in a beaker containing water; the water rises in the capillary tube to a height h due to surface tension (Figure 7.31).

The surface tension force F_T , acts along the tangent at the point of contact downwards and its reaction force upwards. Surface tension T , is resolved into two components
i) Horizontal component $T \sin \theta$ and
ii) Vertical component $T \cos \theta$ acting upwards, all along the whole circumference of the meniscus.

$$\begin{aligned} \text{Total upward force} \\ = (T \cos \theta) (2\pi r) = 2\pi r T \cos \theta \end{aligned}$$

where θ is the angle of contact, r is the radius of the tube. Let ρ be the density of water and h be the height to which the liquid rises inside the tube. Then,

$$\left(\begin{array}{l} \text{the volume of} \\ \text{liquid column in} \\ \text{the tube, } V \end{array} \right) = \left(\begin{array}{l} \text{volume of the} \\ \text{liquid column of radius } r \\ \text{height } h \end{array} \right) + \left(\begin{array}{l} \text{volume of liquid of radius } r \\ \text{and height } r - \text{Volume of the} \\ \text{hemisphere of radius } r \end{array} \right)$$

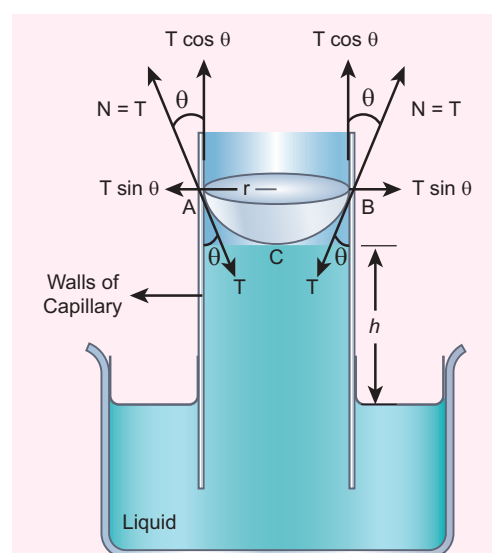


Figure 7.31 Capillary rise by surface tension



$$V = \pi r^2 h + \left(\pi r^2 \times r - \frac{2}{3} \pi r^3 \right) \Rightarrow V = \pi r^2 h + \frac{1}{3} \pi r^3$$

The upward force supports the weight of the liquid column above the free surface, therefore,

$$2\pi r T \cos\theta = \pi r^2 \left(h + \frac{1}{3} r \right) \rho g \Rightarrow T = \frac{r \left(h + \frac{1}{3} r \right) \rho g}{2 \cos\theta}$$

If the capillary is a very fine tube of radius (i.e., radius is very small) then $\frac{r}{3}$ can be neglected when it is compared to the height h . Therefore,

$$T = \frac{r \rho g h}{2 \cos\theta} \quad (7.33)$$

Liquid rises through a height h

$$h = \frac{2T \cos\theta}{r \rho g} \Rightarrow h \propto \frac{1}{r} \quad (7.34)$$

This implies that the capillary rise (h) is inversely proportional to the radius (r) of the tube. i.e., the smaller the radius of the tube greater will be the capillarity.

EXAMPLE 7.12

Water rises in a capillary tube to a height of 2.0cm. How much will the water rise through another capillary tube whose radius is one-third of the first tube?

Solution

From equation (7.34), we have

$$h \propto \frac{1}{r} \Rightarrow hr = \text{constant}$$

Consider two capillary tubes with radius r_1 and r_2 which on placing in a liquid, capillary rises to height h_1 and h_2 , respectively. Then,

$$h_1 r_1 = h_2 r_2 = \text{constant}$$

$$\Rightarrow h_2 = \frac{h_1 r_1}{r_2} = \frac{(2 \times 10^{-2} \text{ m}) \times r}{\frac{r}{3}} \Rightarrow h_2 = 6 \times 10^{-2} \text{ m}$$

EXAMPLE 7.13

Mercury has an angle of contact equal to 140° with soda lime glass. A narrow tube of radius 2 mm, made of this glass is dipped in a trough containing mercury. By what amount does the mercury dip down in the tube relative to the liquid surface outside? Surface tension of mercury $T = 0.456 \text{ N m}^{-1}$; Density of mercury $\rho = 13.6 \times 10^3 \text{ kg m}^{-3}$

Solution

Capillary descent, $\cos 140 = \cos(90 + 50)$
 $-\sin 50 = -0.7660$

$$h = \frac{2T \cos\theta}{r \rho g} = \frac{2 \times (0.456 \text{ N m}^{-1}) (\cos 140^\circ)}{(2 \times 10^{-3} \text{ m}) (13.6 \times 10^3) (9.8 \text{ m s}^{-2})}$$

$$= \frac{2 \times 0.456 \times (-0.7660)}{2 \times 13.6 \times 9.8}$$

$$= \frac{-0.6986}{266.56} = -2.62 \times 10^{-3} \text{ m}$$

where, negative sign indicates that there is fall of mercury (mercury is depressed) in glass tube.

7.5.8 Applications of surface tension

- Mosquitoes lay their eggs on the surface of water. To reduce the surface tension of water, a small amount of oil is poured. This breaks the elastic film of water surface and eggs are killed by drowning.
- Chemical engineers must finely adjust the surface tension of the liquid, so it forms droplets of designed size and so it adheres to the surface without smearing. This is used in desktop printing, to paint automobiles and decorative items.
- Specks of dirt get removed when detergents are added to hot water while washing clothes because surface tension is reduced.
- A fabric can be made waterproof, by adding suitable waterproof material (wax) to the fabric. This increases the angle of contact.

7.6

BERNOULLI'S THEOREM

7.6.1 Equation of continuity

In order to discuss the mass flow rate through a pipe, it is necessary to assume that the flow of fluid is steady, the flow of the fluid is said to be steady if at any given point, the velocity of each passing fluid particle remains constant with respect to time. Under this condition, the path taken by the fluid particle is a streamline.

Consider a pipe AB of varying cross sectional area a_1 and a_2 such that $a_1 > a_2$. A non-viscous and incompressible liquid flows steadily through the pipe, with velocities v_1 and v_2 in area a_1 and a_2 , respectively as shown in Figure 7.32.

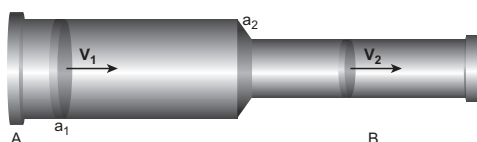


Fig 7.32 A streamlined flow of fluid through a pipe of varying cross sectional area

Let m_1 be the mass of fluid flowing through section A in time Δt , $m_1 = (a_1 v_1 \Delta t) \rho$

Let m_2 be the mass of fluid flowing through section B in time Δt , $m_2 = (a_2 v_2 \Delta t) \rho$

For an incompressible liquid, mass is conserved $m_1 = m_2$

$$a_1 v_1 \Delta t \rho = a_2 v_2 \Delta t \rho$$

$$a_1 v_1 = a_2 v_2 \Rightarrow a v = \text{constant} \quad (7.35)$$

which is called the equation of continuity and it is a statement of conservation of mass in the flow of fluids.

In general, $a v = \text{constant}$, which means that the volume flux or flow rate remains constant throughout the pipe. In other words, the smaller the cross section, greater will be the velocity of the fluid.

EXAMPLE 7.14

In a normal adult, the average speed of the blood through the aorta (radius $r = 0.8$ cm) is 0.33 ms^{-1} . From the aorta, the blood goes into major arteries, which are 30 in number, each of radius 0.4 cm. Calculate the speed of the blood through the arteries.

Solution:

$$a_1 v_1 = 30 a_2 v_2 \Rightarrow \pi r_1^2 v_1 = 30 \pi r_2^2 v_2$$

$$v_2 = \frac{1}{30} \left(\frac{r_1}{r_2} \right)^2 v_1$$

$$\Rightarrow v_2 = \frac{1}{30} \times \left(\frac{0.8 \times 10^{-2} \text{ m}}{0.4 \times 10^{-2} \text{ m}} \right)^2 \times (0.33 \text{ ms}^{-1})$$

$$v_2 = 0.044 \text{ m s}^{-1}$$

7.6.2 Pressure, kinetic and potential energy of liquids

A liquid in a steady flow can possess three kinds of energy. They are (1) Kinetic energy, (2) Potential energy, and (3) Pressure energy, respectively.

- i) **Kinetic energy:** The kinetic energy of a liquid of mass m moving with a velocity v is given by

$$KE = \frac{1}{2} m v^2$$

The kinetic energy per unit mass =

$$\frac{KE}{m} = \frac{\frac{1}{2} m v^2}{m} = \frac{1}{2} v^2$$



Similarly, the kinetic energy per unit volume

$$= \frac{KE}{\text{volume}} = \frac{\frac{1}{2}mv^2}{V} = \frac{1}{2} \left(\frac{m}{V} \right) v^2 = \frac{1}{2} \rho v^2$$

- ii) **Potential energy:** The potential energy of a liquid of mass m at a height h above the ground level is given by

$$PE = mgh$$

The potential energy per unit mass

$$= \frac{PE}{m} = \frac{mgh}{m} = gh$$

Similarly, the potential energy per unit

$$\text{volume} = \frac{PE}{\text{volume}} = \frac{mgh}{V} = \left(\frac{m}{V} \right) gh = \rho gh$$

- iii) **Pressure energy:** The energy acquired by a fluid by applying pressure on the fluid. We know that

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \Rightarrow \text{Force} = \text{Pressure} \times \text{Area}$$

$$F \times d = (P A) \times d = P (A \times d)$$

$$\Rightarrow F \times d = W = P V = \text{pressure energy}$$

Therefore, pressure energy, $E_p = PV$

The pressure energy per unit mass =

$$\frac{E_p}{m} = \frac{PV}{m} = \frac{P}{\frac{m}{V}} = \frac{P}{\rho}$$

Similarly, the pressure energy per unit

$$\text{volume} = \frac{E_p}{\text{volume}} = \frac{PV}{V} = P$$

7.6.3 Bernoulli's theorem and its applications

In 1738, the Swiss scientist Daniel Bernoulli developed a relationship for the flow of fluid through a pipe of varying cross section. He proposed a theorem for the streamline flow of a liquid based on the law of conservation of energy.

Bernoulli's theorem

According to Bernoulli's theorem, the sum of pressure energy, kinetic energy, and potential energy per unit mass of an incompressible, non-viscous fluid in a streamlined flow remains a constant. Mathematically,

$$\frac{P}{\rho} + \frac{1}{2}v^2 + gh = \text{constant} \quad (7.36)$$

This is known as Bernoulli's equation.

Proof:

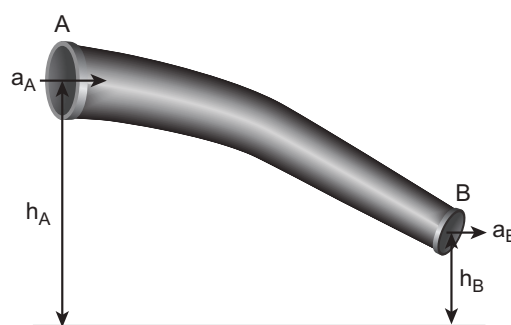


Figure 7.33 Flow of liquid through a pipe AB

Let us consider a flow of liquid through a pipe AB as shown in Figure 7.33. Let V be the volume of the liquid when it enters A in a time t which is equal to the volume of the liquid leaving B in the same time. Let a_A , v_A and P_A be the area of cross section of the tube, velocity of the liquid and pressure exerted by the liquid at A respectively.

Let the force exerted by the liquid at A is

$$F_A = P_A a_A$$

Distance travelled by the liquid in time t is

$$d = v_A t$$

Therefore, the work done is

$$W = F_A d = P_A a_A v_A t$$

But $a_A v_A t = a_A d = V$, volume of the liquid entering at A.

Thus, the work done is the pressure energy (at A), $W = F_A d = P_A V$

Pressure energy per unit volume at A

$$\frac{\text{Pressure energy}}{\text{volume}} = \frac{P_A V}{V} = P_A$$

Pressure energy per unit mass at A

$$\frac{\text{Pressure energy}}{\text{mass}} = \frac{P_A V}{m} = \frac{P_A}{\frac{m}{V}} = \frac{P_A}{\rho}$$

Since m is the mass of the liquid entering at A in a given time, therefore, pressure energy of the liquid at A is

$$E_{PA} = P_A V = P_A V \times \left(\frac{m}{m}\right) = m \frac{P_A}{\rho}$$

Potential energy of the liquid at A,

$$PE_A = mg h_A,$$

Due to the flow of liquid, the kinetic energy of the liquid at A,

$$KE_A = \frac{1}{2} m v_A^2$$

Therefore, the total energy due to the flow of liquid at A, $E_A = E_{PA} + KE_A + PE_A$

$$E_A = m \frac{P_A}{\rho} + \frac{1}{2} m v_A^2 + mg h_A$$

Similarly, let a_B , v_B , and P_B be the area of cross section of the tube, velocity of the liquid, and pressure exerted by the liquid at B. Calculating the total energy at E_B , we get

$$E_B = m \frac{P_B}{\rho} + \frac{1}{2} m v_B^2 + mg h_B$$

From the law of conservation of energy,

$$E_A = E_B$$

$$m \frac{P_A}{\rho} + \frac{1}{2} m v_A^2 + mg h_A = m \frac{P_B}{\rho} + \frac{1}{2} m v_B^2 + mg h_B$$

$$\frac{P_A}{\rho} + \frac{1}{2} v_A^2 + g h_A = \frac{P_B}{\rho} + \frac{1}{2} v_B^2 + g h_B = \text{constant}$$

Thus, the above equation can be written as

$$\frac{P}{\rho g} + \frac{1}{2} \frac{v^2}{g} + h = \text{constant}$$

The above equation is the consequence of the conservation of energy which is true until there is no loss of energy due to friction. But in practice, some energy is lost due to friction. This arises due to the fact that in a fluid flow, the layers flowing with different velocities exert frictional forces on each other. This loss of energy is generally converted into heat energy. Therefore, Bernoulli's relation is strictly valid for fluids with zero viscosity or non-viscous liquids. Notice that when the liquid flows through a horizontal pipe, then $h = 0 \Rightarrow \frac{P}{\rho g} + \frac{1}{2} \frac{v^2}{g} = \text{constant}$.

Applications of Bernoulli's Theorem

(a) Blowing off roofs during wind storm

In olden days, the roofs of the huts or houses were designed with a slope as shown in Figure.7.34. One important scientific reason is that as per the Bernoulli's principle, it will be safeguarded except roof during storm or cyclone.

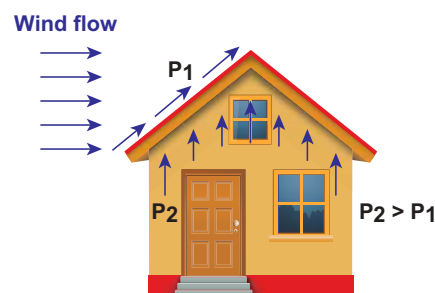


Figure 7.34 Roofs of the huts or houses

During cyclonic condition, the roof is blown off without damaging the other parts of the house. In accordance with the Bernoulli's principle, the high wind blowing over the roof creates a low-pressure P_1 . The pressure under the roof P_2 is greater. Therefore, this pressure difference ($P_2 - P_1$) creates an up thrust and the roof is blown off.

(b) Aerofoil lift

The wings of an airplane (aerofoil) are so designed that its upper surface is more curved than the lower surface and the front edge is broader than the rear edge. As the aircraft moves, the air moves faster above the aerofoil than at the bottom as shown in Figure 7.35.

According to Bernoulli's Principle, the pressure of air below is greater than above, which creates an upthrust called the dynamic lift to the aircraft.

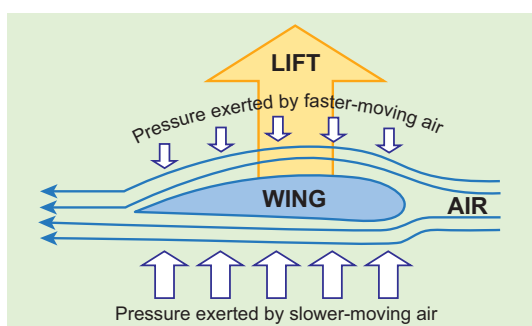


Figure 7.35 Aerofoil lift

(c) Bunsen burner

In this, the gas comes out of the nozzle with high velocity, hence the pressure in the stem decreases. So outside air reaches into the burner through an air vent and the mixture of air and gas gives a blue flame as shown in Figure 7.36.

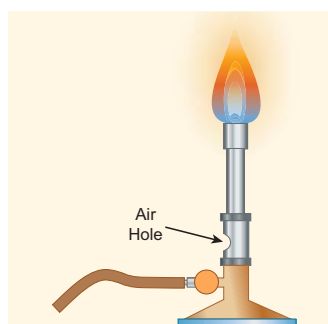


Figure 7.36 Bunsen burner

(d) Venturimeter

This device is used to measure the rate of flow (or say flow speed) of the incompressible

fluid flowing through a pipe. It works on the principle of Bernoulli's theorem. It consists of two wider tubes A and A' (with cross sectional area A) connected by a narrow tube B (with cross sectional area a). A manometer in the form of U-tube is also attached between the wide and narrow tubes as shown in Figure 7.37. The manometer contains a liquid of density ' ρ_m '.

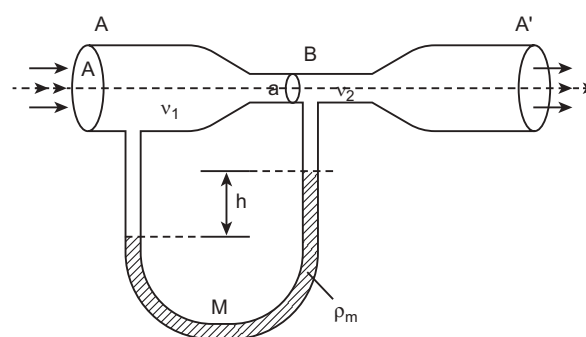


Figure 7.37 A schematic diagram of venturimeter

Let P_1 be the pressure of the fluid at the wider region of the tube A. Let us assume that the fluid of density ' ρ ' flows from the pipe with speed ' v_1 ' and into the narrow region, its speed increases to ' v_2 '. According to the Bernoulli's equation, this increase in speed is accompanied by a decrease in the fluid pressure P_2 at the narrow region of the tube B. Therefore, the pressure difference between the tubes A and B is noted by measuring the height difference ($\Delta P = P_1 - P_2$) between the surfaces of the manometer liquid.

From the equation of continuity, we can say that $Av_1 = av_2$ which means that

$$v_2 = \frac{A}{a} v_1.$$

Using Bernoulli's equation,

$$P_1 + \rho \frac{v_1^2}{2} = P_2 + \rho \frac{v_2^2}{2} = P_2 + \rho \frac{1}{2} \left(\frac{A}{a} v_1 \right)^2$$

From the above equation, the pressure difference

$$\Delta P = P_1 - P_2 = \rho \frac{v_1^2}{2} \frac{(A^2 - a^2)}{a^2}$$

Thus, the speed of flow of fluid at the wide end of the tube A

$$v_1^2 = \frac{2(\Delta P)a^2}{\rho(A^2 - a^2)} \Rightarrow v_1 = \sqrt{\frac{2(\Delta P)a^2}{\rho(A^2 - a^2)}}$$

The volume of the liquid flowing out per second is

$$V = Av_1 = A \sqrt{\frac{2(\Delta P)a^2}{\rho(A^2 - a^2)}} = aA \sqrt{\frac{2(\Delta P)}{\rho(A^2 - a^2)}}$$

(e) Other applications

This Bernoulli's concept is mainly used in the design of carburetor of automobiles, filter pumps, atomizers, and sprayers. For

example, the carburetor has a very fine channel called nozzle through which the air is allowed to flow in larger speed. In this case, the pressure is lowered at the narrow neck and in turn, the required fuel or petrol is sucked into the chamber so as to provide the correct mixture of air and fuel necessary for ignition process.

ACTIVITY

A bottle is filled with thermocol balls. One end of a flexible tube is kept inside the bottle immersed inside the balls. The free end is rotated and we find the balls sprayed all around. This explains the working of an atomizer or sprayer.



A spider web is much stronger than what we think. A single strand of spider silk can stop flying insects which are tens and thousands times its mass. The young's modulus of the spider web is approximately $4.5 \times 10^9 \text{ N m}^{-2}$. Compare this value with Young's modulus of wood.



SUMMARY

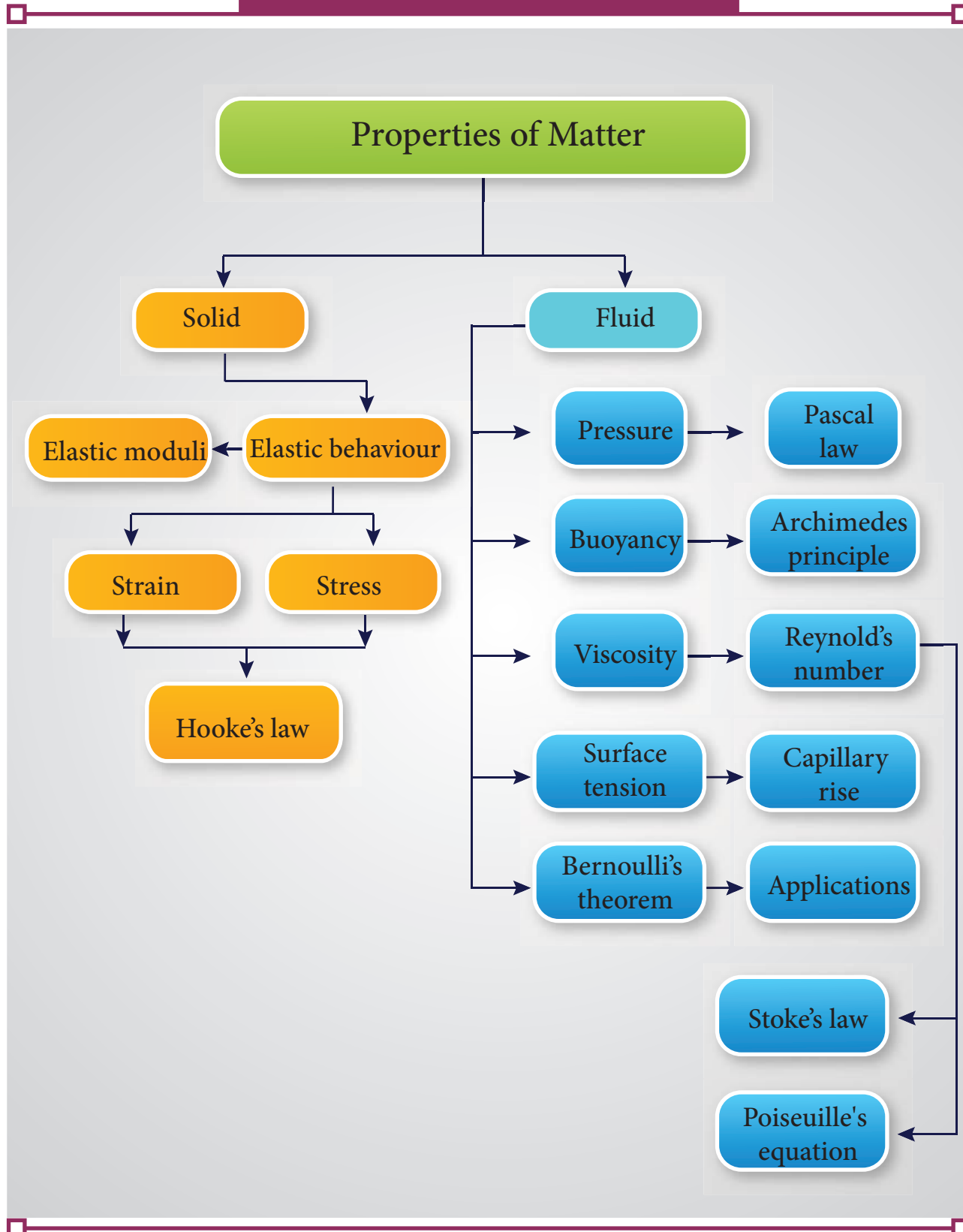
- The force between the atoms of an element is called inter-atomic force whereas the force between the molecules of a compound is called inter-molecular force.
- *Hooke's law*: within the elastic limit, the stress is directly proportional to strain.
- The force per unit area is known as *stress*. If F is the force applied and A is the area of cross section of the body then the magnitude of *stress* is equal to F/A . Tensile or compressional stress can be expressed using a single term called *longitudinal stress*.
- The ratio of change in length to the original length of a cylinder is $\Delta L/L$, which is known as *longitudinal strain*.
- Within the elastic limit, the ratio of longitudinal stress to the longitudinal strain is called the Young's modulus of the material of the wire.
- Within the elastic limit, the ratio of volume stress to the volume strain is called the *bulk modulus*.
- Within the elastic limit, the ratio of shear stress to the shear strain is called the *rigidity modulus*.
- Poisson's ratio = lateral strain/longitudinal strain
- The elastic potential energy stored in the wire per unit volume is

$$U = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{1}{2} \times Y \times (\text{strain})^2$$
, where Y denotes Young's modulus of the material.
- If F is the magnitude of the normal force acting on the surface area A , then the pressure is defined as the '*force acting per unit area*'.
- The total pressure at a depth h below the liquid surface is $P = P_a + \rho gh$, where P_a is the atmospheric pressure which is equal to 1.013×10^5 Pa.
- Pascal's law states that the pressure in a fluid at rest is the same at all points if they are at the same height.
- The law of floatation states that a body will float in a liquid if the weight of the liquid displaced by the immersed part of the body is equal to or greater than the weight of the body.
- The coefficient of viscosity of a liquid is the viscous force acting tangentially per unit area of a liquid layer having a unit velocity gradient in a direction perpendicular to the direction of flow of the liquid.
- When a liquid flows such that each particle of the liquid passing a point moves along the same path and has the same velocity as its predecessor then the flow of liquid is said to be streamlined flow.
- During the flow of fluid, when the critical velocity is exceeded by the moving fluid, the motion becomes *turbulent*.
- Reynold's number has a significance as it decides which decides whether the flow of fluid through a cylindrical pipe is streamlined or turbulent.

SUMMARY (cont.)

- Stokes formula $F = 6\pi\eta av$, where F is the viscous force acting on a sphere of radius a and v is the terminal velocity of the sphere.
- The surface tension of a liquid is defined as the force of tension acting on a unit length of an imaginary line drawn on the free surface of the liquid, the direction of the force being perpendicular to the line so drawn and acting parallel to the surface.
- The angle between tangents drawn at the point of contact to the liquid surface and solid surface inside the liquid is called the *angle of contact* for a pair of solid and liquid.
- The flow of a fluid is said to be steady if, at any given point, the velocity of each passing fluid particle remains constant with respect to time.
- The equation $a_1 v_1 = a_2 v_2$ is called the equation of continuity for a flow of fluid through a tube and it is due to the conservation of mass in the flow of fluids. It states that the sum of pressure energy, kinetic energy, and potential energy per unit mass of an incompressible, non-viscous fluid in a streamlined flow remains constant. i.e., $P/\rho + v^2/2 + gh = \text{constant}$.

CONCEPT MAP





EVALUATION

I. Multiple Choice Questions

1. Consider two wires X and Y. The radius of wire X is 3 times the radius of Y. If they are stretched by the same load then the stress on Y is

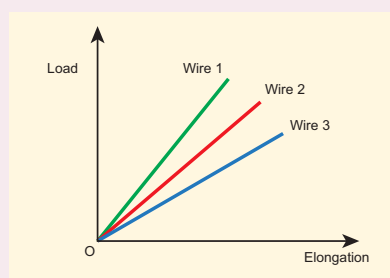
(a) equal to that on X
(b) thrice that on X
(c) nine times that on X
(d) Half that on X



2. If a wire is stretched to double of its original length, then the strain in the wire is

(a) 1 (b) 2
(c) 3 (d) 4

3. The load – elongation graph of three wires of the same material are shown in figure. Which of the following wire is the thickest?



- (a) wire 1
(b) wire 2
(c) wire 3
(d) all of them have same thickness
4. For a given material, the rigidity modulus is $\left(\frac{1}{3}\right)^{rd}$ of Young's modulus. Its Poisson's ratio is
- (a) 0 (b) 0.25
(c) 0.3 (d) 0.5

5. A small sphere of radius 2cm falls from rest in a viscous liquid. Heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity is proportional to

(NEET model 2018)

(a) 2^2 (b) 2^3
(c) 2^4 (d) 2^5

6. Two wires are made of the same material and have the same volume. The area of cross sections of the first and the second wires are A and 2A respectively. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

(NEET model 2018)

(a) 2 F (b) 4 F
(c) 8 F (d) 16 F

7. With an increase in temperature, the viscosity of liquid and gas, respectively will

(a) increase and increase
(b) increase and decrease
(c) decrease and increase
(d) decrease and decrease

8. The Young's modulus for a perfect rigid body is

(a) 0 (b) 1
(c) 0.5 (d) infinity



9. Which of the following is not a scalar?
(a) viscosity
(b) surface tension
(c) pressure
(d) stress
10. If the temperature of the wire is increased, then the Young's modulus will
(a) remain the same
(b) decrease
(c) increase rapidly
(d) increase by very a small amount
11. Copper of fixed volume V is drawn into a wire of length l . When this wire is subjected to a constant force F , the extension produced in the wire is Δl . If Y represents the Young's modulus, then which of the following graphs is a straight line?
(NEET 2014 model)
(a) Δl versus V
(b) Δl versus Y
(c) Δl versus F
(d) Δl versus $\frac{1}{l}$
12. A certain number of spherical drops of a liquid of radius R coalesce to form a single drop of radius R and volume V . If T is the surface tension of the liquid, then
(a) energy $= 4 V T \left(\frac{1}{r} - \frac{1}{R} \right)$ is released
(b) energy $= 3 V T \left(\frac{1}{r} + \frac{1}{R} \right)$ is absorbed
(c) energy $= 3 V T \left(\frac{1}{r} - \frac{1}{R} \right)$ is released
(d) energy is neither released nor absorbed
13. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?
(a) length = 200 cm, diameter = 0.5 mm
(b) length = 200 cm, diameter = 1 mm
(c) length = 200 cm, diameter = 2 mm
(d) length = 200 cm, diameter = 3 mm
14. The wettability of a surface by a liquid depends primarily on
(a) viscosity
(b) surface tension
(c) density
(d) angle of contact between the surface and the liquid
15. In a horizontal pipe of non-uniform cross section, water flows with a velocity of 1 m s^{-1} at a point where the diameter of the pipe is 20 cm. The velocity of water (1.5 m s^{-1}) at a point where the diameter of the pipe is (in cm)
(a) 8 (b) 16
(c) 24 (d) 32
- Answers:**
1) c 2) a 3) a 4) d
5) d 6) b 7) c 8) d
9) d 10) b 11) c 12) c
13) a 14) d 15) b
- ## II. Short Answer Questions
1. Define stress and strain.
 2. State Hooke's law of elasticity.
 3. Define Poisson's ratio.
 4. Explain elasticity using intermolecular forces.
 5. Which one of these is more elastic, steel or rubber? Why?



6. A spring balance shows wrong readings after using for a long time. Why?
7. What is the effect of temperature on elasticity?
8. Write down the expression for the elastic potential energy of a stretched wire.
9. State Pascal's law in fluids.
10. State Archimedes principle.
11. What do you mean by upthrust or buoyancy?
12. State the law of floatation.
13. Define coefficient of viscosity of a liquid.
14. Distinguish between streamlined flow and turbulent flow.
15. What is Reynold's number? Give its significance.
16. Define terminal velocity.
17. Write down the expression for the Stoke's force and explain the symbols involved in it.
18. State Bernoulli's theorem.
19. What are the energies possessed by a liquid? Write down their equations.
20. Two streamlines cannot cross each other. Why?
21. Define surface tension of a liquid. Mention its S.I unit and dimension.
22. How is surface tension related to surface energy?
23. Define angle of contact for a given pair of solid and liquid.
24. Distinguish between cohesive and adhesive forces.
25. What are the factors affecting the surface tension of a liquid?
26. What happens to the pressure inside a soap bubble when air is blown into it?
27. What do you mean by capillarity or capillary action?
28. A drop of oil placed on the surface of water spreads out. But a drop of water placed on oil contracts to a spherical shape. Why?
29. State the principle and usage of Venturimeter.

III. Long Answer Questions

1. State Hooke's law and verify it with the help of an experiment.
2. Explain the different types of modulus of elasticity.
3. Derive an expression for the elastic energy stored per unit volume of a wire.
4. Derive an equation for the total pressure at a depth 'h' below the liquid surface.
5. State and prove Pascal's law in fluids.
6. State and prove Archimedes principle.
7. Derive the expression for the terminal velocity of a sphere moving in a high viscous fluid using Stokes force.
8. Derive Poiseuille's formula for the volume of a liquid flowing per second through a pipe under streamlined flow.
9. Obtain an expression for the excess of pressure inside a i) liquid drop ii) liquid bubble iii) air bubble.
10. What is capillarity? Obtain an expression for the surface tension of a liquid by capillary rise method.
11. Obtain an equation of continuity for a flow of fluid on the basis of conservation of mass.



12. State and prove Bernoulli's theorem for a flow of incompressible, non-viscous, and streamlined flow of fluid.
13. Describe the construction and working of venturimeter and obtain an equation for the volume of liquid flowing per second through a wider entry of the tube.

IV. Exercises

1. A capillary of diameter d_{mm} is dipped in water such that the water rises to a height of $30mm$. If the radius of the capillary is made $\left(\frac{2}{3}\right)$ of its previous value, then compute the height up to which water will rise in the new capillary?
(Answer: 45 mm)
2. A cylinder of length 1.5 m and diameter 4 cm is fixed at one end. A tangential force of 4×10^5 N is applied at the other end. If the rigidity modulus of the cylinder is 6×10^{10} N m^{-2} then, calculate the twist produced in the cylinder.
(Answer: 45.60)
3. A spherical soap bubble A of radius 2 cm is formed inside another bubble B of radius 4 cm. Show that the radius of a single soap bubble which maintains the same pressure difference as inside the smaller and outside the larger soap bubble is lesser than radius of both soap bubbles A and B.
4. A block of Ag of mass x kg hanging from a string is immersed in a liquid of relative density 0.72. If the relative density of Ag is 10 and tension in the string is 37.12 N then compute the mass of Ag block. (Answer: $x = 4$ kg)
5. The reading of pressure meter attached with a closed pipe is 5×10^5 N m^{-2} . On opening the valve of the pipe, the reading of the pressure meter is 4.5×10^5 N m^{-2} . Calculate the speed of the water flowing in the pipe.
(Answer: 10 ms^{-1})

BOOKS FOR REFERENCE

1. Serway and Jewett, Physics for scientist and Engineers with modern physics, Brook/Cooler publishers, Eighth edition
2. Paul Tipler and Gene Mosca, Physics for scientist and engineers with modern physics, Sixth edition, W.H. Freeman and Company
3. H.C. Verma, Concepts of physics volume 1 and Volume 2, Bharati Bhawan Publishers



ICT CORNER

Properties of Matter

Through this activity you will be able to learn about the Viscosity.

STEPS:

- Use the URL or scan the QR code to open 'Viscosity' activity page.
- In the activity window, 'Select Viscosity' by dragging the ball in the meter in the side.
- Select the 'Start' button, sphere falls from top to bottom of the beaker. We can see the changes in Distance and Time.
- Spheres can be reset to the top of each beaker by clicking the 'Reset' button.



Step1



Step2



Step3



Step4



URL:

<http://www.geo.cornell.edu/hawaii/220/PRI/viscosity.html>

- * Pictures are indicative only.
- * If browser requires, allow **Flash Player** or **Java Script** to load the page.



UNIT 8

HEAT AND THERMODYNAMICS

Classical thermodynamics.... is the only physical theory of universal content which I am convinced... will never be overthrown. – Albert Einstein

LEARNING OBJECTIVES

In this unit, a student is exposed to

- meaning of heat, work and temperature
- ideal gas laws
- concept of specific heat capacity
- thermal expansion of solids, liquids and gases
- various states of matter
- Newton's law of cooling
- Stefan's law and Wien's law
- meaning of thermodynamic equilibrium
- meaning of internal energy
- zeroth and first laws of thermodynamics
- various thermodynamic processes
- work done in various thermodynamic processes
- second law of thermodynamics
- working of carnot engine and refrigerator



8.1

HEAT AND TEMPERATURE

8.1.1 Introduction

Temperature and heat play very important role in everyday life. All species can function properly only if its body is maintained at a particular temperature. In fact life on Earth is possible because the Sun maintains its temperature. Understanding the meaning of temperature and heat are very crucial to understand the nature. Thermodynamics

is a branch of physics which explains the phenomena of temperature, heat etc. The concepts presented in this chapter will help us to understand the terms 'hot' and 'cold' and also differentiate heat from temperature. In thermodynamics, heat and temperature are two different but closely related parameters.

8.1.2 Meaning of heat

When an object at higher temperature is placed in contact with another object at lower temperature, there will be a spontaneous flow of energy from the object

at higher temperature to the one at lower temperature. This energy is called heat. This process of energy transfer from higher temperature object to lower temperature object is called heating. Due to flow of heat sometimes the temperature of the body will increase or sometimes it may not increase.



Note There is a misconception that heat is a quantity of energy. People often talk ‘this water has more heat or less heat’. These words are meaningless. Heat is not a quantity. Heat is an energy in transit which flows from higher temperature object to lower temperature object. Once the heating process is stopped we cannot use the word heat. When we use the word ‘heat’, it is the energy in transit but not energy stored in the body.

EXAMPLE 8.1

- ‘A lake has more rain.’
- ‘A hot cup of coffee has more heat.’
What is wrong in these two statements?

Solution

- When it rains, lake receives water from the cloud. Once the rain stops, the lake will have more water than before raining. Here ‘raining’ is a process which brings water from the cloud. Rain is not a quantity rather it is water in transit. So the statement ‘lake has more rain’ is wrong, instead the ‘lake has more water’ will be appropriate.

- When heated, a cup of coffee receives heat from the stove. Once the coffee is taken from the stove, the cup of coffee has more internal energy than before. ‘Heat’ is the energy in transit and which flows from an object at higher temperature to an object at lower temperature. Heat is not a quantity. So the statement ‘A hot cup of coffee has more heat’ is wrong, instead ‘coffee is hot’ will be appropriate.

8.1.3 Meaning of work

When you rub your hands against each other the temperature of the hands increases. You have done some work on your hands by rubbing. The temperature of the hands increases due to this work. Now if you place your hands on the chin, the temperature of the chin increases. This is because the hands are at higher temperature than the chin. In the above example, the temperature of hands is increased due to work and temperature of the chin is increased due to heat transfer from the hands to the chin. It is shown in the Figure 8.1

By doing work on the system, the temperature in the system will increase and sometimes may not. Like heat, work is also



Figure 8.1 Difference between work and heat

not a quantity and through the work energy is transferred to the system. So we cannot use the word 'the object contains more work' or 'less work'.

Either the system can transfer energy to the surrounding by doing work on surrounding or the surrounding may transfer energy to the system by doing work on the system. For the transfer of energy from one body to another body through the process of work, they need not be at different temperatures.

8.1.4 Meaning of temperature

Temperature is the degree of hotness or coolness of a body. Hotter the body higher is its temperature. The temperature will determine the direction of heat flow when two bodies are in thermal contact.

The SI unit of temperature is **kelvin (K)**.

In our day to day applications, **Celsius** ($^{\circ}\text{C}$) and **Fahrenheit** ($^{\circ}\text{F}$) scales are used.

Temperature is measured with a thermometer.

The conversion of temperature from one scale to other scale is given in Table 8.1

Table 8.1 Temperature conversion

Scale	To Kelvin	From Kelvin
Celsius	$K = ^{\circ}\text{C} + 273.15$	$^{\circ}\text{C} = K - 273.15$
Fahrenheit	$K = (^{\circ}\text{F} + 459.67) \div 1.8$	$^{\circ}\text{F} = (K \times 1.8) - 459.67$
Scale	To Fahrenheit	From Fahrenheit
Celsius	$^{\circ}\text{F} = (1.8 \times ^{\circ}\text{C}) + 32$	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1.8$
Scale	To Celsius	From Celsius
Fahrenheit	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1.8$	$^{\circ}\text{F} = (1.8 \times ^{\circ}\text{C}) + 32$

8.2

THERMAL PROPERTIES OF MATTER

8.2.1 Boyle's law, Charles' law and ideal gas law

For a given gas at low pressure (density) kept in a container of volume V , experiments revealed the following information.

- When the gas is kept at constant temperature, the pressure of the gas is inversely proportional to the volume.

$P \propto \frac{1}{V}$. It was discovered by Robert Boyle (1627-1691) and is known as Boyle's law.

- When the gas is kept at constant pressure, the volume of the gas is directly proportional to absolute temperature. $V \propto T$. It was discovered by Jacques Charles (1743-1823) and is known as Charles' law.

- By combining these two equations we have

$PV = CT$. Here C is a positive constant.

We can infer that C is proportional to the number of particles in the gas container by considering the following argument. If we take two containers of same type of gas with same volume V , same pressure P and same temperature T , then the gas in each container obeys the above equation. $PV = CT$. If the two containers of gas is considered as a single system, then the pressure and temperature of this combined system will be same but volume will be twice and number of particles will also be double as shown in Figure 8.2

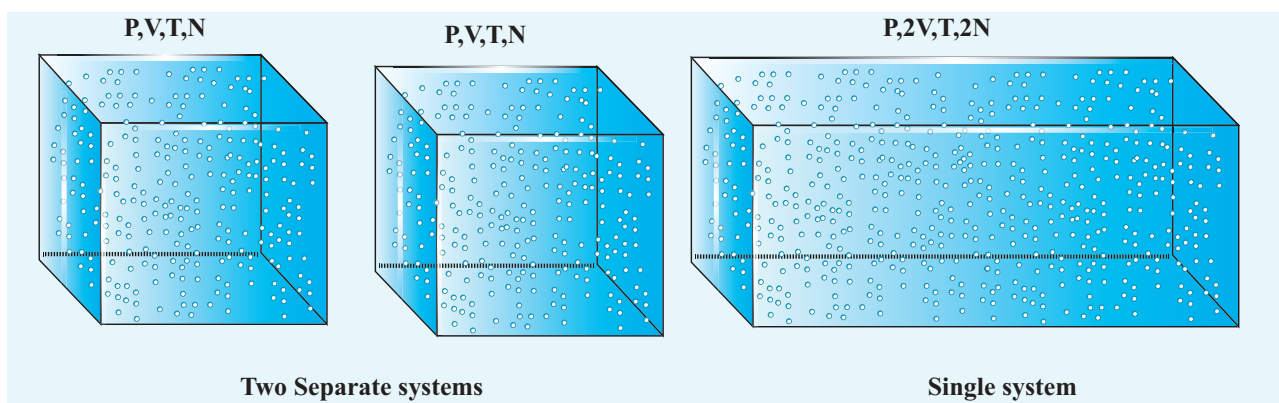


Figure 8.2 Ideal gas law

For this combined system, V becomes $2V$, so C should also double to match with the ideal gas equation $\frac{P(2V)}{T} = 2C$. It implies that C must depend on the number of particles in the gas and also should have the dimension of $\left[\frac{PV}{T}\right] = \text{JK}^{-1}$. So we can write the constant C as k times the number of particles N .

Here k is the Boltzmann constant ($1.381 \times 10^{-23} \text{ JK}^{-1}$) and it is found to be a universal constant.

So the ideal gas law can be stated as follows

$$PV = NkT \quad (8.1)$$

The equation (8.1) can also be expressed in terms of mole.

Mole is the practical unit to express the amount of gas. One mole of any substance is the amount of that substance which contains Avogadro number (N_A) of particles (such as atoms or molecules). The Avogadro's number N_A is defined as the number of carbon atoms contained in exactly 12 g of ^{12}C .

Suppose if a gas contains μ mole of particles then the total number of particles can be written as

$$N = \mu N_A \quad (8.2)$$

where N_A is Avogadro number ($6.023 \times 10^{23} \text{ mol}^{-1}$)

Substituting for N from equation (8.2), the equation (8.1) becomes

$PV = \mu N_A kT$. Here $N_A k = R$ called universal gas constant and its value is $8.314 \text{ J/mol} \cdot \text{K}$.

So the ideal gas law can be written for μ mole of gas as

$$PV = \mu RT \quad (8.3)$$

This is called the equation of state for an ideal gas. It relates the pressure, volume and temperature of thermodynamic system at equilibrium.

EXAMPLE 8.2

A student comes to school by a bicycle whose tire is filled with air at a pressure 240 kPa at 27°C . She travels 8 km to reach the school and the temperature of the bicycle tire increases to 39°C . What is the change in pressure in the tire when the student reaches school?



Solution

We can take air molecules in the tire as an ideal gas. The number of molecules and the volume of tire remain constant. So the air molecules at 27°C satisfies the ideal gas equation $P_1 V_1 = NkT_1$ and at 39°C it satisfies $P_2 V_2 = NkT_2$

But we know

$$V_1 = V_2 = V$$

$$\frac{P_1 V}{P_2 V} = \frac{NkT_1}{NkT_2}$$

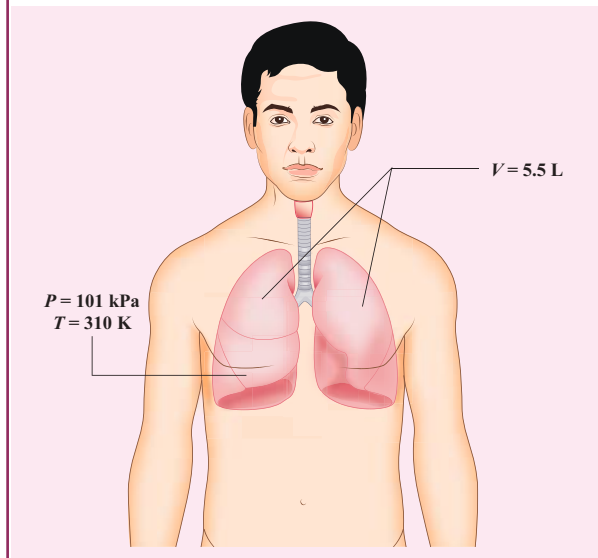
$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$P_2 = \frac{T_2}{T_1} P_1$$

$$P_2 = \frac{312\text{K}}{300\text{K}} \times 240 \times 10^3 \text{ Pa} = 249.6 \text{ kPa}$$

EXAMPLE 8.3

When a person breaths, his lungs can hold up to 5.5 Litre of air at body temperature 37°C and atmospheric pressure ($1 \text{ atm} = 101 \text{ kPa}$). This Air contains 21% oxygen. Calculate the number of oxygen molecules in the lungs.



Solution

We can treat the air inside the lungs as an ideal gas. To find the number of molecules, we can use the ideal gas law.

$$PV = NkT$$

Here volume is given in the Litre. 1 Litre is volume occupied by a cube of side 10 cm.

$$1 \text{ Litre} = 10\text{cm} \times 10\text{cm} \times 10\text{cm} = 10^{-3} \text{ m}^3$$

$$N = \frac{PV}{kT} = \frac{1.01 \times 10^5 \text{ Pa} \times 5.5 \times 10^{-3} \text{ m}^3}{1.38 \times 10^{-23} \text{ JK}^{-1} \times 310\text{K}}$$

$$= 1.29 \times 10^{23} \text{ Molecules}$$

Only 21% of N are oxygen. The total number of oxygen molecules

$$= 1.29 \times 10^{23} \times \frac{21}{100}$$

Number of oxygen molecules

$$= 2.7 \times 10^{22} \text{ molecules}$$

EXAMPLE 8.4

Calculate the volume of one mole of any gas at STP and at room temperature (300K) with the same pressure 1 atm.

Solution:

Here STP means standard temperature ($T=273\text{K}$ or 0°C) and Pressure ($P=1\text{ atm}$ or 101.3 kPa)

We can use ideal gas equation $V = \frac{\mu RT}{P}$.

Here $\mu = 1\text{ mol}$ and $R = 8.314\text{ J/mol.K}$.

By substituting the values

$$V = \frac{(1\text{ mol})(8.314 \frac{\text{J}}{\text{mol.K}})(273\text{K})}{1.013 \times 10^5 \text{ Nm}^{-2}}$$

$$= 22.4 \times 10^{-3} \text{ m}^3$$

We know that $1\text{ Litre (L)} = 10^{-3} \text{ m}^3$. So we can conclude that 1 mole of any ideal gas has volume 22.4 L .

By multiplying 22.4L by $\frac{300\text{K}}{273\text{K}}$ we get the volume of one mole of gas at room temperature. It is 24.6 L .

determine the number of mole. At room temperature 300K , the volume of a gas occupied by any gas is equal to 24.6L .

$$\text{The number of mole } \mu = \frac{120\text{m}^3}{24.6 \times 10^{-3} \text{ m}^3} \approx 4878 \text{ mol.}$$

Air is the mixture of about 20% oxygen, 79% nitrogen and remaining one percent are argon, hydrogen, helium, and xenon. The molar mass of air is 29 gmol^{-1} .

So the total mass of air in the room $m = 4878 \times 29 \times 10^{-3} = 141.4\text{kg}$.

8.2.2 Heat capacity and specific heat capacity

Take equal amount of water and oil at temperature 27°C and heat both of them till they reach the temperature 50°C . Note down the time taken by the water and oil to reach the temperature 50°C . Obviously these times are not same. We can see that water takes more time to reach 50°C than oil. It implies that water requires more heat energy to raise its temperature than oil. Now take twice the amount of water at 27°C and heat it up to 50°C , note the time taken for this rise in temperature. The time taken by the water is now twice compared to the previous case.

We can define 'heat capacity' as the amount of heat energy required to raise the temperature of the given body from T to $T + \Delta T$.

$$\text{Heat capacity } S = \frac{\Delta Q}{\Delta T}$$

Specific heat capacity of a substance is defined as the amount of heat energy required to raise the temperature of 1kg of a substance by 1 Kelvin or 1°C

EXAMPLE 8.5

Estimate the mass of air in your class room at NTP. Here NTP implies normal temperature (room temperature) and 1 atmospheric pressure.



Solution

The average size of a class is 6m length, 5 m breadth and 4 m height. The volume of the room $V = 6 \times 5 \times 4 = 120\text{m}^3$. We can



$$\Delta Q = m s \Delta T$$

Therefore,

$$s = \frac{1}{m} \left(\frac{\Delta Q}{\Delta T} \right)$$



Where s is known as *specific heat capacity* of a substance and its value depends only on the nature of the substance not amount of substance

ΔQ = Amount of heat energy

ΔT = Change in temperature

m = Mass of the substance

The SI unit for specific heat capacity is $\text{J kg}^{-1} \text{K}^{-1}$. Heat capacity and specific heat capacity are always positive quantities.

Table 8.2 Specific heat capacity of some common substances at 1 atm (20°C)

Material	Specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
Air	1005
Lead	130
Copper	390
Iron (steel)	450
Glass	840
Aluminium	900
Human body	3470
Water	4186

From the table 8.2 it is clear that water has the highest value of specific heat capacity. For this reason it is used as a coolant in power stations and reactors.



Note

The term heat capacity or specific heat capacity does not mean that object contains a certain amount of heat. Heat is energy transfer from the object at higher temperature to the object at lower temperature. The correct usage is 'internal energy capacity'. But for historical reason the term 'heat capacity' or 'specific heat capacity' are retained.



When two objects of same mass are *heated* at equal rates, the object with *smaller specific heat capacity* will have a *faster temperature increase*.

When two objects of same mass are *left to cool down*, the temperature of the object with *smaller specific heat capacity* will *drop faster*.

When we study properties of gases, it is more practical to use molar specific heat capacity. Molar specific heat capacity is defined as heat energy required to increase the temperature of one mole of substance by 1K or 1°C . It can be written as follows

$$C = \frac{1}{\mu} \left(\frac{\Delta Q}{\Delta T} \right)$$

Here C is known as *molar specific heat capacity* of a substance and μ is number of moles in the substance.

The SI unit for molar specific heat capacity is $\text{J mol}^{-1} \text{K}^{-1}$. It is also a positive quantity.

8.2.3 Thermal expansion of solids, liquids and gases

Thermal expansion is the tendency of matter to change in shape, area, and volume due to a change in temperature.

All three states of matter (solid, liquid and gas) expand when heated. When a solid is heated, its atoms vibrate with higher amplitude about their fixed points. The relative change in the size of solids is small. Railway tracks are given small gaps so that in the summer, the tracks expand and do not buckle. Railroad tracks and bridges have expansion joints to allow them to expand and contract freely with temperature changes. It is shown in Figure 8.3



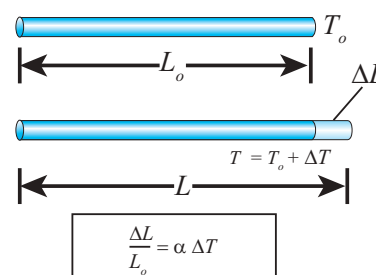
Figure 8.3 Expansion joints for safety

Liquids, have less intermolecular forces than solids and hence they expand more than solids. This is the principle behind the mercury thermometers.

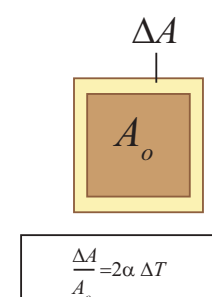
In the case of **gas** molecules, the intermolecular forces are almost negligible and hence they expand much more than solids. For example in hot air balloons when gas particles get heated, they expand and take up more space.

The increase in dimension of a body due to the increase in its temperature is called thermal expansion.

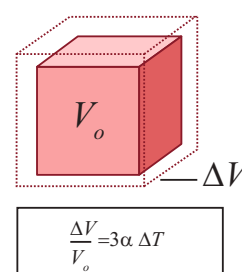
The expansion in length is called **linear expansion**. Similarly the expansion in area is termed as **area expansion** and the



(a) Linear expansion



(b) Area expansion



(c) Volume expansion

Figure 8.4 Thermal expansions

expansion in volume is termed as **volume expansion**. It is shown in Figure 8.4

Linear Expansion

In solids, for a small change in temperature ΔT , the fractional change in length $\left(\frac{\Delta L}{L_0}\right)$ is directly proportional to ΔT .

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T$$

Therefore, $\alpha_L = \frac{\Delta L}{L_0 \Delta T}$

Where, α_L = coefficient of linear expansion.

ΔL = Change in length

L_0 = Original length

ΔT = Change in temperature.



- When the lid of a glass bottle is tight, keep the lid near the hot water which makes it easier to open. It is because the lid has higher thermal expansion than glass.
- When the hot boiled egg is dropped in cold water, the egg shell can be removed easily. It is because of the different thermal expansions of the shell and egg.

EXAMPLE 8.6

Eiffel tower is made up of iron and its height is roughly 300 m. During winter season (January) in France the temperature is 2°C and in hot summer its average temperature 25°C . Calculate the change in height of Eiffel tower between summer and winter. The linear thermal expansion coefficient for iron $\alpha = 10 \times 10^{-6}$ per $^\circ\text{C}$



Solution

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T$$

$$\Delta L = \alpha_L L_0 \Delta T$$

$$\Delta L = 10 \times 10^{-6} \times 300 \times 23 = 0.069 \text{ m} = 69 \text{ mm}$$

Area Expansion

For a small change in temperature ΔT the fractional change in area $\left(\frac{\Delta A}{A_0}\right)$ of a substance is directly proportional to ΔT and it can be written as

$$\frac{\Delta A}{A_0} = \alpha_A \Delta T$$

Therefore, $\alpha_A = \frac{\Delta A}{A_0 \Delta T}$

Where, α_A = coefficient of area expansion.

ΔA = Change in area

A_0 = Original area

ΔT = Change in temperature

Volume Expansion

For a small change in temperature ΔT the fractional change in volume $\left(\frac{\Delta V}{V_0}\right)$ of a substance is directly proportional to ΔT .

$$\frac{\Delta V}{V_0} = \alpha_V \Delta T$$

Therefore, $\alpha_V = \frac{\Delta V}{V_0 \Delta T}$

Where, α_v = coefficient of volume expansion.

ΔV = Change in volume

V_0 = Original volume

ΔT = Change in temperature

Unit of coefficient of linear, area and volumetric expansion of solids is $^{\circ}\text{C}^{-1}$ or K^{-1}



For a given specimen,

$$\frac{\Delta L}{L_0} = \alpha_L \Delta T \text{ (Linear expansion)}$$

$$\frac{\Delta A}{A_0} \approx 2 \alpha_L \Delta T \text{ (Area expansion } \approx 2 \times \text{ Linear expansion)}$$

$$\frac{\Delta V}{V_0} \approx 3 \alpha_L \Delta T \text{ (Volume expansion } \approx 3 \times \text{ Linear expansion)}$$

8.2.4 Anomalous expansion of water

Liquids expand on heating and contract on cooling at moderate temperatures. But water exhibits an anomalous behavior. It contracts on heating between 0°C and 4°C . The volume of the given amount of water decreases as it is cooled from room temperature, until it reach 4°C . Below 4°C the volume increases and so the density decreases. This means that the water has a maximum density at 4°C . This behavior of water is called anomalous expansion of water. It is shown in the Figure 8.5

In cold countries during the winter season, the surface of the lakes will be at lower temperature than the bottom as shown in the Figure 8.6. Since the solid water (ice) has lower density than its liquid form, below 4°C , the frozen water will be

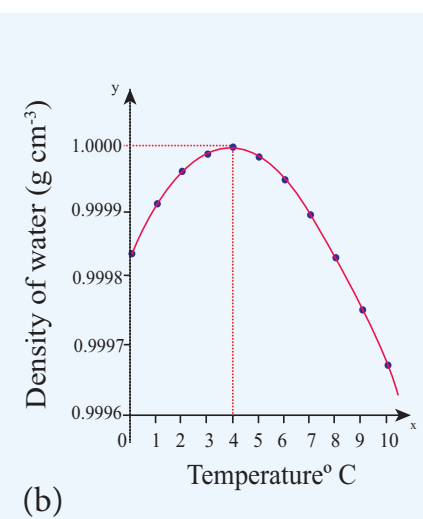
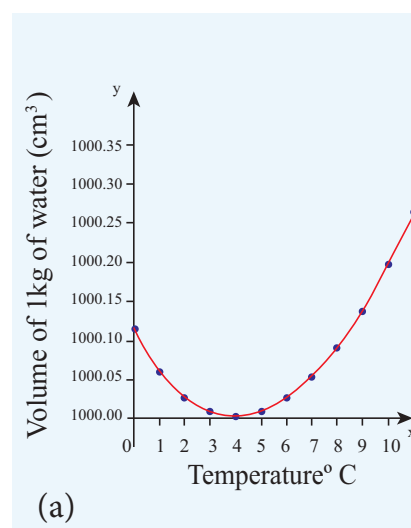


Figure 8.5 Anomalous Expansion of water

on the top surface above the liquid water (ice floats). This is due to the anomalous expansion of water. As the water in lakes and ponds freeze only at the top the species living in the lakes will be safe at the bottom.

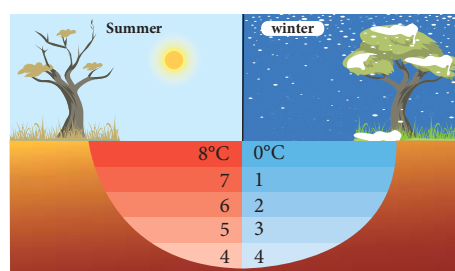


Figure 8.6 Anomalous expansion of water in lakes

8.2.5 Change of state

All matter exists normally in three states as solids, liquids or gases. Matter can be changed from one state to another either by heating or cooling.

Examples:

1. Melting (solid to liquid)
2. Evaporation (liquid to gas)
3. Sublimation (solid to gas)
4. Freezing / Solidification (liquid to solid)
5. Condensation (gas to liquid)

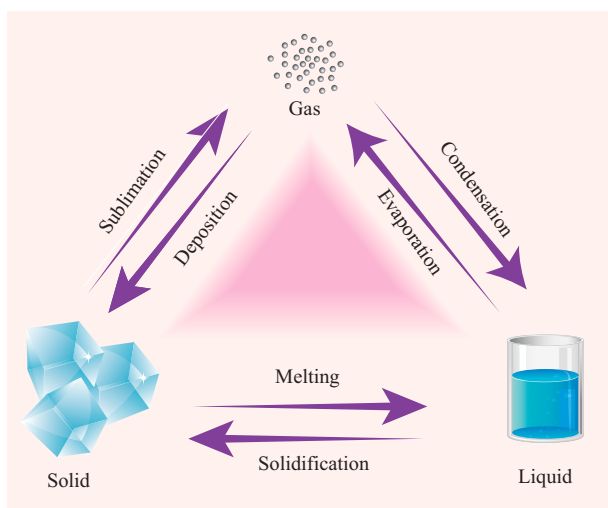


Figure 8.7 Change of states of matter

Latent heat capacity:

While boiling a pot of water, the temperature of the water increases until it reaches 100 °C which is the boiling point of water, and then the temperature remains constant until all the water changes from liquid to gas. During this process heat is continuously added to the water. But the temperature of water does not increase above its boiling point. This is the concept of latent heat capacity.

Latent heat capacity of a substance is defined as the amount of heat energy required to change the state of a unit mass of the material.

$$Q = m \times L$$

Therefore, $L = \frac{Q}{m}$

Where L = Latent heat capacity of the substance

Q = Amount of heat

m = mass of the substance

The SI unit for Latent heat capacity is J kg⁻¹.

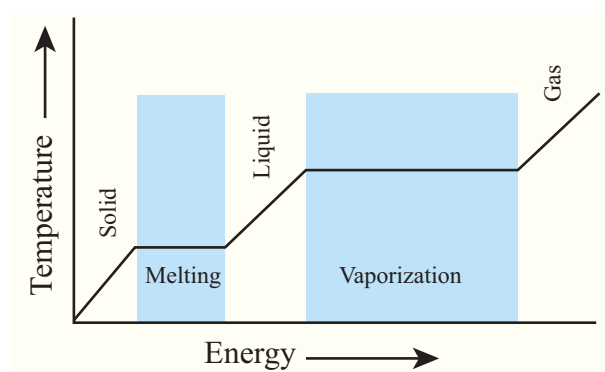


Figure 8.8 Temperature versus heat for water



Note When heat is added or removed during a change of state, the temperature remains constant.

- The latent heat for a solid - liquid state change is called the latent heat of fusion (L_f)
- The latent heat for a liquid - gas state change is called the latent heat of vaporization (L_v)
- The latent heat for a solid - gas state change is called the latent heat of sublimation (L_s)

Triple point

The triple point of a substance is the temperature and pressure at which the three phases (gas, liquid and solid) of that substance coexist in thermodynamic equilibrium.

The triple point of water is at 273.1 K and a partial vapour pressure of 611.657 Pascal.

8.2.6 Calorimetry

Calorimetry means the measurement of the amount of heat released or absorbed by thermodynamic system during the heating process. When a body at higher temperature is brought in contact with another body at lower temperature, the heat lost by the hot body is equal to the heat gained by the cold body. No heat is allowed to escape to the surroundings. It can be mathematically expressed as

$$Q_{\text{gain}} = -Q_{\text{lost}}$$

$$Q_{\text{gain}} + Q_{\text{lost}} = 0$$

Heat gained or lost is measured with a calorimeter. Usually the calorimeter is an insulated container of water as shown in Figure 8.9.

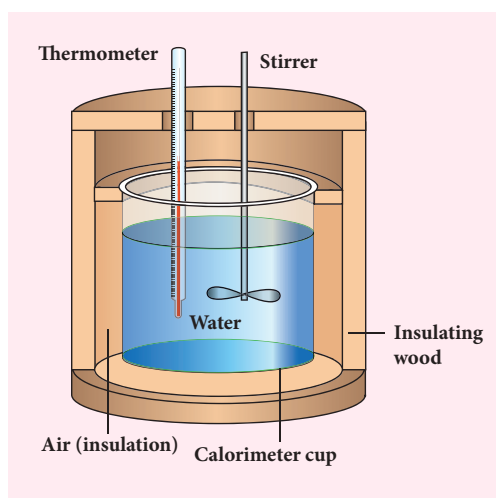


Figure 8.9 Calorimeter

A sample is heated at high temperature (T_1) and immersed into water at room temperature (T_2) in the calorimeter. After some time both sample and water reach a final equilibrium temperature T_f . Since the calorimeter is insulated, heat given by the hot sample is equal to heat gained by the water. It is shown in the Figure 8.10

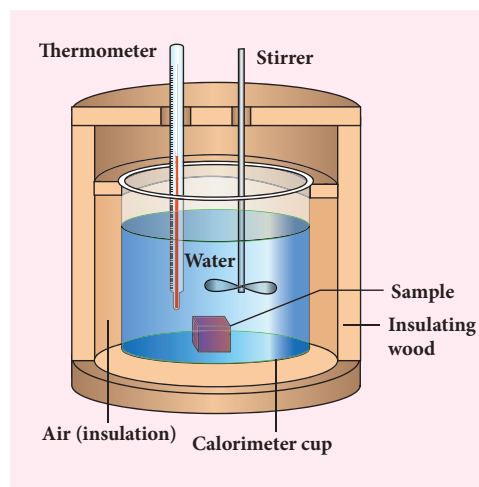


Figure 8.10 Calorimeter with sample of block

$$Q_{\text{gain}} = -Q_{\text{lost}}$$

Note the sign convention. The heat lost is denoted by negative sign and heat gained is denoted as positive.

From the definition of specific heat capacity

$$Q_{\text{gain}} = m_2 s_2 (T_f - T_2)$$

$$Q_{\text{lost}} = m_1 s_1 (T_f - T_1)$$

Here s_1 and s_2 specific heat capacity of hot sample and water respectively.

So we can write

$$m_2 s_2 (T_f - T_2) = -m_1 s_1 (T_f - T_1)$$

$$m_2 s_2 T_f - m_2 s_2 T_2 = -m_1 s_1 T_f + m_1 s_1 T_1$$

$$m_2 s_2 T_f + m_1 s_1 T_f = m_2 s_2 T_2 + m_1 s_1 T_1$$

The final temperature

$$T_f = \frac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2}$$

EXAMPLE 8.7

If 5 L of water at 50°C is mixed with 4L of water at 30°C , what will be the final temperature of water? Take the specific heat capacity of water as $4184 \text{ J kg}^{-1} \text{ K}^{-1}$.

Solution

We can use the equation

$$T_f = \frac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2}$$

$m_1 = 5L = 5\text{kg}$ and $m_2 = 4L = 4\text{kg}$, $s_1 = s_2$
and $T_1 = 50^\circ\text{C} = 323\text{K}$ and $T_2 = 30^\circ\text{C} = 303\text{K}$.

So

$$T_f = \frac{m_1 T_1 + m_2 T_2}{m_1 + m_2} = \frac{5 \times 323 + 4 \times 303}{5 + 4} = 314.11\text{ K}$$

$$T_f = 314.11\text{ K} - 273\text{K} \approx 41^\circ\text{C}.$$

Suppose if we mix equal amount of water ($m_1 = m_2$) with 50°C and 30°C , then the final temperature is average of two temperatures.

$$T_f = \frac{T_1 + T_2}{2} = \frac{323 + 303}{2} = 313\text{K} = 40^\circ\text{C}$$

Suppose if both the water are at 30°C then the final temperature will also 30°C . It implies that they are at equilibrium and no heat exchange takes place between each other.

Note

It is important to note that the final equilibrium temperature of mixing of gas or liquid depends on mass of the substances, their specific heat capacities and their temperatures. Only if we mix the same substances at equal amount, the final temperature will be an average of the individual temperatures.

8.2.7 Heat transfer

As we have seen already heat is a energy in transit which is transferred from one body to another body due to temperature difference. There are three modes of heat transfer: Conduction, Convection and Radiation.

Conduction

Conduction is the process of direct transfer of heat through matter due to temperature difference. When two objects are in direct contact with one another, heat will be transferred from the hotter object to the colder one. The objects which allow heat to travel easily through them are called conductors.

Thermal conductivity

Thermal conductivity is the ability to conduct heat.

The quantity of heat transferred through a unit length of a material in a direction normal to unit surface area due to a unit temperature difference under steady state conditions is known as thermal conductivity of a material.

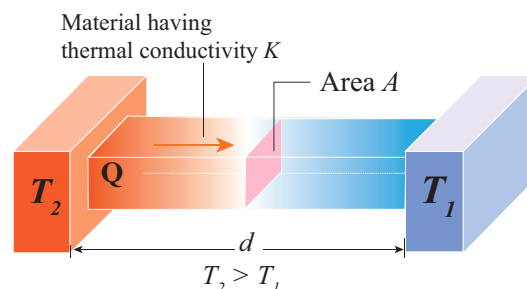


Figure 8.11 Steady state heat flow by conduction.

In steady state, the rate of flow of heat Q is proportional to the temperature difference ΔT and the area of cross section A and is inversely proportional to the length L . So the rate of flow of heat is written as

$$\frac{Q}{t} = \frac{KA\Delta T}{L}$$

Where, K is known as the **coefficient of thermal conductivity**.

(Not to be confused with Kelvin represented by upper case K)

The SI unit of thermal conductivity is

$$\text{J s}^{-1} \text{ m}^{-1} \text{ K}^{-1} \text{ or } \text{W m}^{-1} \text{ K}^{-1}.$$

Table 8.3: Thermal conductivities (in $\text{W m}^{-1} \text{K}^{-1}$) of some materials at 1 atm

Material	Thermal Conductivity	Material	Thermal Conductivity
Diamond	2300	Water	0.56
Silver	420	Human tissue	0.2
Copper	380	Wood	0.17
Aluminum	200	Helium	0.152
Steel	40	Cork	0.042
Glass	0.84	Air	0.023
Brick	0.84		
Ice	2		

Note

Steady state:

The state at which temperature attains constant value everywhere and there is no further transfer of heat anywhere is called steady state.

Thermal conductivity depends on the nature of the material. For example silver and

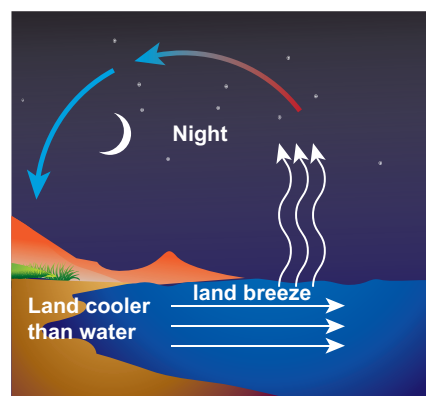
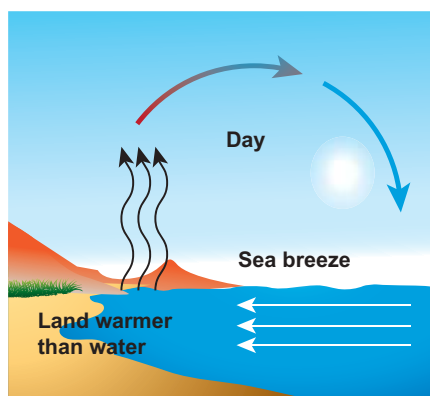
aluminum have high thermal conductivities. So they are used to make cooking vessels.

Convection

Convection is the process in which heat transfer is by actual movement of molecules in fluids such as liquids and gases. In convection, molecules move freely from one place to another. It happens naturally or forcefully.



During the day, sun rays warm up the land more quickly than sea water. It is because land has less specific heat capacity than water. As a result the air above the land becomes less dense and rises. At the same time the cooler air above the sea flows to land and it is called 'sea breeze'. During the night time the land gets cooled faster than sea due to the same reason (specific heat). The air molecules above sea are warmer than air molecules above the land. So air molecules above the sea are replaced by cooler air molecules from the land. It is called 'land breeze'.





Boiling water in a cooking pot is an example of convection. Water at the bottom of the pot receives more heat. Due to heating, the water expands and the density of water decreases at the bottom. Due to this decrease in density, molecules rise to the top. At the same time the molecules at the top receive less heat and become denser and come to the bottom of the pot. This process goes on continuously. The back and forth movement of molecules is called convection current.

To keep the room warm, we use room heater. The air molecules near the heater will heat up and expand. As they expand, the density of air molecules will decrease and rise up while the higher density cold air will come down. This circulation of air molecules is called convection current.

Radiation:

When we keep our hands near the hot stove we feel the heat even though our hands are not touching the hot stove. Here heat transferred from the hot stove to our hands is in the form of radiation. We receive energy from the sun in the form of radiations. These radiations travel through vacuum and reach the Earth. It is the peculiar character of radiation which requires no medium to transfer energy from one object to another. The conduction or convection requires medium to transfer the heat.

Radiation is a form of energy transfer from one body to another by electromagnetic waves.

Example:

1. Solar energy from the Sun.
2. Radiation from room heater.



Note

The parameter temperature is generally thought to be associated with matter (solid, liquid and gas). But radiation is also considered as a thermodynamic system which has well defined temperature and pressure. The visible radiation coming from the Sun is at the temperature of 5700 K and the Earth re emits the radiation in the infrared range into space which is at a temperature of around 300K.

8.2.8 Newton's law of cooling

Newton's law of cooling states that the rate of loss of heat of a object is directly proportional to the difference in the temperature between that object and its surroundings .

$$\frac{dQ}{dt} \propto -(T - T_s) \quad (8.4)$$

The negative sign indicates that the quantity of heat lost by object goes on decreasing with time. Where,

T = Temperature of the object

T_s = Temperature of the surrounding

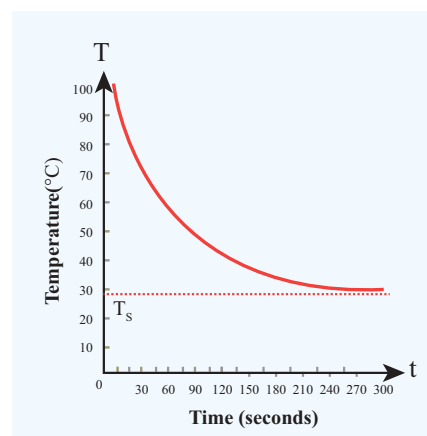


Figure 8.12 Cooling of hot water with time

From the graph in Figure 8.12 it is clear that the rate of cooling is high initially and decreases with falling temperature.

Let us consider an object of mass m and specific heat capacity s at temperature T . Let T_s be the temperature of the surroundings. If the temperature falls by a small amount dT in time dt , then the amount of heat lost is,

$$dQ = msdT \quad (8.5)$$

Dividing both sides of equation (8.5) by dt

$$\frac{dQ}{dt} = \frac{msdT}{dt} \quad (8.6)$$

From Newton's law of cooling

$$\begin{aligned} \frac{dQ}{dt} &\propto -(T - T_s) \\ \frac{dQ}{dt} &= -a(T - T_s) \end{aligned} \quad (8.7)$$

Where a is some positive constant.

From equation (8.6) and (8.7)

$$\begin{aligned} -a(T - T_s) &= ms \frac{dT}{dt} \\ \frac{dT}{T - T_s} &= -\frac{a}{ms} dt \end{aligned} \quad (8.8)$$

Integrating equation (8.8) on both sides,

$$\begin{aligned} \int \frac{dT}{T - T_s} &= -\int \frac{a}{ms} dt \\ \ln(T - T_s) &= -\frac{a}{ms} t + b_1 \end{aligned}$$

Where b_1 is the constant of integration. taking exponential both sides, we get

$$T = T_s + b_2 e^{-\frac{a}{ms}t} \quad (8.9)$$

here $b_2 = e^{b_1} = \text{constant}$

EXAMPLE 8.8

A hot water cools from 92°C to 84°C in 3 minutes when the room temperature is 27°C . How long will it take for it to cool from 65°C to 60°C ?

The hot water cools 8°C in 3 minutes. The average temperature of 92°C and 84°C is 88°C . This average temperature is 61°C

above room temperature. Using equation (8.8)

$$\begin{aligned} \frac{dT}{T - T_s} &= -\frac{a}{ms} dt \quad \text{or} \quad \frac{dT}{dt} = -\frac{a}{ms}(T - T_s) \\ \frac{8^\circ\text{C}}{3\text{min}} &= -\frac{a}{ms}(61^\circ\text{C}) \end{aligned}$$

Similarly the average temperature of 65°C and 60°C is 62.5°C . The average temperature is 35.5°C above the room temperature. Then we can write

$$\frac{5^\circ\text{C}}{dt} = -\frac{a}{ms}(35.5^\circ\text{C})$$

By dividing both the equation, we get

$$\begin{aligned} \frac{8^\circ\text{C}}{3\text{min}} \bigg/ \frac{5^\circ\text{C}}{dt} &= \frac{-\frac{a}{ms}(61^\circ\text{C})}{-\frac{a}{ms}(35.5^\circ\text{C})} \\ \frac{8 \times dt}{3 \times 5} &= \frac{61}{35.5} \\ dt &= \frac{61 \times 15}{35.5 \times 8} = \frac{915}{284} = 3.22\text{min} \end{aligned}$$

8.3

LAWS OF HEAT TRANSFER

8.3.1 Prevost theory of heat exchange

Every object emits heat radiations at all finite temperatures (except 0 K) as well as it absorbs radiations from the surroundings. For example, if you touch someone, they might feel your skin as either hot or cold.

A body at high temperature radiates more heat to the surroundings than it receives from it. Similarly, a body at a lower temperature receives more heat from the surroundings than it loses to it.

Prevost applied the idea of ‘thermal equilibrium’ to radiation. He suggested that all bodies radiate energy but hot bodies radiate more heat than the cooler bodies. At one point of time the rate of exchange of heat from both the bodies will become the same. Now the bodies are said to be in ‘thermal equilibrium’.

Only at absolute zero temperature a body will stop emitting. Therefore *Prevost theory states that all bodies emit thermal radiation at all temperatures above absolute zero irrespective of the nature of the surroundings.*

8.3.2 Stefan Boltzmann law

Stefan Boltzmann law states that, *the total amount of heat radiated per second per unit area of a black body is directly proportional to the fourth power of its absolute temperature.*

$$E \propto T^4 \text{ or } E = \sigma T^4 \quad (8.10)$$

Where, σ is known as Stefan’s constant. Its value is $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$



Note

If a body is not a perfect black body, then

$$E = e \sigma T^4 \quad (8.11)$$

Where ‘e’ is emissivity of surface.

Emissivity is defined as the ratio of the energy radiated from a material’s surface to that radiated from a perfectly black body at the same temperature and wavelength.

8.3.3 Wien’s displacement law

In the universe every object emits radiation. The wavelengths of these radiations depend on the object’s absolute temperature. These

radiations have different wavelengths and all the emitted wavelengths will not have equal intensity.

Wien’s law states that, *the wavelength of maximum intensity of emission of a black body radiation is inversely proportional to the absolute temperature of the black body.*

$$\lambda_m \propto \frac{1}{T} \text{ (or) } \lambda_m = \frac{b}{T} \quad (8.12)$$

Where, b is known as Wien’s constant. Its value is $2.898 \times 10^{-3} \text{ m K}$

It implies that if temperature of the body increases, maximal intensity wavelength (λ_m) shifts towards lower wavelength (higher frequency) of electromagnetic spectrum. It is shown in Figure 8.13

Graphical representation

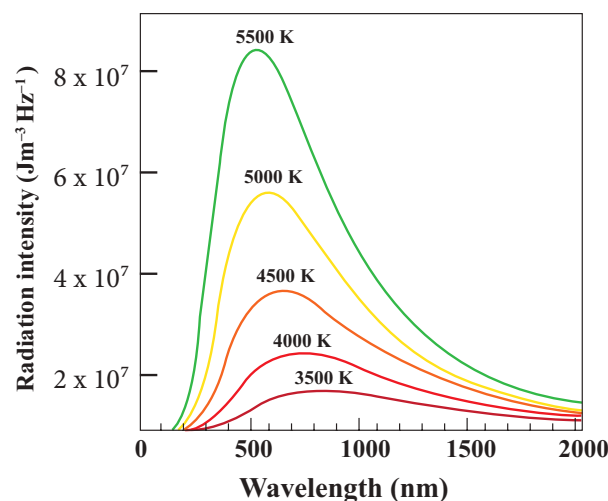


Figure 8.13 Wien’s displacement law

From the graph it is clear that the peak of the wavelengths is inversely proportional to temperature. The curve is known as ‘black body radiation curve’.

Wien’s law and Vision:

Why our eye is sensitive to only visible wavelength (in the range 400 nm to 700nm)?

The Sun is approximately taken as a black body. Since any object above 0 K will emit radiation, Sun also emits radiation. Its surface temperature is about 5700 K. By substituting this value in the equation (8.12),

$$\lambda_m = \frac{b}{T} = \frac{2.898 \times 10^{-3}}{5700} \approx 508 \text{ nm}$$

It is the wavelength at which maximum intensity is 508 nm. Since the Sun's temperature is around 5700 K, the spectrum of radiations emitted by Sun lie between 400 nm to 700 nm which is the visible part of the spectrum. It is shown in Figure 8.14

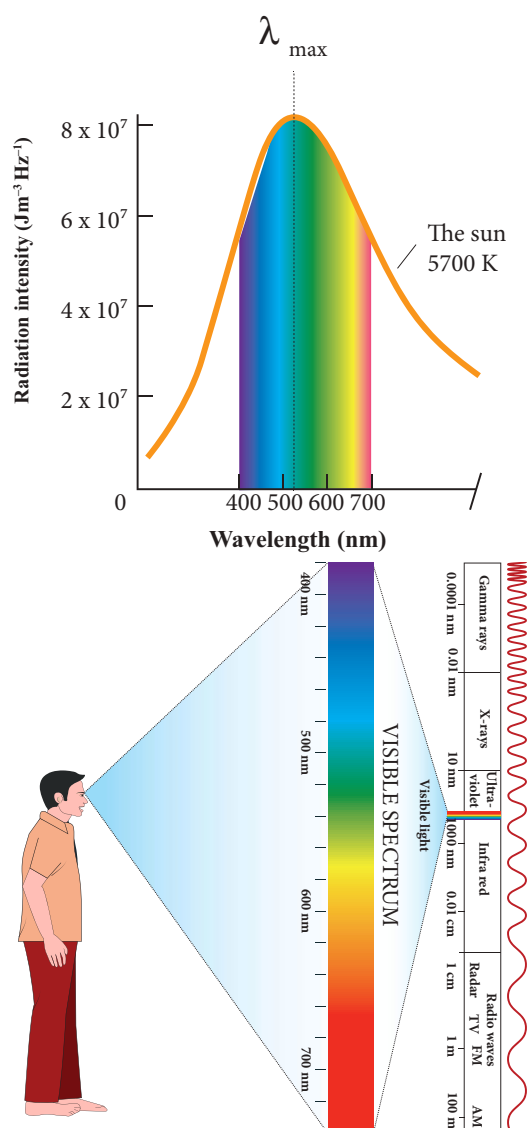


Figure 8.14 Wien's law and Human's vision

The humans evolved under the Sun by receiving its radiations. The human eye is sensitive only in the visible not in infrared or X-ray ranges in the spectrum.

Suppose if humans had evolved in a planet near the star Sirius (9940 K), then they would have had the ability to see the Ultraviolet rays!

EXAMPLE 8.9

The power radiated by a black body A is E_A and the maximum energy radiated was at the wavelength λ_A . The power radiated by another black body B is $E_B = N E_A$ and the radiated energy was at the maximum wavelength, $\frac{1}{2} \lambda_A$. What is the value of N?

According to Wien's displacement law $\lambda_{\text{max}} T = \text{constant}$ for both object A and B

$$\lambda_A T_A = \lambda_B T_B, \text{ Here } \lambda_B = \frac{1}{2} \lambda_A$$

$$\frac{T_B}{T_A} = \frac{\lambda_A}{\lambda_B} = \frac{\lambda_A}{\left(\frac{1}{2}\right)\lambda_A} = 2$$

$$T_B = 2T_A$$

From Stefan-Boltzmann law

$$\frac{E_B}{E_A} = \left(\frac{T_B}{T_A}\right)^4 = (2)^4 = 16 = N$$

Object B has emitted at lower wavelength compared to A. So the object B would have emitted more energetic radiation than A.

8.4

THERMODYNAMICS:

8.4.1 Introduction

In the previous sections we have studied about the heat, temperature and thermal properties of matter. Thermodynamics is

a branch of physics which describes the laws governing the process of conversion of work into heat and conversion of heat into work. The laws of thermodynamics are formulated over three centuries of experimental works of Boyle, Charles, Bernoulli, Joule, Clausius, Kelvin, Carnot and Helmholtz. In our day to day life, the functioning of everything around us and even our body is governed by the laws of thermodynamics. Therefore thermodynamics is one of the essential branches of physics.

Thermodynamic system:

A thermodynamic system is a finite part of the universe. It is a collection of large number of particles (atoms and molecules) specified by certain parameters called pressure (P), Volume (V) and Temperature (T). The remaining part of the universe is called surrounding. Both are separated by a boundary. It is shown in Figure 8.15

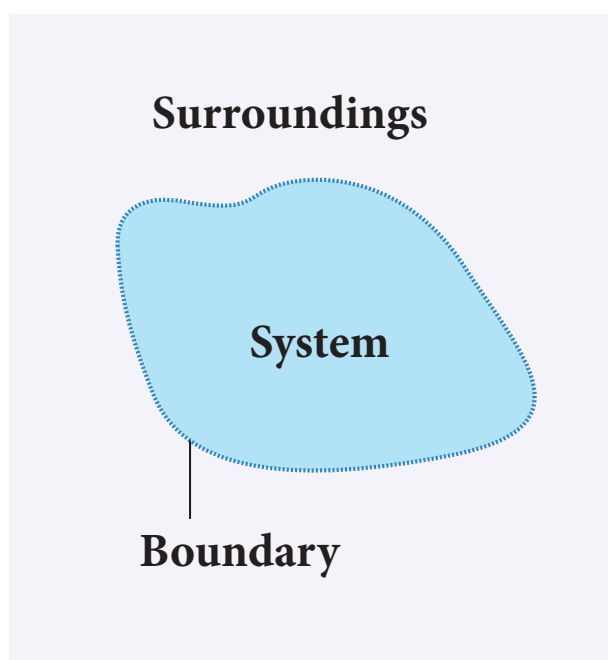


Figure 8.15 Thermodynamic system

Examples:

A thermodynamic system can be liquid, solid, gas and radiation.

Thermodynamic system	Surrounding
Bucket of water	Open atmosphere
Air molecules in the room	Outside air
Human body	Open atmosphere
Fish in the sea	Sea of water

We can classify thermodynamics system into three types: It is given in Figure 8.16

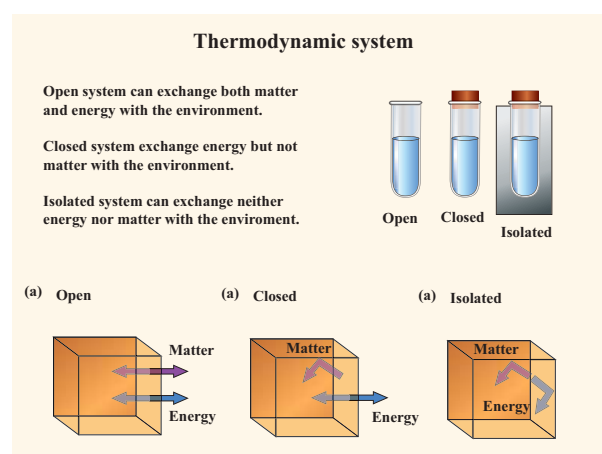


Figure 8.16 Different types of thermodynamic systems

8.4.2 Thermal equilibrium

When a hot cup of coffee is kept in the room, heat flows from coffee to the surrounding air. After some time the coffee reaches the same temperature as the surrounding air and there will be no net heat flow from coffee to air or air to coffee. It implies that the coffee and surrounding air are in thermal equilibrium with each other.

Two systems are said to be in thermal equilibrium with each other if they are at the same temperature, which will not change with time.

Mechanical equilibrium:

Consider a gas container with piston as shown in Figure 8.17. When some mass is placed on the piston, it will move downward due to downward gravitational force and after certain humps and jumps the piston will come to rest at a new position. When the downward gravitational force given by the piston is balanced by the upward force exerted by the gas, the system is said to be in mechanical equilibrium. A system is said to be in mechanical equilibrium if no unbalanced force acts on the thermodynamic system or on the surrounding by thermodynamic system.

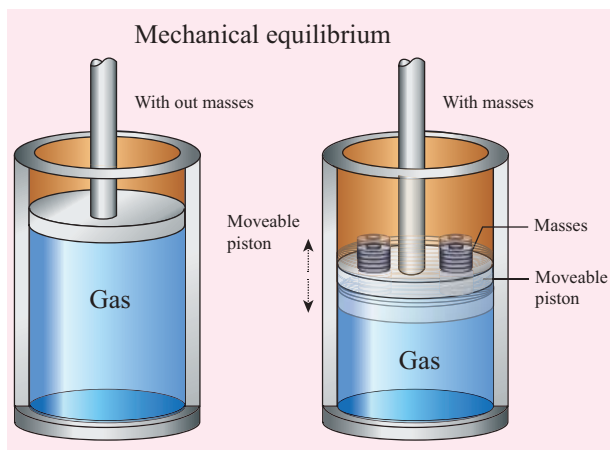


Figure 8.17 Mechanical equilibrium

Chemical equilibrium:

If there is no net chemical reaction between two thermodynamic systems in contact with each other then it is said to be in chemical equilibrium.

Thermodynamic equilibrium:

If two systems are set to be in thermodynamic equilibrium, then the systems are at thermal, mechanical and chemical equilibrium with

each other. In a state of thermodynamic equilibrium the macroscopic variables such as pressure, volume and temperature will have fixed values and do not change with time.

8.4.3 Thermodynamic state variables

In mechanics velocity, momentum and acceleration are used to explain the state of any moving object (which you would have realized in Volume 1). In thermodynamics, the state of a thermodynamic system is represented by a set of variables called thermodynamic variables.

Examples: Pressure, temperature, volume and internal energy etc.

The values of these variables completely describe the equilibrium state of a thermodynamic system. Heat and work are not state variables rather they are process variables.

There are two types of thermodynamic variables: Extensive and Intensive

Extensive variable depends on the size or mass of the system.

Example: Volume, total mass, entropy, internal energy, heat capacity etc.

Intensive variables do not depend on the size or mass of the system.

Example: Temperature, pressure, specific heat capacity, density etc.

Equation of state:

The equation which connects the state variables in a specific manner is called equation of state. A thermodynamic equilibrium is completely specified by these state variables by the equation of state. If the system is not in thermodynamic equilibrium then these equations cannot specify the state of the system.



An ideal gas obeys the equation $PV = NkT$ at thermodynamic equilibrium. Since all four macroscopic variables (P, V, T and N) are connected by this equation, we cannot change one variable alone. For example, if we push the piston of a gas container, the volume of the gas will decrease but pressure will increase or if heat is supplied to the gas, its temperature will increase, pressure and volume of the gas may also increase.

There is another example of equation of state called van der Waals equation. Real gases obey this equation at thermodynamic equilibrium. The air molecules in the room truly obey van der Waals equation of state. But at room temperature with low density we can approximate it into an ideal gas.

8.5

ZEROth LAW OF THERMODYNAMICS

The zeroth law of thermodynamics states that if two systems, A and B , are in thermal equilibrium with a third system, C ,

then A and B are in thermal equilibrium with each other.

Consider three systems A , B and C which are initially at different temperatures. Assume that A and B are not in thermal contact with each other as shown in Figure 8.18 (a) but each of them is in thermal contact with a third system C . After a lapse of time, system A will be in thermal equilibrium with C and B also will be in thermal equilibrium with C . In this condition, if the systems A and B are kept in thermal contact as shown in the Figure 8.18 (b), there is no flow of heat between the systems A and B . It implies that the system A and B are also in thermal equilibrium with each other. Once the three systems are at thermal equilibrium, there will be no heat flow between them as they are at the same temperature. This can be mathematically expressed as if, $T_A = T_C$ and $T_B = T_C$, it implies that $T_A = T_B$, where T_A , T_B and T_C are the temperatures of the systems A , B , and C respectively.

Temperature is the property which determines whether the system is in

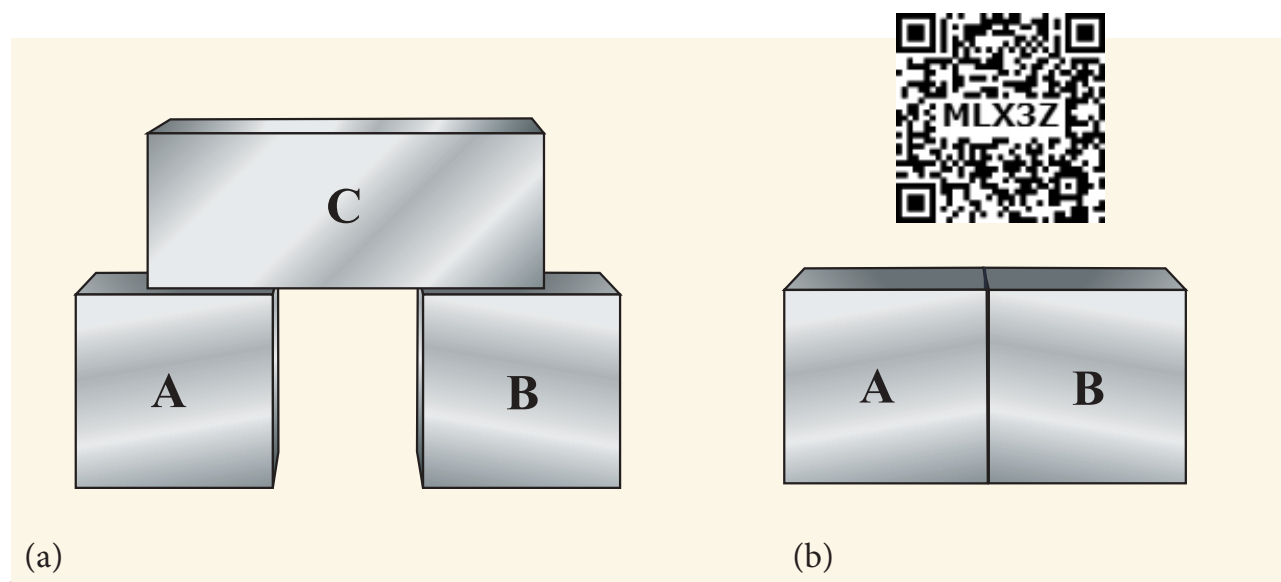


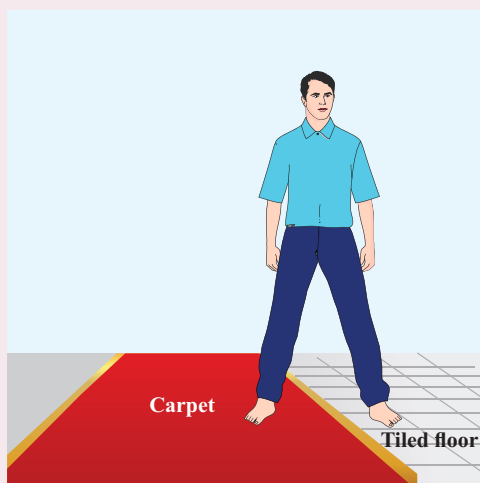
Figure 8.18 (a) Two systems A and B in thermal contact with object C separately (b) If systems A and B are in thermal contact, they are also in thermal equilibrium with each other.

thermal equilibrium with other systems or not. Zeroth law enables us to determine the temperature. For example, when a thermometer is kept in contact with a human body, it reaches thermal equilibrium with the body. At this condition, the temperature of the thermometer will be same as the human body. This principle is used in finding the body temperature.

ACTIVITY

We often associate the temperature as a measure of how hot or cold an object is while touching it. Can we use our sensory organs to determine the temperature of an object?

When you stand bare feet with one foot on the carpet and the other on the tiled floor, your foot on tiled floor feels cooler than the foot on the carpet even though both the tiled floor and carpet are at the same room temperature. It is because the tiled floor transfers the heat energy to your skin at higher rate than the carpet. So the skin is not measuring the actual temperature of the object; instead it measures the rate of heat energy transfer. But if we place a thermometer on the tiled floor or carpet it will show the same temperature.



8.6

INTERNAL ENERGY (U)

The internal energy of a thermodynamic system is the sum of kinetic and potential energies of all the molecules of the system with respect to the center of mass of the system.

The energy due to molecular motion including translational, rotational and vibrational motion is called internal kinetic energy (E_k)

The energy due to molecular interaction is called internal potential energy (E_p). Example: Bond energy.

$$U = E_k + E_p$$

- Since ideal gas molecules are assumed to have no interaction with each other the internal energy consists of only kinetic energy part (E_k) which depends on the temperature, number of particles and is independent of volume. However this is not true for real gases like Van der Waals gases.
- Internal energy is a state variable. It depends only on the initial and final states of the thermodynamic system. For example, if the temperature of water is raised from 30°C to 40°C either by heating or by stirring, the final internal energy depends only on the final temperature 40°C and not the way it is arrived at.

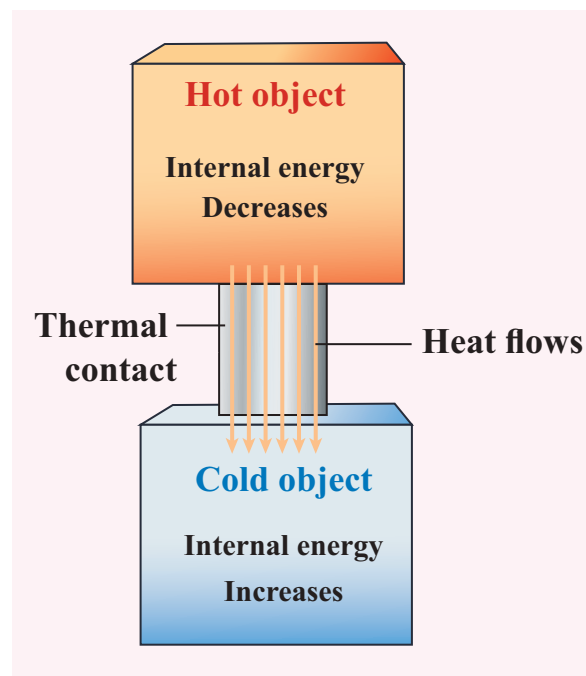
Note

It is very important to note that the internal energy of a thermodynamic system is associated with only the kinetic energy of the individual molecule due to its random motion and the potential energy of molecules which depends on their chemical nature. The bulk kinetic energy of the entire system or gravitational potential energy of the system should not be mistaken as a part of internal energy. For example

**Note**

(a) Consider two gas containers at the same temperature having the same internal energy, one is kept at rest on the ground and the other is kept in a moving train. Even though the gas container in the train is moving with the speed of the train, the internal energy of the gas in it will not increase.

(b) Consider two gas containers at the same temperature having the same internal energy, one is kept on the ground and the other is kept at some height h . Even though the container at height h is having higher gravitational potential energy, this has no influence on internal energy of the gas molecules.

**EXAMPLE 8.10**

When you mix a tumbler of hot water with one bucket of normal water, what will be the direction of heat flow? Justify.

The water in the tumbler is at a higher temperature than the bucket of normal water. But the bucket of normal water has larger internal energy than the hot water in the tumbler. This is because the internal energy is an extensive variable and it depends on the size or mass of the system.

Even though the bucket of normal water has larger internal energy than the tumbler of hot water, heat will flow from water in the tumbler to the water in the bucket. This is because heat flows from a body at higher temperature to the one at lower temperature and is independent of internal energy of the system.

Once the heat is transferred to an object it becomes internal energy of the object. The right way to say is 'object has certain amount of internal energy'. Heat is one of the ways to increase the internal energy of a system as shown in the Figure.

Note

It is to be noted that heat does not always increases the internal energy. Later we shall see that in ideal gases during isothermal process the internal energy will not increase even though heat flows in to the system.

8.6.1 Joule's mechanical equivalent of heat

The temperature of an object can be increased by heating it or by doing some work on it. In the eighteenth century, James Prescott Joule showed that mechanical energy can be converted into internal energy and vice versa. In his experiment, two masses were attached with a rope and a paddle wheel as shown in Figure 8.19. When these masses fall through a distance h due to gravity, both the masses lose potential energy equal to $2mgh$. When the masses fall, the paddle wheel turns. Due to the turning of wheel inside water, frictional force comes in between the water and the paddle wheel. This causes

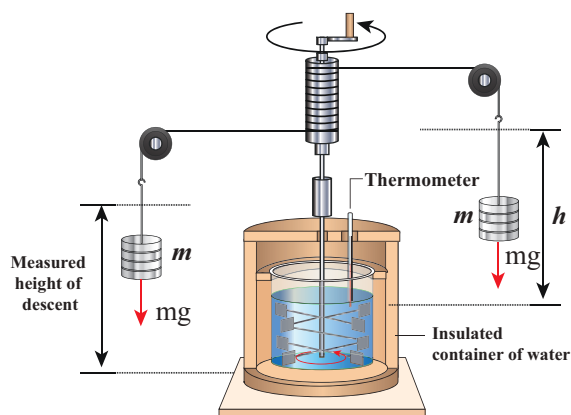


Figure.8.19 Joule's experiment for determining the mechanical equivalent of heat energy.

a rise in temperature of the water. This implies that gravitational potential energy is converted to internal energy of water. The temperature of water increases due to the work done by the masses. In fact, Joule was able to show that the mechanical work has the same effect as giving heat. He found that to raise 1 g of an object by 1°C , 4.186 J of energy is required. In earlier days the heat was measured in calorie.

$$1 \text{ cal} = 4.186 \text{ J}$$

This is called Joule's mechanical equivalent of heat.



Before James Prescott Joule, people thought that heat was a kind of fluid called caloric fluid which flows from an object at high temperature to that in low temperature. According to caloric fluid idea, the hot object contains more caloric fluid and the cold object contains less caloric fluid since heat was treated as a quantity. Now we understand that heat is not a quantity but it is an energy in transit. So the word 'mechanical equivalent of heat' is wrong terminology. Because mechanical energy is a quantity and any object can contain more or less

mechanical energy, but this is not so with heat as it is not a quantity. However this terminology is retained for historical reasons. A correct name would be 'Joule's mechanical equivalence of internal energy'. Joule Essentially converted mechanical energy to internal energy. In his experiment potential energy is converted to rotational kinetic energy of paddle wheel and this rotational kinetic energy is converted to internal energy of water.

EXAMPLE 8.11

A student had a breakfast of 200 food calories. He thinks of burning this energy by drawing water from the well and watering the trees in his school. Depth of the well is about 25 m. The pot can hold 25L of water and each tree requires one pot of water. How many trees can he water? (Neglect the mass of the pot and the energy spent by walking. Take $g = 10 \text{ m s}^{-2}$)



Solution

To draw 25 L of water from the well, the student has to do work against gravity by burning his energy.

$$\text{Mass of the water} = 25 \text{ L} = 25 \text{ kg} \quad (1\text{L}=1\text{kg})$$

The work required to draw 25 kg of water = gravitational potential energy gained by water.

$$W = mgh = 25 \times 10 \times 25 = 6250 \text{ J}$$



The total energy gained from the food = 200 food cal = 200 kcal.

$$= 200 \times 10^3 \times 4.186 \text{ J} = 8.37 \times 10^5 \text{ J}$$

If we assume that by using this energy the student can draw 'n' pots of water from the well, the total energy spent by him = $8.37 \times 10^5 \text{ J} = nmgh$

$$n = \frac{8.37 \times 10^5 \text{ J}}{6250 \text{ J}} \approx 134.$$

This n is also equal to the number of trees that he can water.

Is it possible to draw 134 pots of water from the well just by having breakfast? No. Actually the human body does not convert entire food energy into work. It is only approximately 20% efficient. It implies that only 20% of 200 food calories is used to draw water from the well. So 20 % of the 134 is only 26 pots of water. It is quite meaningful. So he can water only 26 trees.

The remaining energy is used for blood circulation and other functions of the body. It is to be noted that some energy is always 'wasted'. Why is it that the body cannot have 100% efficiency? You will find the answer in section 8.9

8.6.2 First law of thermodynamics

The first law of thermodynamics is a statement of the law of conservation of energy. In Newtonian mechanics conservation of energy involves kinetic and potential energies of bulk objects. But the first law of thermodynamics includes heat also. This law states that 'Change in internal energy (ΔU) of the system is equal to heat supplied to the system (Q) minus the work done by the system (W) on the surroundings'. Mathematically it is written as

$$\Delta U = Q - W \quad (8.13)$$

The internal energy of a thermodynamic system can be changed either by heating or by work as shown below.

Heat flows into the system	Internal energy increases
Heat flows out of the system	Internal energy decreases
Work is done on the system	Internal energy increases
Work is done by the system	Internal energy decreases

Based on the above table the sign convention is introduced to use first law of thermodynamics appropriately. It is shown in the following table and the Figure 8.20.

System gains heat	Q is positive
System loses heat	Q is negative
Work done on the system	W is negative
Work done by the system	W is positive

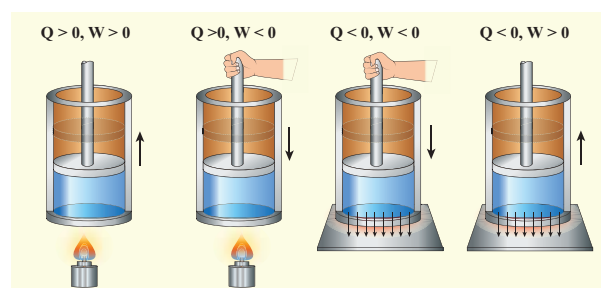


Figure 8.20 The Sign convention for heat and work

Even though we often explain first law of thermodynamics using gases, this law is universal and applies to liquids and solids also.



Some book presents the first law of thermodynamics as $\Delta U = Q + W$. Here the work done by the system is taken as negative and work done on the system is positive. However both the conventions are correct and we can follow any one of the convention.

EXAMPLE 8.12

A person does 30 kJ work on 2 kg of water by stirring using a paddle wheel. While stirring, around 5 kcal of heat is released from water through its container to the surface and surroundings by thermal conduction and radiation. What is the change in internal energy of the system?

Solution

Work done on the system (by the person while stirring), $W = -30 \text{ kJ} = -30,000 \text{ J}$

Heat flowing out of the system, $Q = -5 \text{ kcal} = -5 \times 4184 \text{ J} = -20920 \text{ J}$

Using First law of thermodynamics

$$\Delta U = Q - W$$

$$\Delta U = -20,920 \text{ J} - (-30,000) \text{ J}$$

$$\Delta U = -20,920 \text{ J} + 30,000 \text{ J} = 9080 \text{ J}$$

Here, the heat lost is less than the work done on the system, so the change in internal energy is positive.

EXAMPLE 8.13

Jogging every day is good for health. Assume that when you jog a work of 500 kJ is done and 230 kJ of heat is given off. What is the change in internal energy of your body?

Solution



Work done by the system (body),

$$W = +500 \text{ kJ}$$

Heat released from the system (body),

$$Q = -230 \text{ kJ}$$

The change in internal energy of a body
 $= \Delta U = -230 \text{ kJ} - 500 \text{ kJ} = -730 \text{ kJ}$

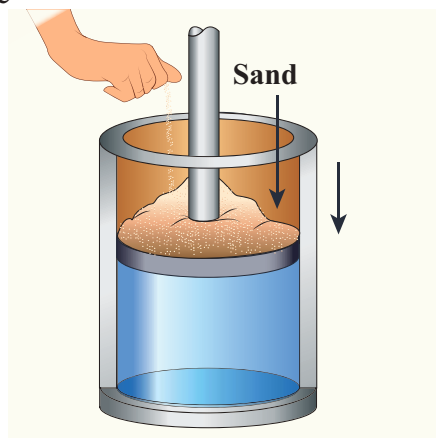
8.6.3 Quasi-static process

Consider a system of an ideal gas kept in a cylinder of volume V at pressure P and temperature T . When the piston attached to the cylinder moves outward the volume of the gas will change. As a result the temperature and pressure will also change because all three variables P , T and V are related by the equation of state $PV = NkT$. If a block of some mass is kept on the piston, it will suddenly push the piston downward. The pressure near the piston will be larger than other parts of the system. It implies that the gas is in non-equilibrium state. We cannot determine pressure, temperature or internal energy of the system until it reaches another equilibrium state. But if the piston is pushed very slowly such that at every stage it is still in equilibrium with surroundings, we can use the equation of state to calculate the internal energy, pressure or temperature. This kind of process is called quasi-static process.

A quasi-static process is an infinitely slow process in which the system changes its variables (P, V, T) so slowly such that it remains in thermal, mechanical and chemical equilibrium with its surroundings throughout. By this infinite slow variation, the system is always almost close to equilibrium state.

EXAMPLE 8.14

Give an example of a quasi-static process. Consider a container of gas with volume V , pressure P and temperature T . If we add sand particles one by one slowly on the top of the piston, the piston will move inward very slowly. This can be taken as almost a quasi-static process. It is shown in the figure



Sand particles added slowly- quasi-static process

8.6.4 Work done in volume changes

Consider a gas contained in the cylinder fitted with a movable piston. Suppose the gas is expanded quasi-statically by pushing the piston by a small distance dx as shown in Figure 8.21. Since the expansion occurs quasi-statically the pressure, temperature and internal energy will have unique values at every instant.

The small work done by the gas on the piston

$$dW = Fdx \quad (8.14)$$

The force exerted by the gas on the piston $F = PA$. Here A is area of the piston and P is pressure exerted by the gas on the piston.

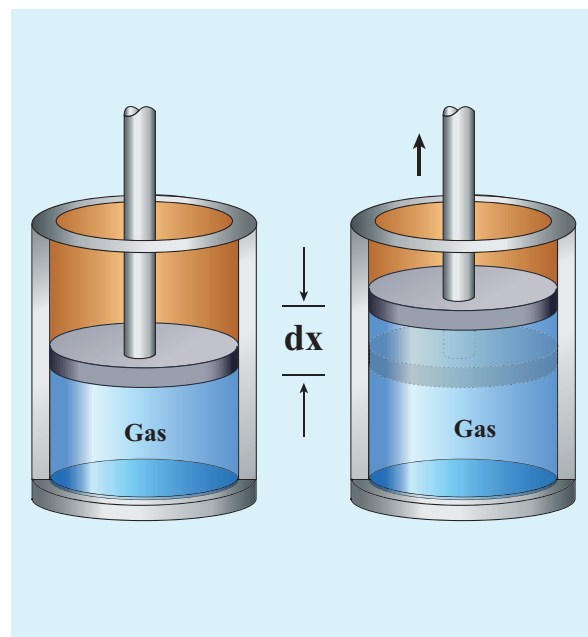


Figure 8.21 Work done by the gas

Equation (8.14) can be rewritten as

$$dW = PA \, dx \quad (8.15)$$

But $A dx = dV =$ change in volume during this expansion process.

So the small work done by the gas during the expansion is given by

$$dW = P dV \quad (8.16)$$

Note here that dV is positive since the volume is increased. So dW is positive.

In general the work done by the gas by increasing the volume from V_i to V_f is given by

$$W = \int_{V_i}^{V_f} P dV \quad (8.17)$$

Suppose if the work is done on the system, then $V_i > V_f$. Then, W is negative.

Note here the pressure P is inside the integral in equation (8.17). It implies that while the system is doing work, the pressure need not be constant. To evaluate the integration we need to first express the pressure as a function of volume and temperature using the equation of state.

8.6.5 PV diagram

PV diagram is a graph between pressure P and volume V of the system. The P-V diagram is used to calculate the amount of work done by the gas during expansion or on the gas during compression. In Unit 2, we have seen that the area under the curve will give integration of the function from lower limit to upper limit. The area under the PV diagram will give the work done during expansion or compression which is shown in Figure 8.22

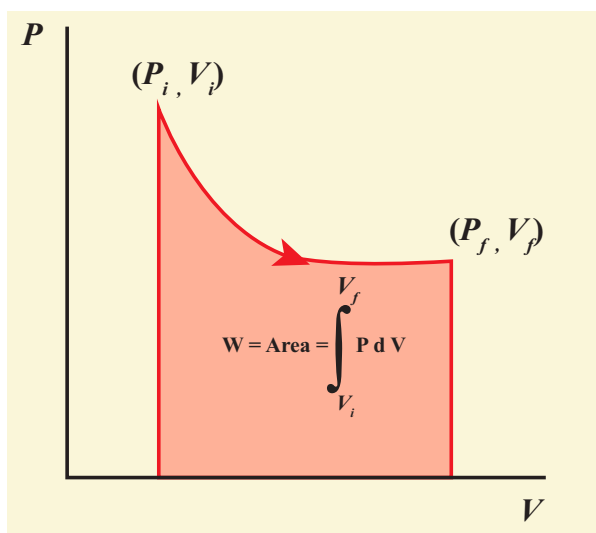


Figure 8.22 Work done by the gas during expansion

The shape of PV diagram depends on the nature of the thermodynamic process.

EXAMPLE 8.15

A gas expands from volume 1 m^3 to 2 m^3 at constant atmospheric pressure.

- Calculate the work done by the gas.
- Represent the work done in PV diagram.

Solution

- The pressure $P = 1 \text{ atm} = 101 \text{ kPa}$, $V_f = 2 \text{ m}^3$ and $V_i = 1 \text{ m}^3$

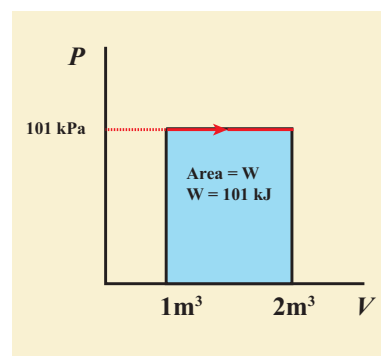
From equation (8.17)

$$W = \int_{V_i}^{V_f} P dV = P \int_{V_i}^{V_f} dV$$

Since P is constant. It is taken out of the integral.

$$W = P (V_f - V_i) = 101 \times 10^3 \times (2 - 1) = 101 \text{ kJ}$$

- Since the pressure is kept constant, PV diagram is straight line as shown in the figure. The area is equal to work done by the gas.



Note the arrow mark in the curve. Suppose the work is done on the system, then volume will decrease and the arrow will point in the opposite direction.

8.7

SPECIFIC HEAT CAPACITY OF A GAS

Specific heat capacity of a given system plays a very important role in determining

the structure and molecular nature of the system. Unlike solids and liquids, gases have two specific heats: specific heat capacity at constant pressure (s_p) and specific heat capacity at constant volume (s_v).

8.7.1 Specific heat capacity

Specific heat capacity at constant pressure (s_p):

The amount of heat energy required to raise the temperature of one kg of a substance by 1 K or 1°C by keeping the pressure constant is called specific heat capacity at constant pressure. When the heat energy is supplied to the gas, it expands to keep the pressure constant as shown in Figure 8.23

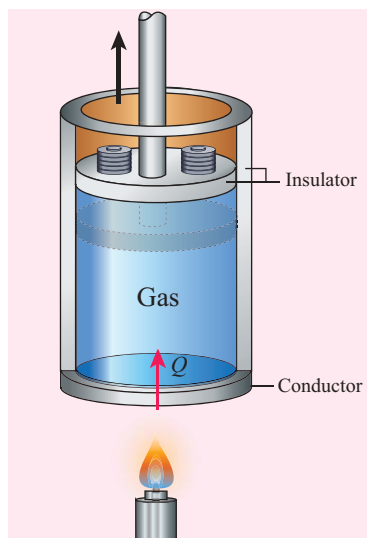


Figure 8.23 Specific heat capacity at constant pressure

In this process a part of the heat energy is used for doing work (expansion) and the remaining part is used to increase the internal energy of the gas.

Specific heat capacity at constant volume (s_v):

The amount of heat energy required to raise the temperature of one kg of a substance by 1 K or 1°C by keeping the volume constant

is called specific heat capacity at constant volume.

If the volume is kept constant, then the supplied heat is used to increase only the internal energy. No work is done by the gas as shown in Figure 8.24.

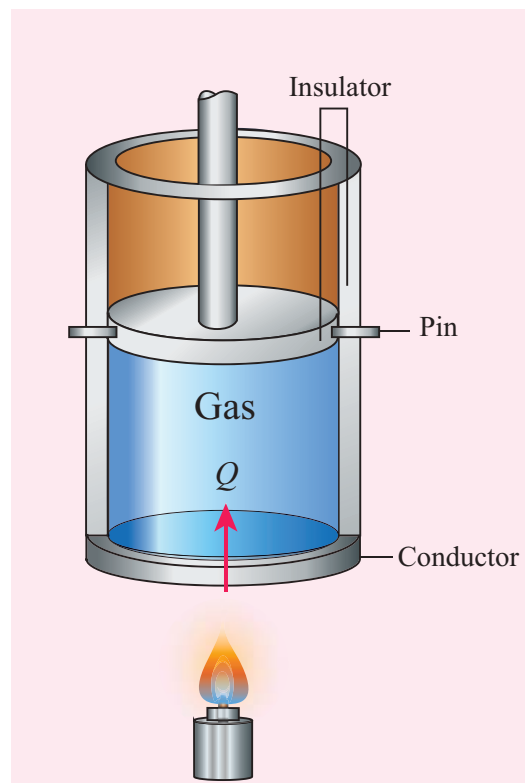


Figure 8.24 Specific heat capacity at constant volume

It implies that to increase the temperature of the gas at constant volume requires less heat than increasing the temperature of the gas at constant pressure. In other words s_p is always greater than s_v .

Molar Specific heat capacities

Sometimes it is useful to calculate the molar heat capacities C_p and C_v . The amount of heat required to raise the temperature of one mole of a substance by 1K or 1°C at constant volume is called molar specific heat capacity at constant volume (C_v). If pressure is kept constant, it is called molar specific heat capacity at constant pressure (C_p).



If Q is the heat supplied to μ mole of a gas at constant volume and if the temperature changes by an amount ΔT , we have

$$Q = \mu C_v \Delta T. \quad (8.18)$$

By applying the first law of thermodynamics for this constant volume process ($W=0$, since $dV=0$), we have

$$Q = \Delta U - 0 \quad (8.19)$$

By comparing the equations (8.18) and (8.19),

$$\Delta U = \mu C_v \Delta T \text{ or } C_v = \frac{1}{\mu} \frac{\Delta U}{\Delta T}$$

If the limit ΔT goes to zero, we can write

$$C_v = \frac{1}{\mu} \frac{dU}{dT} \quad (8.20)$$

Since the temperature and internal energy are state variables, the above relation holds true for any process.

8.7.2 Meyer's relation

Consider μ mole of an ideal gas in a container with volume V , pressure P and temperature T .

When the gas is heated at constant volume the temperature increases by dT . As no work is done by the gas, the heat that flows into the system will increase only the internal energy. Let the change in internal energy be dU .

If C_v is the molar specific heat capacity at constant volume, from equation (8.20)

$$dU = \mu C_v dT \quad (8.21)$$

Suppose the gas is heated at constant pressure so that the temperature increases by dT . If ' Q ' is the heat supplied in this process and ' dV ' the change in volume of the gas.

$$Q = \mu C_p dT \quad (8.22)$$

If W is the workdone by the gas in this process, then

$$W = PdV \quad (8.23)$$

But from the first law of thermodynamics,

$$Q = dU + W \quad (8.24)$$

Substituting equations (8.21), (8.22) and (8.23) in (8.24), we get,

$$\mu C_p dT = \mu C_v dT + PdV \quad (8.25)$$

For μ mole of ideal gas, the equation of state is given by

$$PV = \mu RT \Rightarrow PdV + VdP = \mu R dT \quad (8.26)$$

Since the pressure is constant, $dP=0$

$$\therefore C_p dT = C_v dT + R dT$$

$$\therefore C_p = C_v + R \quad (\text{or}) \quad C_p - C_v = R \quad (8.27)$$

This relation is called Meyer's relation

It implies that the molar specific heat capacity of an ideal gas at constant pressure is greater than molar specific heat capacity at constant volume.

The relation shows that specific heat at constant pressure (s_p) is always greater than specific heat at constant volume (s_v).

8.8

THERMODYNAMIC PROCESSES

8.8.1 Isothermal process

It is a process in which the temperature remains constant but the pressure and volume of a thermodynamic system will change. The ideal gas equation is

$$PV = \mu RT$$

Here, T is constant for this process

So the equation of state for isothermal process is given by

$$PV = \text{constant} \quad (8.28)$$

This implies that if the gas goes from one equilibrium state (P_1, V_1) to another equilibrium state (P_2, V_2) the following relation holds for this process

$$P_1 V_1 = P_2 V_2 \quad (8.29)$$

Since $PV = \text{constant}$, P is inversely proportional to V ($P \propto \frac{1}{V}$). This implies that PV graph is a hyperbola. The pressure-volume graph for constant temperature is also called isotherm.

Figure 8.25 shows the PV diagram for quasi-static isothermal expansion and quasi-static isothermal compression.

We know that for an ideal gas the internal energy is a function of temperature only. For an isothermal process since temperature is constant, the internal energy is also constant. This implies that dU or $\Delta U = 0$.

For an isothermal process, the first law of thermodynamics can be written as follows,

$$Q = W \quad (8.30)$$

From equation (8.30), we infer that the heat supplied to a gas is used to do only external work. It is a common misconception that when there is flow of heat to the system, the temperature will increase. For isothermal process this is not true. The isothermal compression takes place when the piston of the cylinder is pushed. This will increase the internal energy which will flow out of the system through thermal contact. This is shown in Figure 8.26.

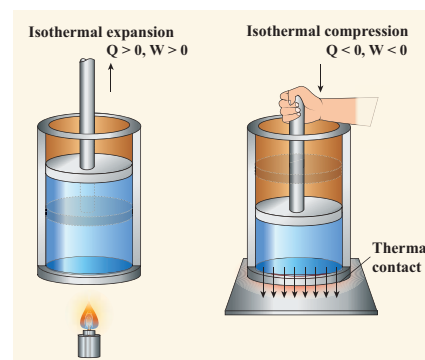


Figure 8.26 Isothermal expansion and isothermal compression

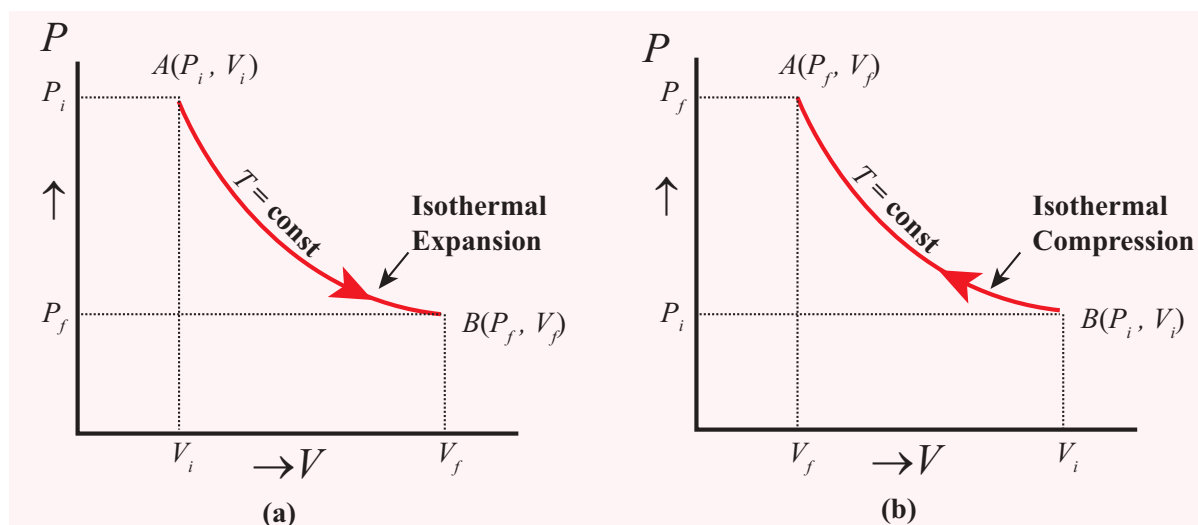


Figure 8.25 (a) Quasi-static isothermal expansion (b) Quasi-static isothermal compression

Examples:

- (i) When water is heated, at the boiling point, even when heat flows to water, the temperature will not increase unless the water completely evaporates. Similarly, at the freezing point, when the ice melts to water, the temperature of ice will not increase even when heat is supplied to ice.
- (ii) All biological processes occur at constant body temperature (37°C).

Work done in an isothermal process:

Consider an ideal gas which is allowed to expand quasi-statically at constant temperature from initial state (P_i, V_i) to the final state (P_f, V_f) . We can calculate the work done by the gas during this process. From equation (8.17) the work done by the gas,

$$W = \int_{V_i}^{V_f} P dV \quad (8.31)$$

As the process occurs quasi-statically, at every stage the gas is at equilibrium with the surroundings. Since it is in equilibrium at every stage the ideal gas law is valid. Writing pressure in terms of volume and temperature,

$$P = \frac{\mu RT}{V} \quad (8.32)$$

Substituting equation (8.32) in (8.31) we get

$$\begin{aligned} W &= \int_{V_i}^{V_f} \frac{\mu RT}{V} dV \\ W &= \mu RT \int_{V_i}^{V_f} \frac{dV}{V} \end{aligned} \quad (8.33)$$

In equation (8.33), we take μRT out of the integral, since it is constant throughout the isothermal process.

By performing the integration in equation (8.33), we get

$$W = \mu RT \ln \left(\frac{V_f}{V_i} \right) \quad (8.34)$$

Since we have an isothermal expansion, $\frac{V_f}{V_i} > 1$, so $\ln \left(\frac{V_f}{V_i} \right) > 0$. As a result the work done by the gas during an isothermal expansion is positive.

The above result in equation (8.34) is true for isothermal compression also. But in an isothermal compression $\frac{V_f}{V_i} < 1$. So $\ln \left(\frac{V_f}{V_i} \right) < 0$.

As a result the work done on the gas in an isothermal compression is negative.

In the PV diagram the work done during the isothermal expansion is equal to the area under the graph as shown in Figure 8.27

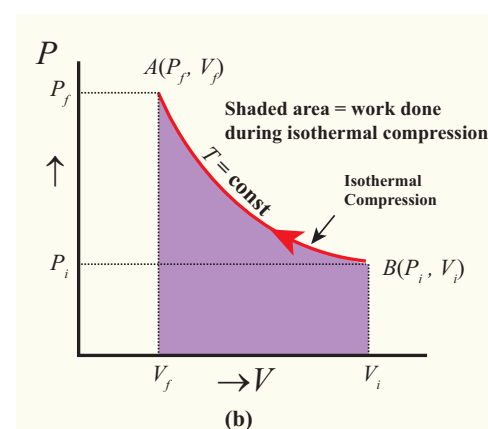
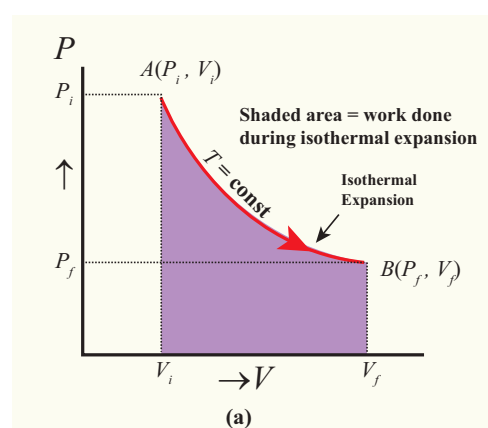


Figure 8.27 Work done in an isothermal process.

Similarly for an isothermal compression, the area under the PV graph is equal to the

work done on the gas which turns out to be the area with a negative sign.

Note

To calculate the work done in an isothermal process, we assume that the process is quasi-static. If it is not quasi-static we cannot substitute $P = \frac{\mu RT}{V}$ in equation (8.31). It is because the ideal gas law is not valid for non equilibrium states. But equation (8.34) is valid even when the isothermal process is not quasi-static. This is because the state variables like pressure and volume depend on initial and final state alone of an ideal gas and do not depend on the way the final state is reached. The assumption of 'quasi-static' requires to do the integration.

EXAMPLE 8.16

A 0.5 mole of gas at temperature 300 K expands isothermally from an initial volume of 2 L to 6 L

- What is the work done by the gas?
- Estimate the heat added to the gas?
- What is the final pressure of the gas? (The value of gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

Solution

- We know that work done by the gas in an isothermal expansion

Since $\mu = 0.5$

$$W = 0.5 \text{ mol} \times \frac{8.31 \text{ J}}{\text{mol.K}} \times 300 \text{ K} \ln \left(\frac{6\text{L}}{2\text{L}} \right)$$

$$W = 1.369 \text{ kJ}$$

Note that W is positive since the work is done by the gas.

- From the First law of thermodynamics, in an isothermal process the heat supplied is spent to do work.

Therefore, $Q = W = 1.369 \text{ kJ}$. Thus Q is also positive which implies that heat flows in to the system.

- For an isothermal process

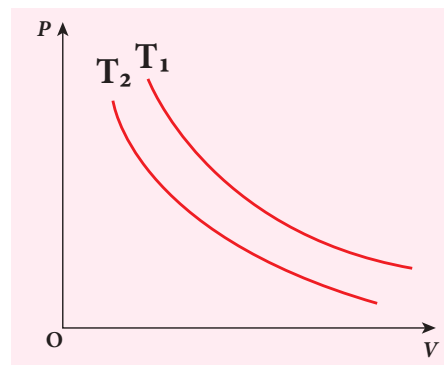
$$P_i V_i = P_f V_f = \mu RT$$

$$P_f = \frac{\mu RT}{V_f} = 0.5 \text{ mol} \times \frac{8.31 \text{ J}}{\text{mol.K}} \times \frac{300 \text{ K}}{6 \times 10^{-3} \text{ m}^3}$$

$$= 207.75 \text{ kPa}$$

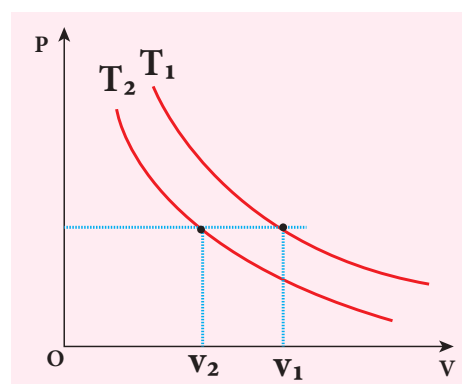
EXAMPLE 8.17

The following PV curve shows two isothermal processes for two different temperatures and. Identify the higher temperature of these two.



Solution

To determine the curve corresponding to higher temperature, draw a horizontal line parallel to x axis as shown in the figure. This is the constant pressure line. The volumes V_1 and V_2 belong to same pressure as the vertical lines from V_1 and V_2 meet the constant pressure line.



At constant pressure, higher the volume of the gas, higher will be the temperature. From the figure, as $V_1 > V_2$ we conclude $T_1 > T_2$. In general the isothermal curve closer to the origin, has lower temperature.

8.8.2 Adiabatic process

This is a process in which no heat flows into or out of the system ($Q=0$). But the gas can expand by spending its internal energy or gas can be compressed through some external work. So the pressure, volume and temperature of the system may change in an adiabatic process.

For an adiabatic process, the first law becomes $\Delta U = -W$.

This implies that the work is done by the gas at the expense of internal energy or work is done on the system which increases its internal energy.

The adiabatic process can be achieved by the following methods

- (i) Thermally insulating the system from surroundings so that no heat flows into or out of the system; for example, when thermally insulated cylinder of gas is compressed (adiabatic compression) or expanded (adiabatic expansion) as shown in the Figure 8.28

- (ii) If the process occurs so quickly that there is no time to exchange heat with surroundings even though there is no thermal insulation. A few examples are shown in Figure 8.29.

Examples:



Figure 8.29 (a) When the tyre bursts the air expands so quickly that there is no time to exchange heat with the surroundings.

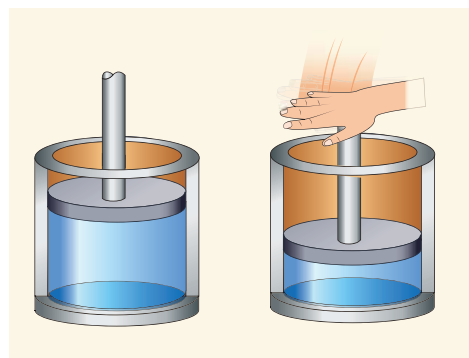


Figure 8.29 (b): When the gas is compressed or expanded so fast, the gas cannot exchange heat with surrounding even though there is no thermal insulation.

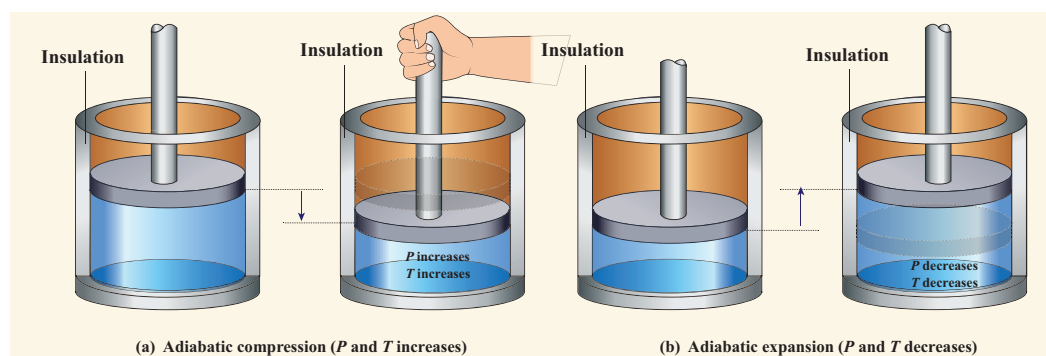


Figure 8.28 Adiabatic compression and expansion



Figure 8.29 (c): When the warm air rises from the surface of the Earth, it adiabatically expands. As a result the water vapor cools and condenses into water droplets forming a cloud.

The equation of state for an adiabatic process is given by

$$PV^\gamma = \text{constant} \quad (8.35)$$

Here γ is called adiabatic exponent ($\gamma = C_p/C_v$) which depends on the nature of the gas.

The equation (8.35) implies that if the gas goes from an equilibrium state (P_i, V_i) to another equilibrium state (P_f, V_f) adiabatically then it satisfies the relation

$$P_i V_i^\gamma = P_f V_f^\gamma \quad (8.36)$$

The PV diagram of an adiabatic expansion and adiabatic compression process are shown in Figure 8.30. The PV diagram for an adiabatic process is also called *adiabat*. Note that the PV diagram for isothermal (Figure 8.25) and adiabatic (Figure 8.30) processes look similar. But actually the adiabatic curve is steeper than isothermal curve.

We can also rewrite the equation (8.35) in terms of T and V. From ideal gas equation, the pressure $P = \frac{\mu RT}{V}$. Substituting this equation in the equation (8.35), we have

$$\frac{\mu RT}{V} V^\gamma = \text{constant (or)} \frac{T}{V} V^\gamma = \frac{\text{constant}}{\mu R}$$

Note here that is another constant. So it can be written as

$$TV^{\gamma-1} = \text{constant}. \quad (8.37)$$

The equation (8.37) implies that if the gas goes from an initial equilibrium state (T_i, V_i) to final equilibrium state (T_f, V_f) adiabatically then it satisfies the relation

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1} \quad (8.38)$$

The equation of state for adiabatic process can also be written in terms of T and P as

$$TP^{1/\gamma} = \text{constant}. \quad (8.39)$$

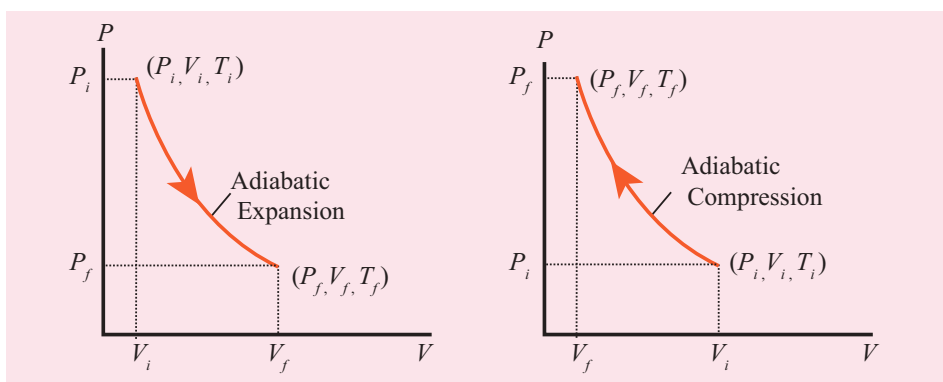


Figure 8.30 PV diagram for adiabatic expansion and adiabatic compression

(The proof of equation (8.39) left as an exercise).

EXAMPLE 8.18



We often have the experience of pumping air into bicycle tyre using hand pump.

Consider the air inside the pump as a thermodynamic system having volume V at atmospheric pressure and room temperature, 27°C . Assume that the nozzle of the tyre is blocked and you push the pump to a volume $1/4$ of V . Calculate the final temperature of air in the pump? (For air, since the nozzle is blocked air will not flow into tyre and it can be treated as an adiabatic compression). Take γ for air = 1.4

Solution

Here, the process is adiabatic compression. The volume is given and temperature is to be found. we can use the equation (8.38)

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}.$$

$$T_i = 300 \text{ K} \quad (273 + 27^\circ\text{C} = 300 \text{ K})$$

$$V_i = V \text{ \& } V_f = \frac{V}{4}$$

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} = 300 \text{ K} \times 4^{1.4-1} = 300 \text{ K} \times 1.741$$

$$T_2 \approx 522 \text{ K or } 249^\circ\text{C}$$

This temperature is higher than the boiling point of water. So it is very dangerous to touch the nozzle of blocked pump when you pump air.



When the piston is compressed so quickly that there is no time to exchange heat to the surrounding, the temperature of the gas increases rapidly. This is shown in the figure. This principle is used in the diesel engine. The air-gasoline mixer is compressed so quickly (adiabatic compression) that the temperature increases enormously, which is enough to produce a spark.



Work done in an adiabatic process:

Consider μ moles of an ideal gas enclosed in a cylinder having perfectly non conducting walls and base. A frictionless and insulating piston of cross sectional area A is fitted in the cylinder as shown in Figure 8.31.

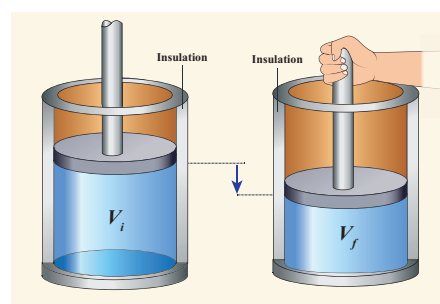


Figure 8.31 Work done in an adiabatic process

Let W be the work done when the system goes from the initial state (P_i, V_i, T_i) to the final state (P_f, V_f, T_f) adiabatically.



$$W = \int_{V_i}^{V_f} P dV \quad (8.40)$$

By assuming that the adiabatic process occurs quasi-statically, at every stage the ideal gas law is valid. Under this condition, the adiabatic equation of state is $PV^\gamma = \text{constant}$ (or)

$P = \frac{\text{constant}}{V^\gamma}$ can be substituted in the equation (8.40), we get

$$\begin{aligned} \therefore W_{\text{adia}} &= \int_{V_i}^{V_f} \frac{\text{constant}}{V^\gamma} dV \\ &= \text{constant} \int_{V_i}^{V_f} V^{-\gamma} dV \\ &= \text{constant} \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_i}^{V_f} \\ &= \frac{\text{constant}}{1-\gamma} \left[\frac{1}{V_f^{\gamma-1}} - \frac{1}{V_i^{\gamma-1}} \right] \\ &= \frac{1}{1-\gamma} \left[\frac{\text{constant}}{V_f^{\gamma-1}} - \frac{\text{constant}}{V_i^{\gamma-1}} \right] \end{aligned}$$

But, $P_i V_i^\gamma = P_f V_f^\gamma = \text{constant}$.

$$\begin{aligned} \therefore W_{\text{adia}} &= \frac{1}{1-\gamma} \left[\frac{P_f V_f^\gamma}{V_f^{\gamma-1}} - \frac{P_i V_i^\gamma}{V_i^{\gamma-1}} \right] \\ W_{\text{adia}} &= \frac{1}{1-\gamma} [P_f V_f - P_i V_i] \end{aligned} \quad (8.41)$$

From ideal gas law,

$$P_f V_f = \mu R T_f \text{ and } P_i V_i = \mu R T_i$$

Substituting in equation (8.41), we get

$$\therefore W_{\text{adia}} = \frac{\mu R}{\gamma-1} [T_i - T_f] \quad (8.42)$$

In adiabatic expansion, work is done by the gas. i.e., W_{adia} is positive. As $T_i > T_f$, the gas cools during adiabatic expansion.

In adiabatic compression, work is done on the gas. i.e., W_{adia} is negative. As $T_i < T_f$, the temperature of the gas increases during adiabatic compression.



Note Even though we have derived equations (8.41) and (8.42) by assuming that the adiabatic process is quasi-static, both the equations are valid even if the process is not quasi-static. This is because P and V are state variables and are independent of how the state is arrived.

In the adiabatic PV diagram shown in the Figure 8.32, the area under the adiabatic curve from initial state to final state will give the total work done in adiabatic process.

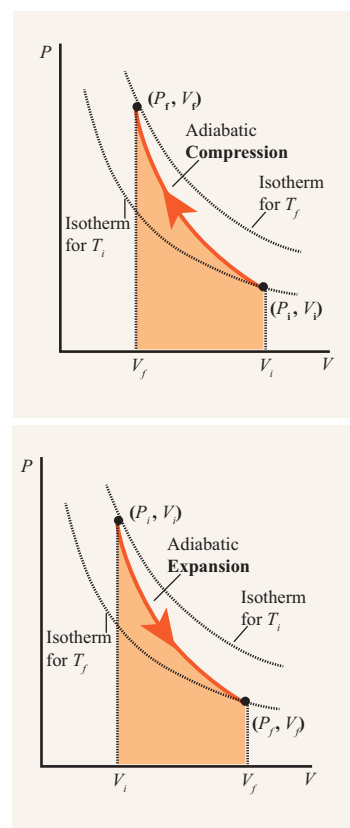


Figure 8.32 PV diagram -Work done in the adiabatic process

To differentiate between isothermal and adiabatic curves in (Figure 8.32) the adiabatic curve is drawn along with isothermal curve for T_f and T_i . Note that adiabatic curve is steeper than isothermal curve. This is because $\gamma > 1$ always.

8.8.3 Isobaric process

This is a thermodynamic process that occurs at constant pressure. Even though pressure is constant in this process, temperature, volume and internal energy are not constant. From the ideal gas equation, we have

$$V = \left(\frac{\mu R}{P} \right) T \quad (8.43)$$

$$\text{Here } \frac{\mu R}{P} = \text{constant}$$

In an isobaric process the temperature is directly proportional to volume.

$$V \propto T \quad (\text{Isobaric process}) \quad (8.44)$$

This implies that for a isobaric process, the V-T graph is a straight line passing through the origin.

If a gas goes from a state (V_i, T_i) to (V_f, T_f) at constant pressure, then the system satisfies the following equation

$$\frac{T_f}{V_f} = \frac{T_i}{V_i} \quad (8.45)$$

Examples for Isobaric process:

- (i) When the gas is heated and pushes the piston so that it exerts a force equivalent to atmospheric pressure plus the force due to gravity then this process is isobaric. This is shown in Figure 8.33

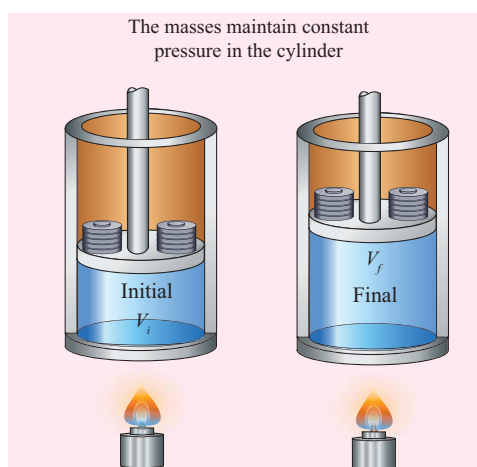


Figure 8.33 Isobaric process

- (ii) Most of the cooking processes in our kitchen are isobaric processes. When the food is cooked in an open vessel, the pressure above the food is always at atmospheric pressure. This is shown in Figure 8.34



Figure 8.34 Isobaric process

The PV diagram for an isobaric process is a horizontal line parallel to volume axis as shown in Figure 8.35.

Figure 8.35 (a) represents isobaric process where volume decreases

Figure 8.35 (b) represents isobaric process where volume increases

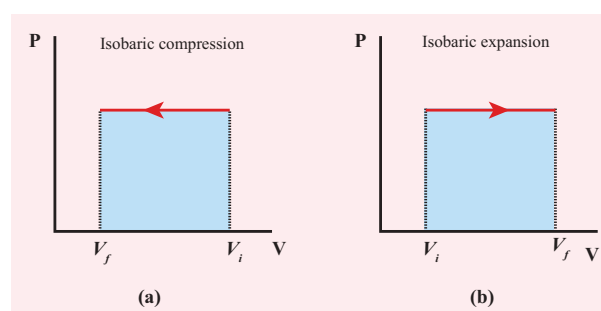


Figure 8.35 PV diagram for an isobaric process

The work done in an isobaric process:
Work done by the gas

$$W = \int_{V_i}^{V_f} P dV \quad (8.46)$$

In an isobaric process, the pressure is constant, so P comes out of the integral,

$$W = P \int_{V_i}^{V_f} dV \quad (8.47)$$

$$W = P[V_f - V_i] = P\Delta V \quad (8.48)$$

Where ΔV denotes change in the volume. If ΔV is negative, W is also negative. This implies that the work is done on the gas. If ΔV is positive, W is also positive, implying that work is done by the gas.

The equation (8.48) can also be rewritten using the ideal gas equation.

From ideal gas equation

$$PV = \mu RT \text{ and } V = \frac{\mu RT}{P}$$

Substituting this in equation (8.48) we get

$$W = \mu RT_f \left(1 - \frac{T_i}{T_f} \right) \quad (8.49)$$

In the PV diagram, area under the isobaric curve is equal to the work done in isobaric process. The shaded area in the following Figure 8.36 is equal to the work done by the gas.

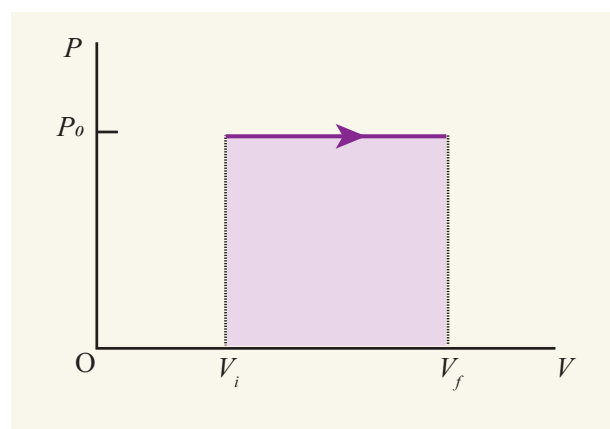


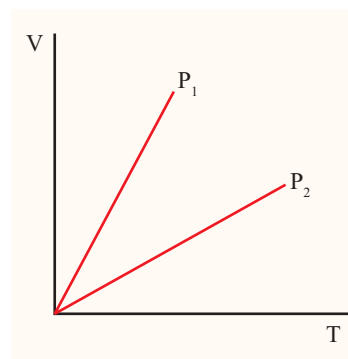
Figure 8.36 Work done in an isobaric process

The first law of thermodynamics for isobaric process is given by

$$\Delta U = Q - P\Delta V \quad (8.50)$$

EXAMPLE 8.19

The following graph shows a V-T graph for isobaric processes at two different pressures. Identify which one occurs at higher pressure.



Solution

From the ideal gas equation, $V = \left(\frac{\mu R}{P} \right) T$

V-T graph is a straight line passing the origin.

The slope = $\frac{\mu R}{P}$

The slope of V-T graph is inversely proportional to the pressure. If the slope is greater, lower is the pressure.

Here P_1 has larger slope than P_2 . So $P_2 > P_1$.

Suppose the graph is drawn between T and V (Temperature along the x-axis and Volume along the y-axis) then will we still have $P_2 > P_1$?

EXAMPLE 8.20

One mole of an ideal gas initially kept in a cylinder at pressure 1 MPa and temperature 27°C is made to expand until its volume is doubled.

- (a) How much work is done if the expansion is (i) adiabatic (ii) isobaric (iii) isothermal?



- (b) Identify the processes in which work done is least and is maximum.
- (c) Show each process on a PV diagram.
- (d) Name the processes in which the heat transfer is maximum and minimum.

(Take $\gamma = \frac{5}{3}$ and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

Solution

- (a) (i) In an adiabatic process the work done by the system is

$$W_{\text{adia}} = \frac{\mu R}{\gamma - 1} [T_i - T_f]$$

To find the final temperature T_f , we can use adiabatic equation of state.

$$T_f V_f^{\gamma-1} = T_i V_i^{\gamma-1}$$
$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} = 300 \times \left(\frac{1}{2} \right)^{\frac{2}{3}}$$

$$= 0.63 \times 300 \text{ K} = 189.8 \text{ K}$$

$$W = 1 \times 8.3 \times \frac{3}{2} (300 - 189.8) = 1.37 \text{ kJ}$$

- (ii) In an isobaric process the work done by the system

$$W = P \Delta V = P(V_f - V_i)$$

and $V_f = 2V_i$ so $W = 2PV_i$

To find V_i , we can use the ideal gas law for initial state. $P_i V_i = RT_i$

$$V_i = \frac{RT_i}{P_i} = 8.3 \times \frac{300}{1} \times 10^{-6} = 24.9 \times 10^{-4} \text{ m}^3$$

The work done during isobaric process,
 $W = 2 \times 10^6 \times 24.9 \times 10^{-4} = 4.9 \text{ kJ}$

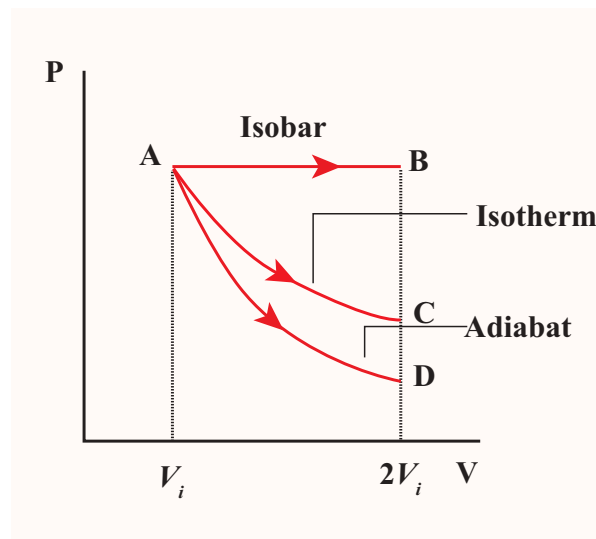
- (iii) In an isothermal process the work done by the system,

$$W = \mu RT \ln \left(\frac{V_f}{V_i} \right)$$

In an isothermal process the initial room temperature is constant.

$$W = 1 \times 8.3 \times 300 \times \ln(2) = 1.7 \text{ kJ}$$

- (b) Comparing all three processes, we see that the work done in the isobaric process is the greatest, and work done in the adiabatic process is the least.
- (c) The PV diagram is shown in the Figure.



The area under the curve AB = Work done during the isobaric process

The area under the curve AC = Work done during the isothermal process

The area under the curve AD = Work done during the adiabatic process

From the PV diagram the area under the curve AB is more, implying that the work done in isobaric process is highest and work done in adiabatic process is least.

- (d) In an adiabatic process no heat enters into the system or leaves from the system. In an isobaric process the work done is more so heat supplied should be more compared to an isothermal process.

8.8.4 Isochoric process

This is a thermodynamic process in which the volume of the system is kept constant. But pressure, temperature and internal energy continue to be variables.

The pressure - volume graph for an isochoric process is a vertical line parallel to pressure axis as shown in Figure 8.37.

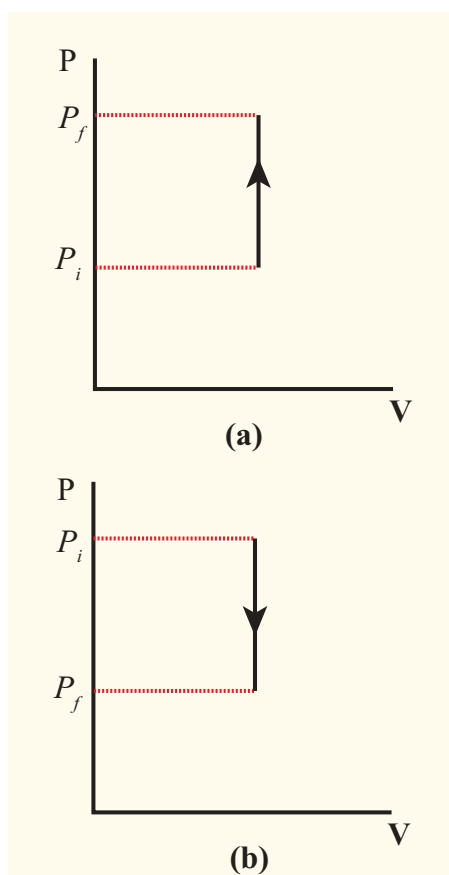


Figure 8.37 Isochoric process with (a) increased pressure and (b) decreased pressure

The equation of state for an isochoric process is given by

$$P = \left(\frac{\mu R}{V} \right) T \quad (8.51)$$

where $\left(\frac{\mu R}{V} \right) = \text{constant}$

We can infer that the pressure is directly proportional to temperature. This implies that the P-T graph for an isochoric process is a straight line passing through origin.

If a gas goes from state (P_i, T_i) to (P_f, T_f) at constant volume, then the system satisfies the following equation

$$\frac{P_i}{T_i} = \frac{P_f}{T_f} \quad (8.52)$$

For an isochoric processes, $\Delta V=0$ and $W=0$. Then the first law becomes

$$\Delta U = Q \quad (8.53)$$

Implying that the heat supplied is used to increase only the internal energy. As a result the temperature increases and pressure also increases. This is shown in Figure 8.38

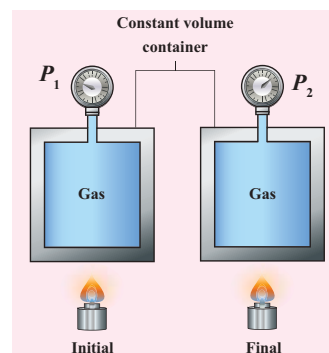


Figure 8.38 Isochoric process

Suppose a system loses heat to the surroundings through conducting walls by keeping the volume constant, then its internal energy decreases. As a result the temperature decreases; the pressure also decreases.

Examples:

1. When food is cooked by closing with a lid as shown in figure.



**Table 8.4:** Summary of various thermodynamic processes

S. No.	Process		Heat Q	Temperature & internal energy	Pressure	
1	Isothermal	Expansion	$Q > 0$	Constant	decreases	
		Compression	$Q < 0$	Constant	Increases	
2	Isobaric	Expansion	$Q > 0$	increases	Constant	
		Compression	$Q < 0$	decreases	Constant	
3	Isochoric		$Q > 0$	Increases	increases	
			$Q < 0$	Decreases	decreases	
4	Adiabatic	Expansion	$Q = 0$	Decreases	Decreases	
		Compression	$Q = 0$	Increases	increases	



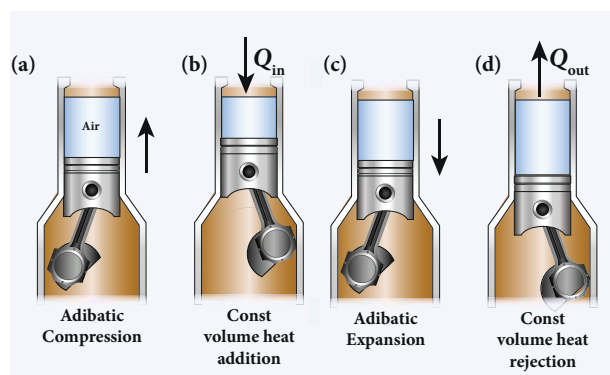


	Volume	Equation of state	Work done (ideal gas)	Indicator diagram (PV diagram)
	increases	$PV = \text{Constant}$	$W = \mu RT \ln \left(\frac{V_f}{V_i} \right) > 0$	
	Decreases		$W = \mu RT \ln \left(\frac{V_f}{V_i} \right) < 0$	
	increases	$\frac{V}{T} = \text{Constant}$	$W = P [V_f - V_i] = P\Delta V > 0$	
	decreases		$W = P [V_f - V_i] = P\Delta V < 0$	
	Constant	$\frac{P}{T} = \text{Constant}$	Zero	
	increases	$PV^\gamma = \text{Constant}$	$W = \frac{\mu R}{\gamma - 1} (T_i - T_f) > 0$	
	Decreases		$W = \frac{\mu R}{\gamma - 1} (T_i - T_f) < 0$	



When food is being cooked in this closed position, after a certain time you can observe the lid is being pushed upwards by the water steam. This is because when the lid is closed, the volume is kept constant. As the heat continuously supplied, the pressure increases and water steam tries to push the lid upwards.

2. In automobiles the petrol engine undergoes four processes. First the piston is adiabatically compressed to some volume as shown in the Figure (a). In the second process (Figure (b)), the volume of the air-fuel mixture is kept constant and heat is being added. As a result the temperature and pressure are increased. This is an isochoric process. For a third stroke (Figure (c)) there will be an adiabatic expansion, and fourth stroke again isochoric process by keeping the piston immovable (Figure (d)).



The summary of various thermodynamic processes is given in the Table 8.4.

EXAMPLE 8.21

500 g of water is heated from 30°C to 60°C. Ignoring the slight expansion of water, calculate the change in internal energy of the water? (specific heat of water 4184 J/kg.K)

Solution

When the water is heated from 30°C to 60°C, there is only a slight change in its volume. So we can treat this process as isochoric. In an isochoric process the work done by the system is zero. The given heat supplied is used to increase only the internal energy.

$$\Delta U = Q = ms_v \Delta T$$

The mass of water = 500 g = 0.5 kg

The change in temperature = 30K

The heat $Q = 0.5 \times 4184 \times 30 = 62.76 \text{ kJ}$

8.8.5 Cyclic processes

This is a thermodynamic process in which the thermodynamic system returns to its initial state after undergoing a series of changes. Since the system comes back to the initial state, the change in the internal energy is zero. In a cyclic process, heat can flow in to the system and heat flow out of the system. From the first law of thermodynamics, the net heat transferred to the system is equal to the work done by the gas.

$$Q_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = W \quad (\text{for a cyclic process}) \quad (8.54)$$

8.8.6 PV diagram for a cyclic process

In the PV diagram the cyclic process is represented by a closed curve. Let the gas undergo a cyclic process in which it returns to the initial stage after an expansion and compression as shown in Figure 8.39

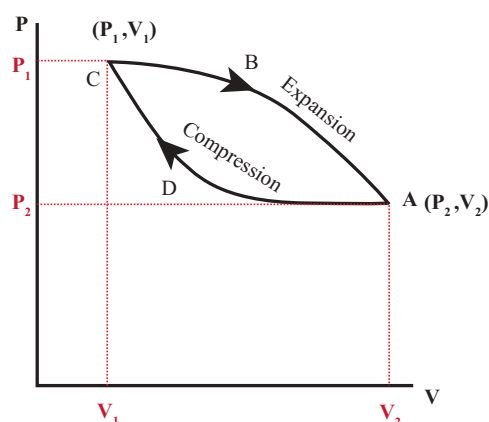


Figure 8.39 PV diagram for cyclic process

Let W_1 be the work done by the gas during expansion from volume V_1 to volume V_2 . It is equal to area under the graph CBA as shown in Figure 8.40 (a).

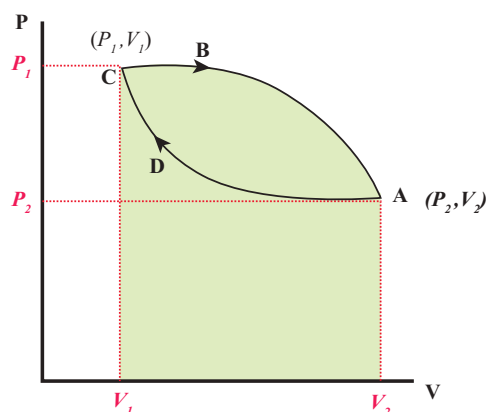


Figure 8.40 (a) W for path CBA

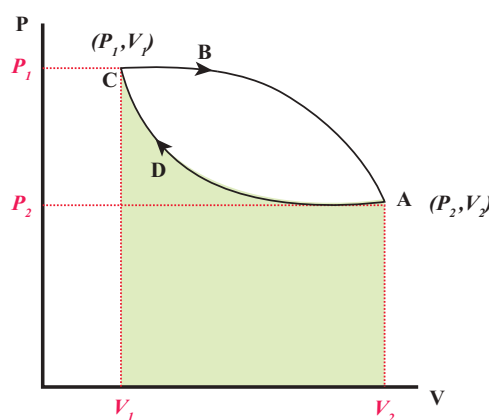


Figure 8.40 (b) W for path ADC

Let W_2 be the work done on the gas during compression from volume V_2 to volume V_1 . It is equal to the area under the graph ADC as shown in Figure 8.40 (b)

The total work done in this cyclic process $= W_1 - W_2 =$ Green shaded area inside the loop, as shown in Figure 8.41.

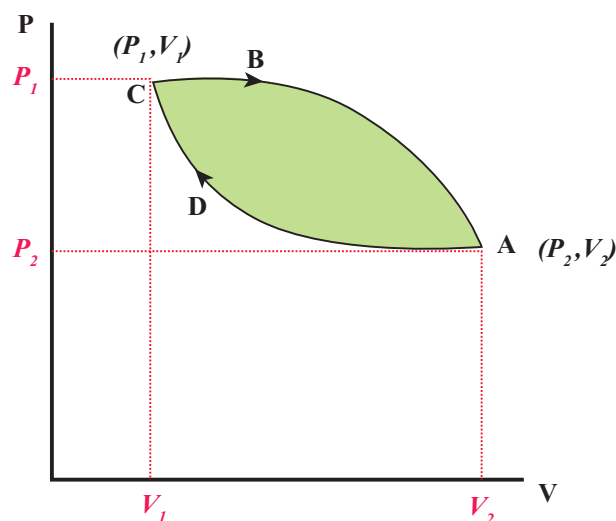


Figure 8.41 Net work done in a cyclic process

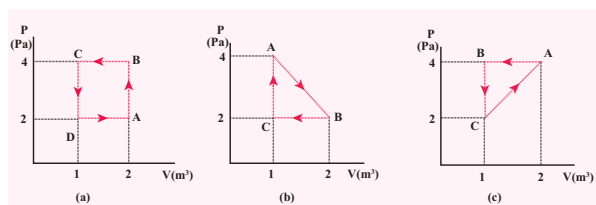
Thus the net work done during the cyclic process shown above is not zero. In general the net work done can be positive or negative. If the net work done is positive, then work done by the system is greater than the work done on the system. If the net work done is negative then the work done by the system is less than the work done on the system.



Note Further, in a cyclic process the net work done is positive if the process goes clockwise and net work done is negative if the process goes anti-clockwise. In Figure 8.41 the process goes clockwise.

EXAMPLE 8.22

The PV diagrams for a thermodynamical system is given in the figure below. Calculate the total work done in each of the cyclic processes shown.



Solution

In the case (a) the closed curve is anticlockwise. So the net work done is negative, implying that the work done on the system is greater than the work done by the system. The area under the curve BC will give work done on the gas (isobaric compression) and area under the curve DA (work done by the system) will give the total work done by the system.

Area under the curve BC = Area of rectangle BC12 = $1 \times 4 = -4J$

Area under the curve DA = $1 \times 2 = +2J$

Net work done in cyclic process
= $-4 + 2 = -2J$

In the case (b) the closed curve is clockwise. So the net work done is positive, implying that the work done on the system is less than the work done by the system. Area under the curve BC will give work done on the gas (isobaric compression) and area under the curve AB will give the total work done by the system.

Area under the curve AB = rectangle area + triangle area = $(1 \times 2) + \frac{1}{2} \times 1 \times 2 = +3J$

Area under the curve BC = rectangle area
= $1 \times 2 = -2J$

Net work done in the cyclic process = $1J$, which is positive.

In the case (c) the closed curve is anticlockwise. So the net work done is negative, implying that the work done on the system is greater than work done by the system. The area under the curve AB will give the work done on the gas (isobaric compression) and area under the curve CA (work done by the system) will give the total work done by the system.

The area under the curve AB = Rectangle of area = $4 \times 1 = -4J$

The area under the curve CA = Rectangle area + triangle area = $(1 \times 2) + \frac{1}{2} \times 1 \times 2 = +3J$
The total work in the cyclic process = $-1J$. It is negative

8.8.7 Limitations of first law of thermodynamics

The first law of thermodynamics explains well the inter convertibility of heat and work. But it does not indicate the direction of change.

For example,

- When a hot object is in contact with a cold object, heat always flows from the hot object to cold object but not in the reverse direction. According to first law, it is possible for the energy to flow from hot object to cold object or from cold object to hot object. But in nature the direction of heat flow is always from higher temperature to lower temperature.
- When brakes are applied, a car stops due to friction and the work done against friction is converted into heat. But this



heat is not reconverted to the kinetic energy of the car.

So the first law is not sufficient to explain many of natural phenomena.

8.8.8 Reversible process

A thermodynamic process can be considered reversible only if it possible to retrace the path in the opposite direction in such a way that the system and surroundings pass through the same states as in the initial, direct process.

Example: A quasi-static isothermal expansion of gas, slow compression and expansion of a spring.

Conditions for reversible process:

1. The process should proceed at an extremely slow rate.
2. The system should remain in mechanical, thermal and chemical equilibrium state at all the times with the surroundings, during the process.
3. No dissipative forces such as friction, viscosity, electrical resistance should be present.



Note All reversible processes are quasi-static but all quasi-static processes need not be reversible. For example when we push the piston very slowly, if there is friction between cylinder wall and piston some amount of energy is lost to surroundings, which cannot be retrieved back.

Irreversible process:

All natural processes are irreversible. Irreversible process cannot be plotted in a PV diagram, because these processes cannot have unique values of pressure, temperature at every stage of the process.

The first law of thermodynamics is the statement about conservation of energy in a thermodynamic process. For example, if a hotter object is placed on a colder object, heat flows from hotter to colder object. Why does heat not flow from the colder object to hotter object? Even if energy flows from colder object to hotter object, the first law of thermodynamics is not violated. For example, if 5 J of heat flows from hotter to colder or from colder to hotter objects the total internal energy of this combined system remains the same. But 5 J of heat never flows from the colder object to hotter object. In nature all such process occur only in one direction but not in the reverse direction, even if the energy is conserved in both the processes. Thus the first law of thermodynamics has no explanation for this irreversibility. When the scientists of the eighteenth century tried to explain this irreversibility, they discovered a new law of nature. This is called the second law of thermodynamics. According to second law of thermodynamics

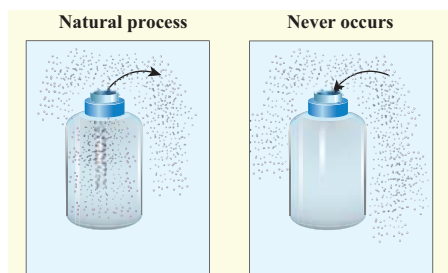
“Heat always flows from hotter object to colder object spontaneously”. This is known as the Clausius form of second law of thermodynamics.

EXAMPLE 8.23

Give some examples of irreversible processes.

All naturally occurring processes are irreversible. Here we give some interesting examples.

- (a) When we open a gas bottle, the gas molecules slowly spread into the entire room. These gas molecules can never get back in to the bottle.



- (b) Suppose one drop of an ink is dropped in water, the ink droplet slowly spreads in the water. It is impossible to get the ink droplet back.
- (c) When an object falls from some height, as soon as it hits the earth it comes to rest. All the kinetic energy of the object is converted to kinetic energy of molecules of the earth surface, molecules of the object and small amount goes as sound energy. The spreaded kinetic energy to the molecules never collected back and object never goes up by itself.

Note that according to first law of thermodynamics all the above processes are possible in both directions. But second law of thermodynamics forbids The processes to occur in the reverse direction. The second law of thermodynamics is one of the very important laws of nature. It controls the way the natural processes occur.

8.9

HEAT ENGINE

In the modern technological world, the role of automobile engines plays a vital role in for transportation. In motor bikes and cars there are engines which take in petrol or diesel as input, and do work by rotating wheels. Most of these automobile engines have efficiency not greater than 40%. The second law of thermodynamics

puts a fundamental restriction on efficiency of engines. Therefore understanding heat engines is very important.

Reservoir:

It is defined as a thermodynamic system which has very large heat capacity. By taking in heat from reservoir or giving heat to reservoir, the reservoir's temperature does not change.

Example: Pouring a tumbler of hot water in to lake will not increase the temperature of the lake. Here the lake can be treated as a reservoir.

When a hot cup of coffee attains equilibrium with the open atmosphere, the temperature of the atmosphere will not appreciably change. The atmosphere can be taken as a reservoir.

We can define heat engine as follows.

Heat engine is a device which takes heat as input and converts this heat in to work by undergoing a cyclic process.

A heat engine has three parts:

- (a) Hot reservoir
- (b) Working substance
- (c) Cold reservoir

A Schematic diagram for heat engine is given below in the figure 8.42.

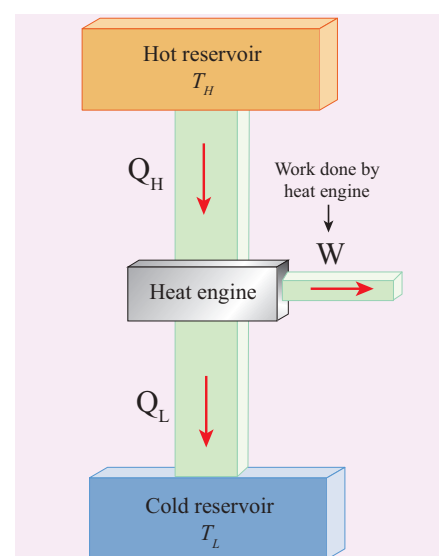


Figure 8.42 Heat Engine





1. Hot reservoir (or) Source: It supplies heat to the engine. It is always maintained at a high temperature T_H
2. Working substance: It is a substance like gas or water, which converts the heat supplied into work.

A simple example of a heat engine is a steam engine. In olden days steam engines were used to drive trains. The working substance in these is water which absorbs heat from the burning of coal. The heat converts the water into steam. This steam does work by rotating the wheels of the train, thus making the train move.

3. Cold reservoir (or) Sink: The heat engine ejects some amount of heat (Q_L) in to cold reservoir after it doing work. It is always maintained at a low temperature T_L .

For example, in the automobile engine, the cold reservoir is the surroundings at room temperature. The automobile ejects heat to these surroundings through a silencer.

The heat engine works in a cyclic process. After a cyclic process it returns to the same state. Since the heat engine returns to the same state after it ejects heat, the change in the internal energy of the heat engine is zero.

The efficiency of the heat engine is defined as the ratio of the work done (output) to the heat absorbed (input) in one cyclic process.

Let the working substance absorb heat Q_H units from the source and reject Q_L units to the sink after doing work W units, as shown in the Figure 8.43.

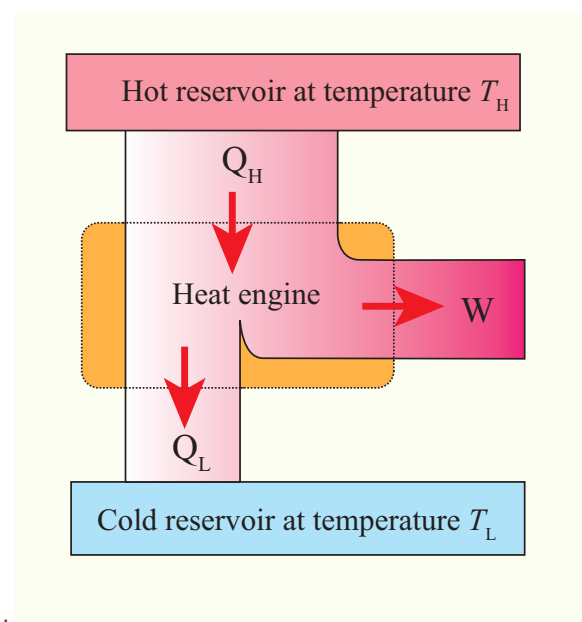


Figure 8.43 Heat engine

We can write

Input heat = Work done + ejected heat

$$Q_H = W + Q_L$$

$$W = Q_H - Q_L$$

Then the efficiency of heat engine

$$\eta = \frac{\text{output}}{\text{input}} = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H}$$
$$\eta = \frac{\text{output}}{\text{input}} = \frac{W}{Q_H} = 1 - \frac{Q_L}{Q_H} \quad (8.55)$$

Note here that Q_H , Q_L and W all are taken as positive, a sign convention followed in this expression.

Since $Q_L < Q_H$, the efficiency (η) always less than 1. This implies that heat absorbed is not completely converted into work. The second law of thermodynamics placed fundamental restrictions on converting heat completely into work.

We can state the heat engine statement of second law of thermodynamics. This is also called Kelvin-Planck's statement.

Kelvin-Planck statement:

It is impossible to construct a heat engine that operates in a cycle, whose sole effect is to convert the heat completely into work. This implies that no heat engine in the universe can have 100% efficiency.



Note According to first law of thermodynamics, in an isothermal process the given heat is completely converted into work ($Q = W$). Is it a violation of the second law of thermodynamics? No. For non-cyclic process like an isothermal expansion, the heat can be completely converted into work. But Second law of thermodynamics implies that 'In a cyclic process only a portion of the heat absorbed is converted into work'. All heat engines operate in a cyclic process.

EXAMPLE 8.24

During a cyclic process, a heat engine absorbs 500 J of heat from a hot reservoir, does work and ejects an amount of heat 300 J into the surroundings (cold reservoir). Calculate the efficiency of the heat engine?

Solution

The efficiency of heat engine is given by

$$\eta = 1 - \frac{Q_L}{Q_H}$$

$$\eta = 1 - \frac{300}{500} = 1 - \frac{3}{5}$$

$$\eta = 1 - 0.6 = 0.4$$

The heat engine has 40% efficiency, implying that this heat engine converts only 40% of the input heat into work.

8.9.1 Carnot's ideal heat engine

In the previous section we have seen that the heat engine cannot have 100% efficiency. What is the maximum possible efficiency can a heat engine have?. In the year 1824 a young French engineer Sadi Carnot proved that a certain reversible engine operated in cycle between hot and cold reservoir can have maximum efficiency. This engine is called Carnot engine.

A reversible heat engine operating in a cycle between two temperatures in a particular way is called a Carnot Engine.

The carnot engine has four parts which are given below.

- Source:** It is the source of heat maintained at constant high temperature T_H . Any amount of heat can be extracted from it, without changing its temperature.
- Sink:** It is a cold body maintained at a constant low temperature T_L . It can absorb any amount of heat.
- Insulating stand:** It is made of perfectly non-conducting material. Heat is not conducted through this stand.
- Working substance:** It is an ideal gas enclosed in a cylinder with perfectly non-conducting walls and perfectly conducting bottom. A non-conducting and frictionless piston is fitted in it.

The four parts are shown in the following Figure 8.44



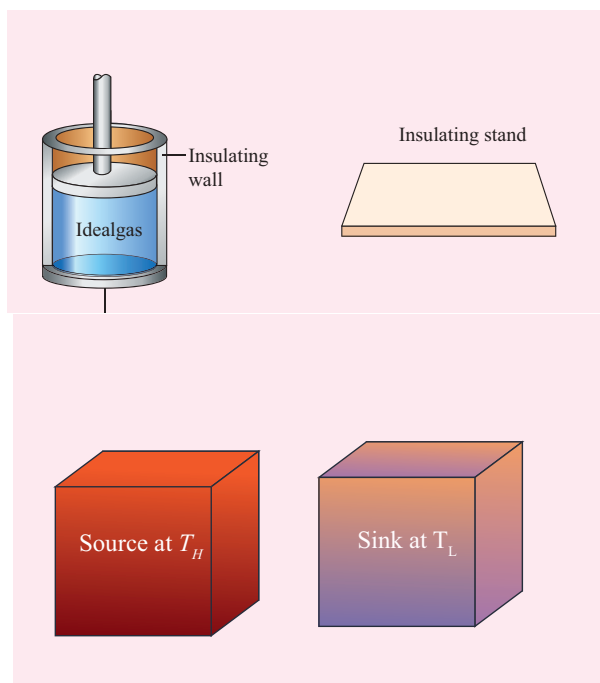


Figure 8.44 Carnot engine

Carnot's cycle:

The working substance is subjected to four successive reversible processes forming what is called Carnot's cycle.

Let the initial pressure, volume of the working substance be P_1, V_1 .

Step A to B: Quasi-static isothermal expansion from (P_1, V_1, T_H) to (P_2, V_2, T_H) :

The cylinder is placed on the source. The heat (Q_H) flows from source to the working substance (ideal gas) through the bottom of the cylinder. Since the process is isothermal, the internal energy of the working substance will not change. The input heat increases the volume of the gas. The piston is allowed to move out very slowly (quasi-statically). It is shown in the figure 8.47(a).

W_1 is the work done by the gas in expanding from volume V_1 to volume V_2 with a decrease of pressure from P_1 to P_2 . This is represented by the P-V diagram along the path AB as shown in the Figure 8.45.

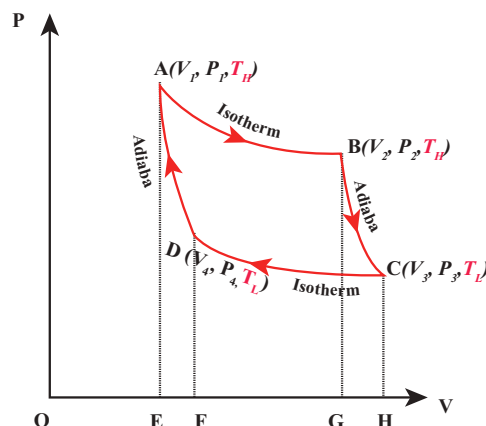


Figure 8.45 PV diagram for Carnot cycle

Then the work done by the gas (working substance) is given by

$$\therefore Q_H = W_{A \rightarrow B} = \int_{V_1}^{V_2} P dV$$

Since the process occurs quasi-statically, the gas is in equilibrium with the source till it reaches the final state. The work done in the isothermal expansion is given by the equation (8.34)

$$W_{A \rightarrow B} = \mu R T_H \ln \left(\frac{V_2}{V_1} \right) = \text{Area under the curve AB} \quad (8.56)$$

This is shown in Figure 8.46 (a)

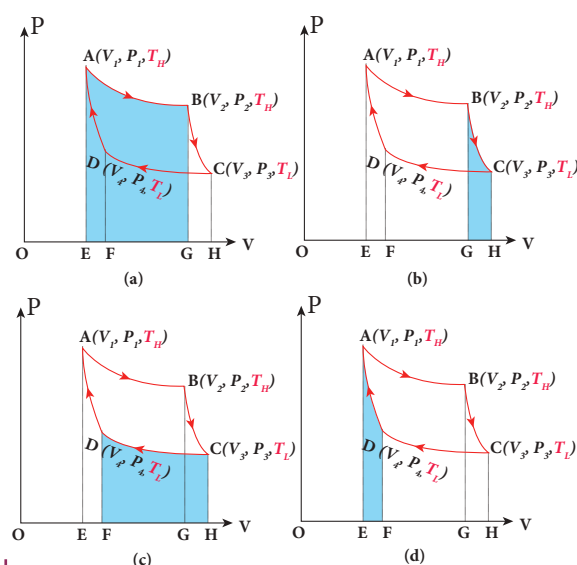


Figure 8.46 Work done in Carnot cycle

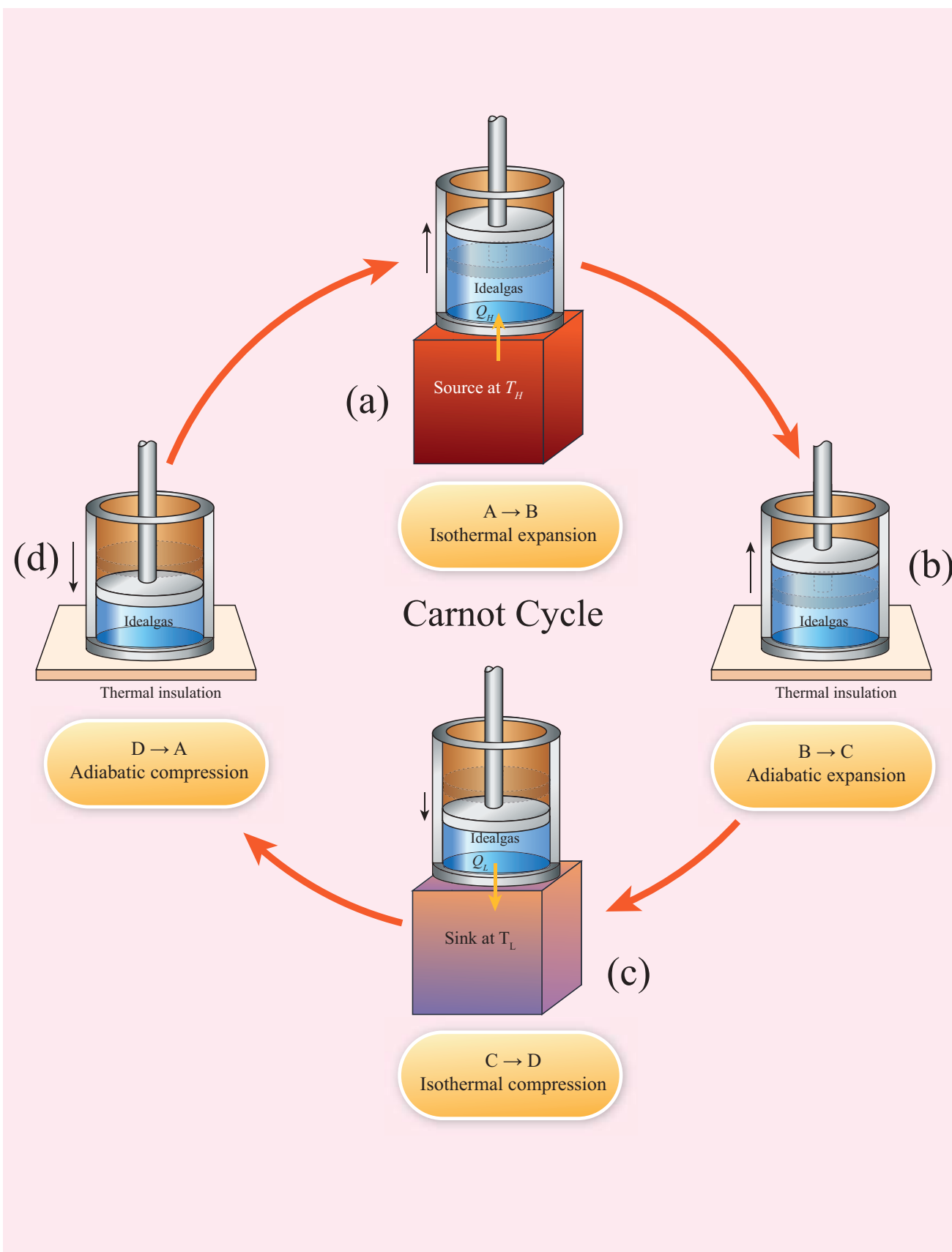


Figure 8.47 Carnot cycle





Step B to C: Quasi-static adiabatic expansion from (P_2, V_2, T_H) to (P_3, V_3, T_L)

The cylinder is placed on the insulating stand and the piston is allowed to move out. As the gas expands adiabatically from volume V_2 to volume V_3 the pressure falls from P_2 to P_3 . The temperature falls to T_L . This adiabatic expansion is represented by curve BC in the P-V diagram. This adiabatic process also occurs quasi-statically and implying that this process is reversible and the ideal gas is in equilibrium throughout the process. It is shown in the figure 8.47(b). From the equation (8.42)

The work done by the gas in an adiabatic expansion is given by,

$$W_{B \rightarrow C} = \int_{V_2}^{V_3} P dV = \frac{\mu R}{\gamma - 1} [T_H - T_L] =$$

Area under the curve BC (8.57)

This is shown in Figure 8.46 (b)

Step C $\rightarrow$ D: Quasi-static isothermal compression from (P_3, V_3, T_L) to (P_4, V_4, T_L) : It is shown in the figure 8.47(c)

The cylinder is placed on the sink and the gas is isothermally compressed until the pressure and volume become P_4 and V_4 respectively. This is represented by the curve CD in the PV diagram as shown in Figure 8.45. Let $W_{C \rightarrow D}$ be the work done on the gas. According to first law of thermodynamics

$$Q_L = W_{C \rightarrow D} = \int_{V_3}^{V_4} P dV = \mu R T_L \ln \left(\frac{V_4}{V_3} \right) = -\mu R T_L \ln \left(\frac{V_3}{V_4} \right)$$

= -Area under the curve CD (8.58)

This is shown in Figure 8.46 (c)

Here V_3 is greater than V_4 . So the work done is negative, implying work is done on the gas.

Step D $\rightarrow$ A: Quasi-static adiabatic compression from (P_4, V_4, T_L) to (P_1, V_1, T_H) : It is shown in the figure 8.47(d)

The cylinder is placed on the insulating stand again and the gas is compressed adiabatically till it attains the initial pressure P_1 , volume V_1 and temperature T_H . This is shown by the curve DA in the P-V diagram.

$$\therefore W_{D \rightarrow A} = \int_{V_4}^{V_1} P dV = -\frac{\mu R}{\gamma - 1} [T_H - T_L] =$$

-Area under the curve DA (8.59)

In the adiabatic compression also work is done on the gas so it is negative, as is shown in Figure 8.46 (d)

Let 'W' be the net work done by the working substance in one cycle

$$= W_{A \rightarrow B} + W_{B \rightarrow C} + W_{C \rightarrow D} + W_{D \rightarrow A}$$

since $W_{D \rightarrow A} = -W_{B \rightarrow C}$

$$= W_{A \rightarrow B} + W_{C \rightarrow D}$$

The net work done by the Carnot engine in one cycle $W = |W|_{A \rightarrow B} - |W|_{C \rightarrow D}$ (8.60)

Equation (8.60) shows that the net work done by the working substance in one cycle is equal to the area (enclosed by ABCD) of the P-V diagram (Figure 8.48)

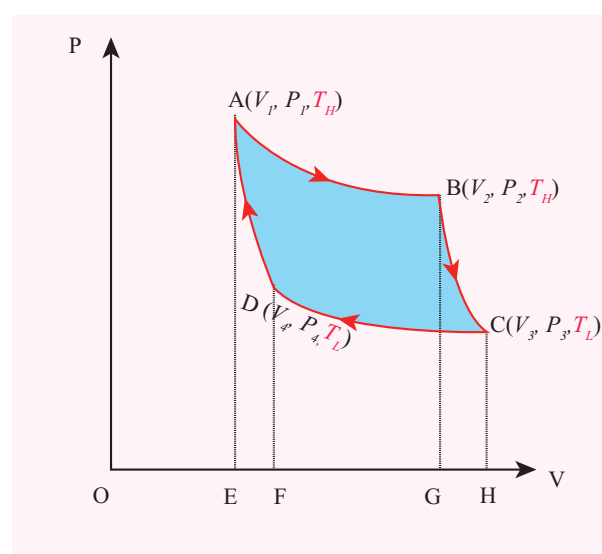


Figure 8.48 Net work done in Carnot cycle

It is very important to note that after one cycle the working substance returns to the initial temperature T_H . This implies that the change in internal energy of the working substance after one cycle is zero.

8.9.2 Efficiency of a Carnot engine

Efficiency is defined as the ratio of work done by the working substance in one cycle to the amount of heat extracted from the source.

$$\eta = \frac{\text{work done}}{\text{Heat extracted}} = \frac{W}{Q_H} \quad (8.61)$$

From the first law of thermodynamics,
 $W = Q_H - Q_L$

$$\therefore \eta = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H} \quad (8.62)$$

Applying isothermal conditions, we get,

$$\begin{aligned} Q_H &= \mu R T_H \ln \left(\frac{V_2}{V_1} \right) \\ Q_L &= \mu R T_L \ln \left(\frac{V_3}{V_4} \right) \end{aligned} \quad (8.63)$$

Here we omit the negative sign. Since we are interested in only the amount of heat (Q_L) ejected into the sink, we have

$$\therefore \frac{Q_L}{Q_H} = \frac{T_L \ln \left(\frac{V_3}{V_4} \right)}{T_H \ln \left(\frac{V_2}{V_1} \right)} \quad (8.64)$$

By applying adiabatic conditions, we get,

$$\begin{aligned} T_H V_2^{\gamma-1} &= T_L V_3^{\gamma-1} \\ T_H V_1^{\gamma-1} &= T_L V_4^{\gamma-1} \end{aligned}$$

By dividing the above two equations, we get

$$\left(\frac{V_2}{V_1} \right)^{\gamma-1} = \left(\frac{V_3}{V_4} \right)^{\gamma-1}$$

$$\text{Which implies that } \frac{V_2}{V_1} = \frac{V_3}{V_4} \quad (8.65)$$

Substituting equation (8.65) in (8.64), we get

$$\frac{Q_L}{Q_H} = \frac{T_L}{T_H} \quad (8.66)$$

$$\therefore \text{The efficiency } \eta = 1 - \frac{T_L}{T_H} \quad (8.67)$$

Note : T_L and T_H should be expressed in Kelvin scale.

Important results:

1. η is always less than 1 because T_L is less than T_H . This implies the efficiency cannot be 100%. It can be 1 or 100% only when $T_L = 0K$ (absolute zero of temperature) which is impossible to attain practically.
2. The efficiency of the Carnot's engine is independent of the working substance. It depends only on the temperatures of the source and the sink. The greater the difference between the two temperatures, higher the efficiency.
3. When $T_H = T_L$ the efficiency $\eta = 0$. No engine can work having source and sink at the same temperature.
4. The entire process is reversible in the Carnot engine cycle. So Carnot engine is itself a reversible engine and has maximum efficiency. But all practical heat engines like diesel engine, petrol engine and steam engine have cycles which are not perfectly reversible. So their efficiency is always less than the Carnot efficiency. This can be stated in the form of the Carnot theorem. It is stated as follows 'Between two constant temperatures reservoirs, only Carnot engine can have maximum efficiency. All real heat engines will have efficiency less than the Carnot engine'

EXAMPLE 8.25

- (a) A steam engine boiler is maintained at 250°C and water is converted into steam. This steam is used to do work and heat is ejected to the surrounding air at temperature 300K . Calculate the maximum efficiency it can have?

Solution

The steam engine is not a Carnot engine, because all the process involved in the steam engine are not perfectly reversible. But we can calculate the maximum possible efficiency of the steam engine by considering it as a Carnot engine.

$$\eta = 1 - \frac{T_L}{T_H} = 1 - \frac{300\text{K}}{523\text{K}} = 0.43$$

The steam engine can have maximum possible 43% of efficiency, implying this steam engine can convert 43% of input heat into useful work and remaining 57% is ejected as heat. In practice the efficiency is even less than 43%.

EXAMPLE 8.26

There are two Carnot engines A and B operating in two different temperature regions. For Engine A the temperatures of the two reservoirs are 150°C and 100°C . For engine B the temperatures of the reservoirs are 350°C and 300°C . Which engine has lesser efficiency?

Solution

The efficiency for engine A $= 1 - \frac{373}{423} = 0.11$. Engine A has 11% efficiency

The efficiency for engine B $= 1 - \frac{573}{623} = 0.08$
Engine B has only 8% efficiency.

Even though the differences between the temperature of hot and cold reservoirs in both engines is same, the efficiency is not same. The efficiency depends on the ratio of the two temperature and not on the difference in the temperature. The engine which operates in lower temperature has highest efficiency.



Diesel engines used in cars and petrol engines used in our motor bikes are all real heat engines.

The efficiency of diesel engines has maximum up to 44% and the efficiency of petrol engines are maximum up to 30%. Since these engines are not ideal heat engines (Carnot engine), their efficiency is limited by the second law of thermodynamics. Now a days typical bikes give a mileage of 50 km per Liter of petrol. This implies only 30% of 1 Liter of petrol is converted into mechanical work and the remaining 70% goes out as wasted heat and ejected into the surrounding atmosphere!

8.9.3 Entropy and second law of thermodynamics

We have seen in the equation (8.66) that the quantity $\frac{Q_H}{T_H}$ is equal to $\frac{Q_L}{T_L}$. The quantity $\frac{Q}{T}$ is called entropy. It is a very important thermodynamic property of a system. It is also a state variable. $\frac{Q_H}{T_H}$ is the entropy received by the Carnot engine from hot reservoir and $\frac{Q_L}{T_L}$ is entropy given out by the Carnot engine to the cold reservoir. For reversible engines (Carnot Engine) both entropies should be same, so that the change

in entropy of the Carnot engine in one cycle is zero. This is proved in equation (8.66). But for all practical engines like diesel and petrol engines which are not reversible engines, they satisfy the relation $\frac{Q_L}{T_L} > \frac{Q_H}{T_H}$. In fact we can reformulate the second law of thermodynamics as follows

“For all the processes that occur in nature (irreversible process), the entropy always increases. For reversible process entropy will not change”. Entropy determines the direction in which natural process should occur.

We now come back to the question: Why does heat always flows from a state of higher temperature to one of lower temperature and not in the opposite direction? Because entropy increases when heat flows from hot object to cold object. If heat were to flow from a cold to a hot object, entropy will decrease leading to violation of second law thermodynamics.

Entropy is also called ‘measure of disorder’. All natural process occur such that the disorder should always increases.

Consider a bottle with a gas inside. When the gas molecules are inside the bottle it has less disorder. Once it spreads into the entire room it leads to more disorder. In other words when the gas is inside the bottle the entropy is less and once the gas spreads into entire room, the entropy increases. From the second law of thermodynamics, entropy always increases. If the air molecules go back in to the bottle, the entropy should decrease, which is not allowed by the second law of thermodynamics. The same explanation applies to a drop of ink diffusing into water. Once the drop of ink spreads, its entropy is increased. The diffused ink can never become a drop again. So the natural

processes occur in such a way that entropy should increase for all irreversible process.

8.10

REFRIGERATOR

A refrigerator is a Carnot’s engine working in the reverse order. It is shown in the figure 8.49.

Working Principle:

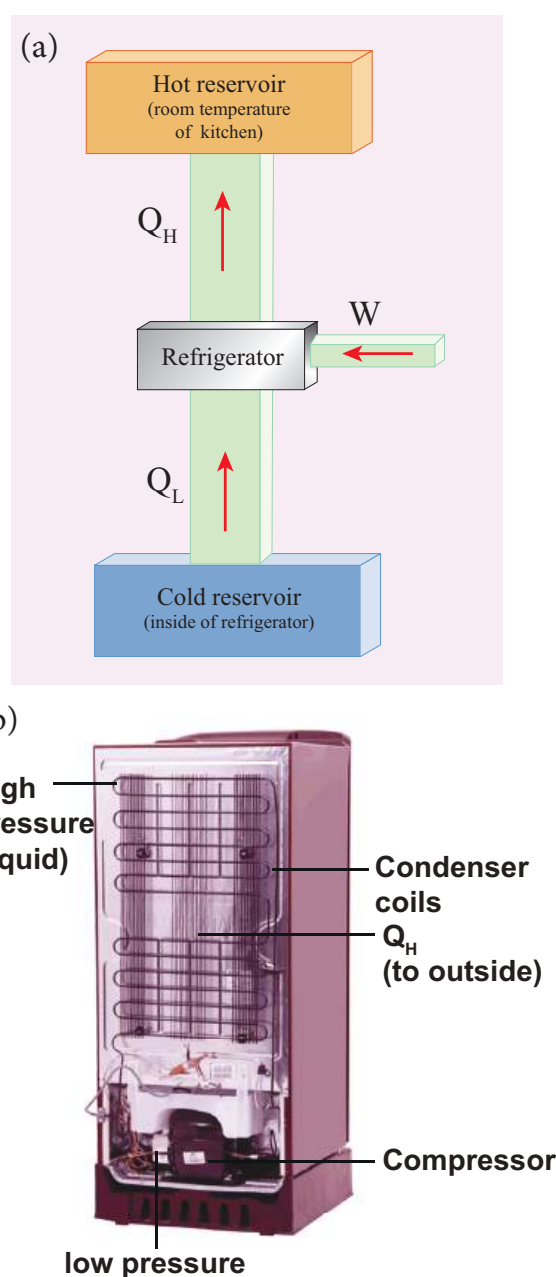


Fig 8.49 (a) Schematic diagram of a refrigerator (b) Actual refrigerator

The working substance (gas) absorbs a quantity of heat Q_L from the cold body (sink) at a lower temperature T_L . A certain amount of work W is done on the working substance by the compressor and a quantity of heat Q_H is rejected to the hot body (source) ie, the atmosphere at T_H . When you stand beneath of refrigerator, you can feel warmth air. From the first law of thermodynamics, we have

$$Q_L + W = Q_H \quad (8.68)$$

As a result the cold reservoir (refrigerator) further cools down and the surroundings (kitchen or atmosphere) gets hotter.

Coefficient of performance (COP) (β):

COP is a measure of the efficiency of a refrigerator. It is defined as the ratio of heat extracted from the cold body (sink) to the external work done by the compressor W .

$$\text{COP} = \beta = \frac{Q_L}{W} \quad (8.69)$$

From the equation (8.68)

$$\begin{aligned} \beta &= \frac{Q_L}{Q_H - Q_L} \\ \beta &= \frac{1}{\frac{Q_H}{Q_L} - 1} \end{aligned} \quad (8.70)$$

But we know that $\frac{Q_H}{Q_L} = \frac{T_H}{T_L}$

Substituting this equation into equation (8.70) we get

$$\beta = \frac{1}{\frac{T_H}{T_L} - 1} = \frac{T_L}{T_H - T_L}$$

Inferences:

1. The greater the COP, the better is the condition of the refrigerator. A typical refrigerator has COP around 5 to 6.
2. Lesser the difference in the temperatures of the cooling chamber and the atmosphere, higher is the COP of a refrigerator.
3. In the refrigerator the heat is taken from cold object to hot object by doing external work. Without external work heat cannot flow from cold object to hot object. It is not a violation of second law of thermodynamics, because the heat is ejected to surrounding air and total entropy of (refrigerator + surrounding) is always increased.

EXAMPLE 8.27

A refrigerator has COP of 3. How much work must be supplied to the refrigerator in order to remove 200 J of heat from its interior?

$$\text{COP} = \beta = \frac{Q_L}{W}$$

$$W = \frac{Q_L}{\text{COP}} = \frac{200}{3} = 66.67 \text{ J}$$





Green House Effect

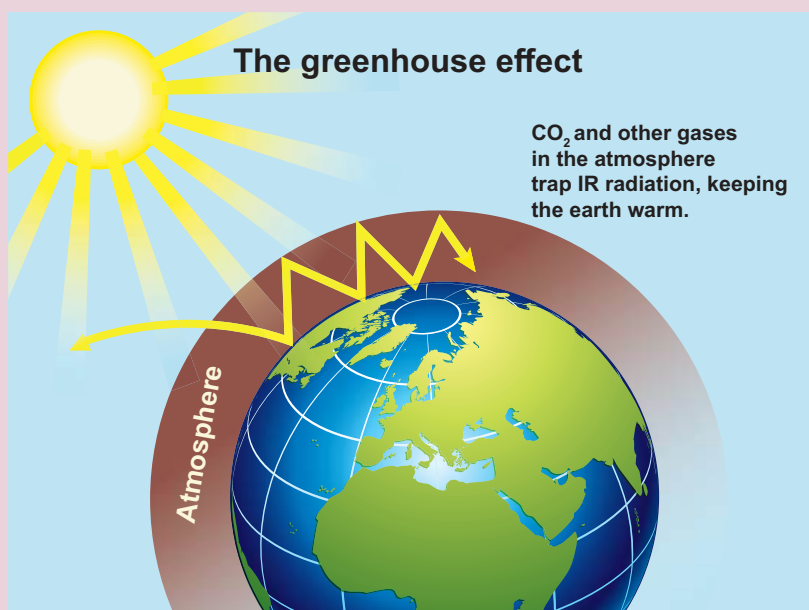
The presence of atmosphere in the earth plays very important role in human lives. Top of the atmosphere is at -19°C and bottom of the atmosphere is at $+14^{\circ}\text{C}$. The increase in 33°C from top to bottom is due to some gases present in the atmosphere. These gases are called Greenhouse gases and this effect is called Greenhouse effect.

The greenhouse gases are mainly CO_2 , water vapour, Ne, He, NO_2 , CH_4 , Xe, Kr, ozone and NH_3 . Except CO_2 and water vapor, all others are present only in very small amount in the atmosphere. The radiation from the Sun is mainly in the visible region of the spectrum. The earth absorbs these radiations and reradiate in the infrared region. Carbon dioxide and water Vapour are good absorbers of infrared radiation since they have more vibrational degree of freedom compared to nitrogen and oxygen (you will learn in unit 9) which keeps earth warmer as shown in Figure.

The amount of CO_2 present in the atmosphere is increased from 20% to 40% due to human activities since 1900s. The major emission of CO_2 comes from burning of fossil fuels. The increase in automobile usage worldwide causes this damage. Due to this increase in the CO_2 content in the atmosphere, the average temperature of the earth increases by 1°C . This effect is called global warming. It has serious influence and alarming effect on ice glaciers on Arctic and Antarctic regions. In addition, the CO_2 content is also increasing in ocean which is very dangerous to species in the oceans.

In addition to CO_2 , another very important greenhouse gas is Chloro fluoro carbon(CFC) which is used as coolant in refrigerators worldwide. In the human made greenhouse gases CO_2 is 55%, CFCs are 24%. Nitrogen oxide is 6% and methane is 15%. CFCs also has made huge damage to ozone layer.

Lot of efforts are taken internationally to reduce the emission of CO_2 and CFCs in various countries. Nowadays a lot of research is going to replace non fossil fuels to replace the fossil-fuels in automobile industry. The major emission of CO_2 comes from developed countries like USA and European countries. Various treaties are formed between countries to reduce the emission of CO_2 to considerable level before 2020s. But still global warming is not taken seriously in various countries.





In hot summer, we use earthen pots to drink cold water. The pot reduces the temperature of water inside it. Does the earthen pot act as a refrigerator? No. cyclic process is the basic necessity for heat engine or refrigerator. In earthen pot, the cooling process is not due to any cyclic process. The cooling occurs due to evaporation of water molecules which oozes out through pores of the pot. Once the water molecules evaporate, they never come back to the pot. Even though the heat flows from cold water to open atmosphere, it is not a violation of second law of thermodynamics. The water inside the pot is an open thermodynamic system, so the entropy of water + surrounding always increases.



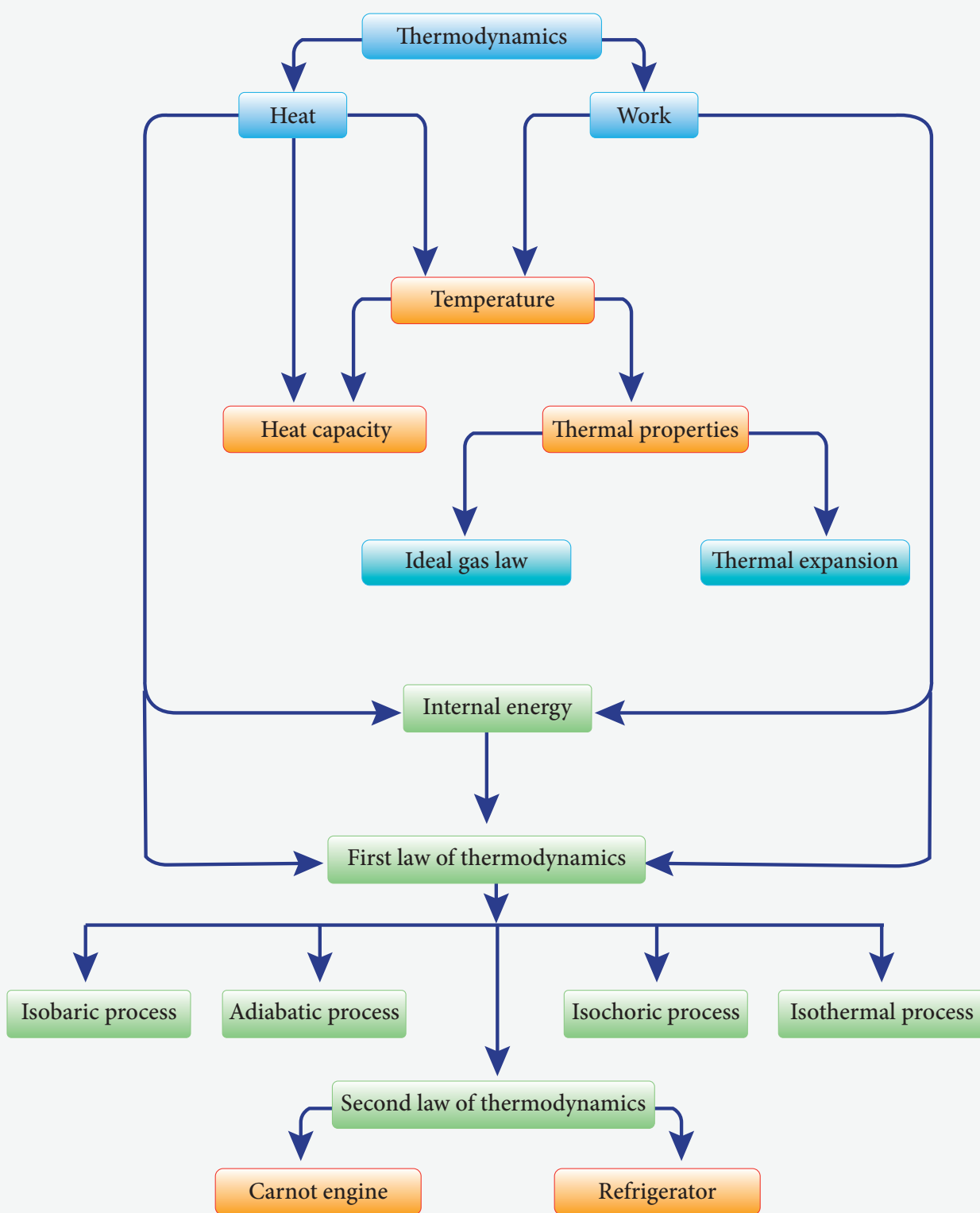
SUMMARY

- Heat is energy in transit which flows from hot object to cold object. However it is not a quantity.
- Work is a process to transfer energy from one object to another object.
- Temperature is a measure of hotness of the object. It determines the direction of the flow of heat.
- The ideal gas law is $PV = NkT$ or $PV = \mu RT$. The Ideal gas law holds for only at thermodynamic equilibrium. For non-equilibrium process, it is not valid.
- Heat capacity is the amount of heat energy required to increase the object's temperature by 1°C or 1K . It is denoted by S .
- Specific heat capacity is the amount of heat energy required to increase the 1 kg of object's temperature by 1°C or 1K . It is denoted by s .
- Molar specific heat capacity is the amount of heat energy requires to increase the 1 mole of substance's temperature by 1°C or 1K . It is denoted by C .
- Thermal expansion is a tendency of an object to change its shape, area, and volume due to change in temperature.
- Water has an anomalous behavior of expansion.
- Latent heat capacity is the amount of heat energy required to change the phase of the substance.
- Calorimetry is the measurement of the amount of heat energy released or absorbed by a thermodynamic system during the heating process.
- Heat transfers in three different modes: conduction, convection and radiation
- Stefan-Boltzmann law: $E = \sigma T^4$ and Wien's law: $\lambda_{\text{max}} T = b$
- Thermodynamic equilibrium: thermal, mechanical and chemical equilibrium
- Thermodynamic variables : Pressure, temperature, volume, internal energy and entropy



- Zeroth law of thermodynamics: If two objects are separately in thermal equilibrium with the third object, then these two are in thermal equilibrium. Temperature is a property which is the same for both the systems.
- Internal energy is the sum of kinetic and potential energies of molecules in the thermodynamic system.
- Joule converted mechanical energy to internal energy of the thermodynamic system
- First law of thermodynamics is a statement of conservation of energy. It included heat energy of the thermodynamic system.
- A quasi-static process is an infinitely slow process in which the system is always at equilibrium with the surrounding.
- When the volume of the system changes, the work done $W = \int P dV$
- The area under the PV diagram gives the work done by the system or work done on the system.
- Specific heat capacity at constant volume is always less than specific heat capacity at constant pressure.
- Isothermal process: $T = \text{constant}$, Isobaric process: $P = \text{constant}$, Isochoric process: $V = \text{constant}$, Adiabatic process $Q = 0$
- Work done in the isobaric process is most and work done in the adiabatic process is least
- In a cyclic process, change in internal energy is zero.
- The total work done in the cyclic process is given by a closed area in PV diagram
- A reversible process is an ideal process.
- All natural processes are irreversible.
- Heat engine takes input from the hot reservoir, performs work and rejects some amount of heat energy into sink.
- Carnot engine is a reversible engine. It has the highest efficiency. No real heat engine can have the efficiency of that of a Carnot engine.
- A refrigerator is reverse of a Carnot engine. COP (coefficient of performance) of the practical refrigerator is always less than ideal refrigerator.

CONCEPT MAP

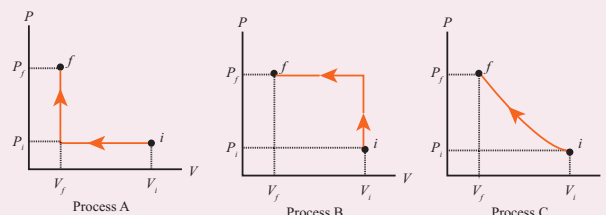




EVALUATION

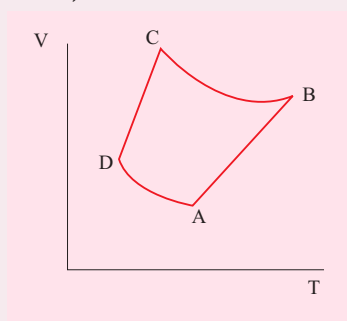
I. Multiple choice questions:

- In hot summer after a bath, the body's
 - internal energy decreases
 - internal energy increases
 - heat decreases
 - no change in internal energy and heat
- The graph between volume and temperature in Charles' law is
 - an ellipse
 - a circle
 - a straight line
 - a parabola
- When a cycle tyre suddenly bursts, the air inside the tyre expands. This process is
 - isothermal
 - adiabatic
 - isobaric
 - isochoric
- An ideal gas passes from one equilibrium state (P_1, V_1, T_1, N) to another equilibrium state $(2P_1, 3V_1, T_2, N)$. Then
 - $T_1 = T_2$
 - $T_1 = \frac{T_2}{6}$
 - $T_1 = 6T_2$
 - $T_1 = 3T_2$
- When a uniform rod is heated, which of the following quantity of the rod will increase
 - mass
 - weight
 - center of mass
 - moment of inertia
- When food is cooked in a vessel by keeping the lid closed, after some time the steam pushes the lid outward. By considering the steam as a thermodynamic system, then in the cooking process
 - $Q > 0, W > 0$,
 - $Q < 0, W > 0$,
 - $Q > 0, W < 0$,
 - $Q < 0, W < 0$,
- When you exercise in the morning, by considering your body as thermodynamic system, which of the following is true?
 - $\Delta U > 0, W > 0$,
 - $\Delta U < 0, W > 0$,
 - $\Delta U < 0, W < 0$,
 - $\Delta U = 0, W > 0$,
- A hot cup of coffee is kept on the table. After some time it attains a thermal equilibrium with the surroundings. By considering the air molecules in the room as a thermodynamic system, which of the following is true
 - $\Delta U > 0, Q = 0$
 - $\Delta U > 0, W < 0$
 - $\Delta U > 0, Q > 0$
 - $\Delta U = 0, Q > 0$
- An ideal gas is taken from (P_i, V_i) to (P_f, V_f) in three different ways. Identify the process in which the work done on the gas the most.

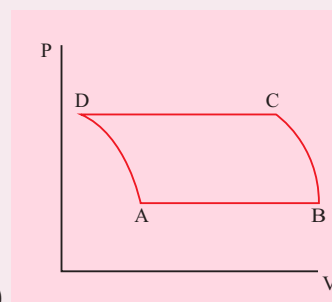
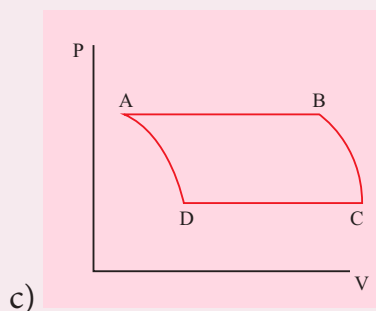
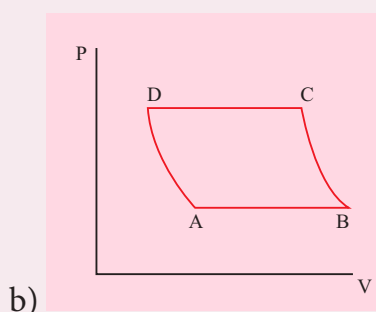
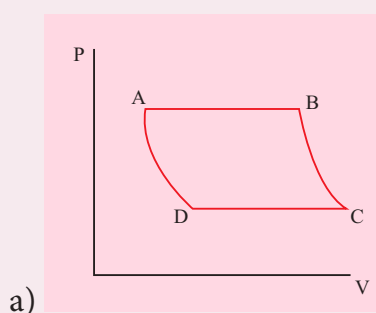


- a) Process A
- b) Process B
- c) Process C
- d) Equal work is done in Process A,B & C

10. The V-T diagram of an ideal gas which goes through a reversible cycle $A \rightarrow B \rightarrow C \rightarrow D$ is shown below. (Processes $D \rightarrow A$ and $B \rightarrow C$ are adiabatic)



The corresponding PV diagram for the process is (all figures are schematic)



11. A distant star emits radiation with maximum intensity at 350 nm. The temperature of the star is
- a) 8280 K
 - b) 5000 K
 - c) 7260 K
 - d) 9044 K
12. Identify the state variables given here?
- a) Q, T, W
 - b) P, T, U
 - c) Q, W
 - d) P, T, Q
13. In an isochoric process, we have
- a) $W = 0$
 - b) $Q = 0$
 - c) $\Delta U = 0$
 - d) $\Delta T = 0$
14. The efficiency of a heat engine working between the freezing point and boiling point of water is
- (NEET 2018)
- a) 6.25%
 - b) 20%
 - c) 26.8%
 - d) 12.5%
15. An ideal refrigerator has a freezer at temperature -12°C . The coefficient of performance of the engine is 5. The temperature of the air (to which the heat ejected) is
- a) 50°C
 - b) 45.2°C
 - c) 40.2°C
 - d) 37.5°C

Answers:

- | | | | |
|-------|-------|-------|-------|
| 1) a | 2) c | 3) b | 4) b |
| 5) d | 6) a | 7) b | 8) c |
| 9) b | 10) a | 11) a | 12) b |
| 13) a | 14) c | 15) c | |

II. Short answer questions:

1. 'An object contains more heat' - is it a right statement? If not why?
2. Obtain an ideal gas law from Boyle's and Charles' law.
3. Define one mole.
4. Define specific heat capacity and give its unit.
5. Define molar specific heat capacity.
6. What is a thermal expansion?
7. Give the expressions for linear, area and volume thermal expansions.
8. Define latent heat capacity. Give its unit.
9. State Stefan-Boltzmann law.
10. What is Wien's law?
11. Define thermal conductivity. Give its unit.
12. What is a black body?
13. What is a thermodynamic system? Give examples.
14. What are the different types of thermodynamic systems?
15. What is meant by 'thermal equilibrium'?
16. What is mean by state variable? Give example.
17. What are intensive and extensive variables? Give examples.
18. What is an equation of state? Give an example.
19. State Zeroth law of thermodynamics.
20. Define the internal energy of the system.
21. Are internal energy and heat energy the same? Explain.
22. Define one calorie.
23. Did joule converted mechanical energy to heat energy? Explain.
24. State the first law of thermodynamics.
25. Can we measure the temperature of the object by touching it?
26. Give the sign convention for Q and W.
27. Define the quasi-static process.
28. Give the expression for work done by the gas.
29. What is PV diagram?
30. Explain why the specific heat capacity at constant pressure is greater than the specific heat capacity at constant volume.
31. Give the equation of state for an isothermal process.
32. Give an expression for work done in an isothermal process.
33. Express the change in internal energy in terms of molar specific heat capacity.
34. Apply first law for (a) an isothermal (b) adiabatic (c) isobaric processes.
35. Give the equation of state for an adiabatic process.
36. Give an equation state for an isochoric process.
37. If the piston of a container is pushed fast inward. Will the ideal gas equation be valid in the intermediate stage? If not, why?
38. Draw the PV diagram for
 - a. Isothermal process
 - b. Adiabatic process
 - c. Isobaric process
 - d. Isochoric process
39. What is a cyclic process?



40. What is meant by a reversible and irreversible processes?
41. State Clausius form of the second law of thermodynamics
42. State Kelvin-Planck statement of second law of thermodynamics.
43. Define heat engine.
44. What are processes involves in a Carnot engine?
45. Can the given heat energy be completely converted to work in a cyclic process? If not, when can the heat can completely converted to work?
46. State the second law of thermodynamics in terms of entropy.
47. Why does heat flow from a hot object to a cold object?
48. Define the coefficient of performance.
9. Discuss the
 - a. thermal equilibrium
 - b. mechanical equilibrium
 - c. Chemical equilibrium
 - d. thermodynamic equilibrium.
10. Explain Joule's Experiment of the mechanical equivalent of heat.
11. Derive the expression for the work done in a volume change in a thermodynamic system.
12. Derive Mayer's relation for an ideal gas.
13. Explain in detail the isothermal process.
14. Derive the work done in an isothermal process
15. Explain in detail an adiabatic process.
16. Derive the work done in an adiabatic process
17. Explain the isobaric process and derive the work done in this process
18. Explain in detail the isochoric process.
19. What are the limitations of the first law of thermodynamics?
20. Explain the heat engine and obtain its efficiency.
21. Explain in detail Carnot heat engine.
22. Derive the expression for Carnot engine efficiency.
23. Explain the second law of thermodynamics in terms of entropy.
24. Explain in detail the working of a refrigerator.

III. Long answer Questions:

1. Explain the meaning of heat and work with suitable examples.
2. Discuss the ideal gas laws.
3. Explain in detail the thermal expansion.
4. Describe the anomalous expansion of water. How is it helpful in our lives?
5. Explain Calorimetry and derive an expression for final temperature when two thermodynamic systems are mixed.
6. Discuss various modes of heat transfer.
7. Explain in detail Newton's law of cooling.
8. Explain Wien's law and why our eyes are sensitive only to visible rays?

IV. Exercises

1. Calculate the number of moles of air is in the inflated balloon at room temperature as shown in the figure.



The radius of the balloon is 10 cm, and pressure inside the balloon is 180 kPa.

Answer: $\mu \cong 0.3 \text{ mol}$

2. In the planet Mars, the average temperature is around -53°C and atmospheric pressure is 0.9 kPa. Calculate the number of moles of the molecules in unit volume in the planet Mars? Is this greater than that in earth?

Answer: $\mu_{\text{Mars}} = 0.49 \text{ mol}$

$\mu_{\text{Earth}} \cong 40 \text{ mol}$

3. An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1 . The other chamber has volume V_2 and contains ideal gas at pressure P_2 and temperature T_2 . If the partition is removed without doing any work on the gases, calculate the final equilibrium temperature of the container.

Answer: $T = \frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$

4. The temperature of a uniform rod of length L having a coefficient of linear expansion α_L is changed by ΔT . Calculate the new moment of inertia of the uniform rod about axis passing through its center and perpendicular to an axis of the rod.

Answer: $I' = I (1 + \alpha_L \Delta T)^2$

5. Draw the TP diagram (P-x axis, T-y axis), VT(T-x axis, V-y axis) diagram for
 - a. Isochoric process
 - b. Isothermal process
 - c. isobaric process
6. A man starts bicycling in the morning at a temperature around 25°C , he checked the pressure of tire which is equal to be 500 kPa. Afternoon he found that the absolute pressure in the tyre is increased to 520 kPa. By assuming the expansion of tyre is negligible, what is the temperature of tyre at afternoon?

Answer: $T = 36.9^\circ\text{C}$

7. Normal human body of the temperature is 98.6°F . During high fever if the temperature increases to 104°F , what is the change in peak wavelength that emitted by our body? (Assume human body is a black body)

Answer: (a) $\lambda_{\text{max}} \approx 9348 \text{ nm}$ at 98.6°F

(b) $\lambda_{\text{max}} \approx 9258 \text{ nm}$ at 104°F

8. In an adiabatic expansion of the air, the volume is increased by 4%, what is percentage change in pressure?

(For air $\gamma = 1.4$)

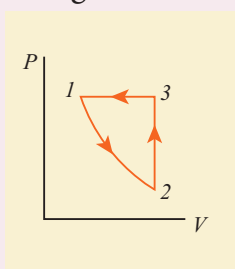
Answer: 5.6%

9. In a petrol engine, (internal combustion engine) air at atmospheric pressure and temperature of 20°C is adiabatically compressed in the cylinder by the piston to $1/8$ of its original volume. Calculate the temperature of the compressed air.

(For air $\gamma = 1.4$)

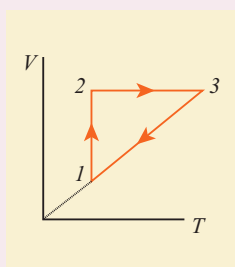
Answer: $T \cong 400^{\circ}\text{C}$

10. Consider the following cyclic process consist of isotherm, isochoric and isobar which is given in the figure.



Draw the same cyclic process qualitatively in the V-T diagram where T is taken along x direction and V is taken along y-direction. Analyze the nature of heat exchange in each process.

Answer:

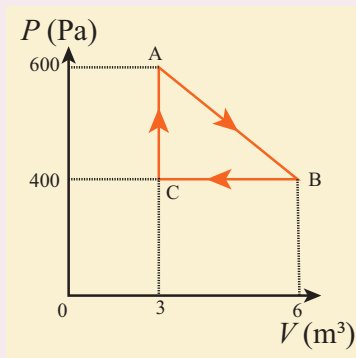


Process 1 to 2 = increase in volume. So heat must be added.

Process 2 to 3 = Volume remains constant. Increase in temperature. The given heat is used to increase the internal energy.

Process 3 to 1 : Pressure remains constant. Volume and Temperature are reduced. Heat flows out of the system. It is an isobaric compression where the work is done on the system.

11. An ideal gas is taken in a cyclic process as shown in the figure. Calculate
- work done by the gas.
 - work done on the gas
 - Net work done in the process



Answer: (a) $W = +1.5\text{kJ}$

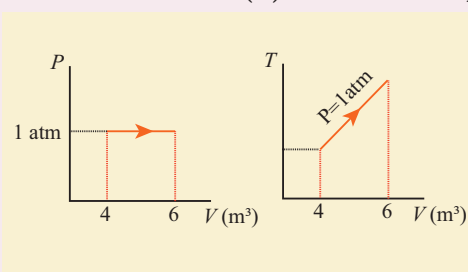
(b) $W = -1.2\text{kJ}$

(c) $W = +300\text{J}$.

12. For a given ideal gas $6 \times 10^5\text{J}$ heat energy is supplied and the volume of gas is increased from 4 m^3 to 6 m^3 at atmospheric pressure. Calculate (a) the work done by the gas (b) change in internal energy of the gas (c) graph this process in PV and TV diagram.

Answer: (a) $W = +202.6\text{ kJ}$

(b) $dU = 397.4\text{ kJ}$



(c)

13. Suppose a person wants to increase the efficiency of the reversible heat engine that is operating between 100°C and 300°C . He had two ways to increase the efficiency. (a) By decreasing the cold reservoir temperature from 100°C to 50°C and keeping the hot reservoir temperature constant (b) by increasing

the temperature of the hot reservoir from 300°C to 350°C by keeping the cold reservoir temperature constant. Which is the suitable method?

Answer: Initial efficiency = 34.9%
Efficiency in method (a) = 43.6%
Efficiency in method (b) = 40.1%
Method (a) is more efficient.

14. A Carnot engine whose efficiency is 45% takes heat from a source maintained at a temperature of 327°C. To have an

engine of efficiency 60% what must be the intake temperature for the same exhaust (sink) temperature?

Answer: 552°C

15. An ideal refrigerator keeps its content at 0°C while the room temperature is 27°C. Calculate its coefficient of performance.

Answer: $\beta = 10.11$

BOOKS FOR REFERENCE

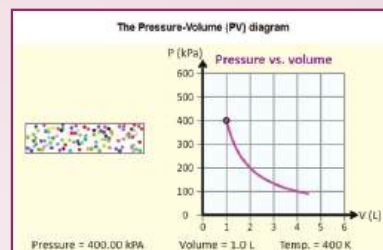
1. Serway and Jewett, Physics for scientist and Engineers with modern physics, Brook/Cole publishers, Eighth edition
2. Paul Tipler and Gene Mosca, Physics for scientist and engineers with modern physics, Sixth edition, W.H. Freeman and Company
3. James Walker, Physics, Addison-Wesley publishers, 4th Edition
4. Douglas C Giancoli, Physics for scientist & Engineers with modern physics, Pearson Prentice Hall, 4th edition.
5. H.C. Verma, Concepts of physics volume 1 and Volume 2, Bharati Bhawan Publishers
6. Tarasov and Tarasova, Question and problems in school physics, Mir Publishers



ICT CORNER

Heat and Thermodynamics

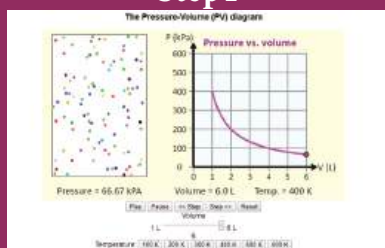
Through this activity you will be able to learn the PV diagrams for various thermodynamic process.



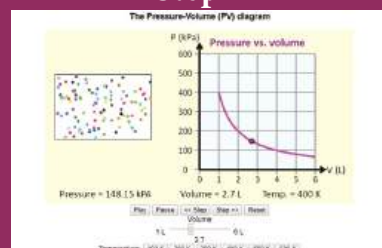
STEPS:

- Use the URL or scan the QR code to open interactive simulation on 'Pressure and Volume Diagram'.
- At selected temperature, change the "Volume" given below the graph and click play button.
- Now select a different temperature, change the 'Volume' again and find the change in the pressure both in the left image and graph.
- Repeat the same with different values and try drawing the graph accordingly. This also helps to understand isothermal process.

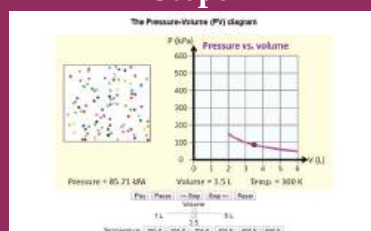
Step1



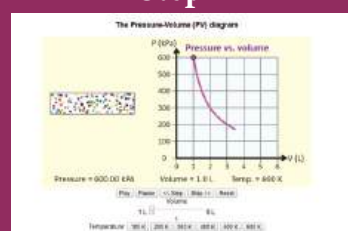
Step2



Step3



Step4



URL:

http://physics.bu.edu/~duffy/HTML5/PV_diagram.html

- * Pictures are indicative only.
- * If browser requires, allow **Flash Player** or **Java Script** to load the page.

